CHAPTER ONE

PROPERTIES, MEASUREMENT AND UNITS

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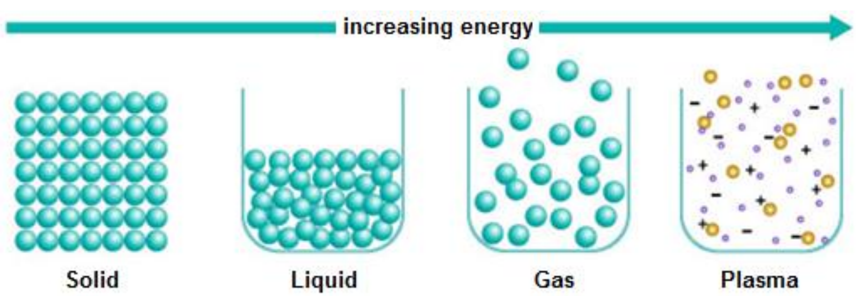
CONTENTS:

- States of Matter
- Physical and Chemical Properties
- Physical and Chemical Changes
- Extensive and Intensive Properties
- Elements, Compound and Mixtures

What is Matter?

- Anything that has mass and takes up space

What are the 4 Physical States of Matter?



- Solids:
  - Definite shape and volume
  - Close packing of molecules
  - Strong molecular forces
- Liquids:
  - Definite volume but not definite shape (assumes shape of container)
  - Slightly weaker molecular forces than solids (this is what allows liquids to flow)
- Gas:
  - No definite shape or volume
  - Very weak molecular forces (which allows gasses to be greatly compressed)
- Plasma:
  - No definite shape or volume

What are Properties?

- Characteristics and behaviors we use to describe matter
- There are 2 types of properties:
  - Physical
  - Chemical

What are Physical Properties:

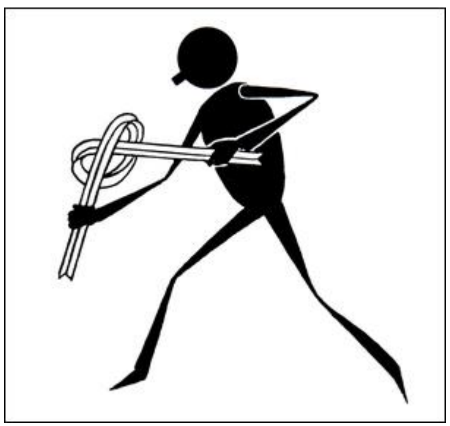
- Can be observed or measured without changing the identity of the matter
- Properties you notice when using one of your five senses or measurement tools
  - Feel – mass, volume, texture, malleability
  - Sight – color, luster, mass, volume, density
  - Hear
  - Smell
  - Taste

Physical Properties

- **Density:** The amount of matter in a given volume.
- D=m/v (mass/volum e)
- Ice cubes float in water because they are less dense than liquid water

Physical Properties

- **Ductility** :The ability to be pulled into a thin strand.
- Wire, Paper clip, Copper wire.



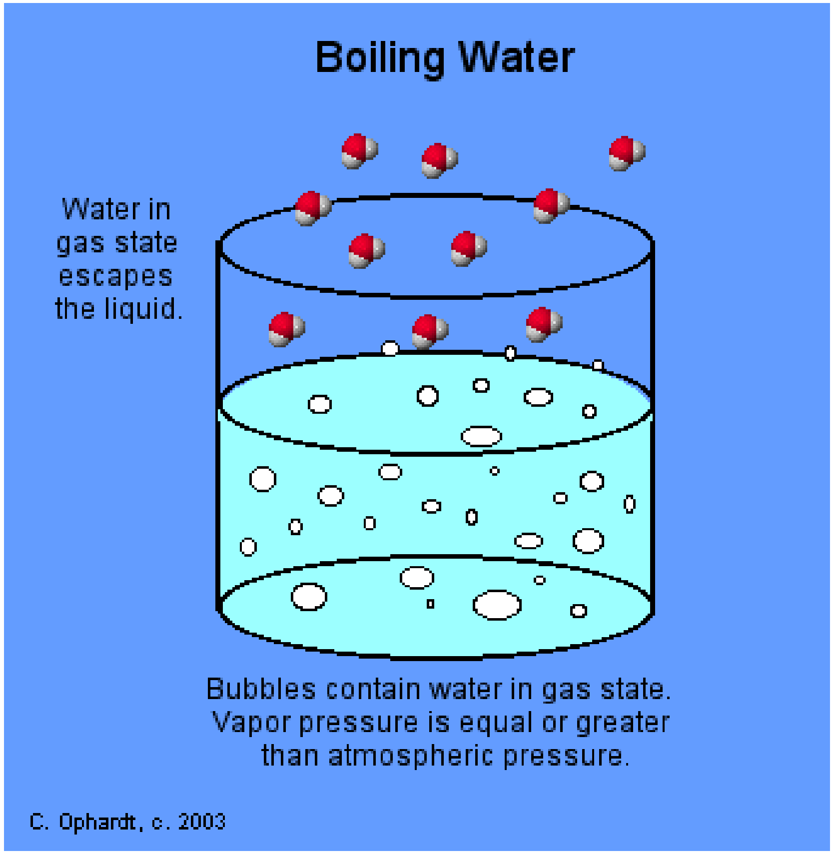
Physical Properties

- **Malleability:** The ability to be pressed or pounded into a thin sheet
    - Tin foil

Physical Properties

- **Boiling Point**: The temperature at which a substance changes from a liquid to a gas.

- Water to steam.



Physical Properties

- **Melting point:** The temperature at which a substance changes from a solid to a liquid
- Ice cube melts to a puddle of water

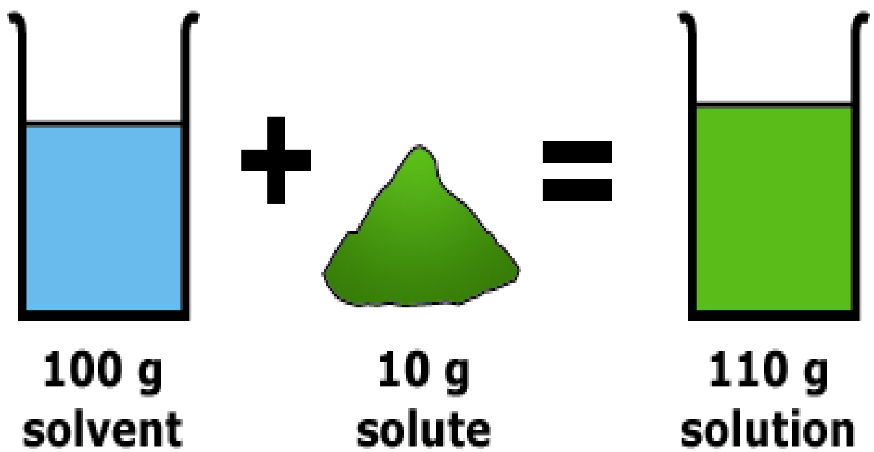
Physical Properties

- **Electrical conductivity:** How well a substance allows electricity to flow through it
- Water conducts electricity so never swim during a lightning storm

Physical Properties

- **Solubility:** *The ability to dissolve in another substance.*

- *Adding sugar to coffee.*



What are Chemical Properties?

- A substances ability to combine with or change into a new substance

- Common chemical properties:

  - Reactivity – how likely a substance it to react with another substance to create something new
    - Reactive to oxygen
    - Reactive to air
    - Reactive to water

  - Flamability – how likely a substance is to catch fire

If it CHanges, it' s CHemical

What is a Chemical Change?

- Changes to a substance that results in an entirely new substance forming.
- Chemical changes are often much more obvious than physical changes.
- Examples:
  - Color change
  - Light, heat or energy released (burning)

Examples of Chemical Changes

Soured milk smells bad because bacteria have formed new substances in the milk.

Effervescent tablets bubble when the citric acid and baking soda in them react in water.

The hot gas formed when hydrogen and oxygen join to make water helps blast the space shuttle into orbit.

The Statue of Liberty is made of shiny, orange-brown copper. But the metal's interaction with carbon dioxide and water has formed a new substance, copper carbonate, and made this landmark lady green over time.

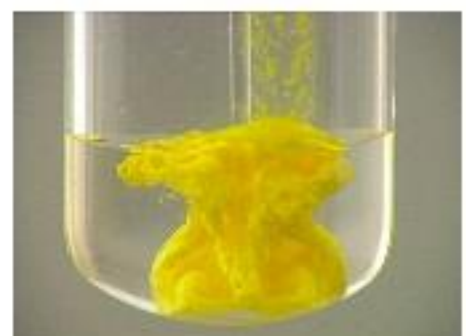
Chemical Changes

Rusting - free Iron oxidizes (rusts), forming a new substance (Iron oxide)

Fireworks - light, heat (energy)

Color change - light, heat (energy)

Precipitate - yellow solid forms when two chemicals are added



5 Signs of a Chemical Change

- The only sure way to know there has been a chemical change is the observance of a new substance formed.

1. Odor Production - this is an odor far different from what it should smell like

Ex: Rotting eggs, decomposing flesh

2. Change in Temperature

a. **Exothermic** - When energy is released do during the chemical change

- ex: wood burning

b. **Endothermic** - Energy is absorbed causing a decrease in temperature of the reactant material.

- ex: cold pack in first aid kit

3. Change in Color:

- Ex: fruit changing color when it ripens, leaves changing color in the Autumn, dying your hair.

4. Formation of Bubbles:

- This can indicate the presence of a gas.
- Bubbles produced when boiling water

5. Formation of a Precipitate:

- When two liquids are combined and a solid is produced

Generally, Physical & Chemical Properties

Properties can be broken down into two types - physical and chemical properties. What's the difference?

Physical: properties of a pure substance, we can see without changing it into a new substance. Examples include:

1. Physical state: solid, liquid, gas
2. Color
3. Shape
4. Mass
5. Texture
6. Melting (0^0C) & boiling point (100^0C)
7. Density
8. Solubility in water – the ability to dissolve in water
9. Odor
10. Luster – shine
11. Malleability – ability to be hammered into thin sheets

Chemical: properties of a pure substance that describe its ability to combine with or change into a new substance. Examples:

1. Flammability
2. Reactivity to water
3. Reactivity to air
4. Reactivity to oxygen

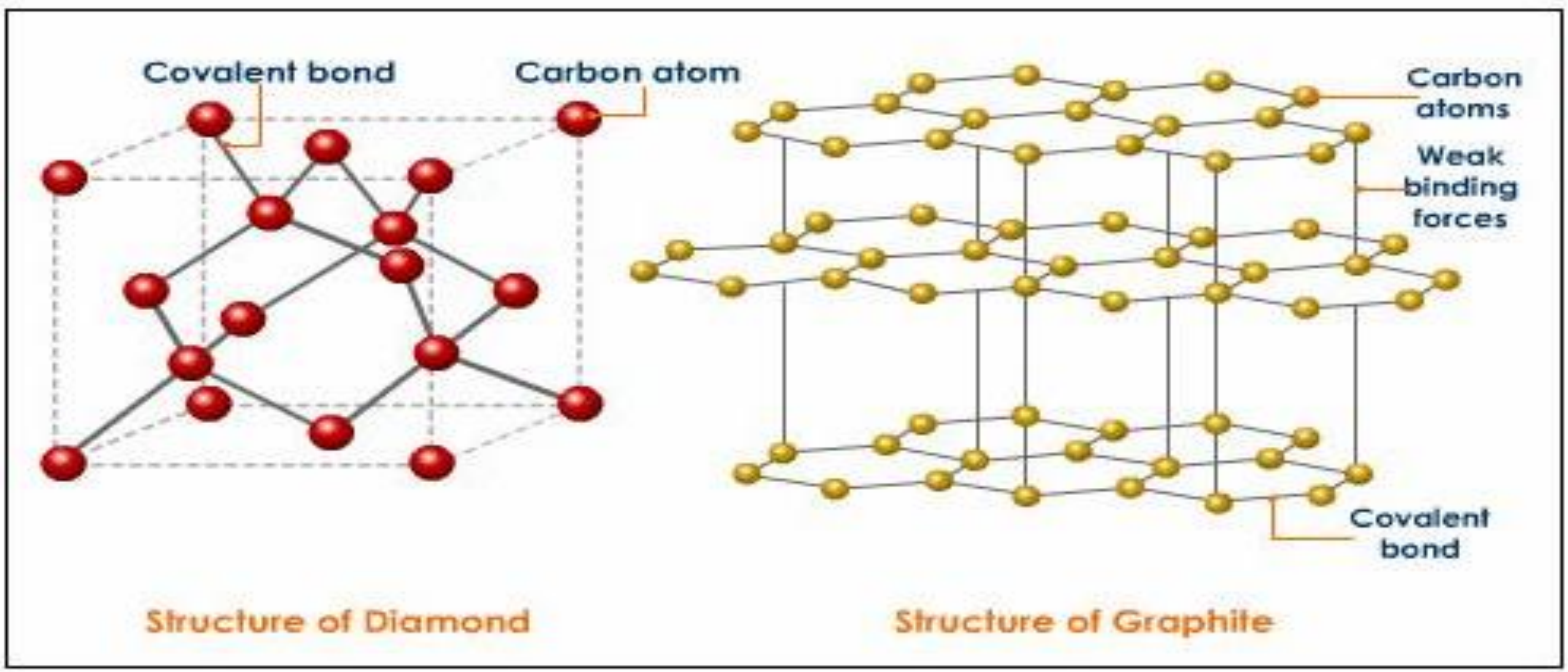
*****Notice that Chemical Properties are not as easy to notice as Physical Properties.**

Intensive and Extensive Properties of Matter

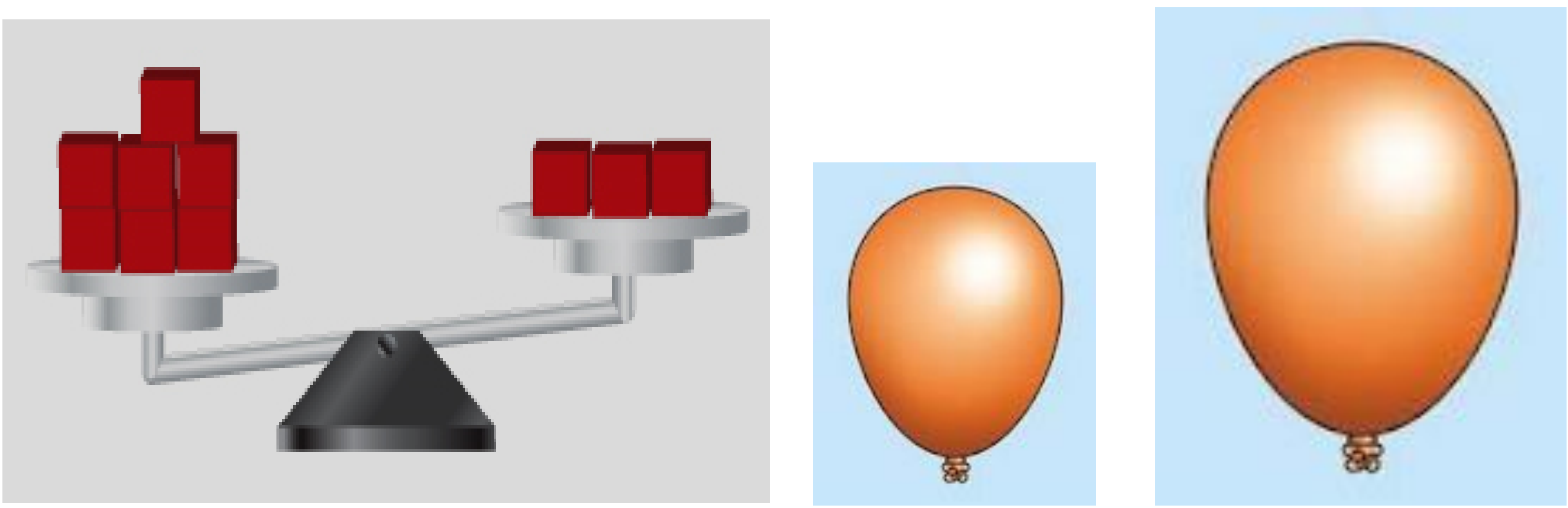
- *Intensive property-* One that DOES NOT depend on the amount of the substance present.

- *Extensive property-* One that DOES depend on the amount of the substance present.

- *Intensive properties* are determined by the chemical composition of the particles and their structure (arrangement).
- *Intensive ~ Internal*



- *Extensive properties*: Depend only on the number of particles, not on their composition or internal arrangement.
- *Extensive ~ External*



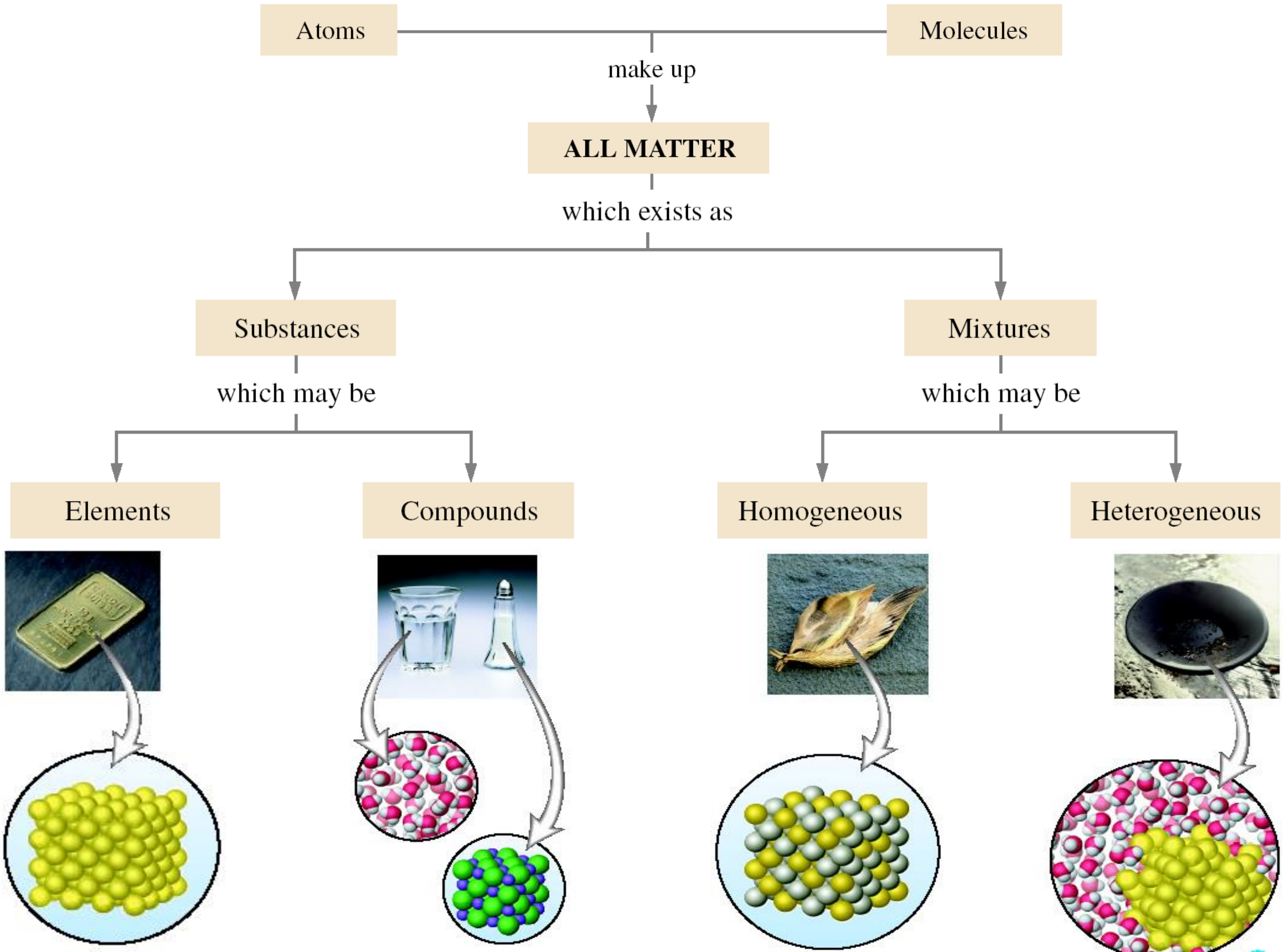
Classifying Matter:

Pure Substances and Mixtures

Classifying Matter by Composition

- Another way to classify matter is to examine its **composition**.
- Composition includes:
  - Types of particles
  - Arrangement of the particles
  - Attractions and attachments between the particles

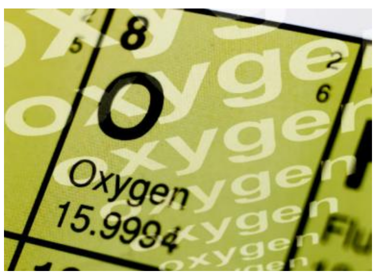
Classifying Matter



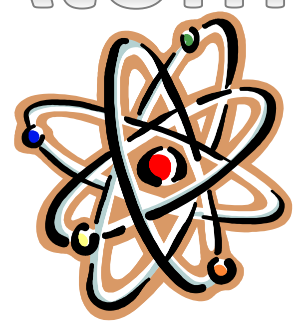
Pure Substances

- *A sample of matter that has definite chemical and physical properties.*

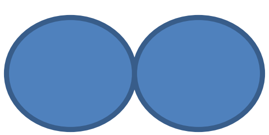
Elements



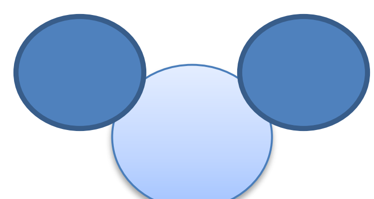
Atoms



Molecules

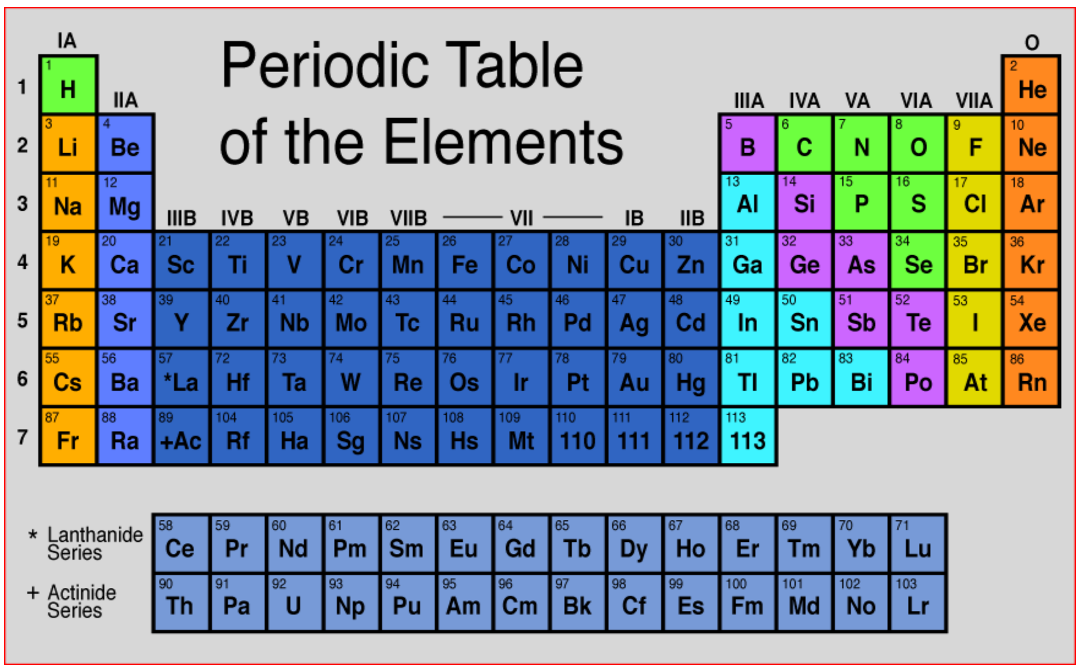


Compounds



Elements

- Pure substance that cannot be separated into simpler substance by physical or chemical means.



Compounds

H_2O

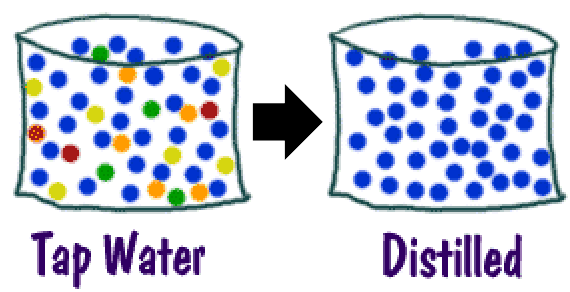
- Pure substance composed of two or more different elements joined by chemical bonds.
  - Made of elements in a specific ratio that is always the same
  - Has a chemical formula
  - Can only be separated by chemical means, not physically

$NaCl$

CO_2

Mixtures

- A combination of two or more pure substances that are not chemically combined.
- Substances held together by physical forces, not chemical.
- No chemical change takes place
- Each item retains its properties in the mixture



MIXTURES

- A mixture is something that CAN be broken down into simpler materials using physical methods.

- There are 3 possible types of mixtures:
  - Element + Another Element
  - Compound + Another Compound
  - Element + Compound

Types of Mixtures

- *There are <u>THREE</u> types of mixtures:*

  a. **Solutions**

  b. **Suspensions**

  c. **Colloids**

Tyndall Effect

- The scattering of light by particles in a mixture.

a. Solutions

- A **solution** is a mixture that appears to be a single substance, but it is actually composed of **2 or more substances** that are distributed evenly amongst each other.

- **Examples:** Salt dissolved in water, sugar dissolved in water, apple juice, tea, copper (II) sulfate solution in water, alloys, carbon dioxide is dissolved in water, rubbing alcohol (ethyl alcohol and water), Air (nitrogen and oxygen).

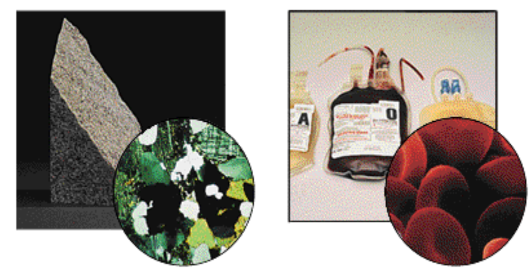
SOLUTIONS ARE:

- HOMOGENEOUS
- No Tyndall effect
- Very small particles
- Particles don't settle
- Well-mixed (uniform) – single phase
- Transparent
- Cannot be separated by filter
- Do not separate on standing

b. Suspensions

- *A __suspension__ is a mixture in which particles of a material are dispersed throughout a liquid or gas, but are **large enough that they settle out.***

- **SUSPENSIONS ARE:**
  - **HETEROGENEOUS**
  - **Tyndall effect**
  - Large particles
  - Does NOT have uniform properties



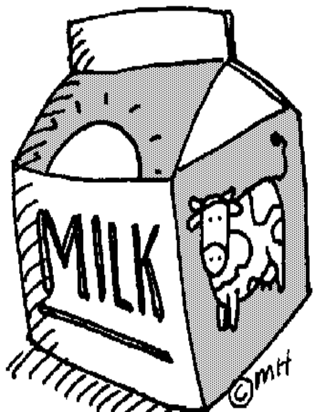
A Granite, a heterogeneous mixture

B Human blood, a heterogeneous mixture

- *__Examples__: muddy water, Sand water, oil and water, sulfur and iron, granite, blood, fresh-squeezed lemonade...*

c. Colloids

- *A __colloid__ is a mixture in which the particles are dispersed throughout but are **not** heavy enough to settle out.*
- *Colloids have properties of both solutions & suspensions.*
- *COLLOIDS ARE:*
  - **HOMOGENEOUS**
  - **Tyndall effect**
  - Medium-sized particles
  - Particles don't settle
  - Non transparent, non uniform, cloudy (milky) but stable system.



- *__Examples__: milk, clouds, smoke…*

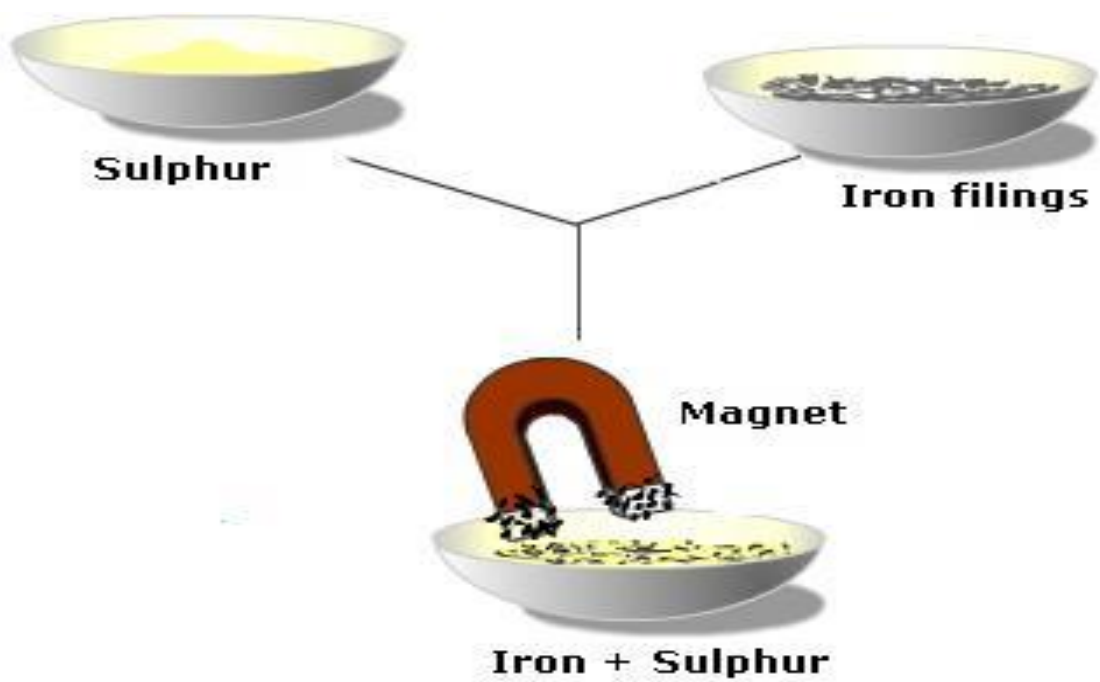
Methods of Mixture Separation

1. **Mechanical Separation**: (often by hand) takes advantage of physical properties such as color and shape.

Example: Recycling Plastic, Paper, Metal

2. Magnetic Separation: takes advantage of the physical property of **magnetism.**

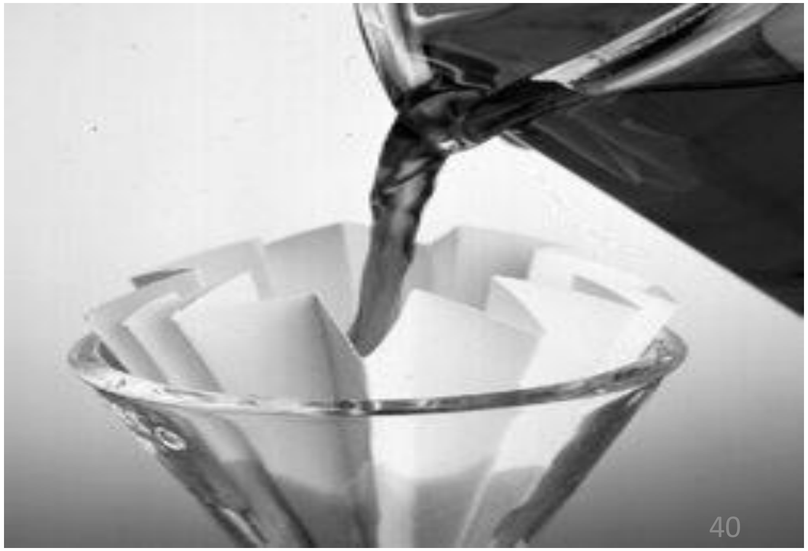
Example: Separating Metals in a Scrap Yard



3. **Filtration:** takes advantage of the physical property of the **state of matter.**

- A screen lets the liquid particles through, but traps the solid particles.
- A filter can also be used to **separate solid particles of different sizes.**

Example: Filtering Coffee, Spaghetti, a window screen, an air filter, a sand sieve

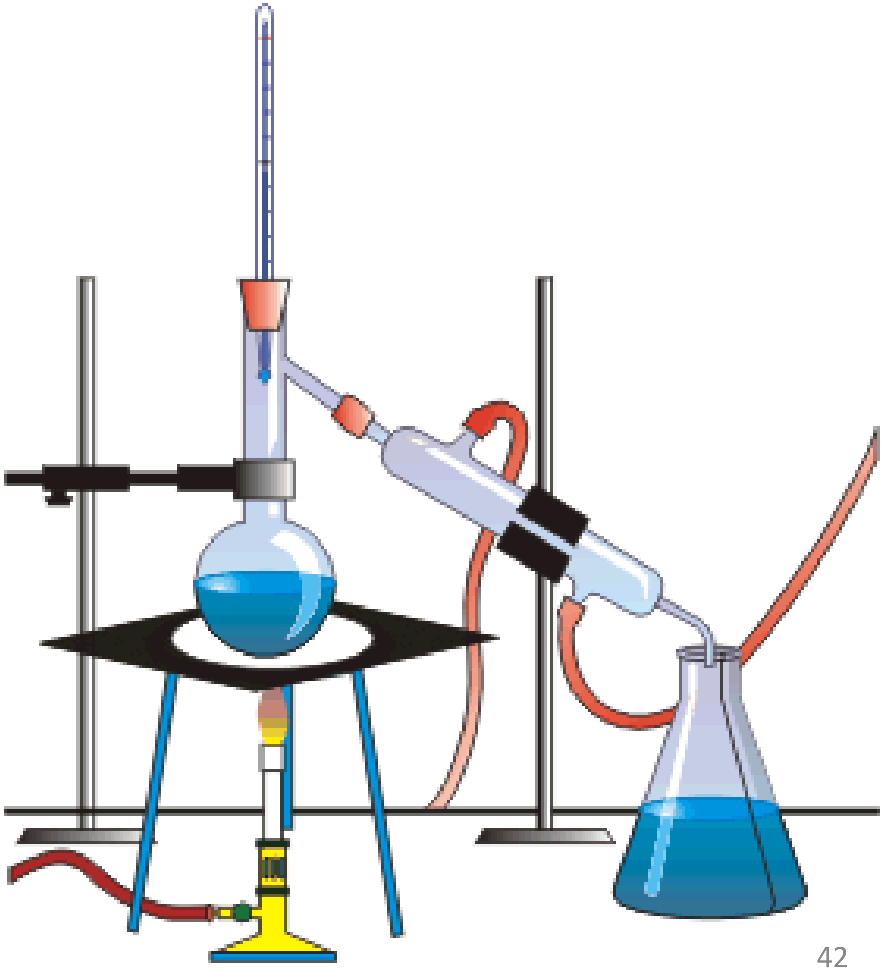


4. Decanting: *To pour off a liquid, leaving another liquid or solid behind. Takes advantage of* ***differences in density.***

Example: To decant a liquid from a precipitate or water from rice.

5. Distillation: The separation of a mixture of liquids based on the physical property of **boiling point.**

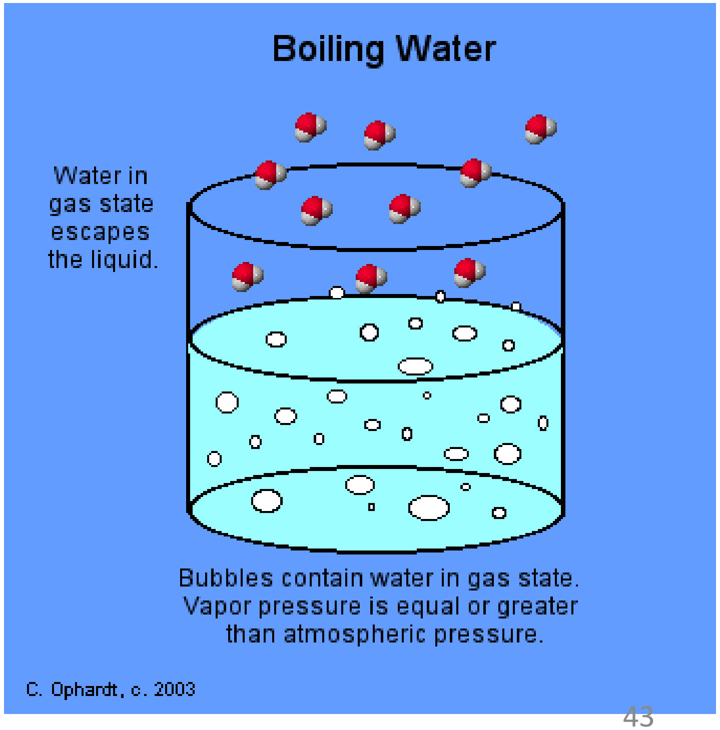
Example: the distillation of alcohol or oil



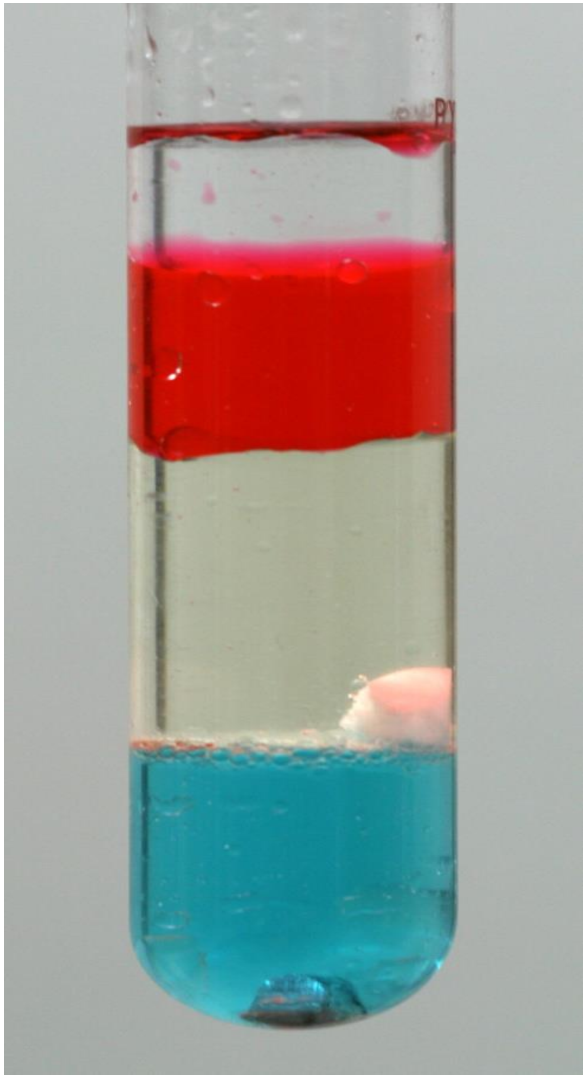
6. Evaporation: Vaporizing a liquid and leaving the dissolved solid(s) behind.

- Used to separate salt solutions.

Example: Obtaining sea salt from sea water



7. **Density Separation**: More dense components sink to the bottom and less dense components float.



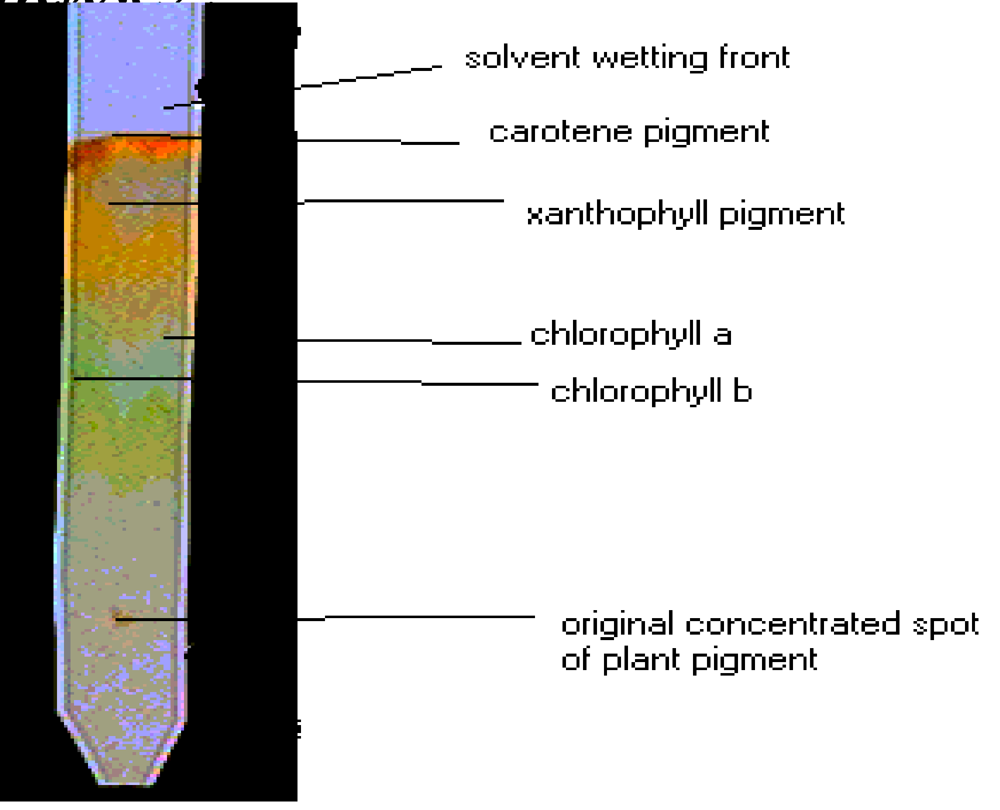
8. Centrifuge: Circular motion helps denser components sink to the bottom faster.

Examples: The separation of blood or DNA from blood

9. **Paper chromatography:** uses the property of molecular attraction to separate a mixture.

- Different molecules have different attractions for the paper (the stationary phase) vs. the solvent (the mobile phase).

Example: the separation of plant pigments and dyes



10. Fractional Crystallization: Dissolved substances crystallize out of a solution once their **solubility limit** is reached as the solution cools.

__Examples__: Growing Rock Candy or the Crystallization of a Magma Chamber

Mixtures vs. Compounds

	Mixture	Compound
Composition	Variable composition – you can vary the amount of each substance in a mixture.	Definite composition – you cannot vary the amount of each element in a compound.
Joined or not	The different substances are not chemically joined together.	The different elements are chemically joined together.
Properties	Each substance in the mixture keeps its own properties.	The compound has properties different from the elements it contains.
Separation	Each substance is easily separated from the mixture.	It can only be separated into its elements using chemical reactions.
Examples	Air, sea water, most rocks.	Water, carbon dioxide, magnesium oxide, sodium chloride.

Review:

- An **element** contains just one type of atom.
- A **compound** contains two or more different atoms joined together.
- A **mixture** contains two or more different substances that are only physically joined together, not chemically.
  - A mixture can contain both elements and compounds.

Measurements

and

Units

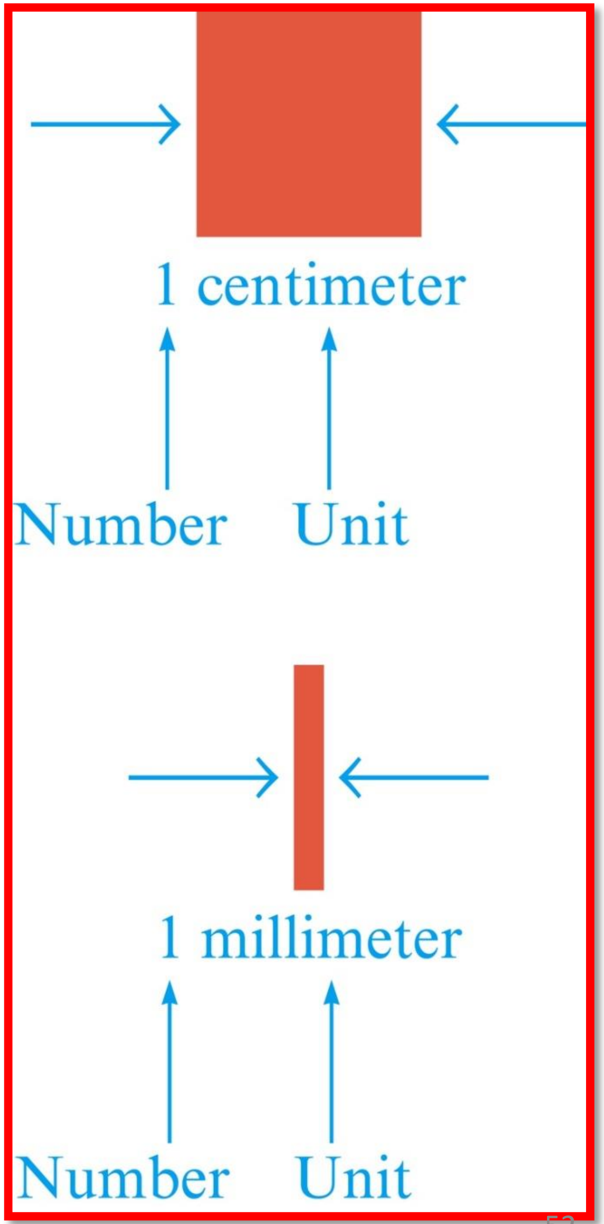
Scientific Notation and Units

Objectives

1. To show how very large or very small numbers can be expressed in scientific notation
2. To learn the English, metric, and SI systems of measurement
3. To use the metric system to measure length, volume and mass

Measurement

- A quantitative observation
- Consists of 2 parts
  - Number
  - Unit – tells the scale being used



A. Scientific Notation

- Very large or very small numbers can be expressed using scientific notation.
  - The number is written as a number between 1 and 10 multiplied by 10 raised to a power.
  - The power of 10 depends on
    - The number of places the decimal point is moved.
    - The direction the decimal point is moved.
- Left ⇒ Positive exponent
- Right ⇒ Negative exponent

Representing Large Numbers

$$93{,}000{,}000 = 9.3 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$$
$$= \underbrace{9.3}_{\text{Number between 1 and 10}} \times \underbrace{10^7}_{\text{Appropriate power of 10 } (10{,}000{,}000 = 10^7)}$$

Representing Small Numbers

- To obtain a number between 1 and 10 we must move the decimal point.

$$0.\underbrace{0}_{1}\underbrace{0}_{2}\underbrace{0}_{3}\underbrace{1}_{4}67$$

i.e: $0.000167 = 1.67 \times 10^{-4}$

B. Units

- Units provide a scale on which to represent the results of a measurement.

Table 5.1 Common Fundamental SI Units

Physical Quality	Name of Unit	Abbreviation
mass	kilogram	kg
length	meter	m
time	second	s
temperature	kelvin	K

- There are THREE commonly used unit systems.
  - English
  - Metric (uses prefixes to change the size of the unit)
  - SI (uses prefixes to change the size of the unit)

Table 5.2 Commonly Used Prefixes in the Metric System

Prefix	Symbol	Meaning	Power of 10 for Scientific Notation
mega	M	1,000,000	10^{6}
kilo	k	1000	10^{3}
deci	d	0.1	10^{-1}
centi	c	0.01	10^{-2}
milli	m	0.001	10^{-3}
micro	μ	0.000001	10^{-6}
nano	n	0.000000001	10^{-9}

C. Measurements of Length, Volume and Mass

- **Length**
  - Fundamental unit is meter
  - 1 meter = 39.37 inches

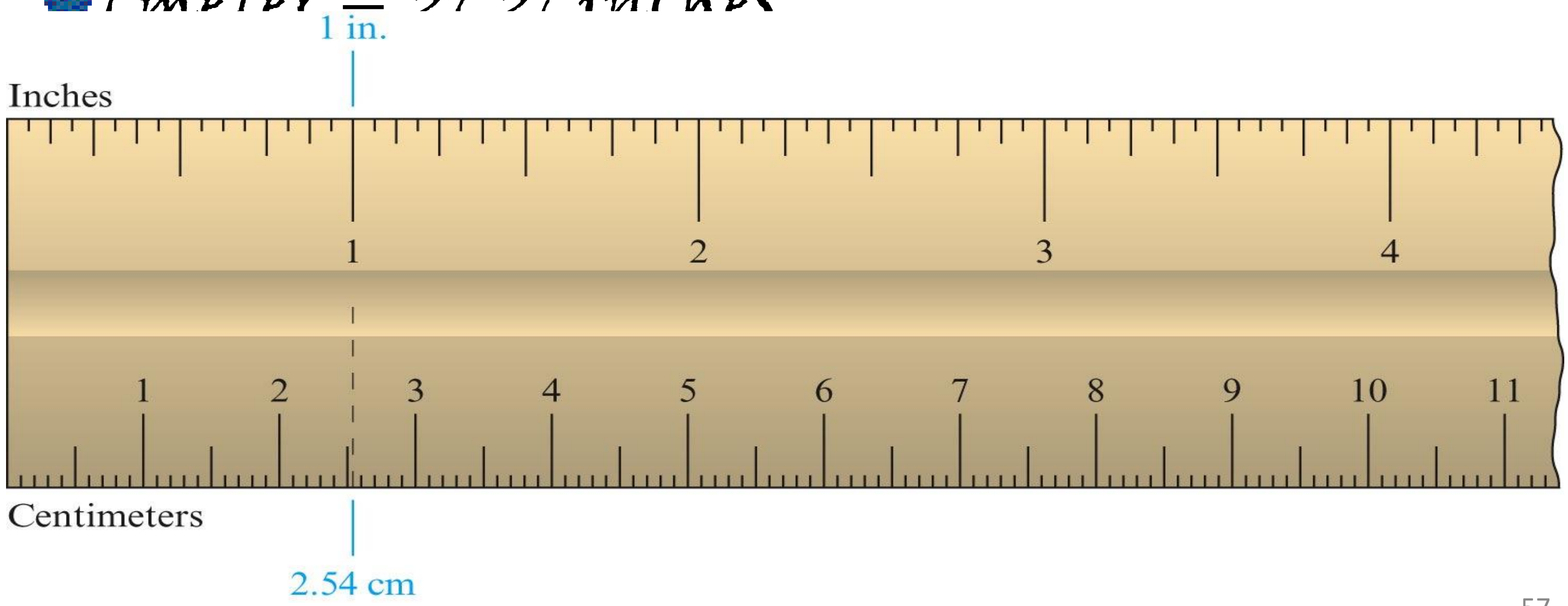


Table 5.3 Metric System for Measuring Length

Unit	Symbol	Meter Equivalent
kilometer	km	1000 m or 10^3 m
meter	m	1 m or 1 m
decimeter	dm	0.1 m or 10^{-1} m
centimeter	cm	0.01 m or 10^{-2} m
millimeter	mm	0.001 m or 10^{-3} m
micrometer	µm	0.000001 m or 10^{-6} m
nanometer	nm	0.000000001 m or 10^{-9} m

Volume

- Amount of 3-D space occupied by a substance
- Fundamental unit is meter^3 (m^3)

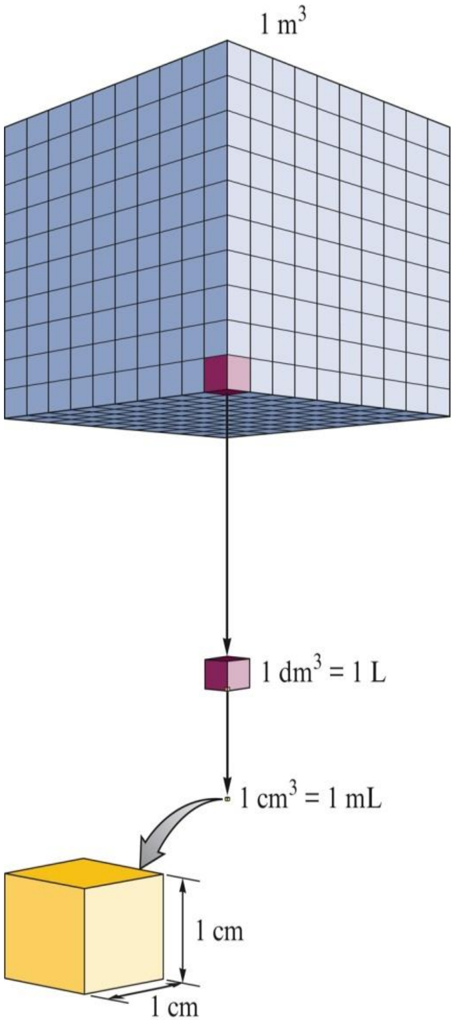


Table 5.4 Relationship Between the Liter and Milliliter

Unit	Symbol	Equivalence
liter	L	1 L = 1000 mL
milliliter	mL	$\frac{1}{1000}$ L = 10^{-3} L = 1 mL

Mass

- Quantity of matter in an object
- Fundamental unit is kilogram

Table 5.5 Most Commonly Used Metric Units for Mass

Unit	Symbol	Gram Equivalent
kilogram	kg	$1000 = 10^3 g = 1$ kg
gram	g	1 g
milligram	mg	0.001 g $= 10^{-3} g = 1$ mg

C. Measurements of Length, Volume and Mass *commonly used*

Table 5.6 Examples of Commonly Used Units

length	A dime is 1 mm thick. A quarter is 2.5 cm in diameter. The average height of an adult man is 1.8 m.
volume	A 12-oz can of soda has a volume of about 360 mL. A half gallon of milk is equal to about 2 L of milk.
mass	A nickel has a mass of about 5 g. A 120-lb woman has a mass of about 55 kg.

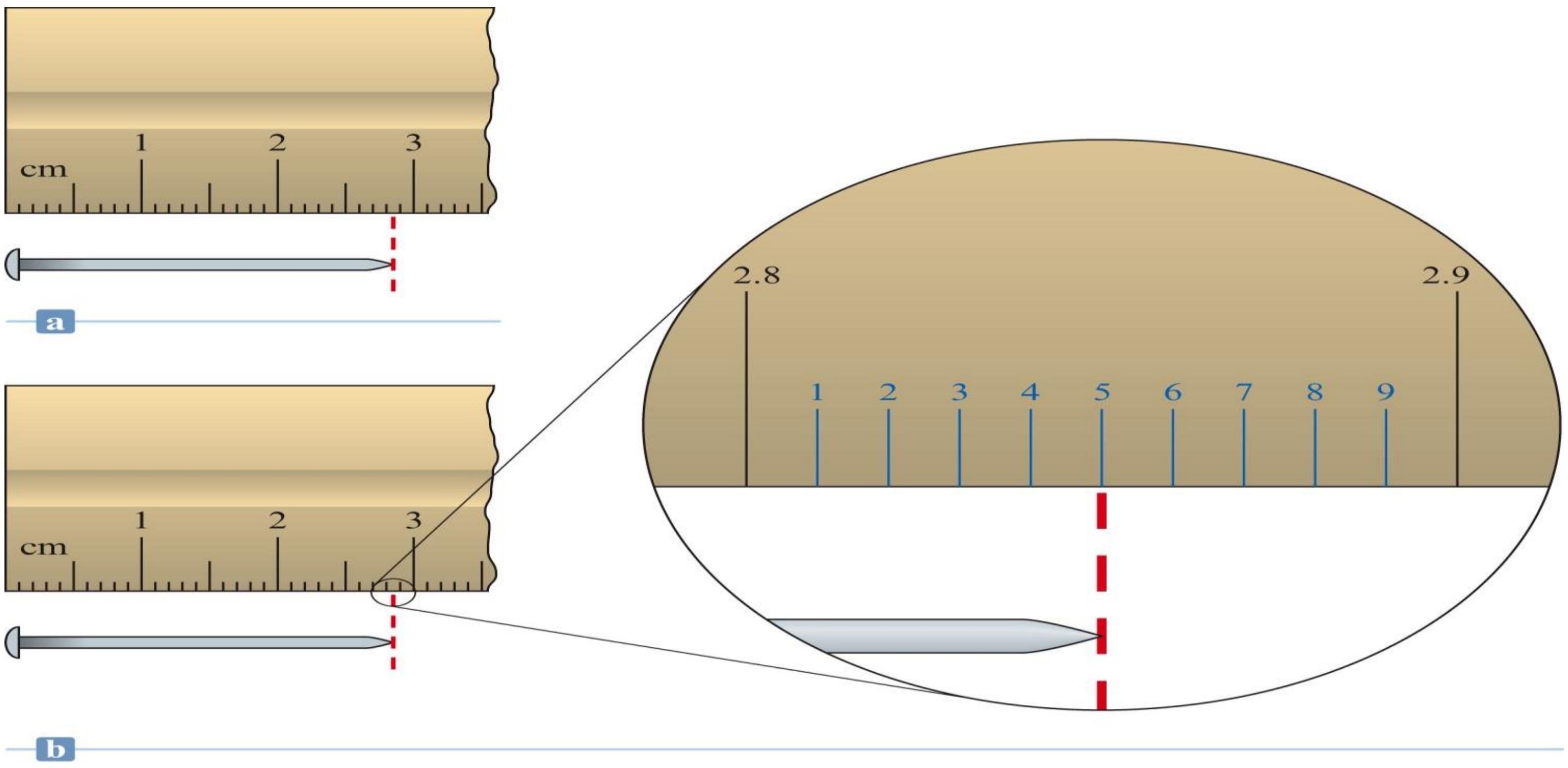
Uncertainty in Measurement and Significant Figures

Objectives

1. To learn how uncertainty in a measurement arises
2. To learn to indicate a measurement's uncertainty by using significant figures
3. To learn to determine the number of significant figures in a calculated result

A. Uncertainty in Measurement

- A measurement always has some degree of uncertainty.



- Different people estimate differently.

Person	Result of Measurement
1	2.85 cm
2	2.84 cm
3	2.86 cm
4	2.85 cm
5	2.86 cm

- Record all **certain** numbers and one **estimated** number.

B. Significant Figures

- *Numbers recorded in a measurement.*
- *All the certain numbers plus first estimated number.*

Rules for Counting Significant Figures

1. Nonzero integers always count as significant figures

Example:-1457   4 significant figures

2. Zeros

a. Leading zeros – never count

<u>Example</u>:- 0.25 2 significant figures

b. Captive zeros – always count

<u>Example</u>:- 1.08 3 significant figures

c. Trailing zeros – count only if the number is written with a decimal point

<u>Example</u>:- 100 1 significant figure

100. 3 significant figures

120.0 4 significant figures

3. *Exact numbers* – unlimited significant figures

- Not obtained by measurement
- Determined by counting
  - <u>Example</u>:- 3 apples
- Determined by definition
  - <u>Example</u>:- 1 in. = 2.54 cm, exactly

Rules for Rounding Off

1. If the digit to be removed
   a. is less than 5, the preceding digit stays the same. For example, 1.33 rounds to 1.3.
   b. is equal to or greater than 5, the preceding digit is increased by 1. For example, 1.36 rounds to 1.4, and 3.15 rounds to 3.2.
2. In a series of calculations, carry the extra digits through to the final result and *then* round off.* This means that you should carry all of the digits that show on your calculator until you arrive at the final number (the answer) and then round off, using the procedures in rule 1.

Rules for Multiplication and Division

- The number of significant figures in the result is the same as in the measurement with the *smallest number of significant figures*.

$$4.56 \times 1.4 = 6.384 \xrightarrow{\text{Round off}} 6.4$$

4.56: Three significant figures

1.4: Limiting (two significant figures)

6.4: Two significant figures

Four significant figures

$$\frac{8.315}{298} = 0.0279027 \xrightarrow{\text{Round off}} 2.79 \times 10^{-2}$$

298: Limiting (three significant figures)

0.0279027: Result shown on calculator

2.79×10^{-2}: Three significant figures

Rules for Addition and Subtraction

- The number of significant figures in the result is the same as in the measurement with the **smallest** number of decimal places.

```
 12.11
 18.0       Limiting term (has one decimal place)
  1.013
 ------
 31.123     Round off  →  31.1
                            ↑
                      One decimal place

   0.6875
  −0.1      Limiting term (one decimal place)
  -------
   0.5875   Round off  →  0.6
```

Problem solving and Unit Conversions

Objectives

1. To learn how dimensional analysis can be used to solve problems
2. To learn the three temperature scales
3. To learn to convert from one temperature scale to another
4. To practice using problem solving techniques
5. To define density and its units

A. Tools for Problem Solving

- Be systematic
- Ask yourself these questions
  - Where do we want to go?
  - What do we know?
  - How do we get there?
  - Does it make sense?

Converting Units of Measurement

- We can convert from one system of units to another by a method called dimensional analysis using conversion factors.

$$\text{Unit}_1 \times \text{conversion factor} = \text{Unit}_2$$

- Conversion factors are ratios of the two parts of the equivalence statement that relate the two units.

- Conversion factors are built from an ***equivalence statement*** which shows the relationship between the units in different systems.

- lb means Pound
- qt means Quarts

Table 5.7 English-Metric and English-English Equivalents

Length	1 m = 1.094 yd
	2.54 cm = 1 in.
	1 mi = 5280. ft
	1 mi = 1760. yd
Mass	1 kg = 2.205 lb
	453.6 g = 1 lb
Volume	1 L = 1.06 qt
	1 ft^3 = 28.32 L

Converting Units of Measure

2.85 cm = ? in.

2.85 cm × conversion factor = ? in.

Equivalence statement 2.54 cm = 1 in.

Possible conversion factors

$$2.85 \text{ cm} \times \frac{1 \text{ in.}}{2.54 \text{ cm}} = \frac{2.85 \text{ in.}}{2.54} = 1.12 \text{ in.}$$

Does this answer make sense?

Tools for Converting from One Unit to Another

Step 1: Find an equivalence statement that relates the two units.

Step 2: Choose the conversion factor by looking at the direction of the required change (cancel the unwanted units).

Step 3: Multiply the original quantity by the conversion factor.

Step 4: Make sure you have the correct number of significant figures.

Example 1: A golfer putted a golf ball 6.8 ft across a green. How many inches does this represent?

- To convert from one unit to another, use the equivalence statement that relates the two units.

$$1 \text{ ft} = 12 \text{ in}$$

- The two conversion factors are:

$$\frac{1 \text{ ft}}{12 \text{ in}} \text{ and } \frac{12 \text{ in}}{1 \text{ ft}}$$

- Derive the appropriate conversion factor by looking at the direction of the required change (to cancel the unwanted units).

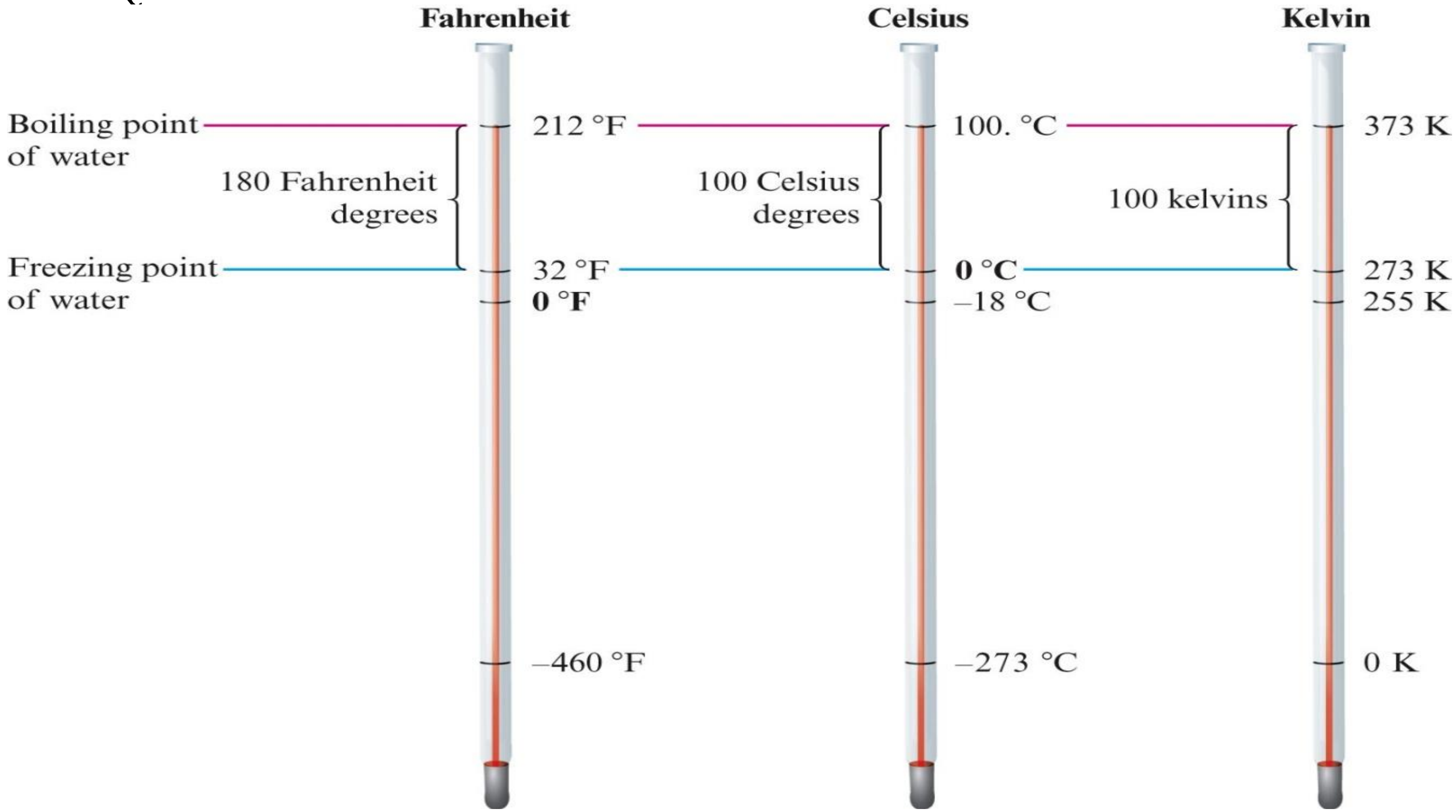
$$6.8\ \cancel{\text{ft}}\ \text{´}\ \frac{12\ \text{in}}{1\ \cancel{\text{ft}}} = \text{------in}$$

- Multiply the quantity to be converted by the conversion factor to give the quantity with the desired units.

$$6.8\ \cancel{\text{ft}}\ \text{´}\ \frac{12\ \text{in}}{1\ \cancel{\text{ft}}} = 82\ \text{in}$$

B. Temperature Conversions

- There are three commonly used temperature scales, Fahrenheit, Celsius and Kelvin.



Converting between the Kelvin and Celsius Scales

- Note that:
  - The temperature unit is the same size.
  - The zero points are different.

- To convert from Celsius to Kelvin, we need to adjust for the difference in zero points.

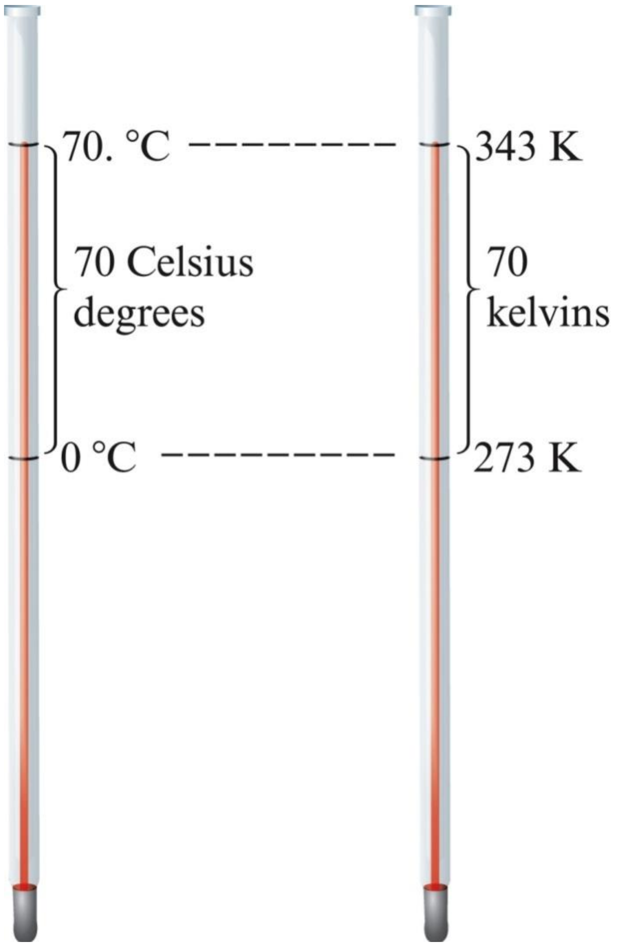
$$T_{\mathrm{K}} = T_{^{\mathrm{o}}\mathrm{C}} + 273$$

Converting between the Kelvin and Celsius Scales

$70.^{o}C = ?\ K$

$T_{^oC} + 273 = T_K$

$70. + 273 = 343\ K$



Converting between the Fahrenheit and Celsius Scales

- Note:
  - The different size units
  - The zero points are different
- To convert between Fahrenheit and Celsius, we need to make 2 adjustments.

$$T_{°F} = 1.80(T_{°C}) + 32$$

Temperature in °F — Temperature in °C

$$T_{^0F} = \frac{9^0\,F}{5^0\,C}\left(T_{^0C}\right) + 32$$

Exercise

At what temperature does °C = °F?

C. Density

- Density is the amount of matter present in a given volume of substance.
- Common units are g/cm^3 or g/mL.

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

Table 5.8 Densities of Various Common Substances at 20°C

Substance	Physical State	Density (g/cm^3)	Substance	Physical State	Density (g/cm^3)
oxygen	gas	0.00133*	aluminum	solid	2.70
hydrogen	gas	0.0000084*	iron	solid	7.87
ethanol	liquid	0.78	copper	solid	8.96
benzene	liquid	0.880	silver	solid	10.5
water	liquid	1.000	lead	solid	11.34
magnesium	solid	1.74	mercury	liquid	13.6
salt (sodium chloride)	solid	2.16	gold	solid	19.32

*At 1 atmosphere pressure

Exercise 1:- A certain mineral has a mass of 17.8 g and a volume of 2.35 cm3. What is the density of this mineral?

CHAPTER TWO

CHEMICAL FOUNDATIONS: ELEMENTS, ATOMS, AND IONS

By: Medhanie G.

Contents:

- The names and symbols of elements
- The periodic table
- Atoms
- Ions and ionic compounds

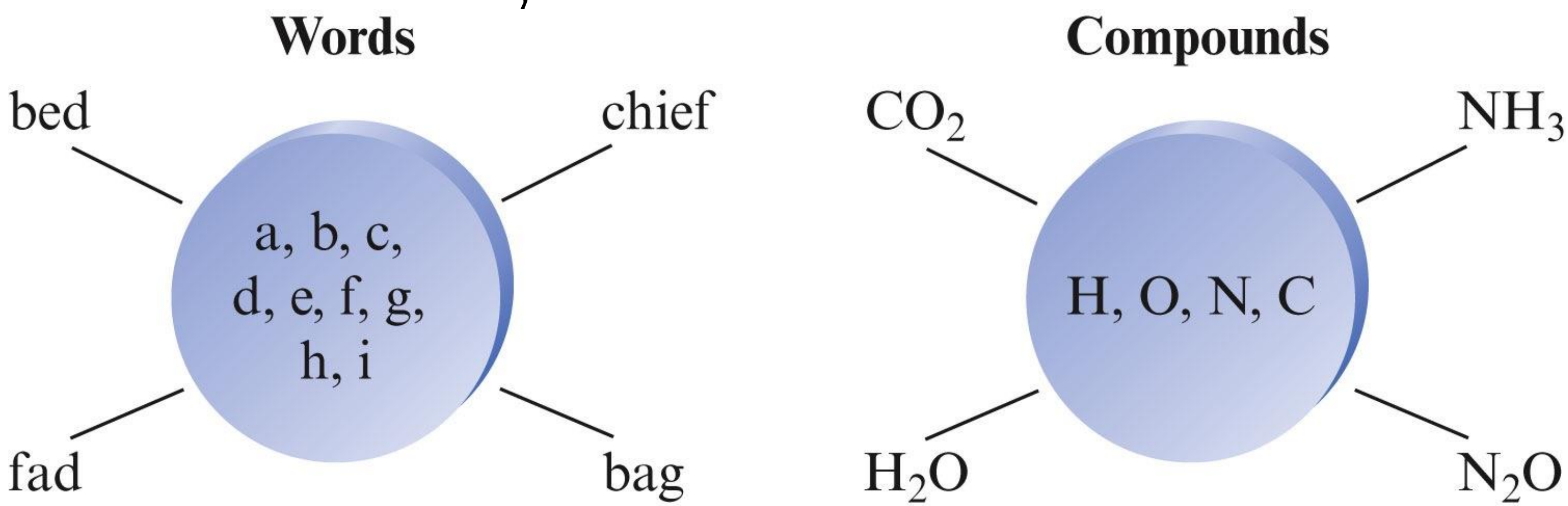
2.1. The Elements

Objectives

i. To learn about the relative abundances of the elements
ii. To learn the names of some elements
iii. To learn the symbols of some elements

The Elements

- All of the materials in the universe can be chemically broken down into about 100 different elements.
- Compounds are made by combining atoms of the elements just as words are constructed from the letters in the alphabet.



A. Abundances of Elements

- Nine elements account for about 98% of the earth's crust, oceans and atmosphere.

Table 3.1 Distribution (Mass Percent) of the 18 Most Abundant Elements in the Earth's Crust, Oceans, and Atmosphere

Element	Mass Percent	Element	Mass Percent
oxygen	49.2	titanium	0.58
silicon	25.7	chlorine	0.19
aluminum	7.50	phosphorus	0.11
iron	4.71	manganese	0.09
calcium	3.39	carbon	0.08
sodium	2.63	sulfur	0.06
potassium	2.40	barium	0.04
magnesium	1.93	nitrogen	0.03
hydrogen	0.87	fluorine	0.03
		all others	0.49

- The elements in living matter are very different from those in the earth's crust.
- In the **human body**, **oxygen, carbon, hydrogen** and **nitrogen** are the most abundant elements.

Table 3.2 Top Ten Elements in the Human Body

Element	Mass Percent
oxygen	65.0
carbon	18.0
hydrogen	10.0
nitrogen	3.0
calcium	1.4
phosphorus	1.0
magnesium	0.50
potassium	0.34
sulfur	0.26
sodium	0.14

B. Names and Symbols for the Elements

- Each element has a **name** and a **symbol.**
  - The symbol usually consists of the first one or two letters of the element's name.
    - <u>Examples</u>: Oxygen ~ O
      Krypton ~ Kr
- Sometimes the symbol is taken from the element's original **Latin** or **Greek** name.
  - <u>Examples</u>: Gold ~ Au ~ Aurum
    Lead ~ Pb ~ Plumbum

Table 3.4 The Names and Symbols of the Most Common Elements*

Element	Symbol	Element	Symbol
aluminum	Al	lithium	Li
arsenic	As	mercury (hydrargyrum)	Hg
barium	Ba	neon	Ne
boron	B	nitrogen	N
bromine	Br	oxygen	O
calcium	Ca	platinum	Pt
carbon	C	potassium (kalium)	K
chromium	Cr	silicon	Si
cobalt	Co	silver (argentium)	Ag
copper (cuprum)	Cu	sodium (natrium)	Na
gold (aurum)	Au	sulfur	S
lead (plumbum)	Pb	zinc	Zn

*Where appropriate, the original name is shown in parentheses so that you can see the sources of some of the symbols.

2.2. Atoms and Compounds

Objectives

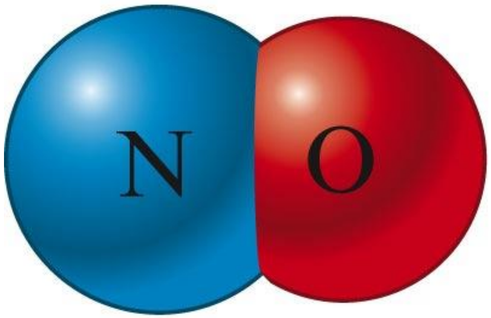
i. To learn about Dalton's theory of atoms
ii. To understand and illustrate the law of constant composition
iii. To learn how a formula describes a compound's composition

Law of Constant Composition

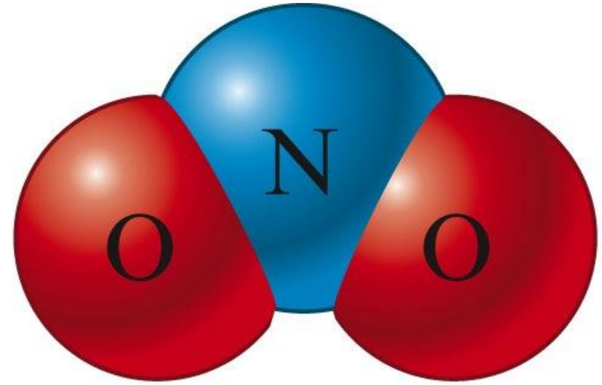
- A given compound always contains the same proportion by mass of the elements of which it is composed.

A. Dalton's Atomic Theory

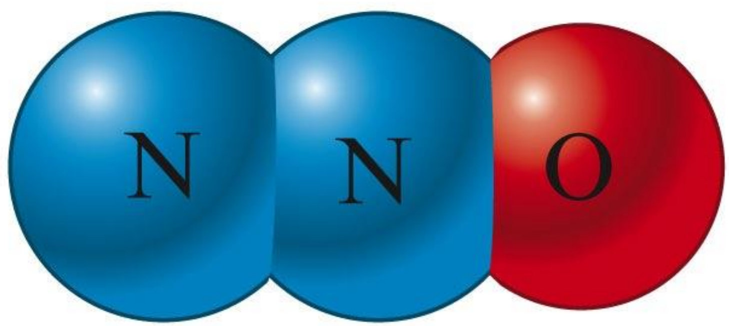
- Dalton's Atomic theory states:
  - All elements are composed of atoms.
  - All atoms of a given element are identical.
  - Atoms of different elements are different.
  - Compounds consist of the atoms of different elements.
  - Atoms are not created or destroyed in a chemical reaction.



NO



NO_2



N_2O

Concept Check

1. Which of the following statements regarding Dalton's atomic theory are still believed to be **true**?

I. Elements are made of tiny particles called atoms.

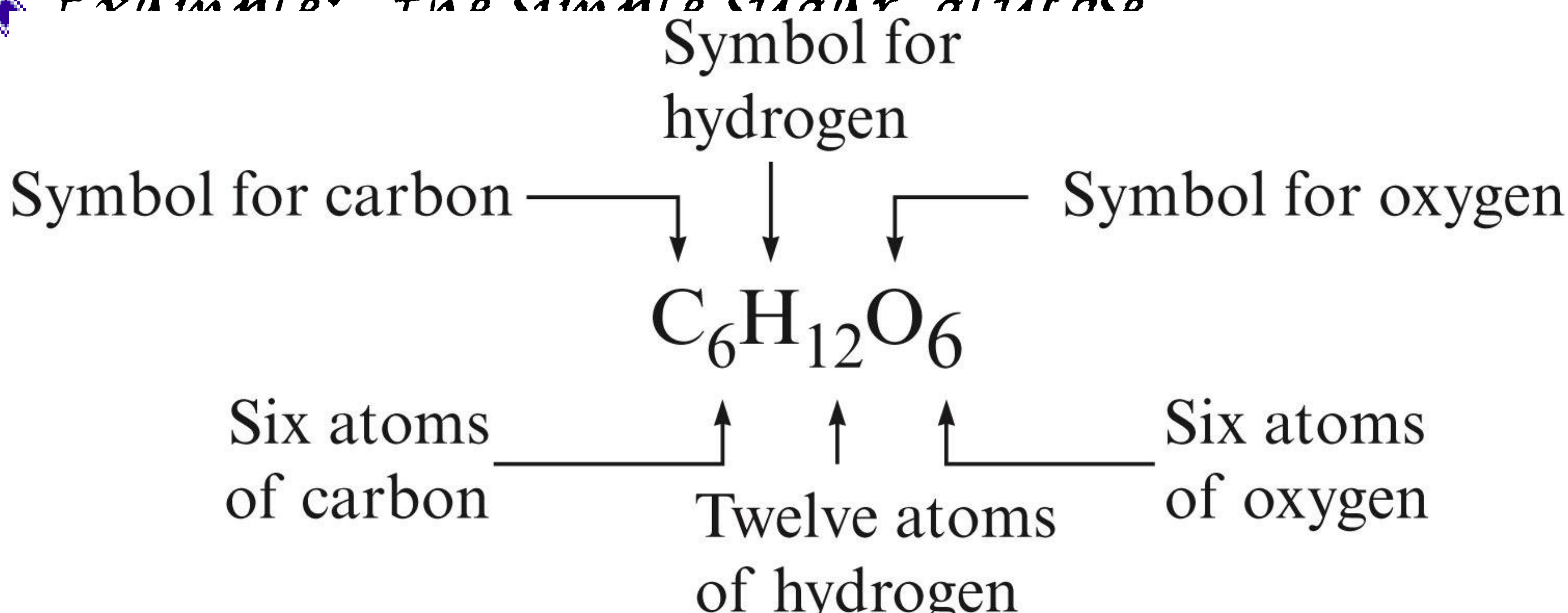
II. All atoms of a given element are identical.

III. A given compound always has the same relative numbers and types of atoms.

IV. Atoms are indestructible.

B. Formulas of Compounds

- A compound is represented by a chemical formula in which the number and kind of atoms present is shown by using the element symbols and subscripts.
- Example: the simple sugar glucose



Tools for Writing Formulas

1. Each atom present is represented by its element symbol.
2. The number of each type of atom is indicated by a subscript written to the right of the element symbol.
3. When only one atom of a given type is present, the subscript 1 is not written.

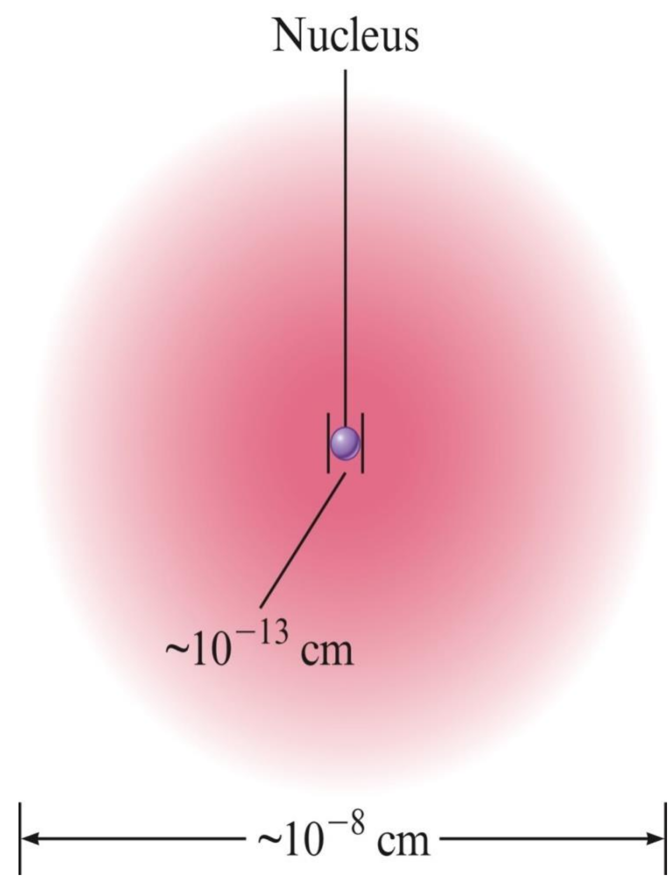
2.3. Atomic Structure

Objectives

i. To learn about the internal parts of an atom
ii. To understand Rutherford' s experiment
iii. To describe some important features of subatomic particles
iv. To learn about the terms isotope, atomic number, and mass number
v. To understand the use of the symbol to describe a given atom

A. Introduction to the Modern Concept of Atomic Structure

- **Ernest Rutherford** showed that atoms have **internal structure.**
  - The **nucleus**, which is at the **center** of the atom, contains **protons** (positively charged) and **neutrons** (uncharged).
  - **Electrons** move around



The **Mass** and **Charge** of the **Electron**, **Proton** and **neutron.**

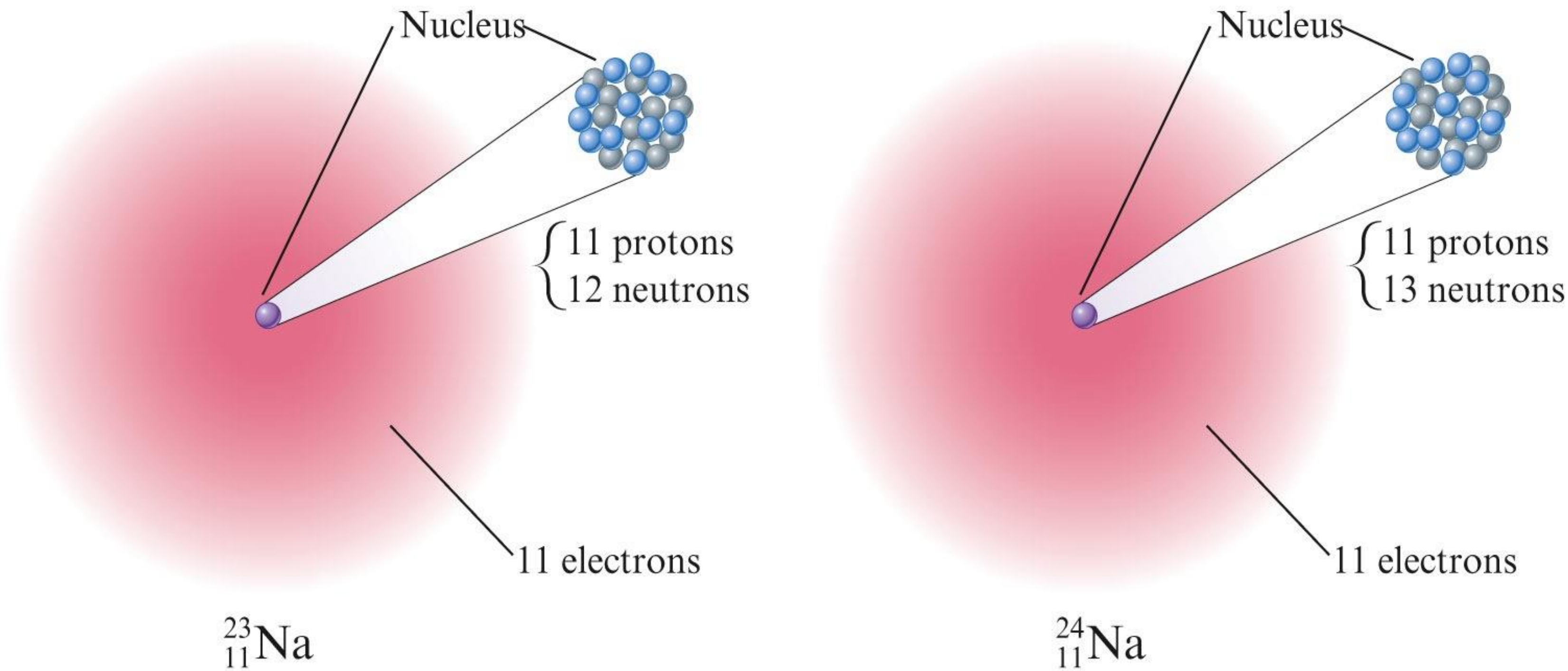
Table 3.5 The Mass and Charge of the Electron, Proton, and Neutron

Particle	Relative Mass*	Relative Charge
electron	1	1−
proton	1836	1+
neutron	1839	none

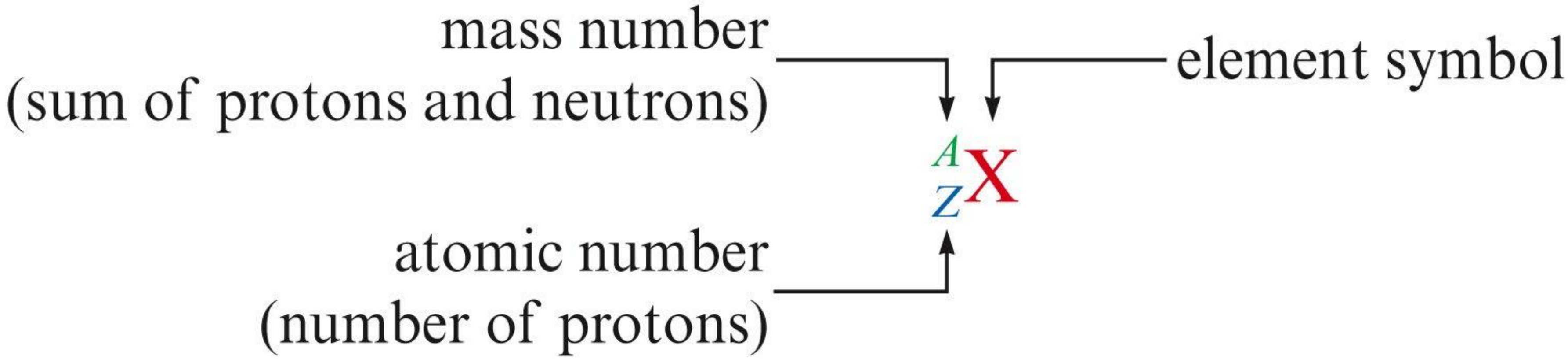
*The electron is arbitrarily assigned a mass of 1 for comparison.

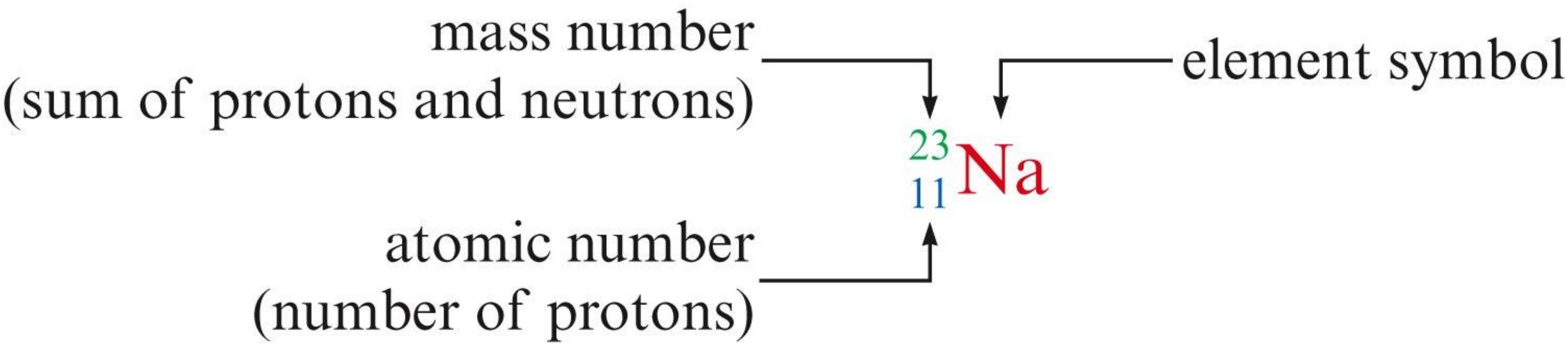
B. Isotopes

- Isotopes are atoms with the same number of protons but different numbers of neutrons.



- A particular isotope is represented by the symb $^{A}_{Z}X$
.





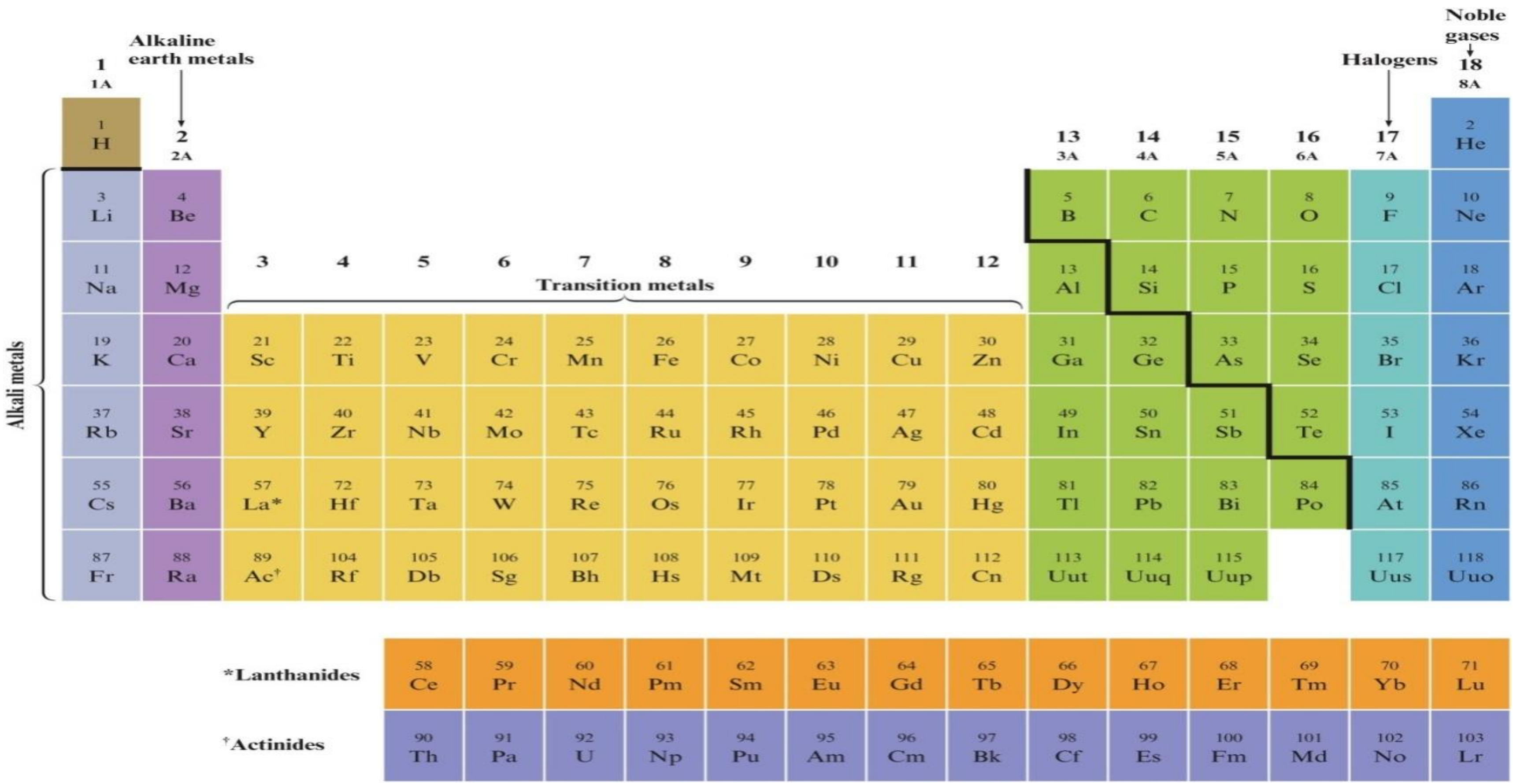
2.4. The Periodic Table

Objectives

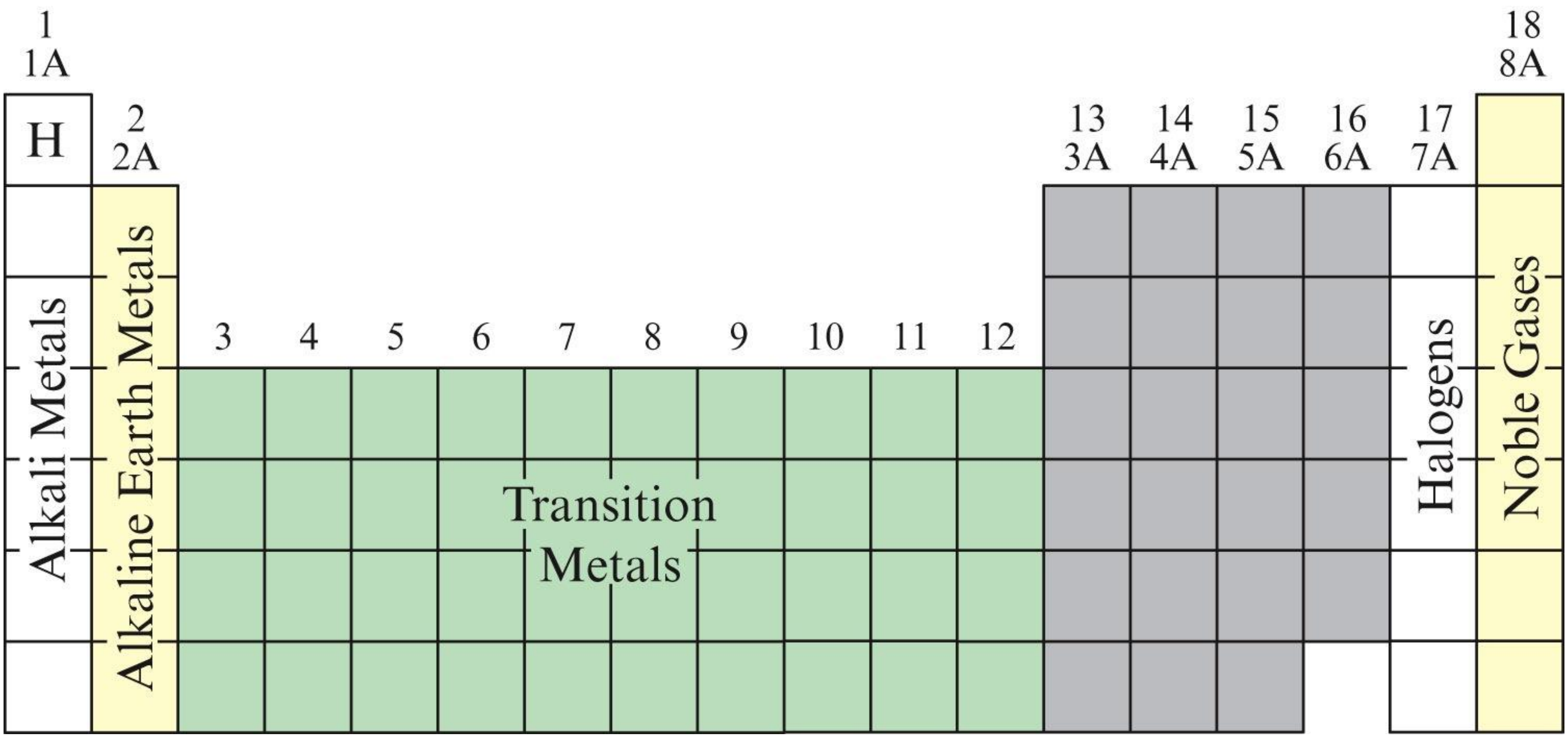
i. To learn the various features of the periodic table
ii. To learn some of the properties of metals, nonmetals and metalloids
iii. To learn the natures of the common elements

A. Introduction to the Periodic Table

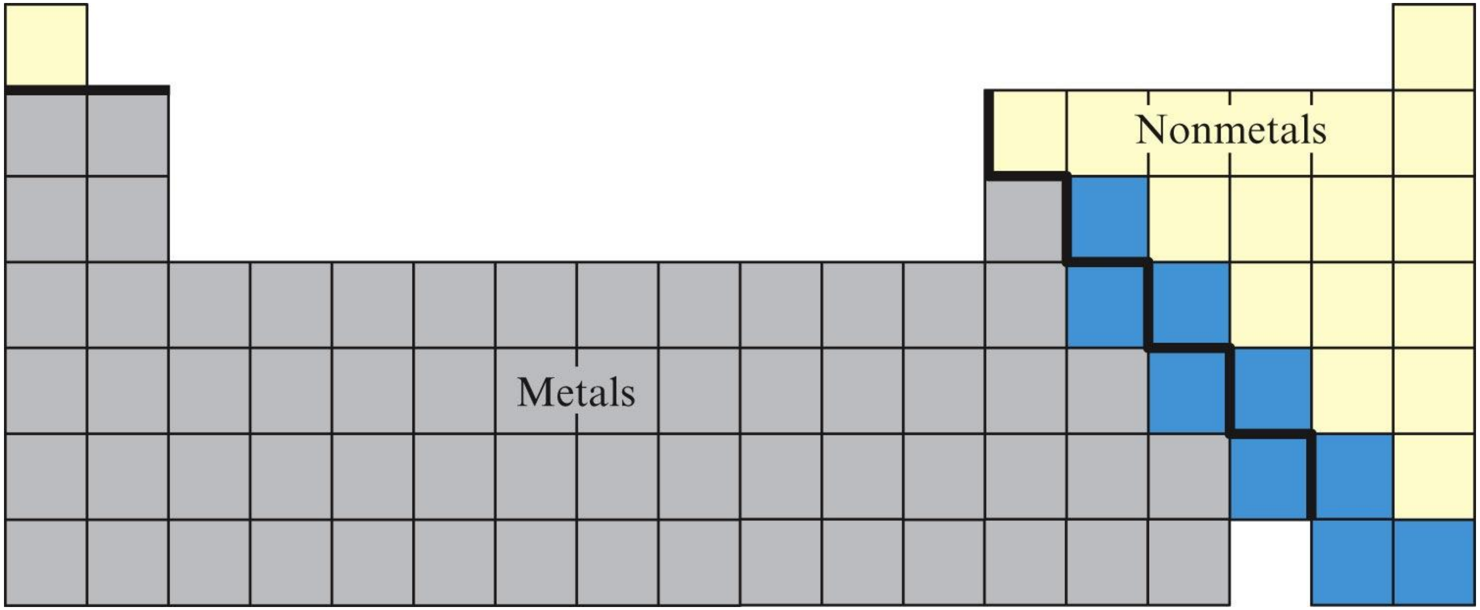
- The periodic table shows all of the known elements in order of increasing atomic number.



- The periodic table is organized to group elements with **similar** properties in vertical columns.



- Most elements are **metals** and occur on the left side.
- The **nonmetals** appear on the right side.
- Metalloids are elements that have some metallic and some nonmetallic properties.



i. Introduction to the Periodic Table

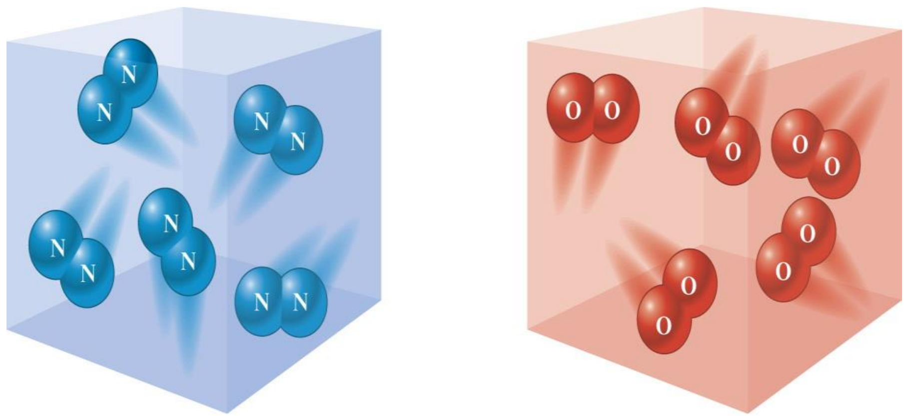
- Physical Properties of Metals
  1. Efficient conduction of heat and electricity
  2. Malleability (can be hammered into thin sheets)
  3. Ductility (can be pulled into wires)
  4. A lustrous (shiny) appearance

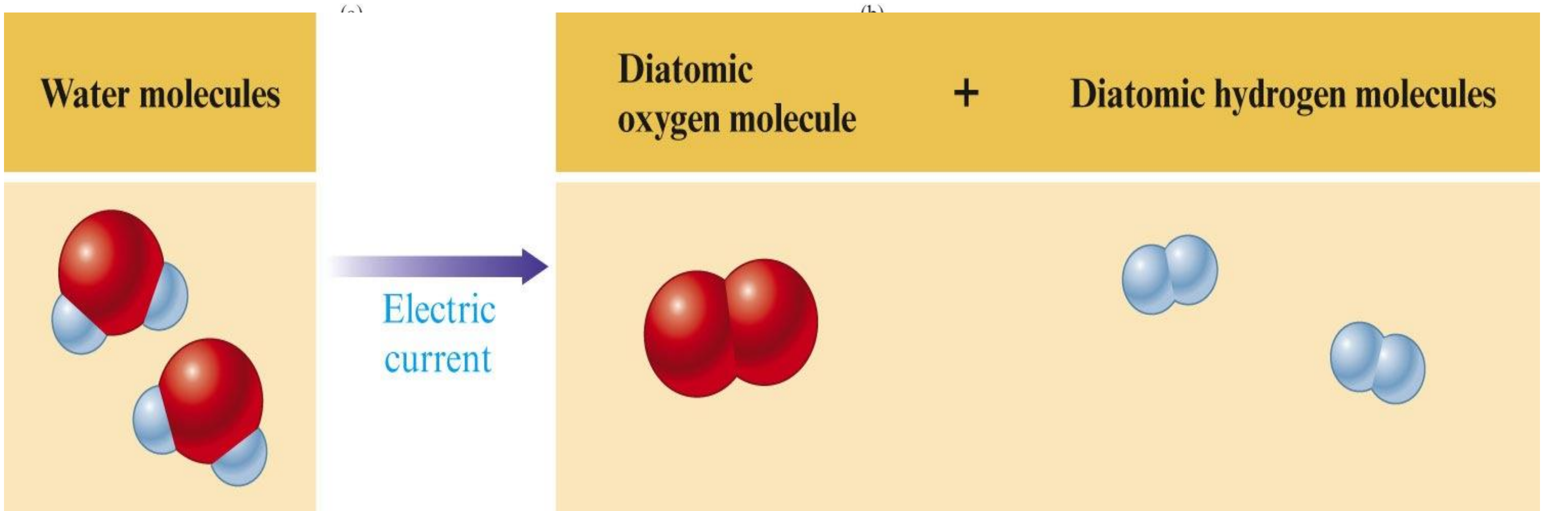
ii. Natural States of the Elements

- Most elements are very reactive.
- Elements are not generally found in uncombined form.
- Exceptions are:
  - Noble metals – Gold, Platinum and Silver

ii. Natural States of the Elements

Diatomic Molecules





Group 7

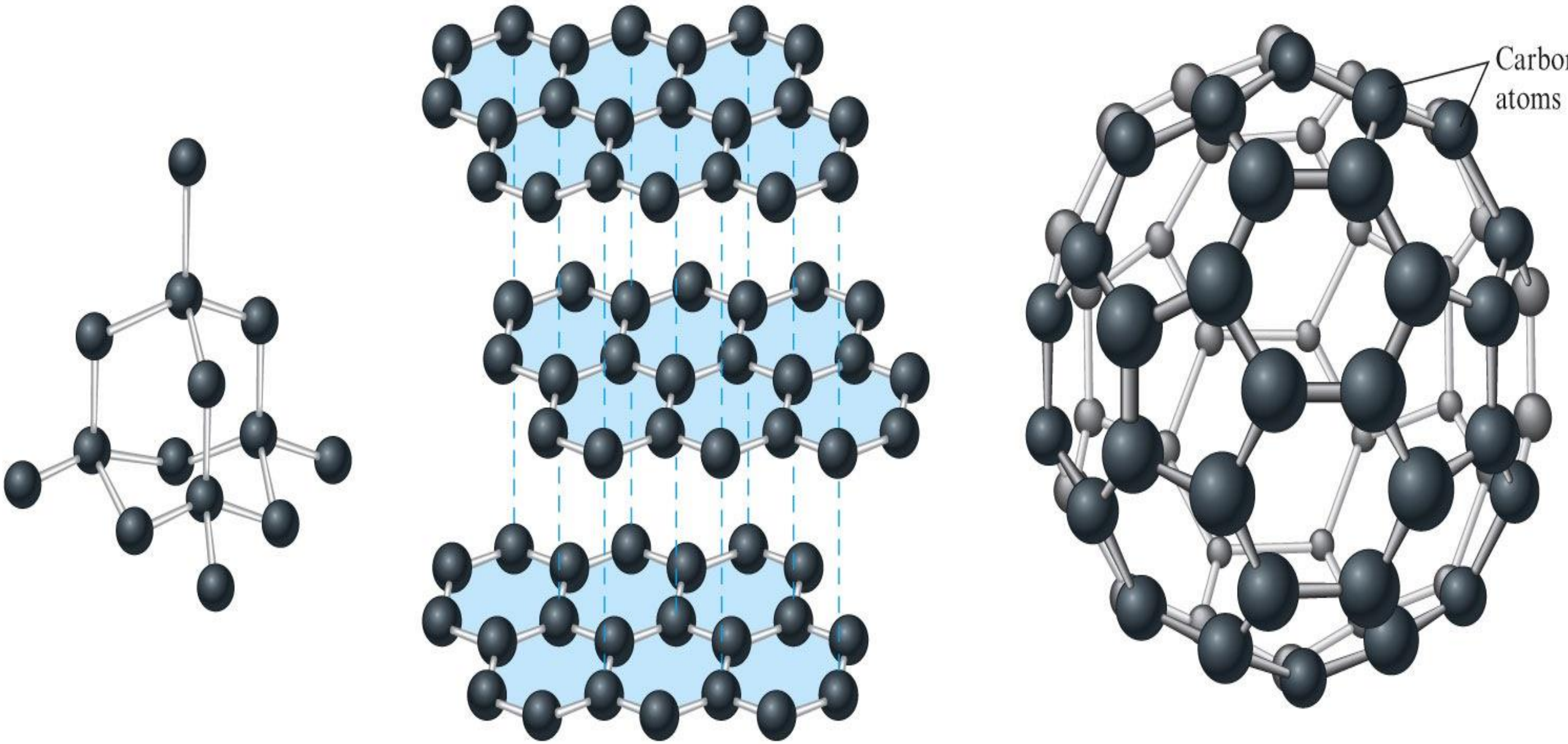
9 F
17 Cl
35 Br
53 I

Diatomic Molecules

Table 3.6 Elements That Exist as Diatomic Molecules in Their Elemental Forms

Element Present	Elemental State at 25°C	Molecule
hydrogen	colorless gas	H_2
nitrogen	colorless gas	N_2
oxygen	pale blue gas	O_2
fluorine	pale yellow gas	F_2
chlorine	pale green gas	Cl_2
bromine	reddish-brown liquid	Br_2
iodine	lustrous, dark purple solid	I_2

Elemental Solids



Diamond Graphite Buckminsterfullerene

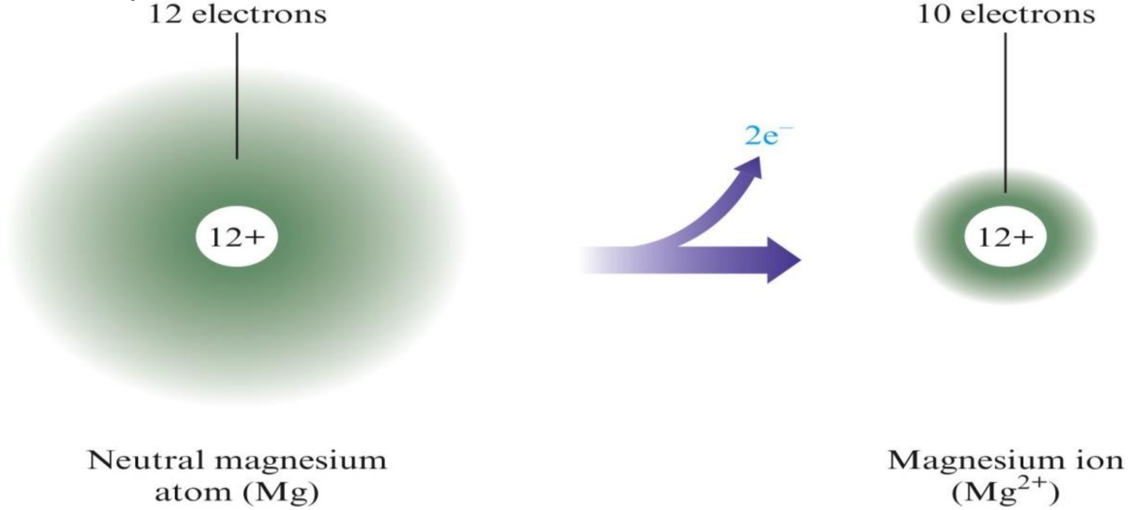
2.5. Ions and their Compounds

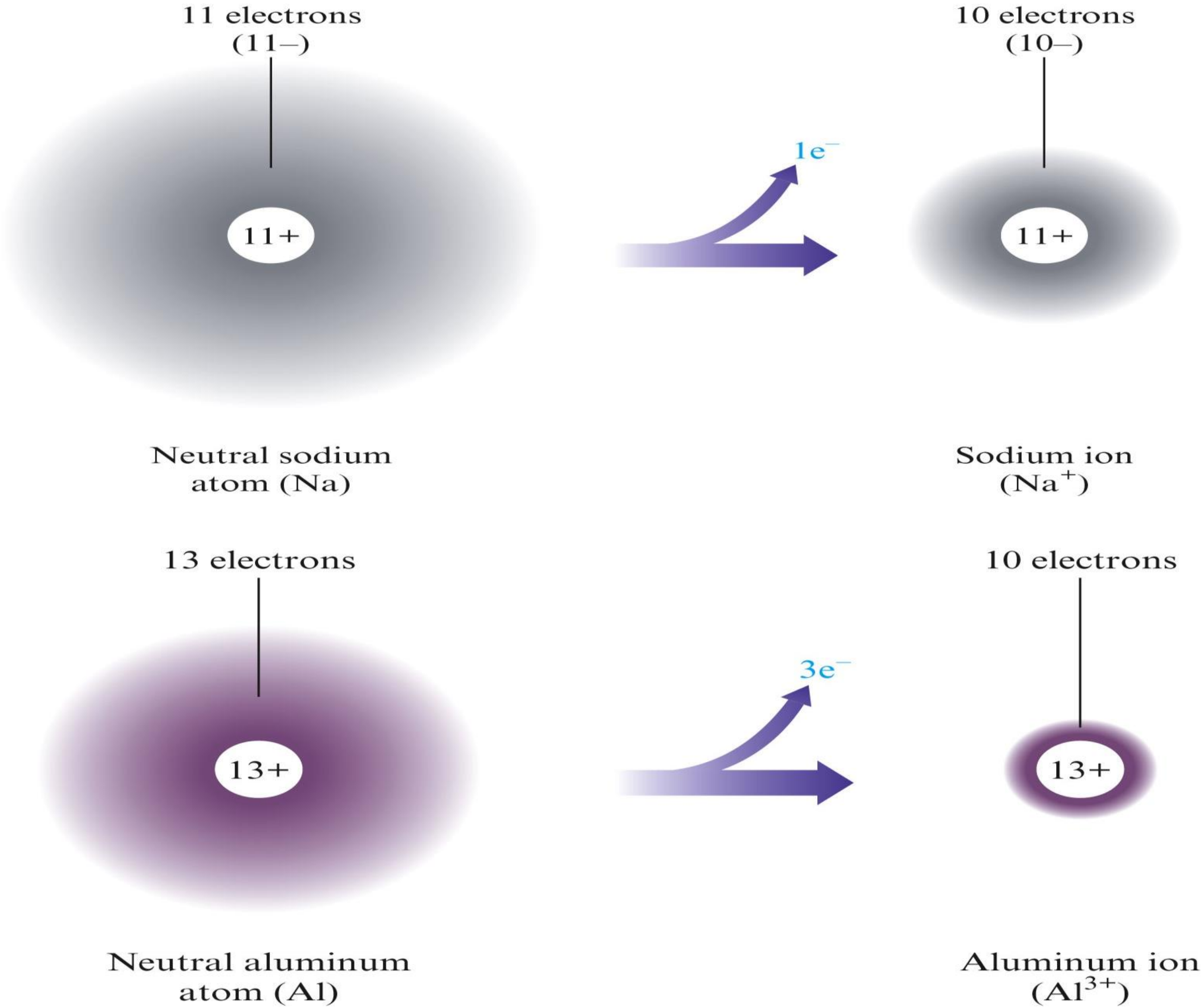
Objectives

i. To describe the formation of ions from their parent atoms
ii. To learn to name ions
iii. To predict which ion a given element forms by using the periodic table
iv. To describe how ions combine to form neutral compounds

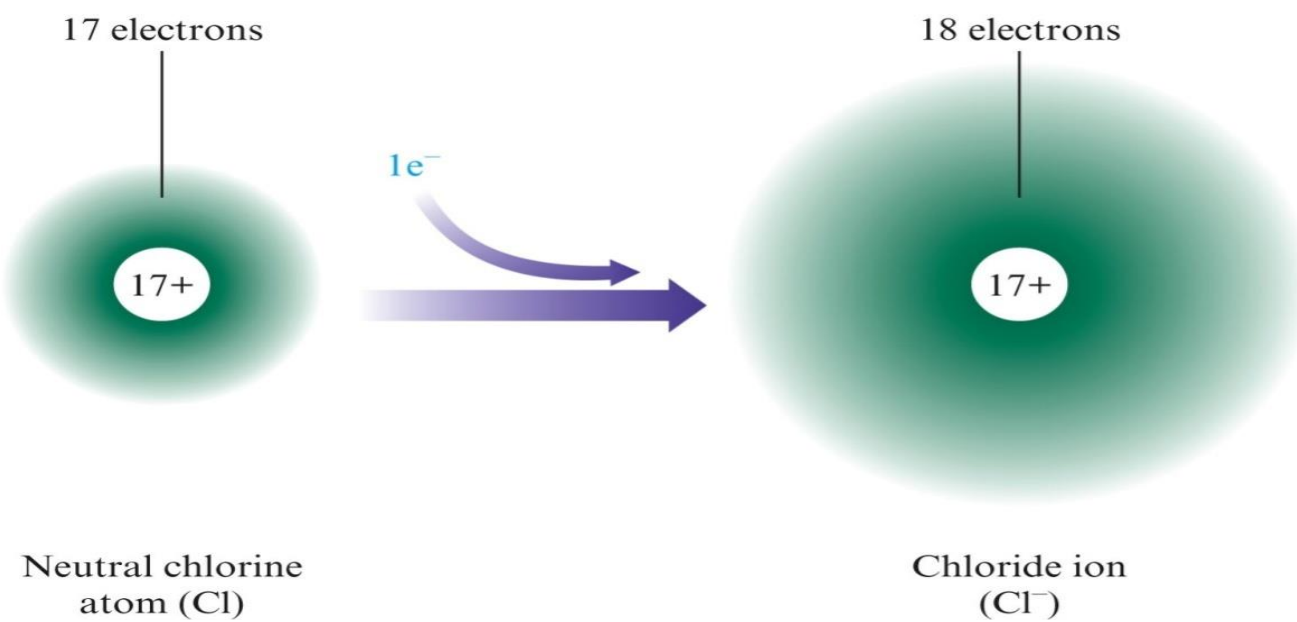
A. Ions

- Atoms can form ions by **gaining** or **losing** electrons.
  - **Metals** tend to lose one or more electrons to form positive ions called **cations.**
  - **Cations** are generally named by using the name of the parent atom.




11 electrons
(11–)
11+
1e⁻
10 electrons
(10–)
11+
Neutral sodium
atom (Na)
Sodium ion
(Na⁺)
13 electrons
13+
3e⁻
10 electrons
13+
Neutral aluminum
atom (Al)
Aluminum ion
(Al³⁺)

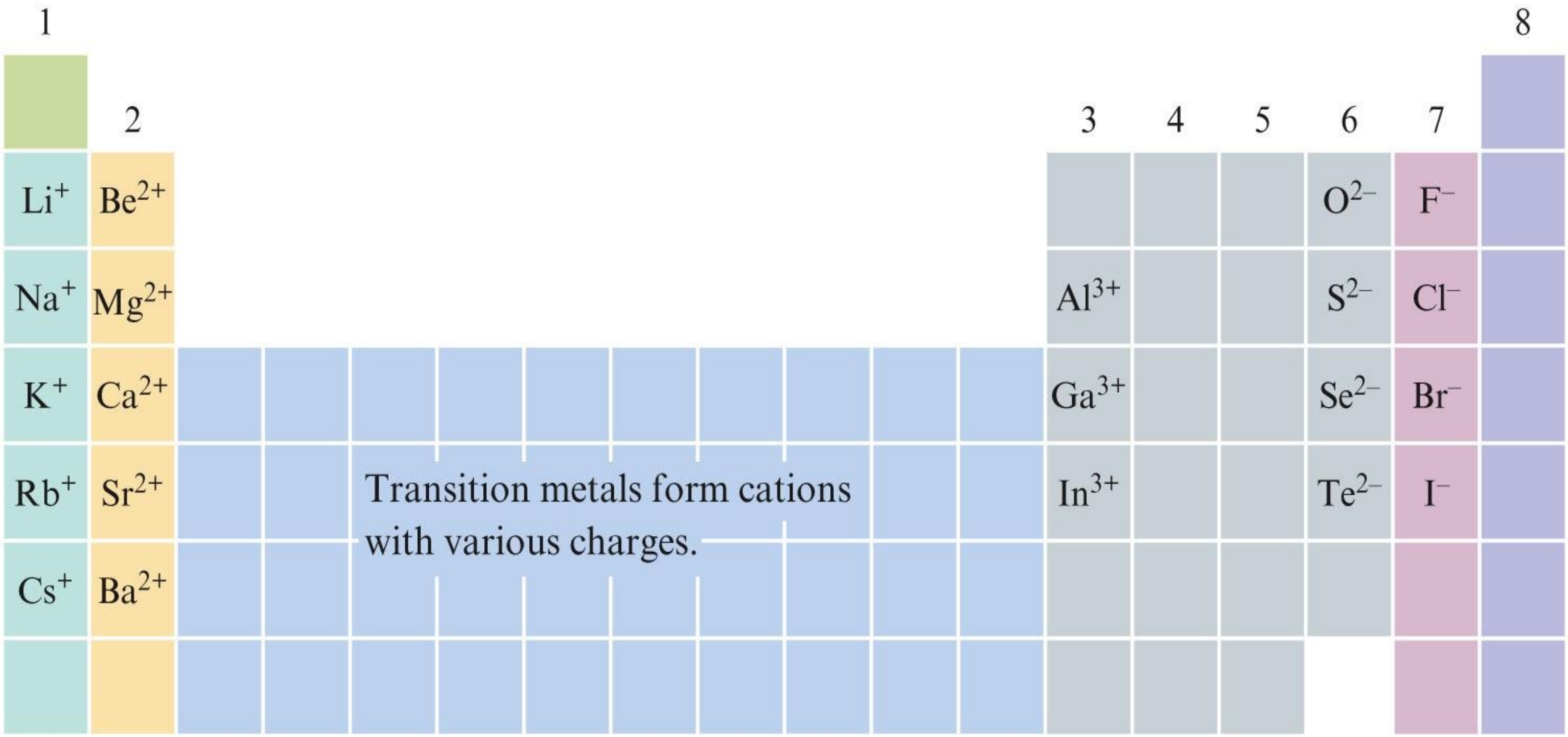
- **Nonmetals** tend to **gain** one or more electrons to form negative ions called **anions.**
- Anions are named by using the root of the atom name followed by the suffix *–ide.*



Ion Charges and the Periodic Table

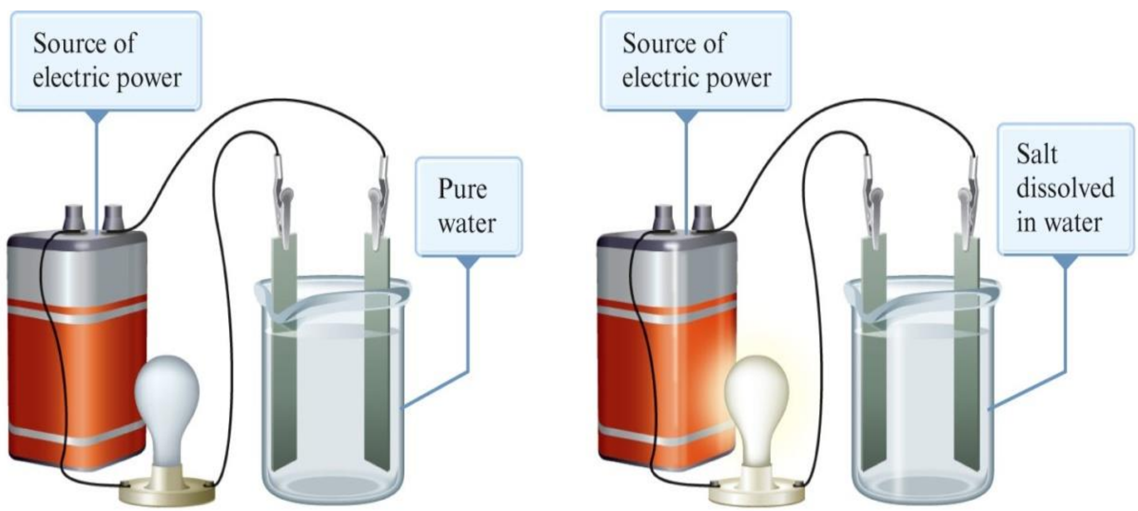
- The ion that a particular atom will form can be predicted from the periodic table.
  - Elements in Group 1 and 2 form 1+ and 2+ ions, respectively
  - Group 7 atoms form anions with 1– charges
  - Group 6 atoms form anions with 2– charges

Ion Charges and the Periodic Table

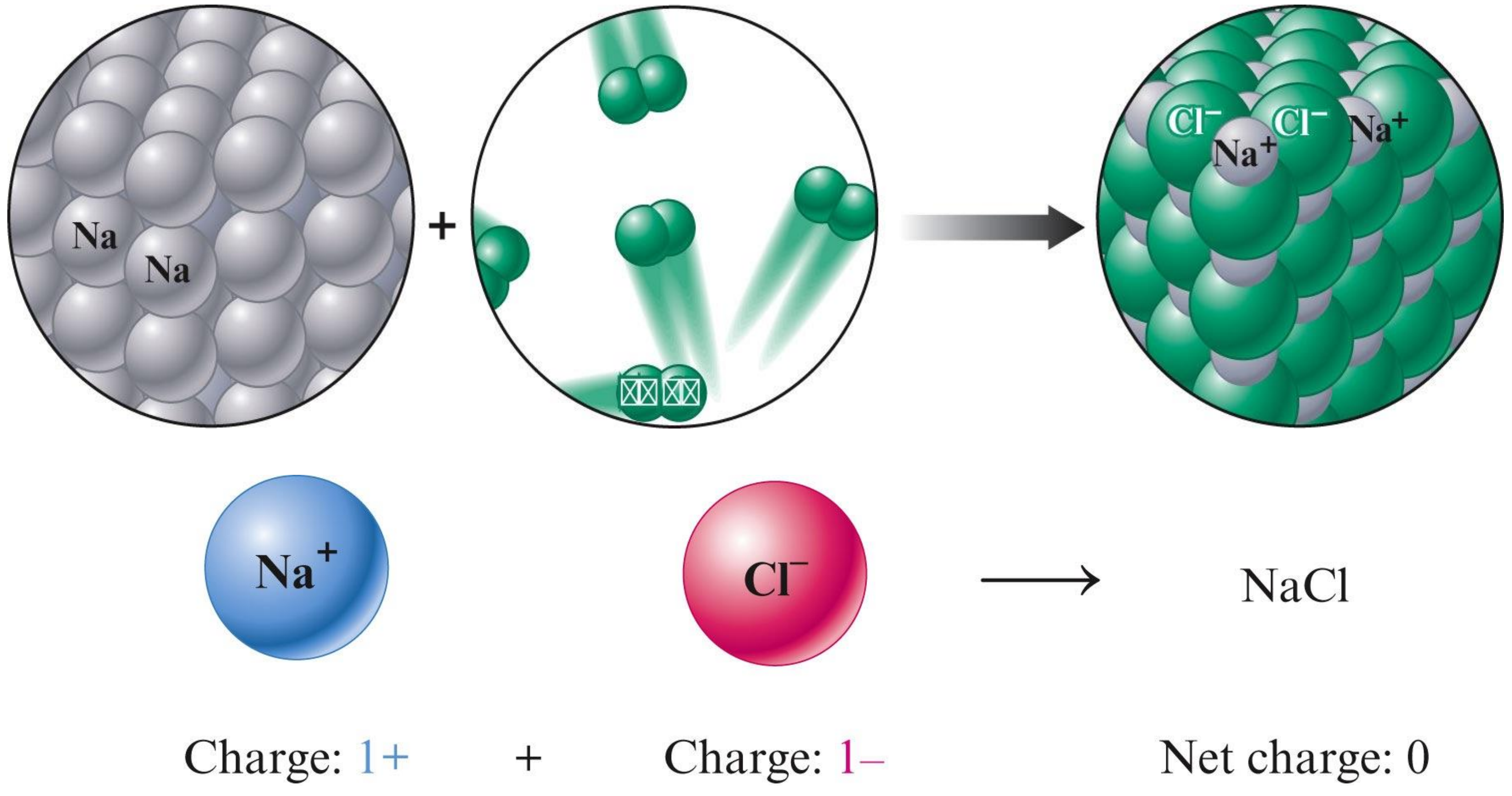


B. Compounds that Contain Ions

- Ions combine to form ionic compounds.
- Properties of ionic compounds.
  - High melting points
  - Conduct electricity (If **melted** & If dissolved in **water**)

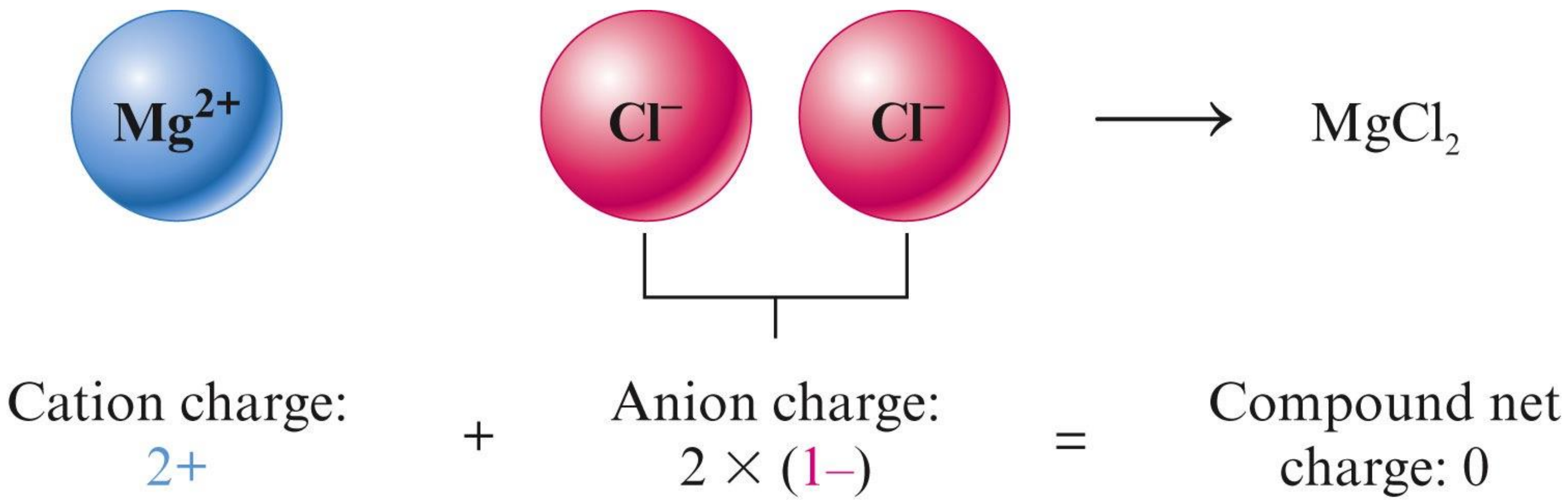


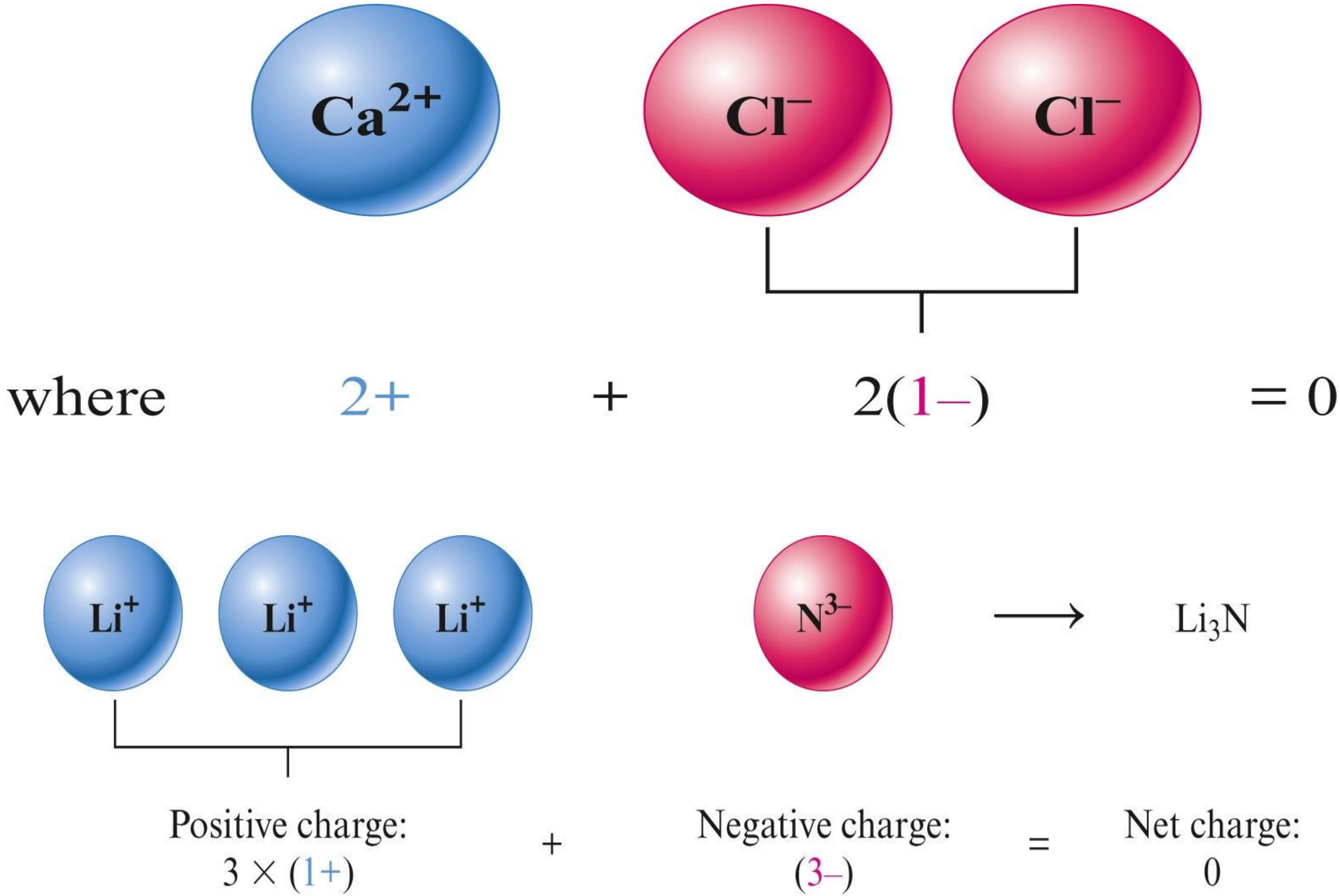
- Ionic compounds are electrically neutral.
- The charges on the anions and cations in the compound must sum to zero.



Formulas for Ionic compounds

- Write the **cation** element symbol followed by the anion element symbol.
- The **number** of **cations** and anions must be correct for their charges to sum to zero.




Ca^{2+}
Cl^-
Cl^-
where $2+$ + $2(1-)$ $= 0$
Li^+
Li^+
Li^+
N^{3-}
$\longrightarrow$ Li_3N
Positive charge: $3 \times (1+)$
+
Negative charge: $(3-)$
=
Net charge: 0

2.6. Chemical Nomenclature

A. Inorganic Nomenclature

- Write the name of the cation.
  - If the cation can have more than one possible charge, write the charge as a Roman numeral in parentheses.
- If the anion is an element, change its ending to -*ide*; if the anion is a polyatomic ion, simply write the name of the polyatomic ion.

Patterns in Oxyanion Nomenclature

- When there are two oxyanions involving the same element.
  - The one with fewer oxygens ends in *-ite.*
  - The one with more oxygens ends in *-ate.*
    - NO_2^-: nitrite; NO_3^-: nitrate
    - SO_3^{2-}: sulfite; SO_4^{2-}: sulfate

Patterns in Oxyanion Nomenclature

Maximum of three O atoms in period 2.

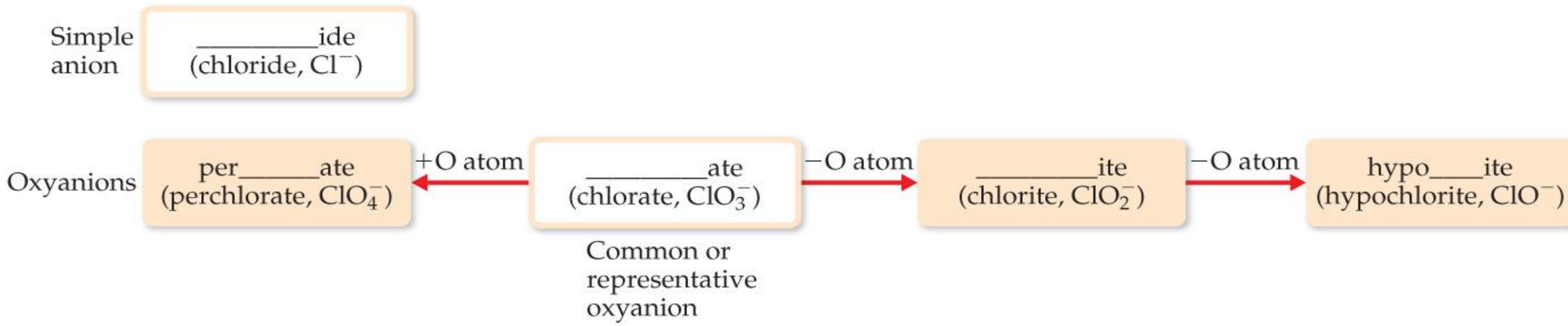
Maximum of four O atoms in period 3.

Charges increase right to left.

	Group 4A	Group 5A	Group 6A	Group 7A
Period 2	CO_3^{2-} **Carbonate ion**	NO_3^- **Nitrate ion**		
Period 3		PO_4^{3-} **Phosphate ion**	SO_4^{2-} **Sulfate ion**	ClO_4^- **Perchlorate ion**

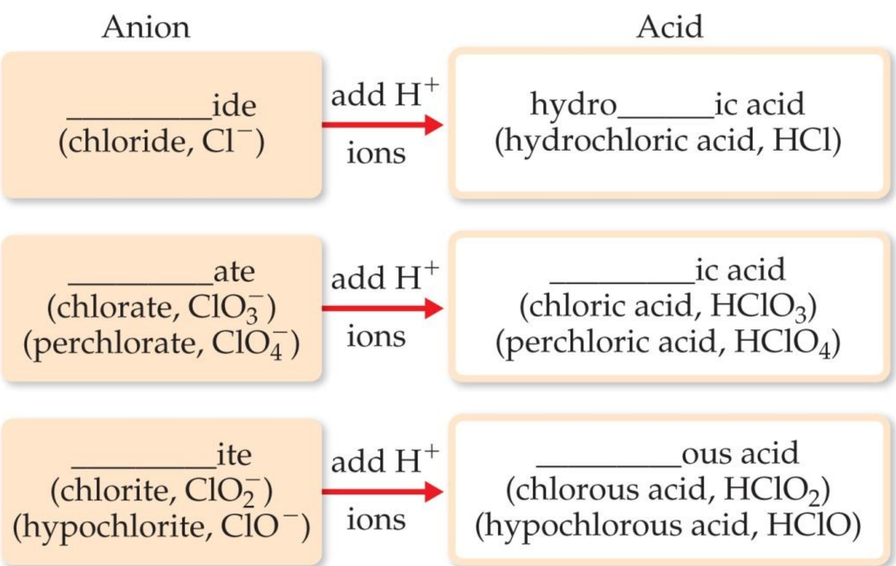
- Central atoms on the second row have a bond to, at most, three oxygens; those on the third row take up to four.
- Charges increase as you go from ***right*** to ***left***.

Patterns in Oxyanion Nomenclature



- The one with the second fewest oxygens ends in -***ite***: ClO_2^- is chlorite.
- The one with the second most oxygens ends in -***ate***: ClO_3^- is chlorate.
- The one with the fewest oxygens has the prefix ***hypo-*** and ends in -***ite***: ClO^- is hypochlorite.
- The one with the most oxygens has the prefix ***per-*** and ends in -***ate***: ClO_4^- is perchlorate.

B. Acid Nomenclature



- If the anion in the acid ends in *-**ide***, change the ending to ***-ic acid*** and add the prefix ***hydro-***.
  - HCl: hydrochloric acid
  - HBr: hydrobromic acid
  - HI: hydroiodic acid
- If the anion ends in *-**ite***, change the ending to ***-ous acid***.
  - HClO: hypochlorous acid
  - $HClO_2$: chlorous acid
- If the anion ends in *-**ate***, change the ending to ***-ic acid***.
  - $HClO_3$: chloric acid

C. Nomenclature of Binary Molecular Compounds

Table 2.6 Prefixes Used in Naming Binary Compounds Formed between Nonmetals

Prefix	Meaning
Mono-	*1*
Di-	*2*
Tri-	*3*
Tetra-	*4*
Penta-	*5*
Hexa-	*6*
Hepta-	*7*
Octa-	*8*
Nona-	*9*
Deca-	*10*

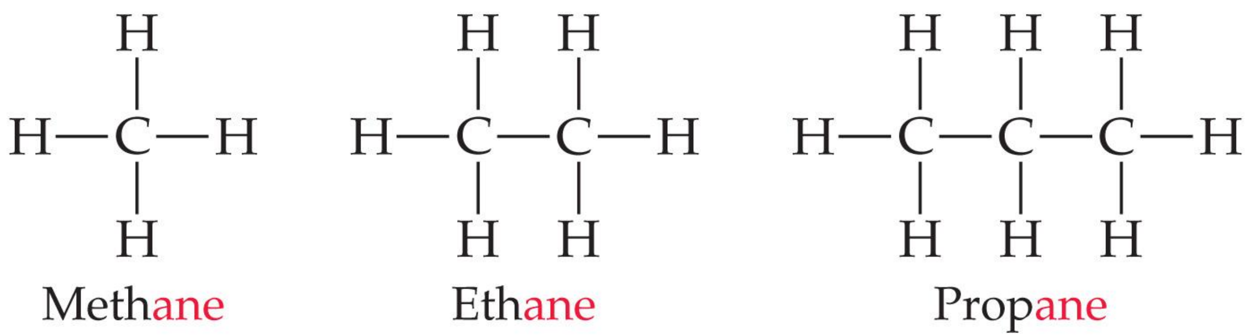
- The name of the element farther to the left in the periodic table (closer to the metals) or lower in the same group is usually written first.

- A prefix is used to denote the number of atoms of each element in the compound (***mono-*** is not used on the first element listed, however).

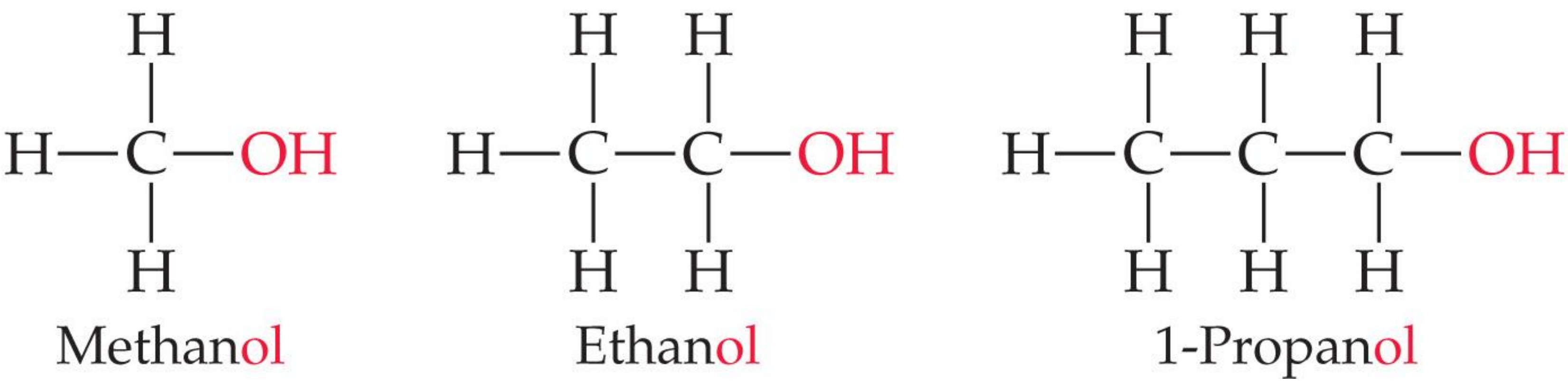
- The ending on the second element is changed to -**ide.**
  - CO_2: carbon dioxide
  - CCl_4: carbon tetrachloride

- If the prefix ends with ***a*** or ***o*** and the name of the element begins with a vowel, the two successive vowels are often elided into one.
  - N_2O_5: dinitrogen pentoxide

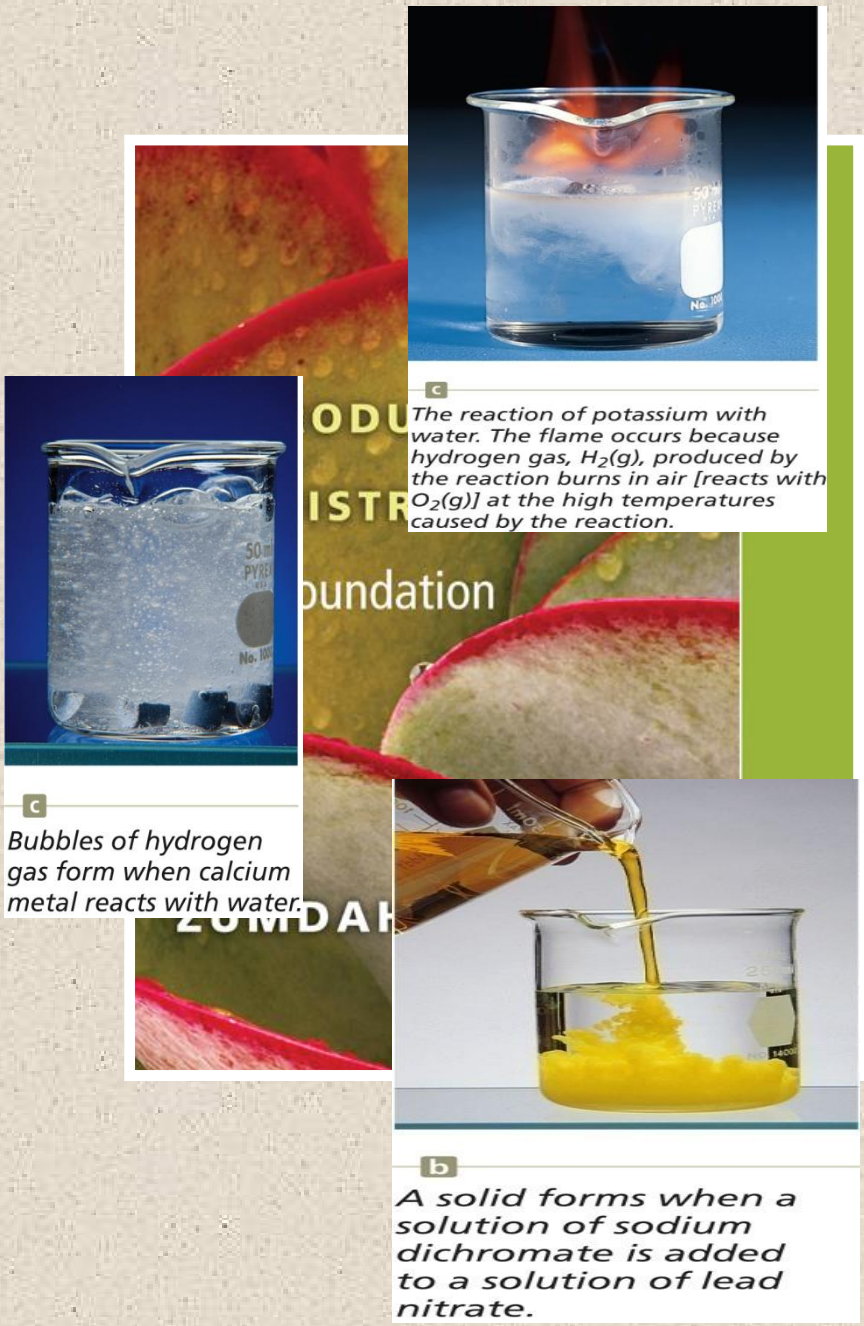
D. Nomenclature of Organic Compounds



- **Organic chemistry** is the study of carbon.
- Organic chemistry has its own system of nomenclature.
- The simplest hydrocarbons (compounds containing only carbon and hydrogen) are **alkanes.**
- The first part of the names just listed correspond to the number of carbons (***meth-*** = 1, ***eth-*** = 2, ***prop-*** = 3, etc.).



- When a hydrogen in an alkane is replaced with something else (a **functional group**, like -OH in the compounds above), the name is derived from the name of the alkane.
- The ending denotes the type of compound.
  - An **alcohol** ends in ***-ol***.



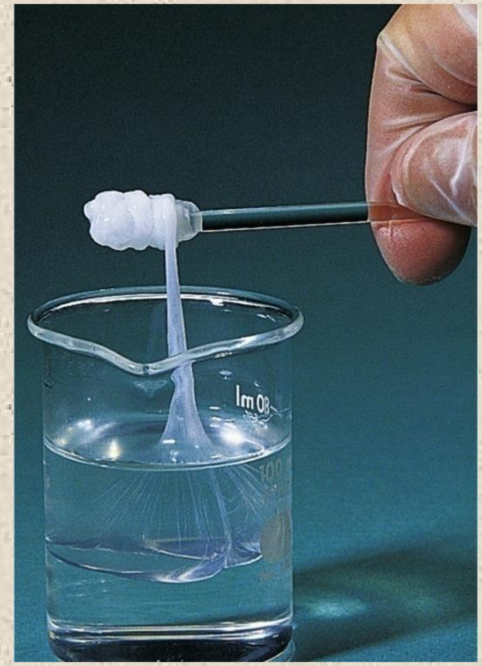
CHAPTER THREE

CHEMICAL REACTIONS

Evidence for a Chemical Reaction

What are the clues that a chemical change has taken place?

- Chemical reactions often give a visual signal.
- But reactions are not always visible.

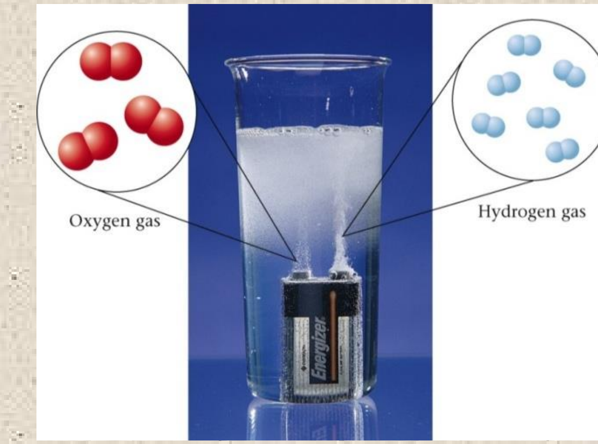


Evidence for a Chemical Reaction

Some Clues That a Chemical Reaction Has Occurred

The color changes

A solid forms



Bubbles are present

"Colorless hydrochloric acid is added to a red solution of cobalt(II) nitrate, turning the solution blue."

"A solid forms when a solution of sodium dichromate is added to a solution of lead nitrate."

Return to TOC

Chemical Equations

- Chemical reactions involve a rearrangement of the ways atoms are grouped together.
- A chemical equation represents a chemical reaction.
  - **Reactants** are shown to the left of the arrow.
  - **Products** are shown to the right of the arrow.

$$\underbrace{\overset{\text{Methane}}{CH_4} + \overset{\text{Oxygen}}{O_2}}_{\text{Reactants}} \rightarrow \underbrace{\overset{\text{Carbon dioxide}}{CO_2} + \overset{\text{Water}}{H_2O}}_{\text{Products}}$$

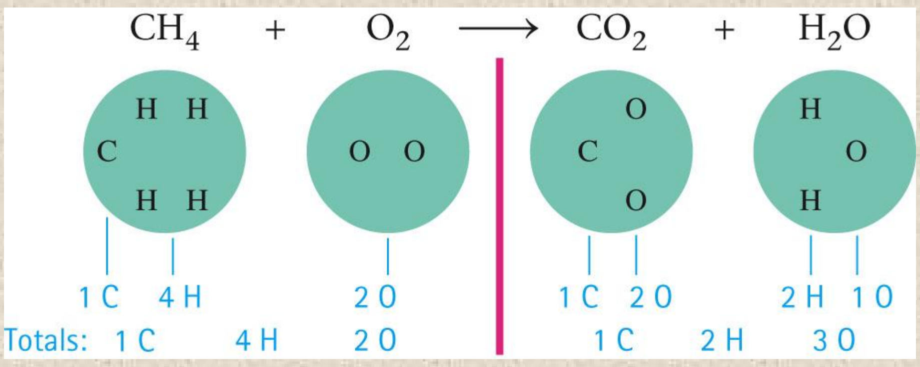
Chemical Equations

- In a chemical reaction atoms are not created or destroyed.
- All atoms present in the reactants must be accounted for in the products.
  - Same number of each type of atom on both sides of the arrow.

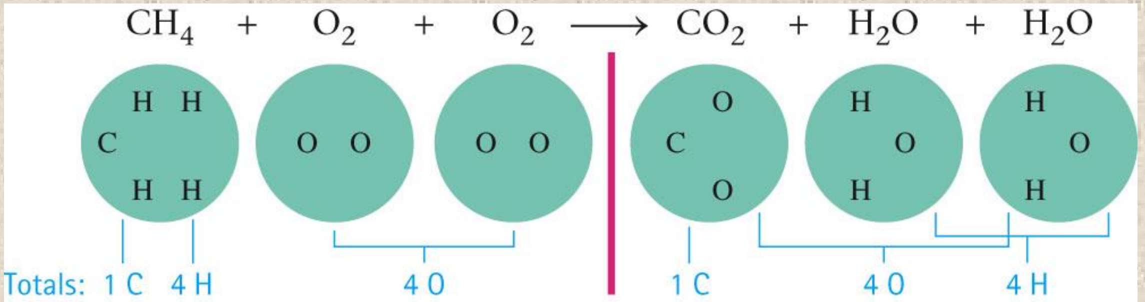
Chemical Equations

Balancing a Chemical Equation

- Unbalanced Equation:



- Balancing the Equation:



- The balanced equation:
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Physical States

- Physical states of compounds are often given in a chemical equation. These are sometimes called descriptors.

Symbol	State
(*s*)	solid
(*l*)	liquid
(*g*)	gas
(*aq*)	dissolved in water (in aqueous solution)

Return to TOC

Chemical Equations

Example

Solid potassium		Water		Hydrogen gas		Potassium hydroxide dissolved in water
$K(s)$	$+$	$H_2O(l)$	$\rightarrow$	$H_2(g)$	$+$	$KOH(aq)$

Exercise

When blue light shines on a mixture of hydrogen and chlorine gas, the elements react explosively to form gaseous hydrochloric acid.

What is the unbalanced equation for this process?

a) $HCl(g) \longrightarrow H(g) + Cl(g)$

b) $H(g) + Cl(g) \longrightarrow HCl(g)$

c) $\boxed{H_2(g) + Cl_2(g) \longrightarrow HCl(g)}$

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Balancing Chemical Equations

- The principle that lies at the heart of the balancing process is that atoms are **conserved** in a chemical reaction.
- Atoms are neither created nor destroyed.
- The same number of each type of atom is found among the reactants and among the products.

Return to TOC

How to Write and Balance Equations

1. Read the description of the chemical reaction. What are the reactants, the products, and their states? Write the appropriate formulas.

   Hydrogen gas (H_2) and oxygen gas (O_2) combine to form liquid water (H_2O).

2. Write the ***unbalanced*** equation that summarizes the information from step 1.

$$H_2(g) + O_2(g) \rightarrow H_2O(l)$$

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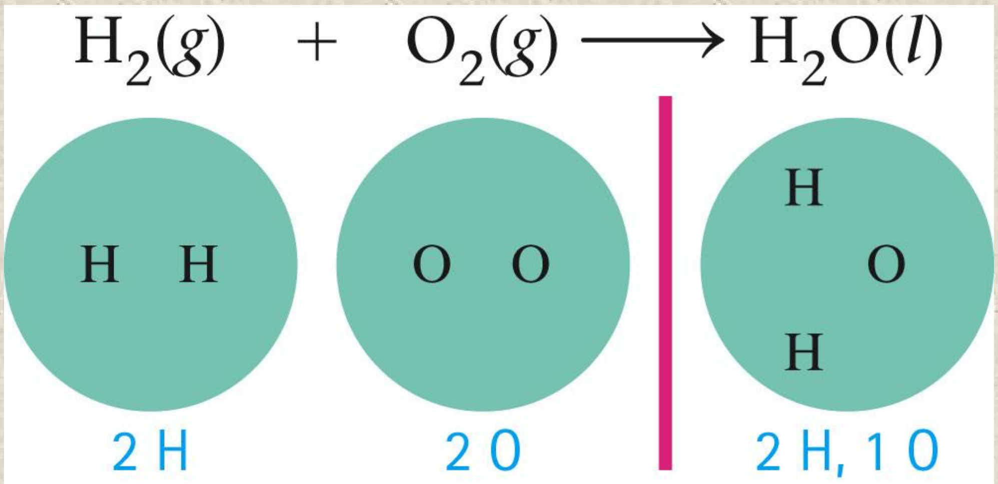
Balancing Chemical Equations

How to Write and Balance Equations

3. Balance the equation by inspection, starting with the most complicated molecule.

   Equation is unbalanced by counting the atoms on both sides of the arrow.



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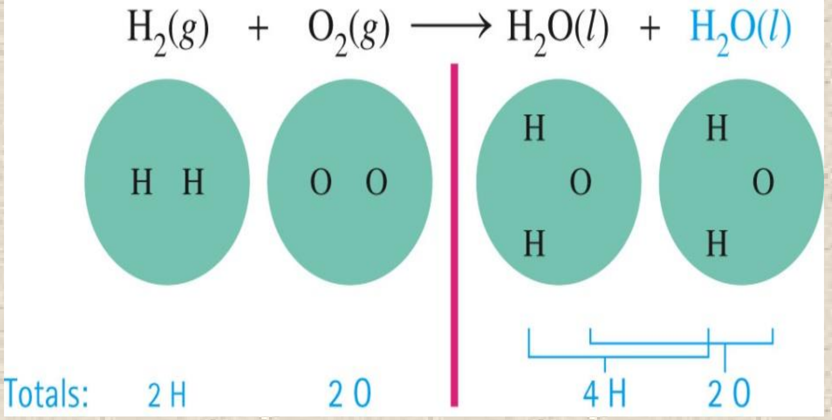
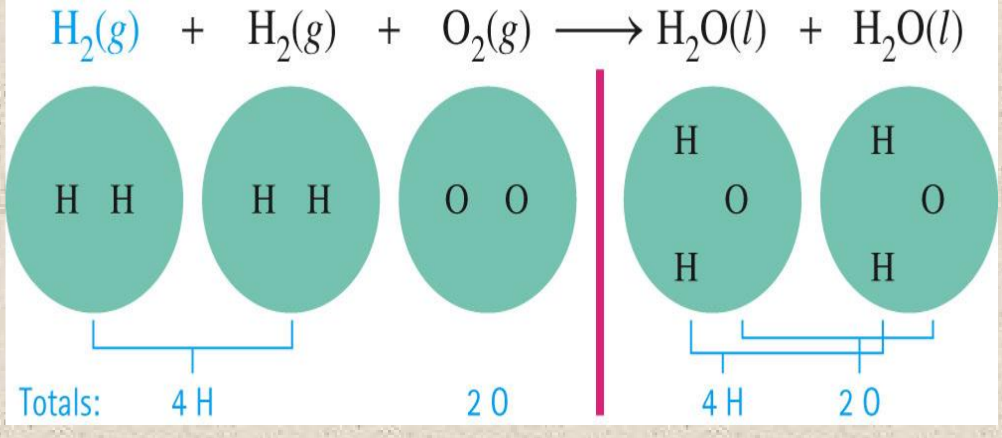
Balancing Chemical Equations

How to Write and Balance Equations

3. Balance the equation by inspection, starting with the most complicated molecule.

   We must balance the equation by adding more molecules of reactants and/or products.





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How to Write and Balance Equations

4. Check to see that the coefficients used give the same number of each type of atom on both sides of the arrow. Also check to see that the coefficients used are the smallest integers that give the balanced equation.

The balanced equation is:

$$\boxed{2H_2(g) + O_2(g) \rightarrow 2H_2O(l)}$$ preferred

or could be:

$$4H_2(g) + 2O_2(g) \rightarrow 4H_2O(l)$$

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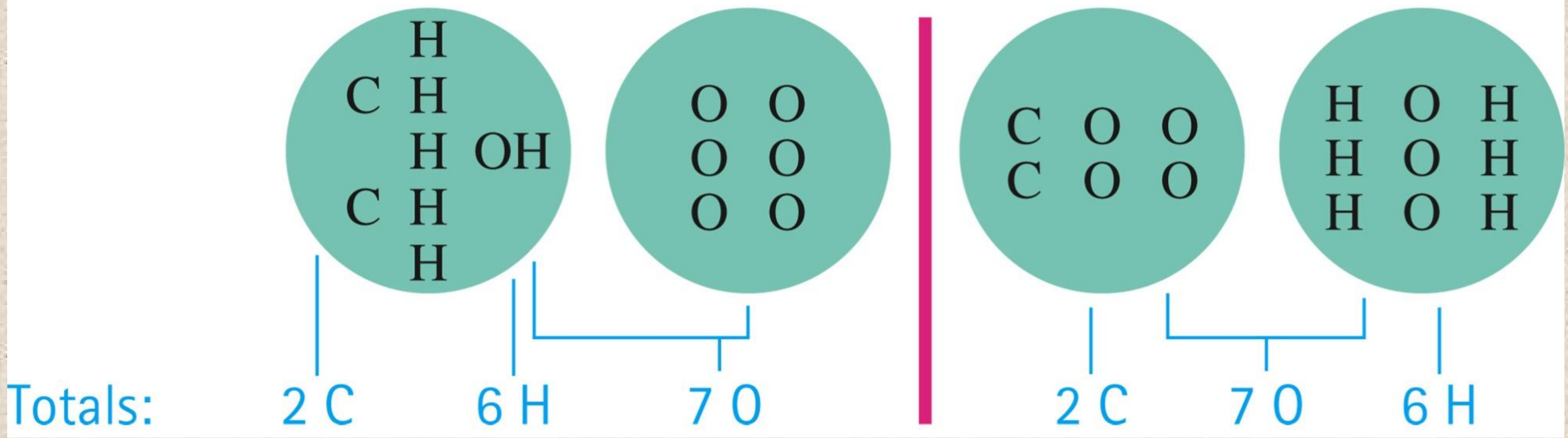
Balancing Chemical Equations

Another Balancing Example:

$$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(g)$$



Balancing Chemical Equations

$$Na_2O(s) + H_2O(l) \longrightarrow 2\,NaOH(aq)$$

$$CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(g)$$

$$4\,Fe(s) + 3\,O_2(g) \longrightarrow 2\,Fe_2O_3(s)$$

$$LiOH(s) + CO_2(g) \longrightarrow LiHCO_3(s)$$

$$2\,KClO_3(s) \xrightarrow[MnO_2]{\Delta} 2\,KCl(s) + 3\,O_2(g)$$

Return to TOC

Balancing Chemical Equations

Exercise

Which of the following correctly balances the chemical equation given below? There may be more than one correct balanced equation. If a balanced equation is incorrect, explain what is incorrect about it.

$$CaO + C \rightarrow CaC_2 + CO_2$$

I. $CaO_2 + 3C \rightarrow CaC_2 + CO_2$

II. $2CaO + 5C \rightarrow 2CaC_2 + CO_2$

III. $CaO + (2.5)C \rightarrow CaC_2 + (0.5)CO_2$

IV. $4CaO + 10C \rightarrow 4CaC_2 + 2CO_2$

Return to TOC

Balancing Chemical Equations

Exercise

Of the three that are correct, which one is preferred most (the most accepted convention)? Why?

$$CaO + C \rightarrow CaC_2 + CO_2$$

I. $CaO_2 + 3C \rightarrow CaC_2 + CO_2$

II. $\boxed{2CaO + 5C \rightarrow 2CaC_2 + CO_2}$

III. $CaO + (2.5)C \rightarrow CaC_2 + (0.5)CO_2$

IV. $4CaO + 10C \rightarrow 4CaC_2 + 2CO_2$

Return to TOC

Balancing Chemical Equations

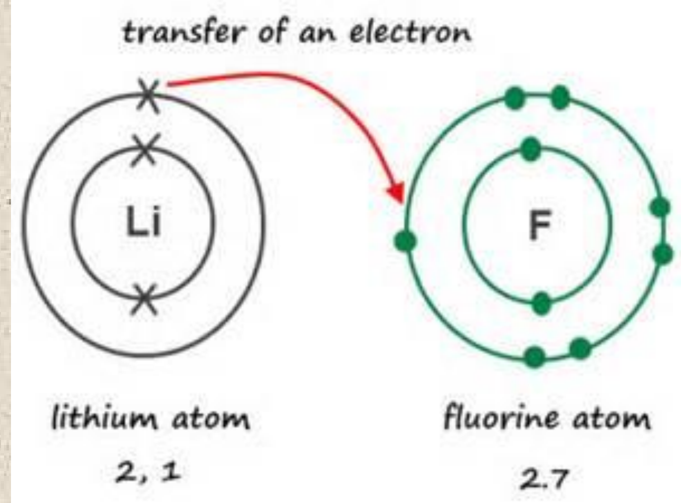
Notice

- The number of atoms of each type of element must be the same on both sides of a balanced equation.
- Subscripts must not be changed to balance an equation.
- A balanced equation tells us the ratio of the number of molecules which react and are produced in a chemical reaction.
- Coefficients can be fractions, although they are usually given as lowest integer multiples.

Return to TOC

Four Driving Forces Favor Chemical Change

1. Formation of a solid
2. Formation of water
3. Transfer of electrons
4. Formation of a gas



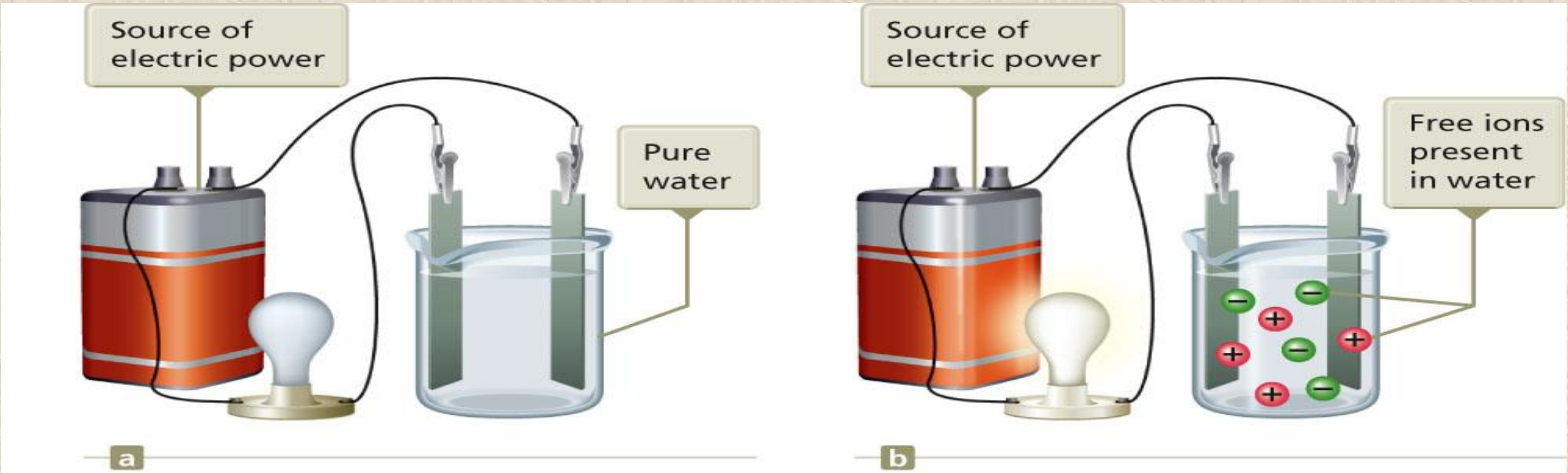
Precipitation

- A reaction in which a solid forms is called a precipitation reaction.
  - Solid = precipitate

Return to TOC

What Happens When an Ionic Compound Dissolves in Water?

- The ions separate and move around independently.
- **Strong electrolyte** – each unit of the substance that dissolves in water produces separated ions.

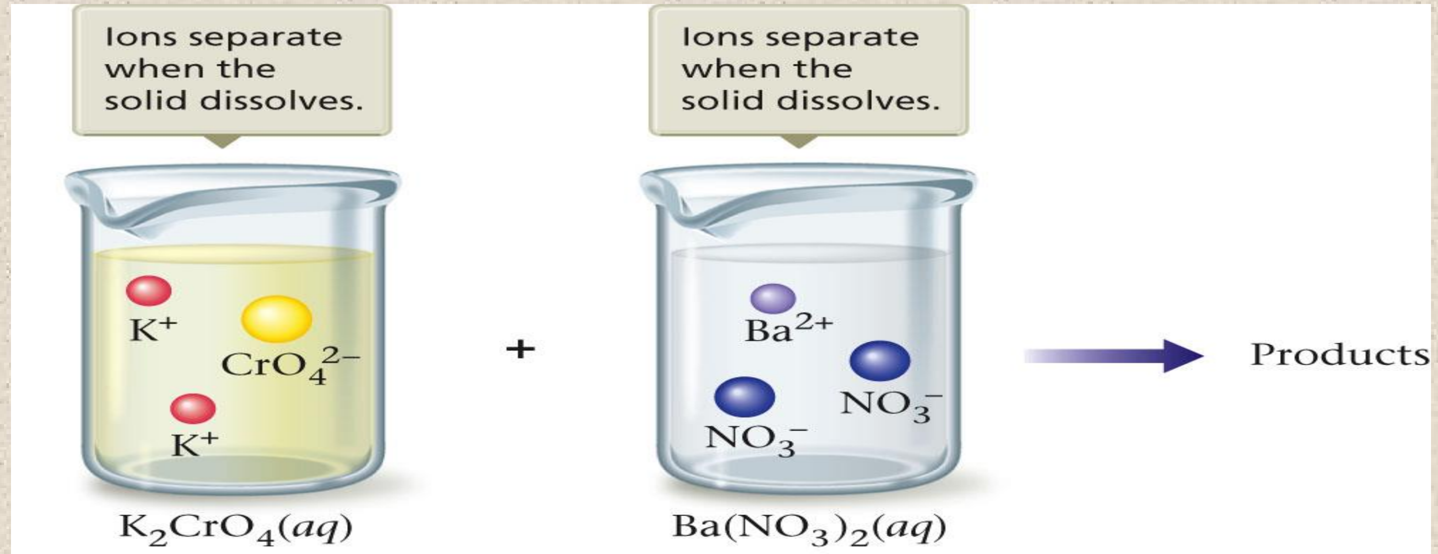


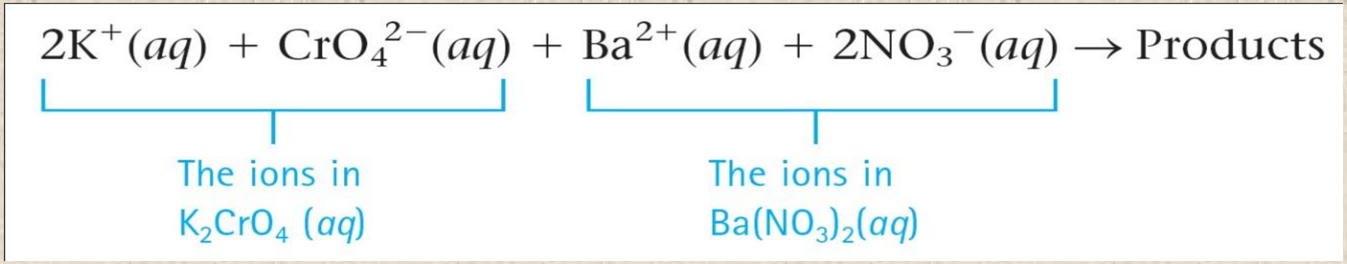
Pure water does not conduct an electric current. The lamp does not light.

When an ionic compound is dissolved in water, current flows and the lamp lights. The result of this experiment is strong evidence that ionic compounds dissolved in water exist in the form of separated ions.

What Happens When an Ionic Compound Dissolves in Water?

- $K_2CrO_4(aq) + Ba(NO_3)_2(aq) \rightarrow$ Products





How to Decide What Products Form

- $K_2CrO_4(aq)$ + $Ba(NO_3)_2(aq)$ $\rightarrow$ Products
- The mixed solution contains four types of ions: K^+, CrO_4^{2-}, Ba^{2+}, and NO_3^-.
- Determine the possible products from the ions in the reactants. The possible ion combinations are:

	NO_3^-	CrO_4^{2-}
K^+	KNO_3	K_2CrO_4
Ba^{2+}	$Ba(NO_3)_2$	$BaCrO_4$

Return to TOC

How to Decide What Products Form

- Decide which is most likely to be the yellow solid formed in the reaction.
- K_2CrO_4*(aq)* reactant
- $Ba(NO_3)_2$*(aq)* reactant
- The possible combinations are KNO_3 and $BaCrO_4$.
  - KNO_3 white solid
  - $BaCrO_4$ yellow solid

Return to TOC

Using Solubility Rules

Table 7.1 General Rules for Solubility of Ionic Compounds (Salts) in Water at 25 °C

1. Most nitrate (NO_3^-) salts are soluble.
2. Most salts of Na^+, K^+, and NH_4^+ are soluble.
3. Most chloride salts are soluble. Notable exceptions are $AgCl$, $PbCl_2$, and Hg_2Cl_2.
4. Most sulfate salts are soluble. Notable exceptions are $BaSO_4$, $PbSO_4$, and $CaSO_4$.
5. Most hydroxide compounds are only slightly soluble.* The important exceptions are NaOH and KOH. $Ba(OH)_2$ and $Ca(OH)_2$ are only moderately soluble.
6. Most sulfide (S^{2-}), carbonate (CO_3^{2-}), and phosphate (PO_4^{3-}) salts are only slightly soluble.*

*The terms *insoluble* and *slightly soluble* really mean the same thing: such a tiny amount dissolves that it is not possible to detect it with the naked eye.

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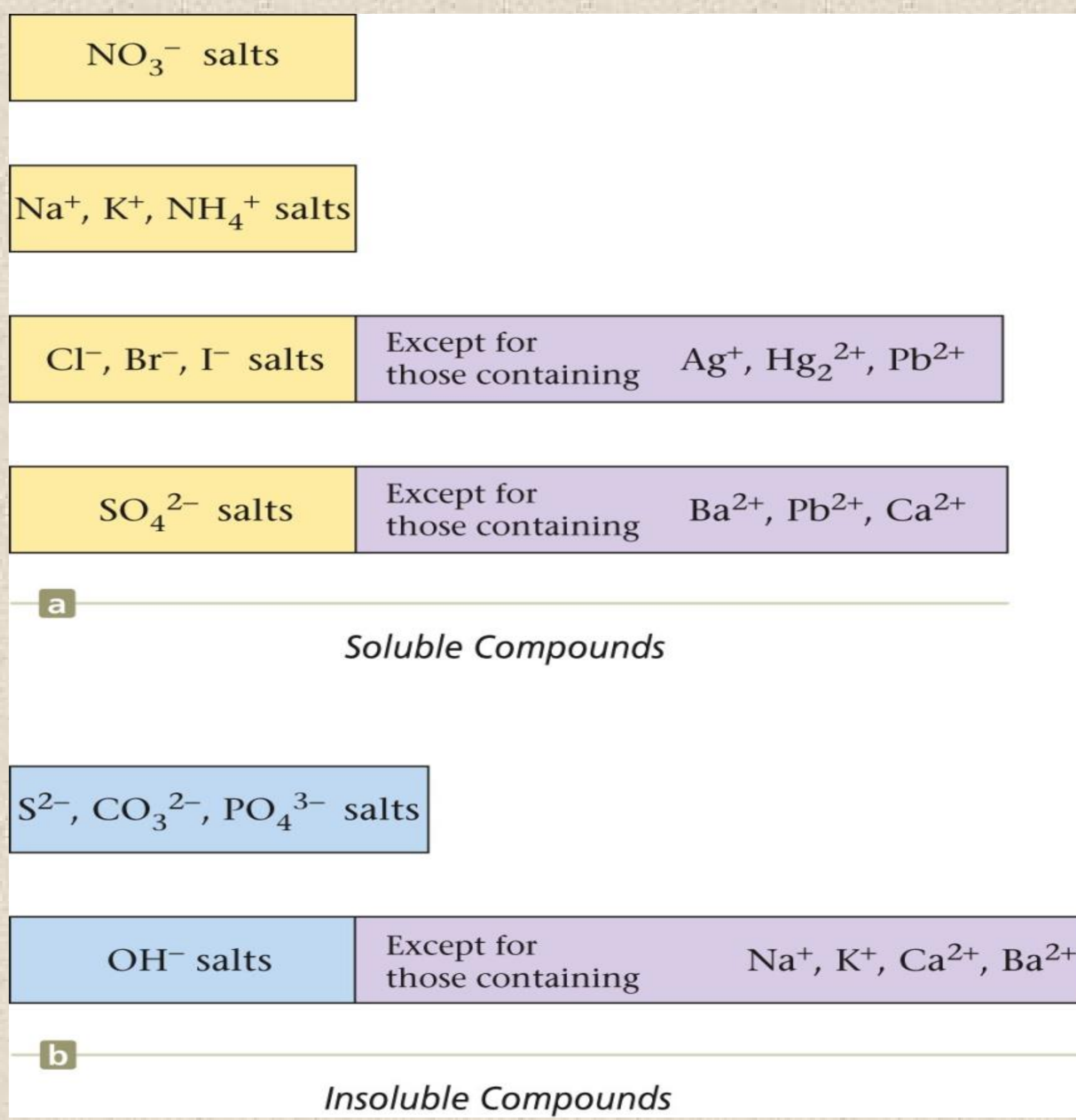
Using Solubility Rules

- Predicting Precipitates
  - Soluble solid
  - Insoluble solid
  - Slightly soluble solid



Let's Practice Determining Solubility

Which of the following are soluble in water?

Na_2CO_3	yes	$Cu(OH)_2$	no
$CaCl_2$	yes	$Ba(OH)_2$	yes
$AgCl$	no	$Ca_3(PO_4)_2$	no
$BaSO_4$	no	$Pb(NO_3)_2$	yes
$(NH_4)_2S$	yes	$PbCl_2$	no

How to Predict Precipitates When Solutions of Two Ionic Compounds Are Mixed

1. Write the reactants as they actually exist before any reaction occurs. Remember that when a salt dissolves, its ions separate.
2. Consider the various solids that could form. To do this, simply exchange the anions of the added salts.
3. Use the solubility rules to decide whether a solid forms and, if so, to predict the identity of the solid.

Return to TOC

Concept Check

Which of the following ions form compounds with Pb^{2+} that are generally soluble in water?

a) S^{2-}
b) Cl^{-}
c) NO_3^{-}
d) SO_4^{2-}
e) Na^{+}

Return to TOC

Concept Check

A sodium phosphate solution reacts with a lead(II) nitrate solution. What precipitate, if any, will form?

a) $Pb_3(PO_4)_2$

b) $NaNO_3$

c) $Pb(NO_3)_2$

d) No precipitate will form.

Return to TOC

Net Ionic Equations

$AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(aq)$

Molecular Equation

1. Divorce $\quad Ag^+ + NO_3^- + Na^+ + Cl^- \longrightarrow$
2. Change Partners $\quad Ag^+ + NO_3^- + Na^+ + Cl^- \longrightarrow AgCl + NaNO_3$
3. Soluble? $\quad Ag^+ + \cancel{NO_3^-} + \cancel{Na^+} + Cl^- \longrightarrow AgCl(s) + \cancel{Na^+} + \cancel{NO_3^-}$

Total Ionic Equation

4. Cross out Spectator Ions $\quad Ag^+ + Cl^- \longrightarrow AgCl(s)$
5. Balance $\quad Ag^+ + Cl^- \longrightarrow AgCl(s)$

Net Ionic Equation

Return to TOC

Net Ionic Equations

$Pb(NO_3)_2(aq) + NaI(aq) \longrightarrow PbI_2(s) + NaNO_3(aq)$

Molecular Equation

1. Divorce — $Pb^{+2} + NO_3^- + Na^+ + I^- \longrightarrow$

2. Change Partners — $Pb^{+2} + NO_3^- + Na^+ + I^- \longrightarrow PbI_2 + NaNO_3$

3. Soluble? — $Pb^{+2} + \cancel{NO_3^-} + \cancel{Na^+} + I^- \longrightarrow PbI_2(s) + \cancel{Na^+} \cancel{NO_3^-}$

Total Ionic Equation

4. Cross out Spectator Ions — $Pb^{+2} + I^- \longrightarrow PbI_2(s)$

5. Balance — $Pb^{+2} + 2\,I^- \longrightarrow PbI_2(s)$

Net Ionic Equation

Return to TOC

Types of Equations for Reactions in Aqueous Solutions

1. **Molecular Equation**
   - Shows the complete formulas of all reactants and products.
   - It does not give a very clear picture of what actually occurs in solution.

$$K_2CrO_4(aq) + Ba(NO_3)_2(aq) \rightarrow BaCrO_4(s) + 2KNO_3(aq)$$

Return to TOC

2. Complete Ionic Equation

- All **strong electrolytes** are shown as ions.

$$\underbrace{2K^+(aq) + CrO_4^{2-}(aq)}_{\text{Ions from } K_2CrO_4} + \underbrace{Ba^{2+}(aq) + 2NO_3^-(aq)}_{\text{Ions from } Ba(NO_3)_2} \rightarrow BaCrO_4(s) + 2K^+(aq) + 2NO_3^-(aq)$$

- Notice: K^+ and NO_3^- ions are present in solution both before and after the reaction.

Return to TOC

3. **Net Ionic Equation**
   - Only those components of the solution that undergo a change.

$$Ba^{2+}(aq) + CrO_4^{2-}(aq) \rightarrow BaCrO_4(s)$$

Return to TOC

Concept Check

Write the correct molecular equation, complete ionic equation, and net ionic equation for the reaction between cobalt(II) chloride and sodium hydroxide.

Molecular Equation:

$$CoCl_2(aq) + 2NaOH(aq) \rightarrow Co(OH)_2(s) + 2NaCl(aq)$$

Complete Ionic Equation:

$$Co^{2+}(aq) + 2Cl^{-}(aq) + 2Na^{+}(aq) + 2OH^{-}(aq) \rightarrow Co(OH)_2(s) + 2Na^{+}(aq) + 2Cl^{-}(aq)$$

Net Ionic Equation:

$$Co^{2+}(aq) + 2OH^{-}(aq) \rightarrow Co(OH)_2(s)$$

Return to TOC

Types of Reactions

1. Combination (one product) $A + B \longrightarrow C$
2. Decompostion (one reactant) $A \longrightarrow B + C$
3. Single Replacement $A + BC \longrightarrow AC + B$
4. Double Replacement $AB + CD \longrightarrow AD(s) + CB(l)$
5. Acid-Base (Neutralization) $HA + BOH \longrightarrow H_2O + BA$
6. Combustion- Organic + $O_2 \longrightarrow CO_2 + H_2O$
7. No Reaction- both products are (aq).

Return to TOC

Arrhenius Acids and Bases

- A strong acid is one in which virtually every molecule dissociates (ionizes) in water to an H^+ ion and an anion.

$$HCl \xrightarrow{H_2O} H^+(aq) + Cl^-(aq)$$

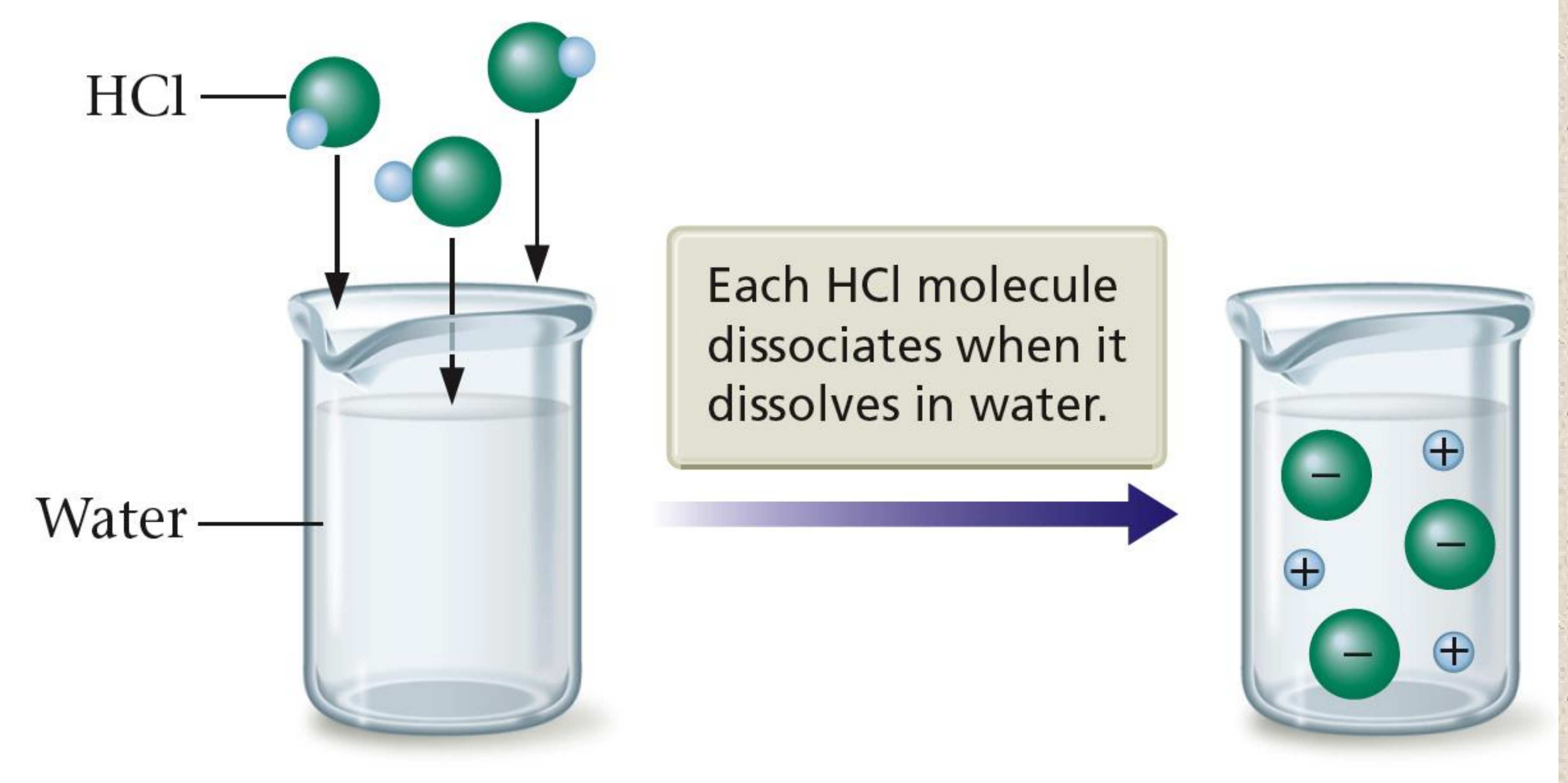
$$HNO_3 \xrightarrow{H_2O} H^+(aq) + NO_3^-(aq)$$

$$H_2SO_4 \xrightarrow{H_2O} H^+(aq) + HSO_4^-(aq)$$

Return to TOC

Arrhenius Acids and Bases

Strong Acids Behave as Strong Electrolytes



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Arrhenius Acids and Bases

- A strong base is a metal hydroxide that is completely soluble in water, giving separate OH^- ions and cations.

  - Most common examples: NaOH and KOH

$$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$$

$$KOH(s) \xrightarrow{H_2O} K^+(aq) + OH^-(aq)$$

Return to TOC

Arrhenius Acids and Bases

- The products of the reaction of a strong acid and a strong base are water and a salt.
  - Salt $\Rightarrow$ Ionic compound
- Net ionic equation
  - $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
- Reaction of H^+ and OH^- is called an acid-base reaction.
  - $H^+ \Rightarrow$ acidic ion
  - $OH^- \Rightarrow$ basic ion

Return to TOC

Arrhenius Acids and Bases

Summary of Strong Acids and Strong Bases

1. The common strong acids are aqueous solutions of HCl, HNO_3, and H_2SO_4.
2. A strong acid is a substance that completely dissociates (ionizes) in water (into H^+ ions and anions).
3. A strong base is a metal hydroxide compound that is very soluble in water (and dissociates into OH^- ions and cations).

Return to TOC

Arrhenius Acids and Bases

Summary of Strong Acids and Strong Bases

4. The net ionic equation for the reaction of a strong acid and a strong base is always the same: it shows the production of water.
5. In the reaction of a strong acid and a strong base, one product is always water and the other is always an ionic compound called a salt, which remains dissolved in the water. This salt can be obtained as a solid by evaporating the water.

Return to TOC

Arrhenius Acids and Bases

Summary of Strong Acids and Strong Bases

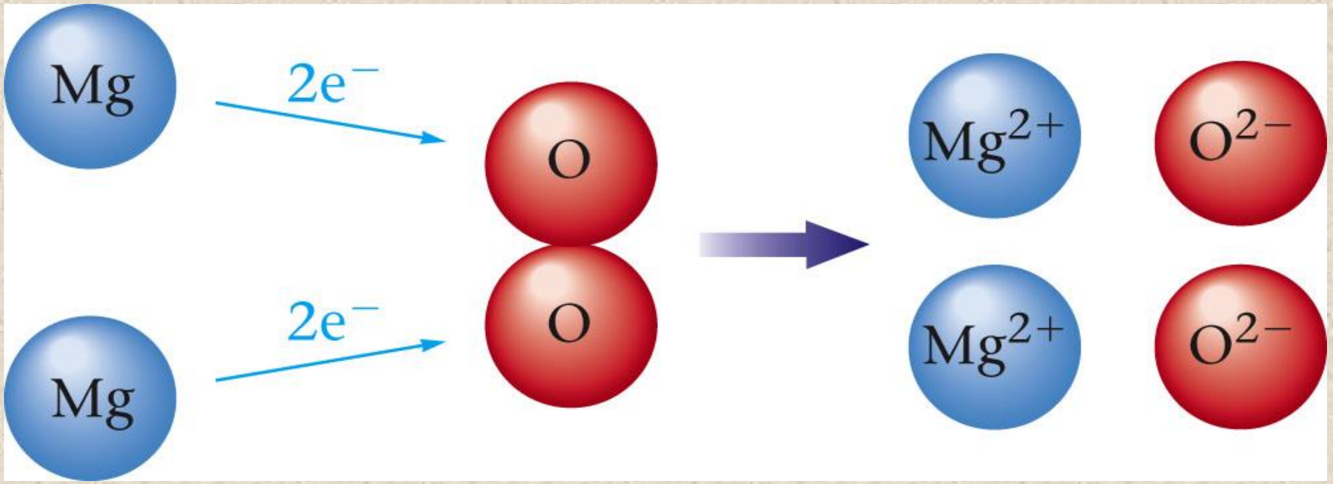
6. The reaction of H^+ and OH^- is often called an acid-base reaction, where H^+ is the acidic ion and OH^- is the basic ion.

Return to TOC

Oxidation-Reduction Reaction

- Reactions between metals and nonmetals involve a transfer of electrons from the metal to the nonmetal.

- A reaction that involves a *transfer of electrons*.
  - $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$



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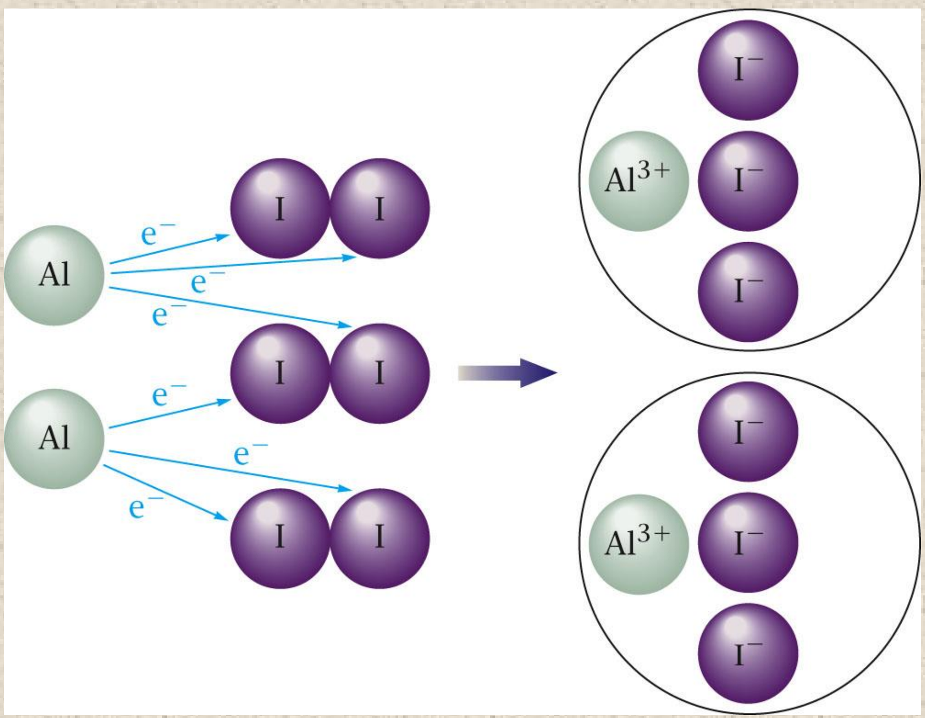
Oxidation-Reduction Reaction

- Transfer of electrons
  - $2Li(s) + F_2(g) \rightarrow 2LiF(s)$



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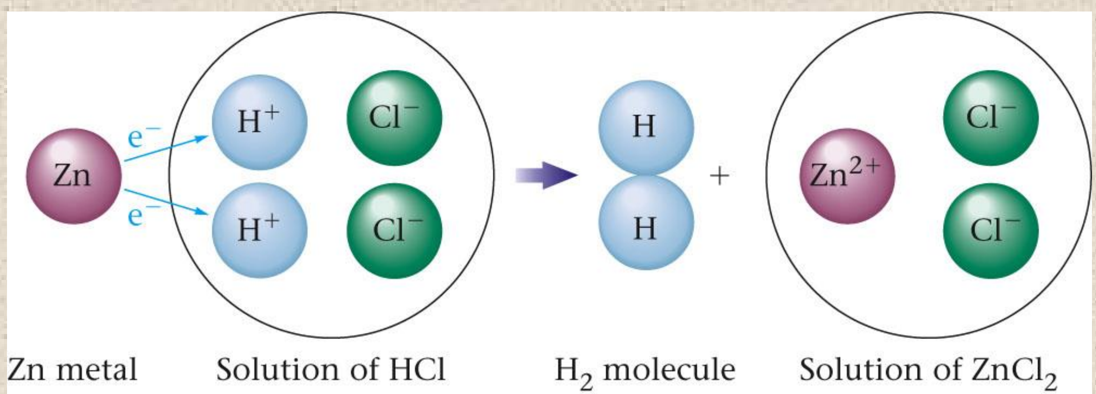
Oxidation-Reduction Reaction

Formation of a Gas

- Oxidation–reduction reaction
- Single–replacement reaction
  - A + BC $\rightarrow$ B + AC

$$\underset{\text{Contains uncharged Zn atoms}}{Zn(s)} + \underbrace{2HCl(aq)}_{\text{Really } 2H^+(aq) + 2Cl^-(aq)} \rightarrow \underset{\text{Contains uncharged H atoms}}{H_2(g)} + \underbrace{ZnCl_2(aq)}_{\text{Really } Zn^{2+}(aq) + 2Cl^-(aq)}$$



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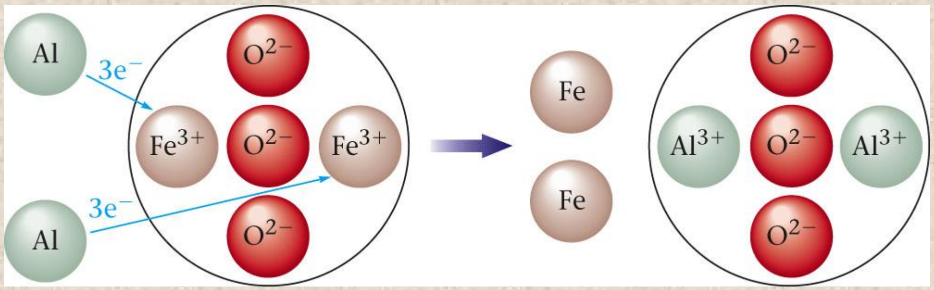
Oxidation-Reduction Reaction

Concept Check

Which of the following best describes what is happening in the following representation of an oxidation–reduction reaction:



a) Metal Al gains 3 e^- and O^{2-} in Fe_2O_3 loses these $3e^-$.

b) Metal Al gains 3 e^- and Fe^{3+} in Fe_2O_3 loses these $3e^-$.

c) Metal Al loses 3 e^- and O^{2-} in Fe_2O_3 gains these $3e^-$.

d) Metal Al loses 3 e^- and Fe^{3+} in Fe_2O_3 gains these $3e^-$.

Return to TOC

Oxidation-Reduction Reaction

Characteristics of Oxidation–Reduction Reactions

1. A metal–nonmetal reaction can always be assumed to be an oxidation–reduction reaction, which involves electron transfer.
2. Two nonmetals can also undergo an oxidation–reduction reaction. At this point we can recognize these cases only by looking for O_2 as a reactant or product. When two nonmetals react, the compound formed is not ionic.

Return to TOC

Precipitation Reaction

- Formation of a solid when two solutions are mixed.

$$K_2CrO_4(aq) + Ba(NO_3)_2(aq) \rightarrow BaCrO_4(s) + 2KNO_3(aq)$$

Solution Solution Solid formed Solution

- Notice this is also a double–displacement reaction.
  - AB + CD $\rightarrow$ AD + CB

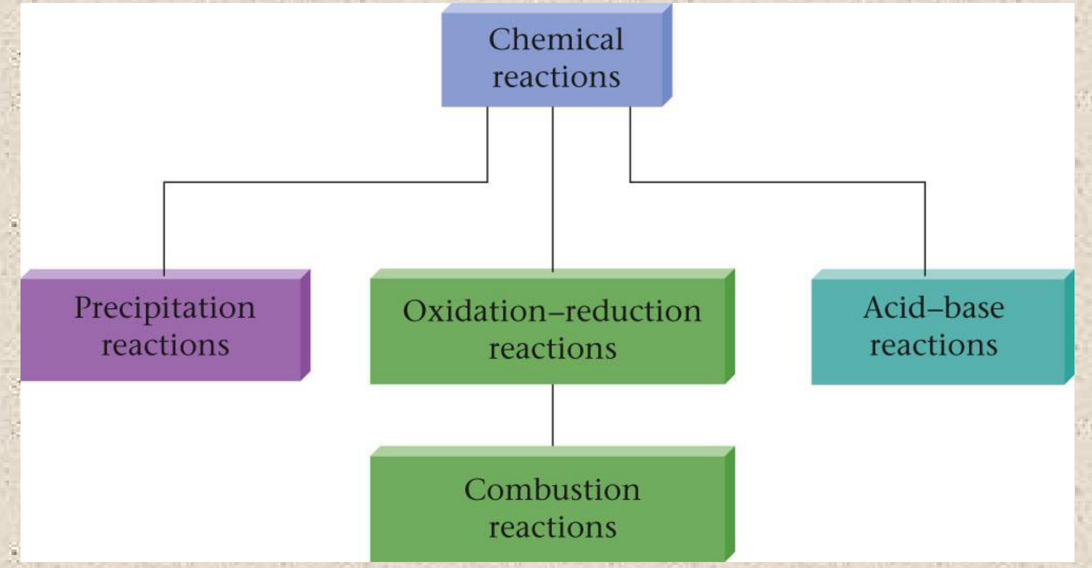
Return to TOC

Acid-Base Reaction

- Involves an H^+ ion that ends up in the product water.
  - $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
  - $HCl(aq) + KOH(aq) \rightarrow H_2O(l) + KCl(aq)$

Combustion Reactions

- Involve oxygen and produce energy (heat) so rapidly that a flame results.
  - $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
  - Special class of oxidation–reduction reactions.



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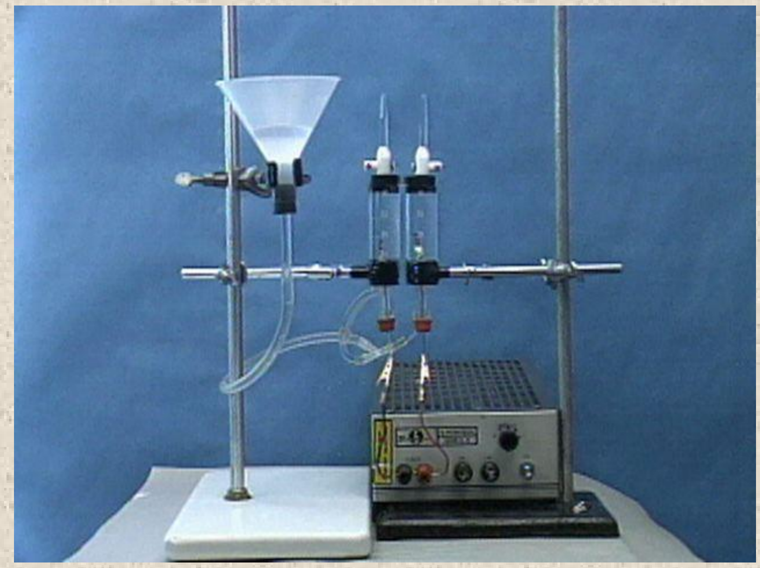
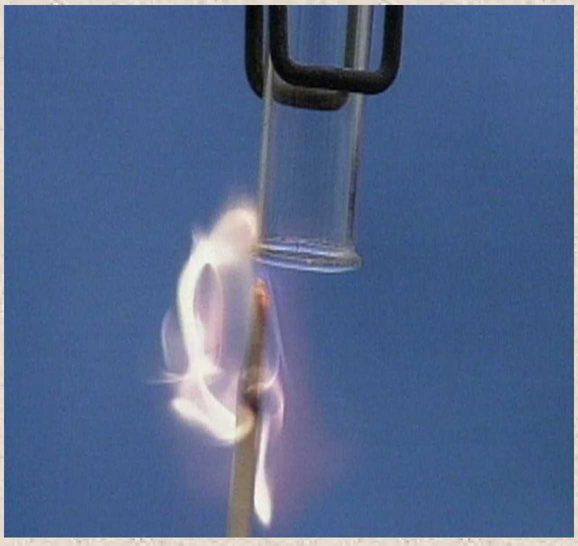
Synthesis (Combination) Reactions

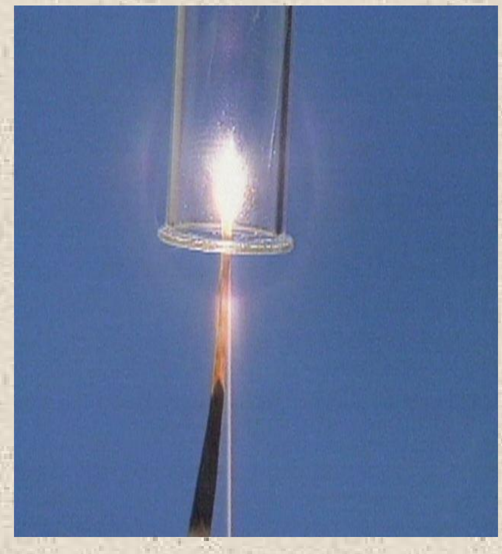
- A compound forms from simpler materials.
  - $C(s) + O_2(g) \rightarrow CO_2(g)$ Only one product!
  - Special class of oxidation–reduction reactions.

Decomposition Reactions

- Occurs when a compound is broken down into simpler substances.
  - $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$ Only one reactant!
  - Special class of oxidation–reduction reactions.







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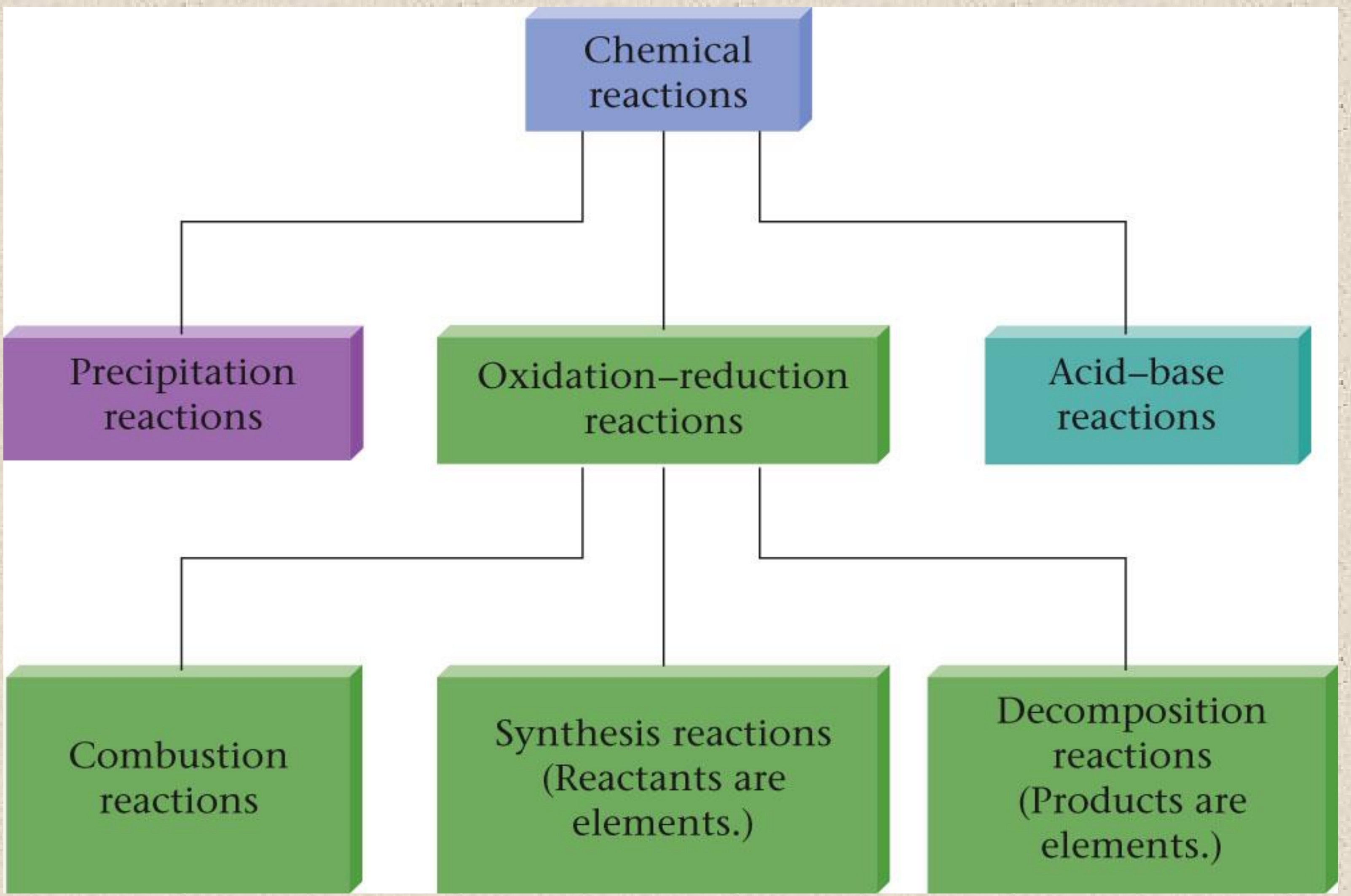
Summary



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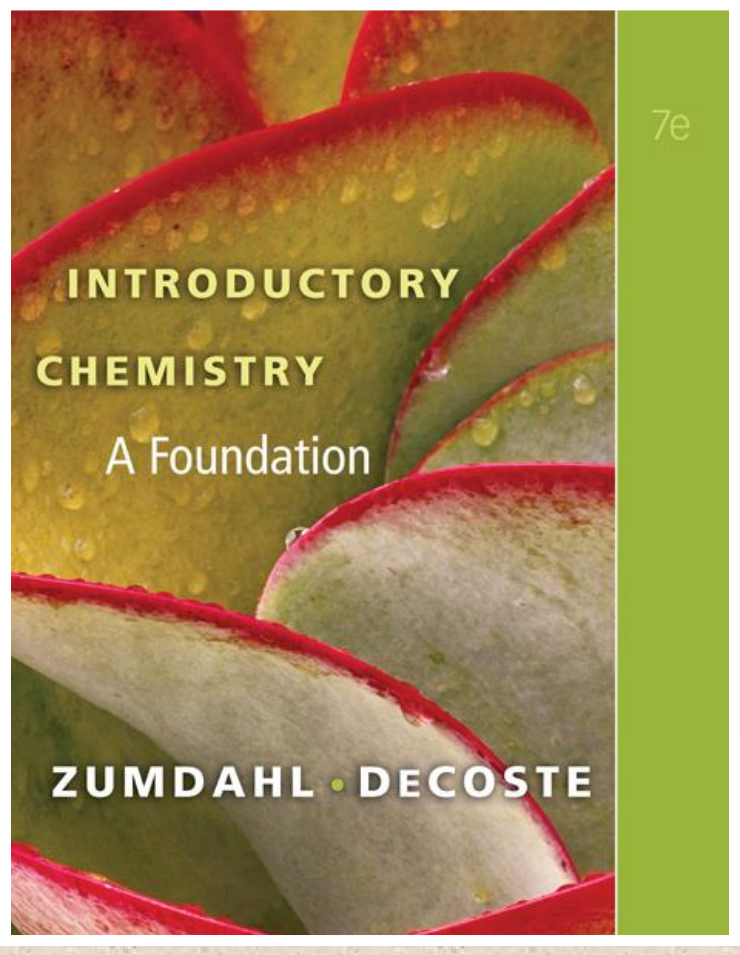


Chapter Four

Reactions Stoichiometry

Counting by Weighing

- Objects do not need to have identical masses to be counted by weighing.
  - All we need to know is the average mass of the objects.
- To count the atoms in a sample of a given element by weighing we must know the mass of the sample and the average mass for that element.

Counting by Weighing

Averaging the Mass of Different Objects

- Two samples containing different types of components (A and B), both contain the same number of components if the ratio of the sample masses is the same as the ratio of the masses of the individual components.

Counting by Weighing

- Elements occur in nature as mixtures of isotopes.
- Carbon = 98.89% ^{12}C

  1.11% ^{13}C

  < 0.01% ^{14}C

Counting by Weighing

Average Atomic Mass for Carbon

98.89% of 12 amu + 1.11% of 13.0034 amu =

exact number

(0.9889)(12 amu) + (0.0111)(13.0034 amu) =

12.01 amu

Atomic Masses: Counting atoms by weighing

- Atoms have very tiny masses so scientists made a unit to avoid using very small numbers.

$$1 \text{ atomic mass unit (amu)} = 1.66 \times 10^{-24} \text{ g}$$

- The average atomic mass for an element is the weighted average of the masses of all the isotopes of an element.

Atomic Masses: Counting atoms by weighing

Average Atomic Mass for Carbon

- Even though natural carbon does not contain a single atom with mass 12.01, for our purposes, we can consider carbon to be composed of only one type of atom with a mass of 12.01.
- This enables us to count atoms of natural carbon by weighing a sample of carbon.

Atomic Masses: Counting atoms by weighing

Example Using Atomic Mass Units

Calculate the mass (in amu) of 431 atoms of carbon.

The mass of 1 carbon atom = 12.01 amu.

Use the relationship as a conversion factor.

$$431\ \text{C}\ \cancel{\text{atoms}} \times \frac{12.01\ \text{amu}}{1\ \text{C}\ \cancel{\text{atom}}} = 5.18 \times 10^{3}\ \text{amu}$$

Atomic Masses: Counting atoms by weighing

Exercise

Calculate the mass (in amu) of 75 atoms of aluminum.

$$75 \text{ atoms Al} \times \frac{26.98 \text{ amu}}{1 \text{ Al atom}} =$$

2.0×10^3 amu Al

The Mole

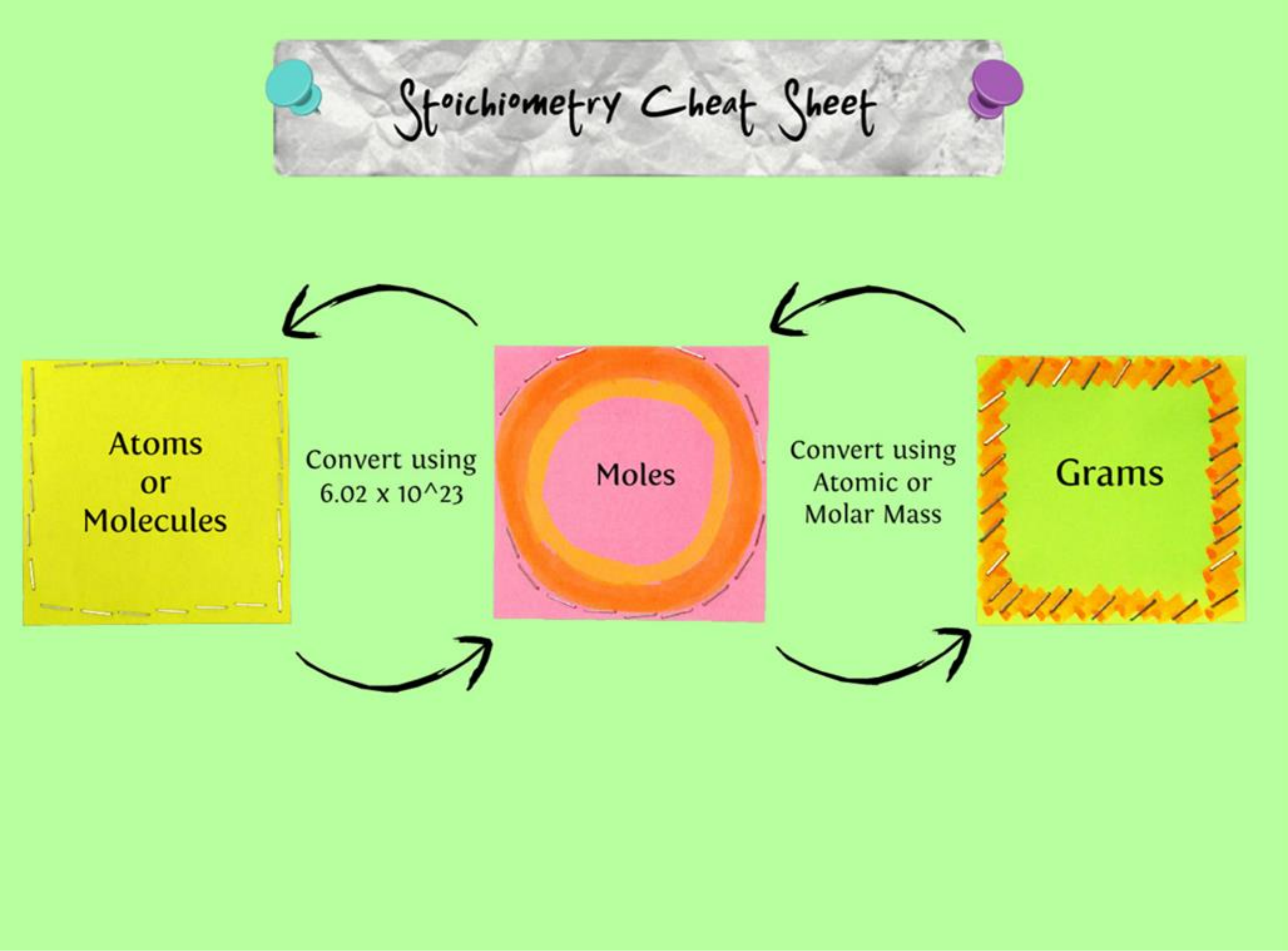
- The number equal to the number of carbon atoms in 12.01 grams of carbon.
- 1 mole of anything = 6.022×10^{23} units of that thing (Avogadro's number).
- 1 mole C = 6.022×10^{23} C atoms = 12.01 g C

$$\frac{1 \text{ mol C}}{6.022\times10^{23}\text{ C atoms}} \text{ or } \frac{6.022\times10^{23}\text{ C atoms}}{1 \text{ mol C}}$$

$$\frac{1 \text{ mol C}}{12.01 \text{ g C}} \text{ or } \frac{12.01 \text{ g C}}{1 \text{ mol C}}$$

$$\frac{6.022\times10^{23}\text{ C atoms}}{12.01 \text{ g C}} \text{ or } \frac{12.01 \text{ g C}}{6.022\times10^{23}\text{ C atoms}}$$

The Mole


Stoichiometry Cheat Sheet
Atoms or Molecules
Convert using 6.02 x 10^23
Moles
Convert using Atomic or Molar Mass
Grams

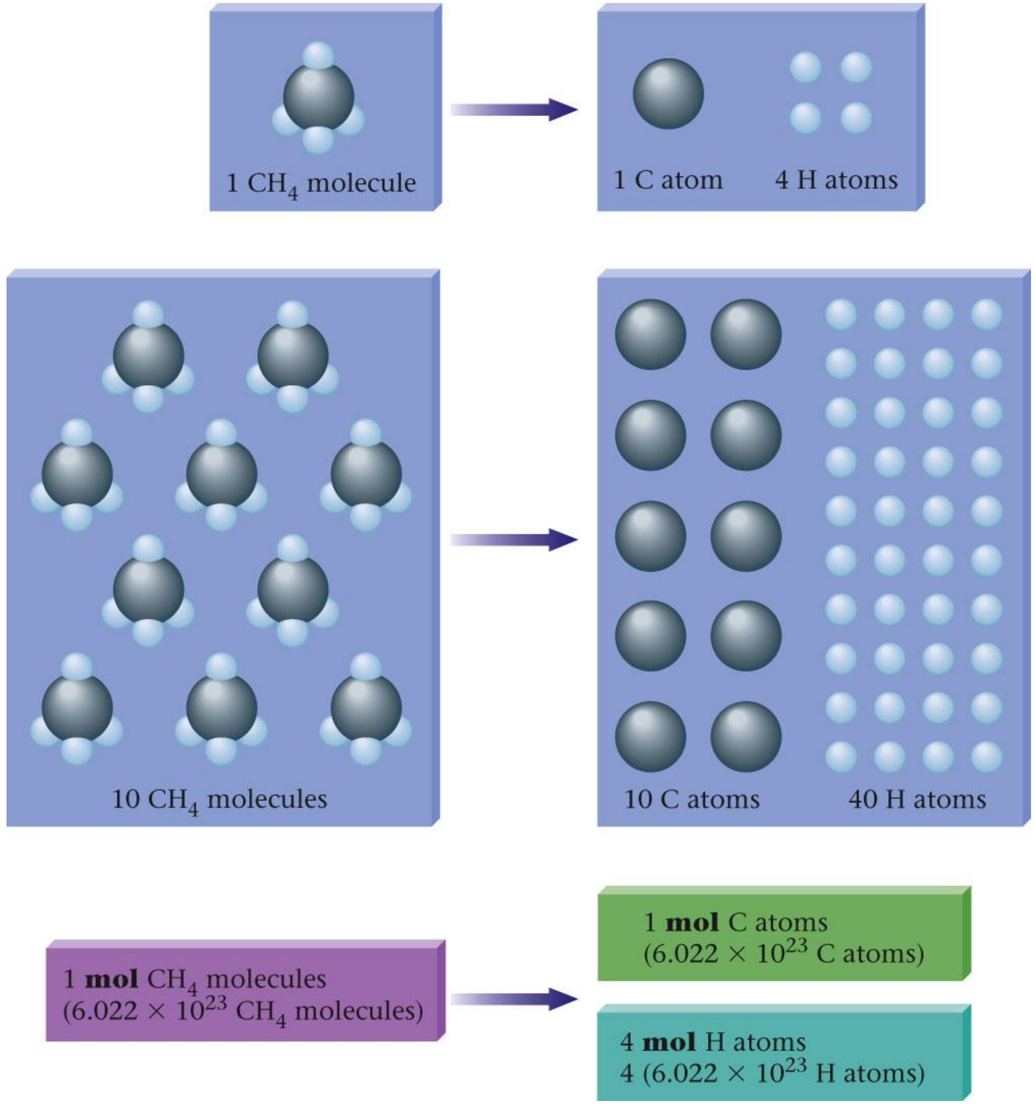
The Mole

Avogadro's Number of various elements





The Mole



1 CH_4 molecule
1 C atom
4 H atoms
10 CH_4 molecules
10 C atoms
40 H atoms
1 **mol** CH_4 molecules
(6.022×10^{23} CH_4 molecules)
1 **mol** C atoms
(6.022×10^{23} C atoms)
4 **mol** H atoms
4 (6.022×10^{23} H atoms)

The Mole

- A sample of an element with a mass equal to that element's average atomic mass (expressed in g) contains one mole of atoms (6.022×10^{23} atoms).
- Comparison of 1-Mol Samples of Various Elements

Table 8.2 Comparison of 1-Mol Samples of Various Elements

Element	Number of Atoms Present	Mass of Sample (g)
Aluminum	6.022×10^{23}	26.98
Gold	6.022×10^{23}	196.97
Iron	6.022×10^{23}	55.85
Sulfur	6.022×10^{23}	32.07
Boron	6.022×10^{23}	10.81
Xenon	6.022×10^{23}	131.3

The Mole

Concept Check

Determine the number of copper atoms in a 63.55 g sample of copper.

$$63.55 \text{ g Cu} \; \frac{1 \text{ mole Cu}}{63.55 \text{ g Cu}} \; \frac{6.022\text{x } 10^{23} \text{ atoms Cu}}{1 \text{ mole Cu atoms}} =$$

6.022×10^{23} Cu atoms

The Mole

Concept Check

Which of the following is closest to the average mass of one atom of copper?

a) 63.55 g
b) 52.00 g
c) 58.93 g
d) 65.38 g
e) $\boxed{1.055 \times 10^{-22}\ \text{g}}$

$$1\ \text{Cu atom} \times \frac{1\ \text{mol Cu}}{6.022 \times 10^{23}\ \text{Cu atoms}} \times \frac{63.55\ \text{g Cu}}{1\ \text{mol Cu}} = 1.055 \times 10^{-22}\ \text{g Cu}$$

The Mole

Concept Check

A sample of 26.98 grams of Al has the same number of atoms as ________ grams of Au.

a) 26.98 g
b) 13.49 g
c) $\boxed{\text{197.0 g}}$
d) 256.5 g

$$26.98\text{ g Al}\ \frac{1\text{ mole Al}}{26.98\text{ g Al}}\ \frac{1\text{ mole Au}}{1\text{ mole Al}}\ \frac{196.97\text{g Au}}{1\text{ mole Au}} = 197.0\text{ g Au}$$

The Mole

Concept Check

Which of the following 100.0 g samples contains the greatest number of atoms?

a) Magnesium
b) Zinc
c) Silver

Information Given by Chemical Equations

- A balanced chemical equation gives relative numbers (or moles) of reactant and product molecules that participate in a chemical reaction.

- The coefficients of a balanced equation give the relative numbers of molecules.

Return to TOC

Information Given by Chemical Equations

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

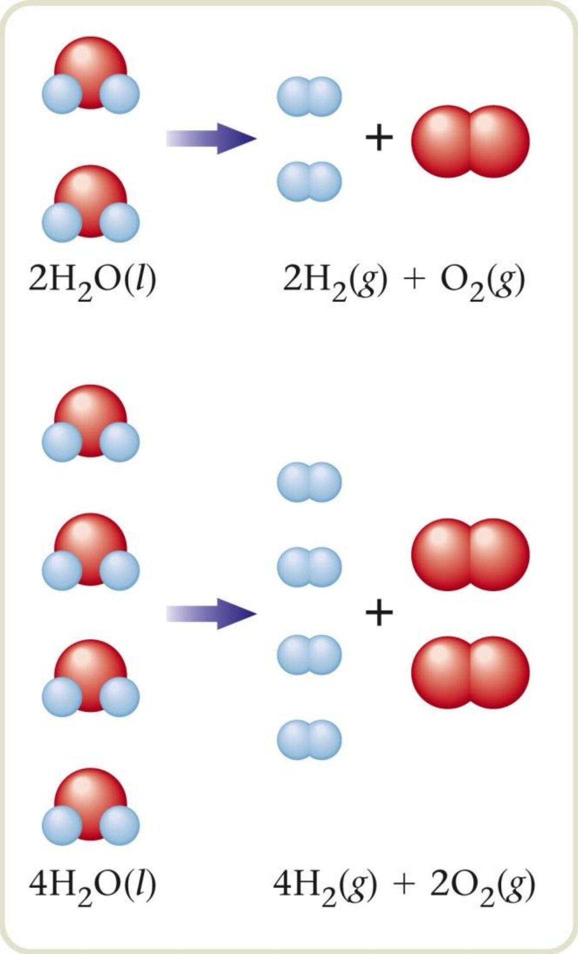
- The equation is balanced.
- All atoms present in the reactants are accounted for in the products.
- 1 *molecule* of ethanol reacts with 3 *molecules* of oxygen to produce 2 *molecules* of carbon dioxide and 3 *molecules* of water.
- 1 *mole* of ethanol reacts with 3 *moles* of oxygen to produce 2 *moles* of carbon dioxide and 3 *moles* of water.

Return to TOC

Mole–Mole Relationships

- A balanced equation can predict the moles of product that a given number of moles of reactants will yield.
- 2 mol of H_2O yields 2 mol of H_2 and 1 mol of O_2.
- 4 mol of H_2O yields 4 mol of H_2 and 2 mol of O_2.

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$



Return to TOC

Mole–Mole Relationships

Mole Ratio

- *The mole ratio allows us to convert from moles of one substance in a balanced equation to moles of a second substance in the equation.*

Return to TOC

Mole–Mole Relationships

Example

Consider the following balanced equation:

$$Na_2SiF_6(s) + 4Na(s) \rightarrow Si(s) + 6NaF(s)$$

How many moles of NaF will be produced if 3.50 moles of Na is reacted with excess Na_2SiF_6?

Where are we going?

- We want to determine the number of moles of NaF produced by Na with excess Na_2SiF_6.

What do we know?

- The balanced equation.
- We start with 3.50 mol Na.

Return to TOC

Mole–Mole Relationships

Example

Consider the following balanced equation:

$$Na_2SiF_6(s) + 4Na(s) \rightarrow Si(s) + 6NaF(s)$$

How many moles of NaF will be produced if 3.50 moles of Na is reacted with excess Na_2SiF_6?

How do we get there?

Mole Ratio

$$3.50 \text{ mol } \cancel{\text{Na}} \times \frac{6 \text{ mol NaF}}{4 \text{ mol } \cancel{\text{Na}}} = 5.25 \text{ mol NaF}$$

↑

Starting Amount

Return to TOC

Mole–Mole Relationships

Exercise

Propane, C_3H_8, is a common fuel used in heating homes in rural areas. Predict how many moles of CO_2 are formed when 3.74 moles of propane are burned in excess oxygen according to the equation:

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

a) 11.2 moles
b) 7.48 moles
c) 3.74 moles
d) 1.25 moles

$$3.74 \text{ moles } C_3H_8 \times \frac{3 \text{ moles } CO_2}{1 \text{ mole } C_3H_8}$$

Return to TOC

Molar Mass

- Mass in grams of one mole of the substance:

  Molar Mass of N = 14.01 g/mol

  Molar Mass of H_2O = 18.02 g/mol

  (2 × 1.008 g) + 16.00 g

  Molar Mass of $Ba(NO_3)_2$ = 261.35 g/mol

  137.33 g + (2 × 14.01 g) + (6 × 16.00 g)

Return to TOC

Molar Mass

Calculations Using Molar Mass

- Moles of a compound = $\dfrac{\text{mass of the sample (g)}}{\text{molar mass of the compound }(\frac{\text{g}}{\text{mol}})}$

$$\text{mol} = \cancel{\text{g}} \times \frac{\text{mol}}{\cancel{\text{g}}}$$

- Mass of a sample (g) = (moles of sample)(molar mass of compound)

$$\text{g} = \cancel{\text{mol}} \times \frac{\text{g}}{\cancel{\text{mol}}}$$

Return to TOC

Molar Mass

Exercise

What is the molar mass of nickel(II) carbonate?

a) $\boxed{118.7 \text{ g/mol}}$
b) 134.7 g/mol
c) 178.71 g/mol
d) 296.09 g/mol

The formula for nickel(II) carbonate is $NiCO_3$. Therefore the molar mass is:

58.69 + 12.01 + 3(16.00) = 118.70 g/mol.

Return to TOC

Molar Mass

Exercise

Consider equal mole samples of N_2O, $Al(NO_3)_3$, and KCN. Rank these from least to most number of nitrogen atoms in each sample.

a) KCN, $Al(NO_3)_3$, N_2O

b) $Al(NO_3)_3$, N_2O, KCN

c) KCN, N_2O, $Al(NO_3)_3$

d) $Al(NO_3)_3$, KCN, N_2O

Return to TOC

Percent Composition of Compounds

- Mass percent of an element:

$$\text{mass \%} = \frac{\text{mass of element in compound}}{\text{mass of compound}} \times 100\%$$

- For iron in iron(III) oxide, (Fe_2O_3):

$$\text{mass \% Fe} = \frac{2(55.85\text{ g})}{2(55.85\text{ g}) + 3(16.00\text{ g})} = \frac{111.70\text{ g}}{159.70\text{ g}} \times 100\% = 69.9\%$$

Return to TOC

Percent Composition of Compounds

Percent Composition

What is the % composition of $C_6H_{12}O_6$?

$$6 \text{ moles C} \ \frac{12.011\text{g C}}{1 \text{ mole C}} = 72.066 \text{ g C} \qquad \frac{72.066}{180.155} \times 100\% = 40.0\% \text{ C}$$

$$12 \text{ moles H} \ \frac{1.0079\text{g H}}{1 \text{ mole H}} = 12.0948 \text{ g H} \qquad \frac{12.0948}{180.155} \times 100\% = 6.7\% \text{ H}$$

$$6 \text{ moles O} \ \frac{15.999\text{g O}}{1 \text{ mole O}} = 95.994\text{g O} \qquad \frac{95.994}{180.155} \times 100\% = 53.3\% \text{ O}$$

$$\overline{180.155}$$

% composition is 40.0% C, 6.7% H, and 53.3% O

Return to TOC

Percent Composition of Compounds

Tetrodotoxin has the empirical formula $C_{11}H_{17}N_3O_8$. Calculate the mass percentages (g/g) of the four element in this compound.

Solution:

1. Calculate molar mass of $C_{11}H_{17}N_3O_8$, by finding the mass contributed by each element

   C : 11x12 = 132 g / mol
   H : 1x17 = 17
   N : 3x14 = 42
   O : 8x16 = 126
   tot = 317 g / mol

2. Calculate the mass of one mole of tetrodotoxin

   317 g / mol x 1 mol = 317 g

3. Divide the mass of each element by the mass of the compound.

   Example C: 132 g / 317 g = 0.42 g/g (42 %)

Return to TOC

Steps for Calculating the Masses of Reactants and Products in Chemical Reactions

1. Balance the equation for the reaction.
2. Convert the masses of reactants or products to moles.
3. Use the balanced equation to set up the appropriate mole ratio(s).
4. Use the mole ratio(s) to calculate the number of moles of the desired reactant or product.
5. Convert from moles back to masses.

Return to TOC

Mass Calculations

Stoichiometry

- Stoichiometry is the process of using a balanced chemical equation to determine the relative masses of reactants and products involved in a reaction.

Return to TOC

Mass Calculations

Example

For the following unbalanced equation:

$$Cr(s) + O_2(g) \rightarrow Cr_2O_3(s)$$

How many grams of chromium(III) oxide can be produced from 15.0 g of solid chromium and excess oxygen gas?

Where are we going?

- We want to determine the mass of Cr_2O_3 produced by Cr with excess O_2.

Return to TOC

Mass Calculations

Example

For the following unbalanced equation:

$$Cr(s) + O_2(g) \rightarrow Cr_2O_3(s)$$

How many grams of chromium(III) oxide can be produced from 15.0 g of solid chromium and excess oxygen gas?

What do we know?

- The unbalanced equation.
- We start with 15.0 g Cr.
- We know the atomic masses of chromium (52.00 g/mol) and oxygen (16.00 g/mol) from the periodic table.

Return to TOC

Mass Calculations

Example

For the following unbalanced equation:

$$Cr(s) + O_2(g) \rightarrow Cr_2O_3(s)$$

How many grams of chromium(III) oxide can be produced from 15.0 g of solid chromium and excess oxygen gas?

What do we need to know?

- We need to know the balanced equation.

  $$4Cr(s) + 3O_2(g) \rightarrow 2Cr_2O_3(s)$$

- We need the molar mass of Cr_2O_3.

  152.00 g/mol

Return to TOC

Mass Calculations

How do we get there?

$$4Cr(s) + 3O_2(g) \rightarrow 2Cr_2O_3(s)$$

- Convert the mass of Cr to moles of Cr.

$$15.0\ \cancel{\text{g Cr}} \times \frac{1\ \text{mol Cr}}{52.00\ \cancel{\text{g Cr}}} = 0.288\ \text{mol Cr}$$

- Determine the moles of Cr_2O_3 produced by using the mole ratio from the balanced equation.

$$0.288\ \cancel{\text{mol Cr}} \times \frac{2\ \text{mol}\ Cr_2O_3}{4\ \cancel{\text{mol Cr}}} = 0.144\ \text{mol}\ Cr_2O_3$$

Return to TOC

Mass Calculations

How do we get there?

$$4Cr(s) + 3O_2(g) \rightarrow 2Cr_2O_3(s)$$

- Convert moles of Cr_2O_3 to grams of Cr_2O_3.

$$0.144 \text{ mol } \cancel{Cr_2O_3} \times \frac{152.00 \text{ g } Cr_2O_3}{1 \text{ mol } \cancel{Cr_2O_3}} = 21.9 \text{ g } Cr_2O_3$$

- Conversion string:

$$15.0 \text{ } \cancel{\text{g Cr}} \times \frac{1 \text{ } \cancel{\text{mol Cr}}}{52.00 \text{ } \cancel{\text{g Cr}}} \times \frac{2 \text{ } \cancel{\text{mol } Cr_2O_3}}{4 \text{ } \cancel{\text{mol Cr}}} \times \frac{152.00 \text{ g } Cr_2O_3}{1 \text{ } \cancel{\text{mol } Cr_2O_3}} = \boxed{21.9 \text{ g } Cr_2O_3}$$

Return to TOC

Mass Calculations

$$C_6H_6 + 3\ H_2 \rightarrow C_6H_{12}$$

How many grams of cyclohexane C_6H_{12} can be made from 4.5 grams of benzene C_6H_6?

Use the table method.

Moles level

Balanced?

Grams level 4.5g ? ?

$$4.5\ g\ C_6H_6 \frac{1\ mole\ C_6H_6}{78.1134\ g} \frac{1\ mole\ C_6H_{12}}{1\ mole\ C_6H_6} \frac{84.174g\ C_6H_{12}}{1\ mole\ C_6H_{12}} = 4.8\ g\ C_6H_{12}$$

How many grams of H_2 will be needed?

$$4.5\ g\ C_6H_6 \frac{1\ mole\ C_6H_6}{78.1134\ g} \frac{3\ moles\ H_2}{1\ mole\ C_6H_6} \frac{2.0158g\ H_2}{1\ mole\ H_2} = 0.35\ g\ H_2$$

Return to TOC

Mass Calculations

Exercise

Tin(II) fluoride is added to some dental products to help prevent cavities. Tin(II) fluoride is made according to the following equation:

$$Sn(s) + 2HF(aq) \rightarrow SnF_2(aq) + H_2(g)$$

How many grams of tin(II) fluoride can be made from 55.0 g of hydrogen fluoride if there is plenty of tin available to react?

a) 431 g

b) 215 g

c) 72.6 g

d) 1.37 g

$$55.0 \text{ g HF} \times \frac{1 \text{ mol HF}}{20.008 \text{ g HF}} \times \frac{1 \text{ mol SnF}_2}{2 \text{ mol HF}} \times \frac{156.71 \text{ g SnF}_2}{1 \text{ mol SnF}_2}$$

Return to TOC

Mass Calculations

Exercise

Consider the following reaction:

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

What mass of H_2O is needed to completely react with 20.0 g of PCl_3?

$\boxed{\text{a) 7.87 g}}$

b) 0.875 g

c) 5.24 g

d) 2.62 g

$$20.0 \text{ g } PCl_3 \times \frac{1 \text{ mol } PCl_3}{137.32 \text{ g } PCl_3} \times \frac{3 \text{ mol } H_2O}{1 \text{ mol } PCl_3} \times \frac{18.016 \text{ g } H_2O}{1 \text{ mol } H_2O}$$

Return to TOC

Exercise

Consider the following reaction where X represents an unknown element:

$$6\ X(s) + 2\ B_2O_3(s) \rightarrow B_4X_3(s) + 3\ XO_2(g)$$

If 175 g of X reacts with diboron trioxide to produce 2.43 moles of B_4X_3, what is the identity of X?

a) Ge

b) Mg

c) Si

$\boxed{\text{d) C}}$

$$2.43 \text{ mol } B_4X_3 \times \frac{6 \text{ mol X}}{1 \text{ mol } B_4X_3} = 14.6 \text{ mol X}$$

$$\text{Molar Mass} = \frac{\#\text{ grams}}{\#\text{ moles}} = \frac{175 \text{ g X}}{14.6 \text{ mol X}} = 12.0 \text{ g/mol}$$

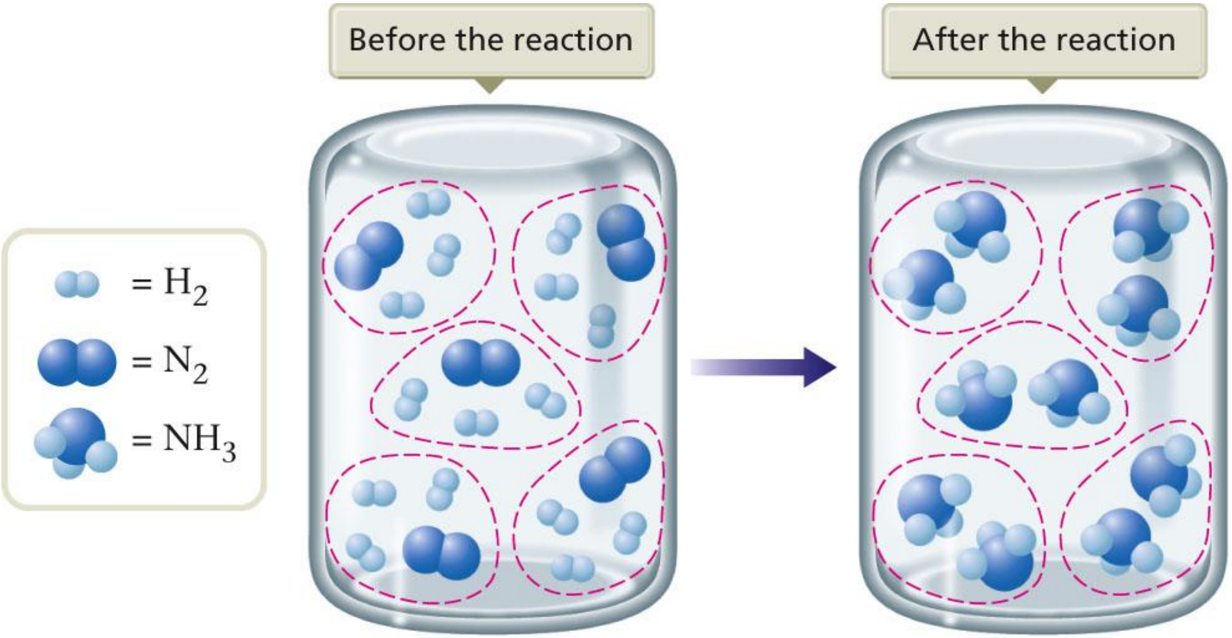
Return to TOC

The Concept of Limiting Reactants

Stoichiometric Mixture

- Contains the relative amounts of reactants that matches the numbers in the balanced equation.

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

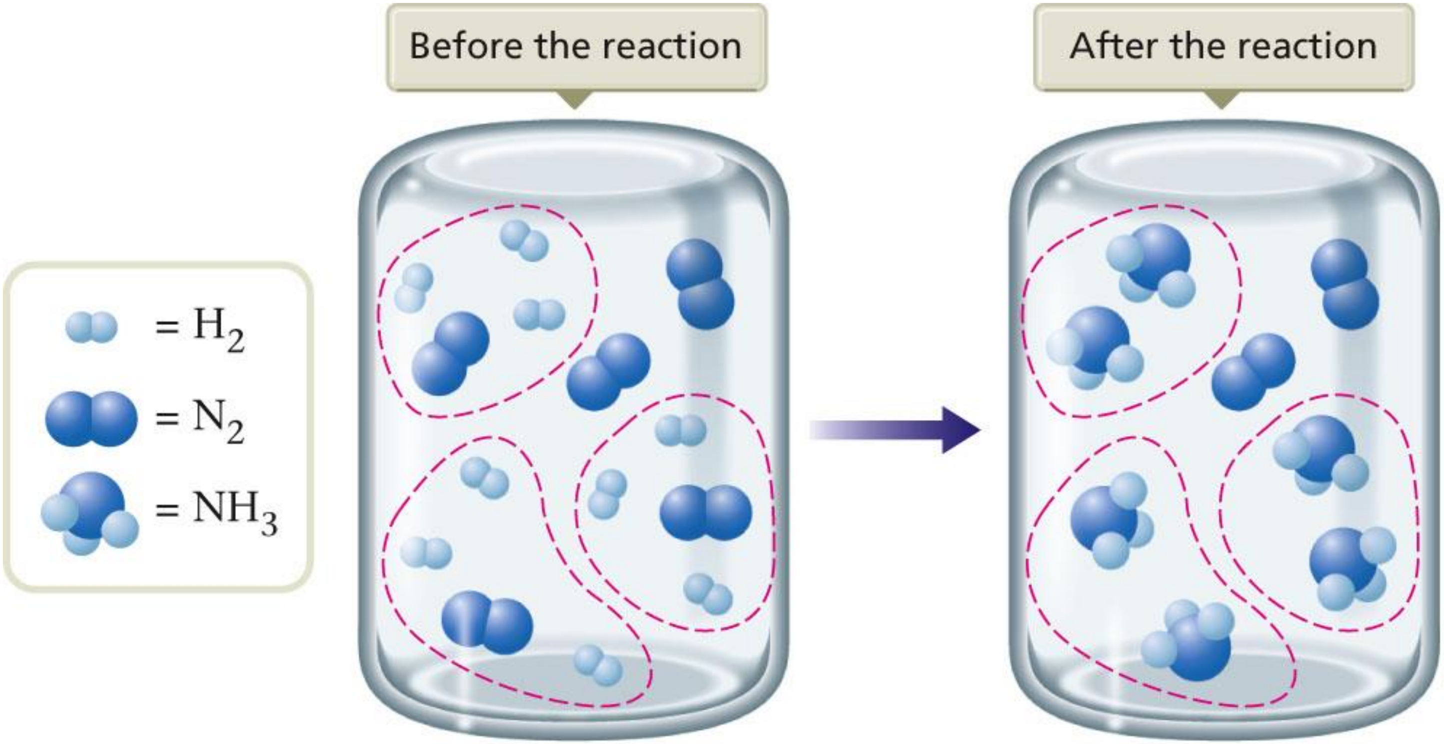


Return to TOC

The Concept of Limiting Reactants

Limiting Reactant Mixture

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$



Return to TOC

The Concept of Limiting Reactants

Limiting Reactant Mixture

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

- Limiting reactant is the reactant that runs out first and thus limits the amounts of product(s) that can form.
  - H_2

Return to TOC

Calculations Involving a Limiting Reactant

- Determine which reactant is limiting to calculate correctly the amounts of products that will be formed.

Return to TOC

Calculations Involving a Limiting Reactant

Limiting Reactants

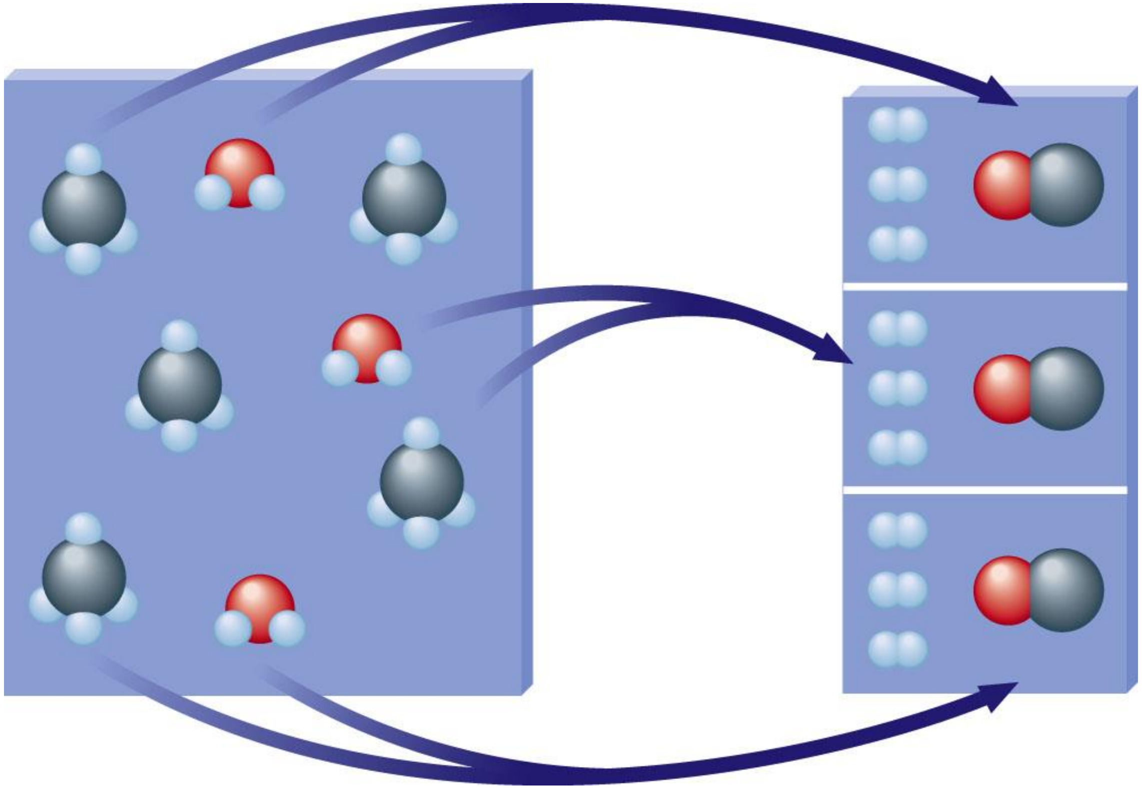
- Methane and water will react to form products according to the equation:

$$CH_4 + H_2O \rightarrow 3H_2 + CO$$

Return to TOC

Calculations Involving a Limiting Reactant

Limiting Reactants



- H_2O molecules are used up first, leaving two CH_4 molecules unreacted.

Return to TOC

Calculations Involving a Limiting Reactant

Limiting Reactants

- The amount of products that can form is limited by the water.
- Water is the limiting reactant.
- Methane is in excess.

Return to TOC

Calculations Involving a Limiting Reactant

Steps for Solving Stoichiometry Problems Involving Limiting Reactants

1. Write and balance the equation for the reaction.
2. Convert known masses of reactants to moles.
3. Using the numbers of moles of reactants and the appropriate mole ratios, determine which reactant is limiting.
4. Using the amount of the limiting reactant and the appropriate mole ratios, compute the number of moles of the desired product.
5. Convert from moles of product to grams of product, using the molar mass (if this is required by the problem).

Return to TOC

Calculations Involving a Limiting Reactant

Example

- *You know that chemical A reacts with chemical B. You react 10.0 g of A with 10.0 g of B.*
  - What information do you need to know in order to determine the mass of product that will be produced?

Return to TOC

Let's Think About It

- **Where are we going?**
  - To determine the mass of product that will be produced when you react 10.0 g of A with 10.0 g of B.
- **What do we need to know?**
  - The mole ratio between A, B, and the product they form. In other words, we need to know the balanced reaction equation.
  - The molar masses of A, B, and the product they form.

Return to TOC

Calculations Involving a Limiting Reactant

Example – Continued

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

$$A + 3B \rightarrow 2C$$

Return to TOC

Calculations Involving a Limiting Reactant

Example – Continued

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

$$A + 3B \rightarrow 2C$$

- **How do we get there?**
  - Convert known masses of reactants to moles.

$$10.0 \text{ g A} \times \frac{1 \text{ mol A}}{10.0 \text{ g A}} = 1.00 \text{ mol A}$$

$$10.0 \text{ g B} \times \frac{1 \text{ mol B}}{20.0 \text{ g B}} = 0.500 \text{ mol B}$$

Return to TOC

Calculations Involving a Limiting Reactant

Example – Continued

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

$$\text{A} + 3\text{B} \rightarrow 2\text{C}$$

- **How do we get there?**
  - Determine which reactant is limiting.

$$1.00 \text{ mol A} \times \frac{3 \text{ mol B}}{1 \text{ mol A}} = 3.00 \text{ mol B required to react with all of the A}$$

  - Only 0.500 mol B is available, so B is the limiting reactant.

Return to TOC

Calculations Involving a Limiting Reactant

Example – Continued

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

$$A + 3B \rightarrow 2C$$

- **How do we get there?**
  - Compute the number of moles of C produced.

$$0.500 \text{ mol B} \times \frac{2 \text{ mol C}}{3 \text{ mol B}} = 0.333 \text{ mol C produced}$$

Return to TOC

Calculations Involving a Limiting Reactant

Example – Continued

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

$$A + 3B \rightarrow 2C$$

- **How do we get there?**
  - Convert from moles of C to grams of C using the molar mass.

$$0.333 \text{ mol C} \times \frac{25.0 \text{ g C}}{1 \text{ mol C}} = 8.33 \text{ g C}$$

Return to TOC

Calculations Involving a Limiting Reactant

Notice

- We cannot simply add the total moles of all the reactants to decide which reactant mixture makes the most product.
- We must always think about how much product can be formed by using what we are given, and the ratio in the balanced equation.

Return to TOC

Calculations Involving a Limiting Reactant

$$H_2 + I_2 \rightarrow 2\,HI$$

How many grams of HI can be formed from 2.00 g H_2 and 2.00 g of I_2?

2.00 g 2.00 g ?

$$2.00\text{ g } H_2 \; \frac{1\text{ mole } H_2}{2.0158\text{ g}} \; \frac{2\text{ mole HI}}{1\text{ mole } H_2} \; \frac{127.9124\text{g HI}}{1\text{ mole HI}} = \cancel{254\text{ g HI}}$$

$$2.00\text{g } I_2 \; \frac{1\text{ mole } I_2}{253.810\text{ g}} \; \frac{2\text{ moles HI}}{1\text{ mole } I_2} \; \frac{127.9124\text{g HI}}{1\text{ mole HI}} = \boxed{2.02\text{ g HI}}$$

Limiting Reactant is always the smallest value!

I_2 is the LIMITING REACTANT and H_2 is in EXCESS.

Return to TOC

Calculations Involving a Limiting Reactant

Exercise

How many grams of NH_3 can be produced from the mixture of 3.00 g each of nitrogen and hydrogen by the Haber process?

$$N_2 + 3H_2 \rightarrow 2NH_3$$

a) 2.00 g
b) 3.00 g
c) $\boxed{3.64\text{ g}}$
d) 1.82 g

3.00 g N_2(1mole/28g N_2)(2 moles NH_3/1 moles N_2)
(17 g NH_3/1 mole NH_3) = 3.64 g NH_3

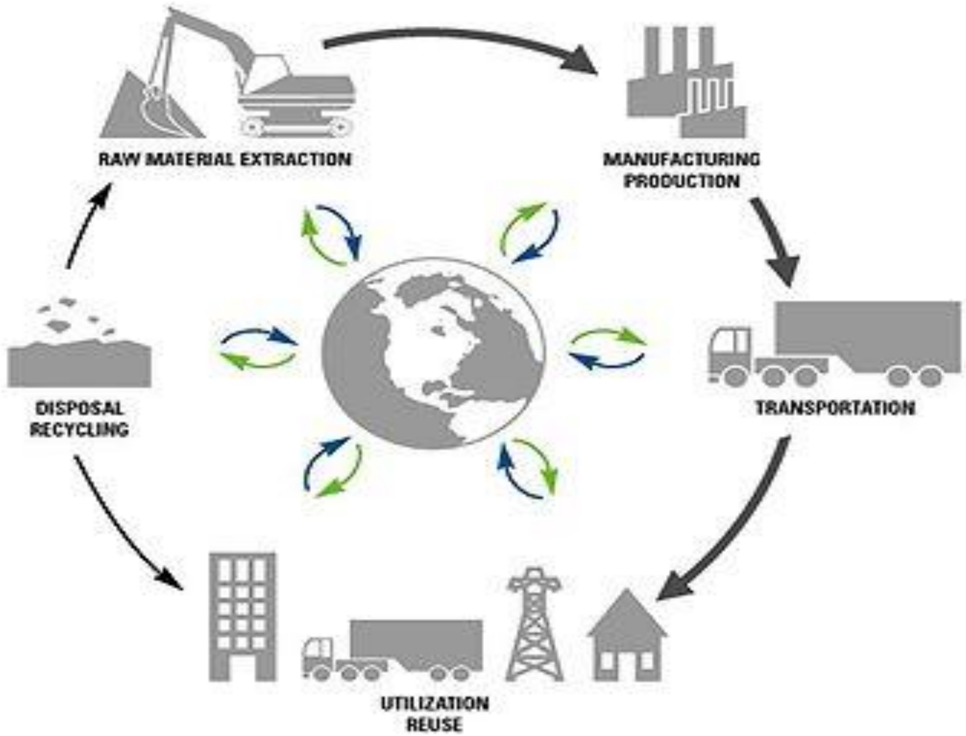
3.00 g H_2(1mole/2g H_2)(2 moles NH_3/3 moles H_2)
(17 g NH_3/1 mole NH_3) = 17.0 g NH_3

Return to TOC

Percent Yield

Percent Yield

- An important indicator of the efficiency of a particular laboratory or industrial reaction.



Return to TOC

Percent Yield

- Theoretical Yield
  - The maximum amount of a given product that can be formed when the limiting reactant is completely consumed.

- The Actual Yield
  - amount actually produced of a reaction is usually less than the maximum expected (theoretical yield).

Return to TOC

Percent Yield

Percent Yield

- The actual amount of a given product as the percentage of the theoretical yield.

$$\frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100\% = \text{percent yield}$$

Return to TOC

Percent Yield

Example – Recall

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

  $$A + 3B \rightarrow 2C$$

  We determined that 8.33 g C should be produced.

  7.23 g of C is what was actually made in the lab. What is the percent yield of the reaction?

- **Where are we going?**
  - We want to determine the percent yield of the reaction.

Return to TOC

Percent Yield

Example – Recall

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

  $$A + 3B \rightarrow 2C$$

  We determined that 8.33 g C should be produced. 7.23 g of C is what was actually made in the lab. What is the percent yield of the reaction?
- **What do we know?**
  - We know the actual and theoretical yields.

Return to TOC

Example – Recall

- *You react 10.0 g of A with 10.0 g of B. What mass of product will be produced given that the molar mass of A is 10.0 g/mol, B is 20.0 g/mol, and C is 25.0 g/mol? They react according to the equation:*

  $$A + 3B \rightarrow 2C$$

  We determined that 8.33 g C should be produced. 7.23 g of C is what was actually made in the lab. What is the percent yield of the reaction?

- **How do we get there?**

$$\frac{7.23\text{ g C}}{8.33\text{ g C}} \times 100\% = 86.8\%$$

Return to TOC

Sample problem

Q - What is the % yield of H_2O if 138 g H_2O is produced from 16 g H_2 and excess O_2?

Step 1: write the balanced chemical equation

$$2H_2 + O_2 \rightarrow 2H_2O$$

Step 2: determine actual and theoretical yield. Actual is given, theoretical is calculated:

$$\# \text{ g } H_2O = 16\ \cancel{\text{g } H_2} \times \frac{1\ \cancel{\text{mol } H_2}}{2.02\ \cancel{\text{g } H_2}} \times \frac{2\ \cancel{\text{mol } H_2O}}{2\ \cancel{\text{mol } H_2}} \times \frac{18.02 \text{ g } H_2O}{1\ \cancel{\text{mol } H_2O}} = 1\underline{4}3 \text{ g}$$

Step 3: Calculate % yield

$$\% \text{ yield} = \frac{\text{actual}}{\text{theoretical}} \times 100\% = \frac{138 \text{ g } H_2O}{1\underline{4}3 \text{ g } H_2O} \times 100\% = 9\underline{6}.7\%$$

Return to TOC

Percent Yield

Exercise

Find the percent yield of product if 1.50 g of SO_3 is produced from 1.00 g of O_2 and excess sulfur via the reaction:

$$2S + 3O_2 \rightarrow 2SO_3$$

a) 40.0%
b) 80.0%
c) $\boxed{\text{89.8\%}}$
d) 92.4%

Return to TOC

Percent Yield

Exercise

Consider the following reaction:

$$P_4(s) + 6F_2(g) \rightarrow 4PF_3(g)$$

What mass of P_4 is needed to produce 85.0 g of PF_3 if the reaction has a 64.9% yield?

a) 29.9 g

b) 46.1 g

c) 184 g

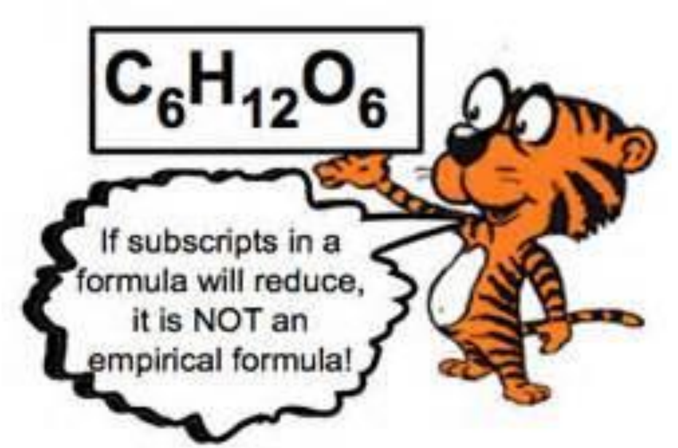
d) 738 g

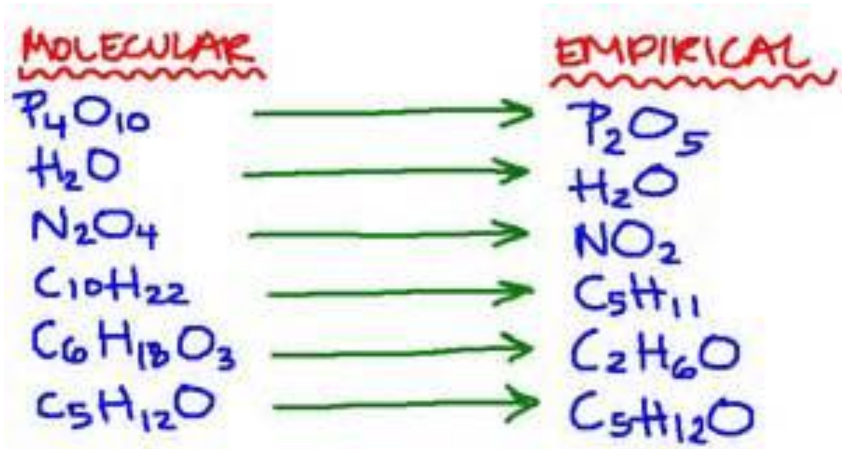
Return to TOC

Formulas of Compounds

Empirical Formulas

- The empirical formula of a compound is the simplest whole number ratio of the atoms present in the compound.
- The empirical formula can be found from the percent composition of the compound.





Return to TOC

Formulas of Compounds

Formulas

- Empirical formula = CH
  - Simplest whole-number ratio
- Molecular formula = (empirical formula)$_n$
  [n = integer]
- Molecular formula = C_6H_6 = $(CH)_6$
  - Actual formula of the compound

Return to TOC

Calculations of Empirical Formulas

Steps for Determining the Empirical Formula of a Compound

A gaseous compound containing carbon and hydrogen was analyzed and found to consist of 83.65% carbon by mass. Determine the empirical formula of the compound.

- Obtain the mass of each element present (in grams).

Assume you have 100 g of the compound.

83.65% C = 83.65 g C

(100.00 – 83.65)

16.35% H = 16.35 g H

Return to TOC

Calculations of Empirical Formulas

Steps for Determining the Empirical Formula of a Compound

2. Determine the number of moles of each type of atom present.

$$83.65 \text{ g C} \times \frac{1 \text{ mol C}}{12.01 \text{ g C}} = 6.965 \text{ mol C}$$

$$16.35 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 16.22 \text{ mol H}$$

Return to TOC

Calculations of Empirical Formulas

Steps for Determining the Empirical Formula of a Compound

3. Divide the number of moles of each element by the smallest number of moles to convert the smallest number to 1. If all of the numbers so obtained are integers, these are the subscripts in the empirical formula. If one or more of these numbers are not integers, go on to step 4.

$$\frac{6.965 \text{ mol C}}{6.965 \text{ mol}} = 1$$

$$\frac{16.22 \text{ mol H}}{6.965 \text{ mol}} = 2.33$$

Return to TOC

Calculations of Empirical Formulas

Steps for Determining the Empirical Formula of a Compound

4. Multiply the numbers you derived in step 3 by the smallest integer that will convert all of them to whole numbers. This set of whole numbers represents the subscripts in the empirical formula.

C: $1 \times 3 = 3$

H: $2.33 \times 3 = 7$

The empirical formula is C_3H_7.

Return to TOC

Empirical Formula from the Percent Composition

What is the Empirical Formula if the % composition is 43.2% K, 39.1% Cℓ, and some O?

$$43.2 \text{ g K} \; \frac{1 \text{ mole K}}{39.098\text{g K}} = 1.10 \text{ moles K} \quad \frac{1.10}{1.10} = 1.0 \text{ mole K}$$

$$39.1 \text{ g C}\ell \; \frac{1 \text{ mole C}\ell}{35.453 \text{ g C}\ell} = 1.10 \text{ moles C}\ell \quad \frac{1.10}{1.10} = 1.0 \text{ mole C}\ell$$

$$17.7 \text{ g O} \; \frac{1 \text{ mole O}}{15.999\text{g O}} = 1.10 \text{ moles O} \quad \frac{1.10}{1.10} = 1.0 \text{ mole O}$$

The Empirical Formula is KCℓO (MW =90.550g/mole)

If MW of the real formula is 90.550, what is the actual formula?

(90.550)/(90.550) = 1 KCℓO x 1 = KCℓO

Return to TOC

Empirical Formula from the Percent Composition

What is the Empirical Formula if the % composition is 40.0% C, 6.7% H, and 53.3% O?

$$40.0 \text{ g C} \frac{1 \text{ mole C}}{12.011\text{g C}} = 3.33 \text{ moles C} \qquad \frac{3.33}{3.33} = 1.0 \text{ mole C}$$

$$6.7 \text{ g H} \frac{1 \text{ mole H}}{1.0079\text{g H}} = 6.64 \text{ moles H} \qquad \frac{6.64}{3.33} = 2.0 \text{ mole H}$$

$$53.3 \text{ g O} \frac{1 \text{ mole O}}{15.999\text{g O}} = 3.33 \text{ moles O} \qquad \frac{3.33}{3.33} = 1.0 \text{ mole O}$$

The Empirical Formula is CH_2O (MW =30.026)

If MW of the real formula is 180.155, what is the actual formula?

(180.155)/(30.026) = 6 $\quad CH_2O \times 6 = C_6H_{12}O_6$

Return to TOC

Molecular Formula

- The molecular formula is the exact formula of the molecules present in a substance.
- The molecular formula is always an integer multiple of the empirical formula.

Molecular formula = (empirical formula)$_n$

where n is a whole number

Return to TOC

Molecular Formula

Continued Example…

A gaseous compound containing carbon and hydrogen was analyzed and found to consist of 83.65% carbon by mass. The molar mass of the compound is 86.2 g/mol. You determined the empirical formula to be C_3H_7. What is the molecular formula of the compound?

Molar mass of C_3H_7 = 43.086 g/mol

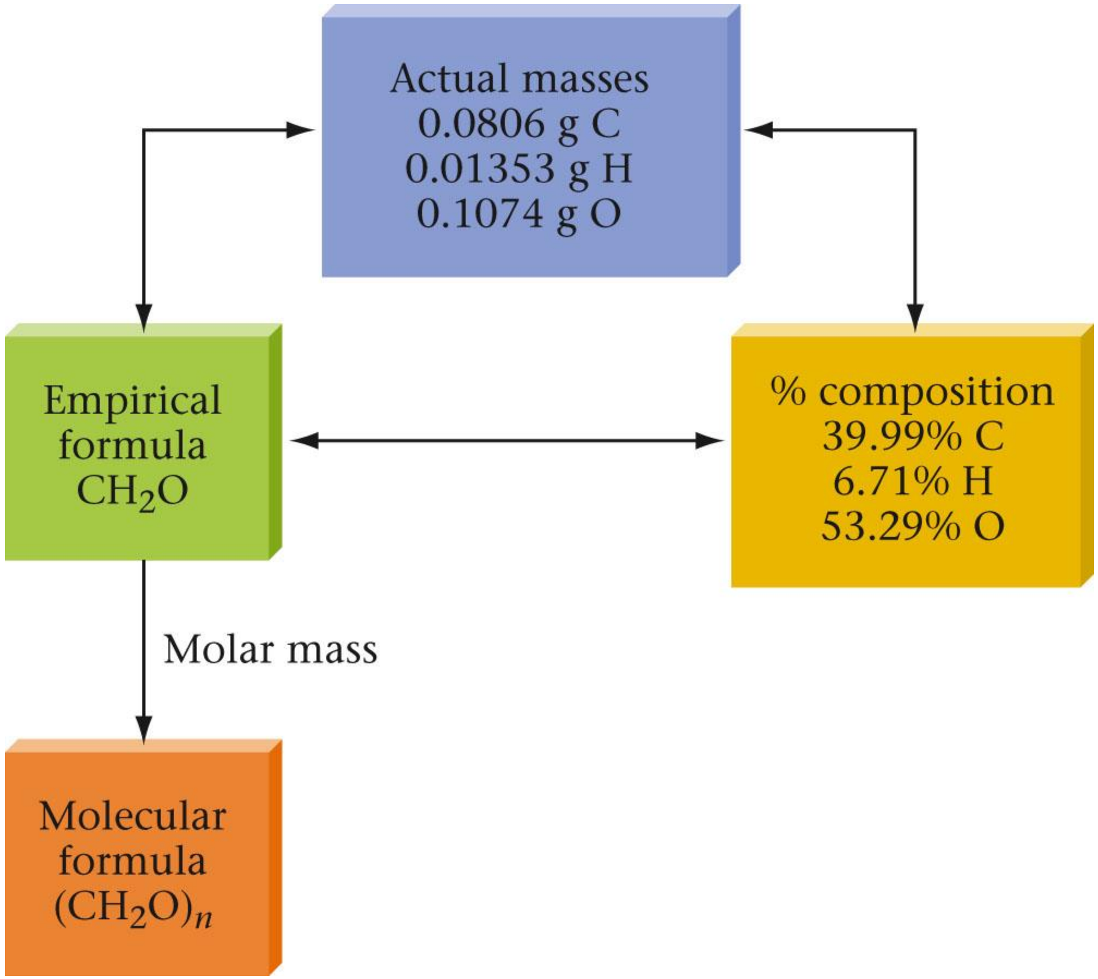
$$\frac{86.2 \text{ g/mol}}{43.086 \text{ g/mol}} = 2$$

$C_3H_7 \times 2 = C_6H_{14}$

Return to TOC

Molecular Formula

Flow Chart of the Relationships of Calculations



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Molecular Formula

Exercise

The composition of adipic acid is 49.3% C, 6.9% H, and 43.8% O (by mass). The molar mass of the compound is about 146 g/mol. The empirical formula is $C_3H_5O_2$. What is the molecular formula?

Mw $C_3H_5O_2$ = 3x12 + 5x1 + 2x16 =73 g/mole

146/73 = 2 2 x $C_3H_5O_2$ = $C_6H_{10}O_4$

Return to TOC

ATOMIC STRUCTURE AND THE PERIODIC TABLE

CHAPTER 5

1.Lights and spectroscopy

- Spectroscopy-It is the study of dispersion of the light of an object into the colors that make it up. It is concerned with the absorption, emission or scattering of electromagnetic radiation by atoms or molecules.

A. The characteristics of light.

- Rectilinear properties of light: light moves in a straight line.
- Its frequency does not change when it passes from one medium to another.
- It passes completely through transparent objects, partially through translucent objects and not at all in opaque objects.
- Its speed changes when it goes from one medium to another

The dual nature of light

- Light has a wave nature and a particle nature.
- When we say wave nature we mean that light acts like a wave. some common terms which are associated with this concept:
- **Amplitude**: - the height of the wave
- **Frequency**: - the number of waves to pass a point per minute
- **Wave speed**: - how fast the wave is moving
- **Wave length**: - the distance between successive waves
- E(energy)=h(Planck's constant)v(velocity) = hC(speed of light)/λ(wave length)
- Where $h=6.62\times10^{-34}$ J.S and $C=3\times10^{8}$ m/s

b. Quantization and photons

- Quantization: process of converting the sampled analogue signal into impulses of discrete values of amplitude.
- Energy is lost in small definite amounts called "quantum"
- $E=nhv$
- Where E is the energy, v is the frequency, n is a positive integer called the quantum number and h is Planck's constant.

Quantum numbers

- The principal quantum number (n) - a positive number having the values; 1, 2, 3…
- The azimuthal quantum number(l)- also known as angular momentum or subsidiary quantum number and having values from 0 to n-1
- The magnetic quantum number(m_l)- also known as the orbital-orientation quantum number and having values from –l to +l
- The electron spin quantum number(m_s) has values of either +1/2 (represented by ↑) or -1/2 (represented by↓)

2. The structure of the hydrogen atom

a. **The spectrum of atomic hydrogen**

- certain properties of the electron in a hydrogen atom are quantized.
- Bohr proposed three postulates for his model:
- The hydrogen atom has stationary states. A stationary state is a certain allowable energy level which is associated with a fixed circular orbit of the electron around the nucleus.
- The atom does not radiate energy in a stationary state.
- $E_{ph}=E_f-E_i=hv$, where the subscripts f and I represent the final and initial states respectively

B. particles and waves

- wave-particle duality- this theory is that matter behaves both as particle or wave depending on the circumstances. What that means is that particles show behaviors like interference or diffraction.
- In reference to what we mentioned above we are not saying that matter displays different behavior in a macroscopic world in comparison to its behavior in a sub atomic world but, rather that as our views shift so does our notice of the behaviors.

3.The structure of many electron atoms

a. **Orbital energies**

- The energy of an electron in a hydrogen atom is determined by the quantum number ‘n’ and increases as follows:
- 1s<2s<=2p<3s=3p=3d<4s=4p=4d=4f<…
- In many electron atoms however, the shielding effect (electron-electron repulsion causes the energy of ‘p’ sub shell to be higher than that of an ‘s’ sub shell of the same shell and the energy of a ‘d’ sub shell a to be higher than that of a ‘p’ sub shell of the same shell. The one with the lower energy is said to be more penetrating than the one with the higher energy.

b. The building up principle

- The principle states that electrons always occupy the lowest energy orbital before occupying a higher one.
- Except for hydrogen, the energies of orbital's with the same quantum number 'n' increases with 'l'.
- When sub shells have the same energy the building up principle is not based on the order of their energies but the total energy.
- Some exceptions to the building up principle are: 24Cr = $[Ar]3d^5 4s^1$ and 29Cu = $[Ar]4s^1 3d^9$

4.A survey of periodic table

a. **Blocks, periods and groups**

- **Periods**- the horizontal rows of elements in the periodic table.
- elements in the same period have the same number of shells.
- Periods 1, 2, 3 are short periods while 4, 5, 6 are long periods. Period 6 is the longest with more than 24 elements. Period 7 elements are radioactive or artificial elements.

Groups/ Families-

- are the vertical columns in the periodic table. There are 18 of them usually assigned to roman numbers.
- IA- VIIIA – main group (A group)
- IB- VIIIB – sub group (B group)

Block

- ***S block elements –*** the last element to enter s- orbital.
- ***p-block elements***- the last electron enters the p- orbital.
- ***d-block (transition)*** - its found between s and p block.
- ***f-block (rare earth metals)***-The last valence electron is being added to the f-orbital

b.Periodicity of physical properties

- The periodicity of some physical properties in the periodic table is listed below:

1. Atomic radius: the distance from the centre of an atom to the outer most edge. It is affected by 2 factors.

a. Number of protons- the greater the number of protons the smaller the atom.

b. Number of electron shells- the greater the number of shells, the greater the atom.

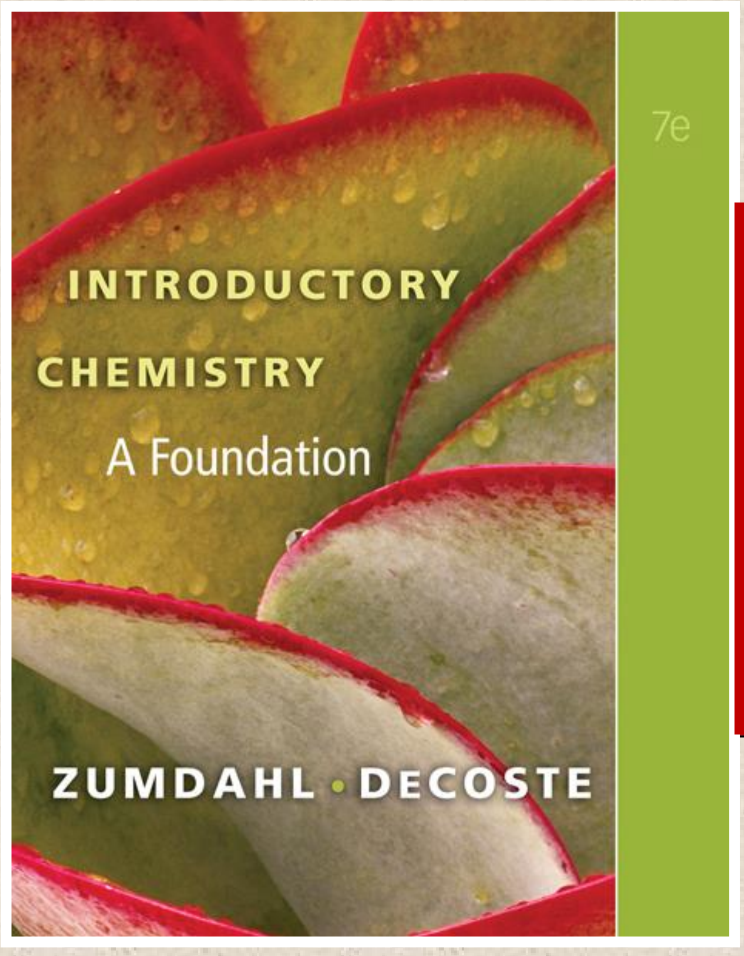
- Atomic radius decreases as we go across a period from left to right.

2. Melting point: the point at which a given solid melts. Melting point decreases across a period from left to right. It increases down a group because of an increase in metallic character.

3. Boiling point: is the temperature at which a liquid starts to vaporize.

c. Trends in chemical properties

- Some of the chemical periodic properties are:
- Metallic character: it depends on the ability of metallic elements to lose their outer valence electrons. It decreases across a period from left to right and decreases down a group.
- Electron negativity: it’s the ability of a substance to attract shared electrons of a covalent bond towards itself. It increases across a period and decreases down a group.
- Ionization energy: it’s the amount of energy require to remove an electron from the valence shell of an atom. It increases across a period and decreases down a group.



CHAPTER SIX

CHEMICAL BONDING

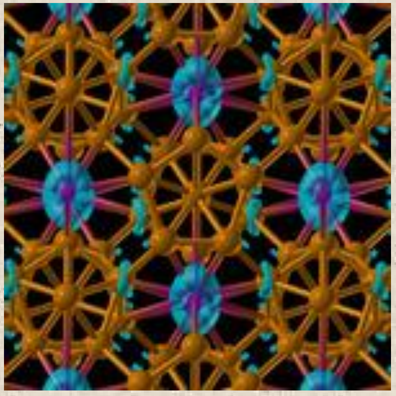


Table of Contents

6.1 Types of Chemical Bonds

6.2 Electronegativity

6.3 Bond Polarity and Dipole Moments

6.4 Stable Electron Configurations and Charges on Ions

6.5 Ionic Bonding and Structures of Ionic Compounds

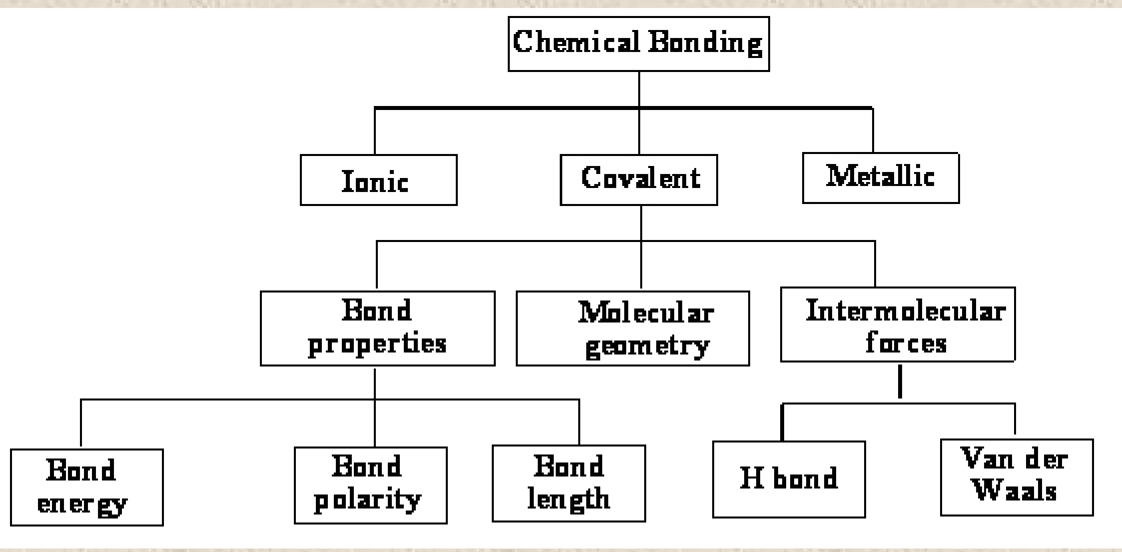
6.6 Lewis Structures

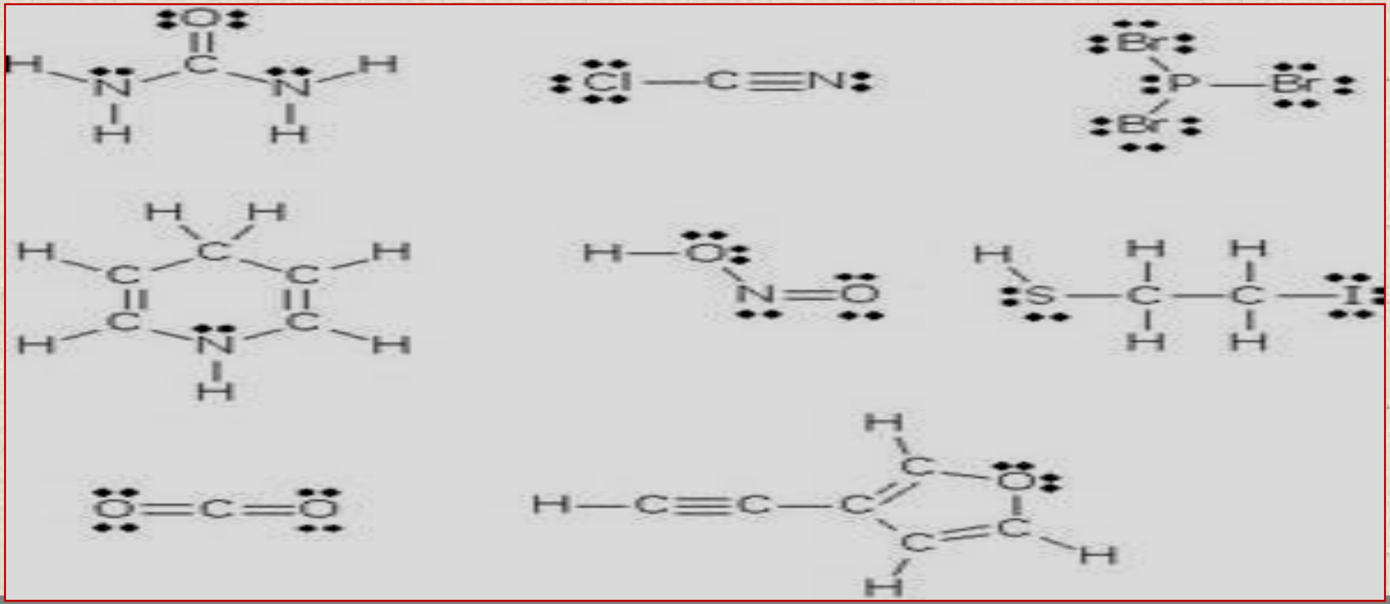
6.7 Lewis Structures of Molecules with Multiple Bonds

6.8 Molecular Structure

6.9 Molecular Structure: The VSEPR Model

6.10 Molecular Structure: Molecules with Double Bonds


Chemical Bonding
Ionic
Covalent
Metallic
Bond properties
Molecular geometry
Intermolecular forces
Bond energy
Bond polarity
Bond length
H bond
Van der Waals



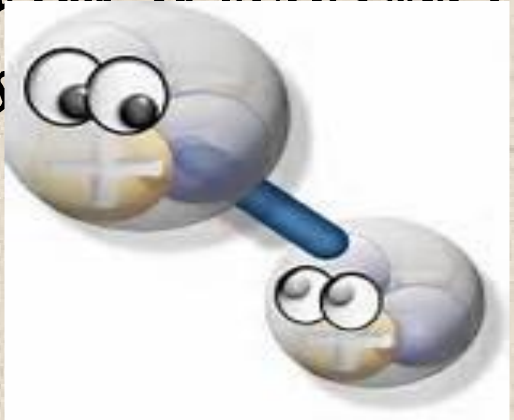
Chemical Bond

- Bonding, the way atoms are attracted to each other to form molecules, determines nearly all of the chemical properties we see.
- And, as we shall see, the number **"8"** is very important to chemical bonding.

8

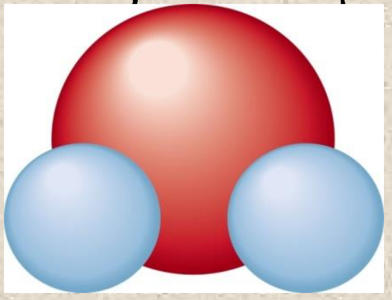
Questions to Consider

- **What** is meant by the term "**chemical bond**"?
- **Why** do atoms bond with each other to form compounds?
- **How** do atoms bond with each other to form compounds?



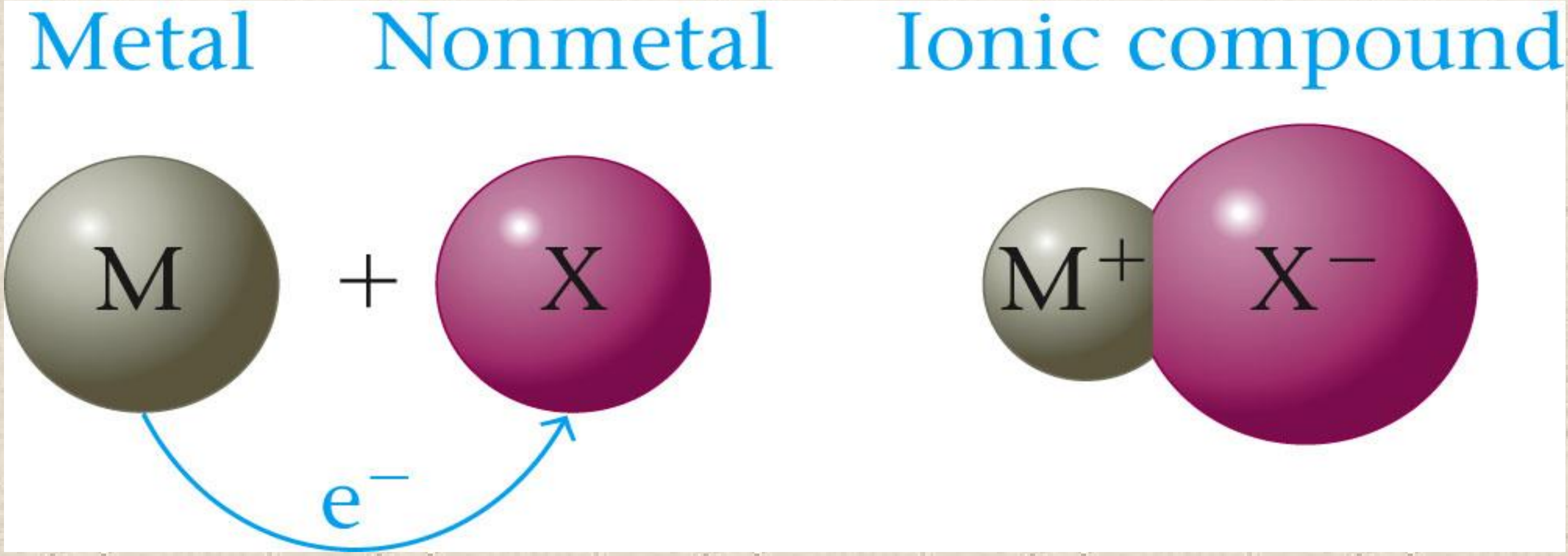
A Chemical Bond

- The forces that hold groups of atoms together and make them function as a unit.
- A bond will form if the energy of the aggregate is lower than that of the separated atoms.
- **Bond energy** – energy required to break a chemical bond.



Ionic Bonding

- Ionic compound results when a metal reacts with a nonmetal.
- Electrons are transferred.



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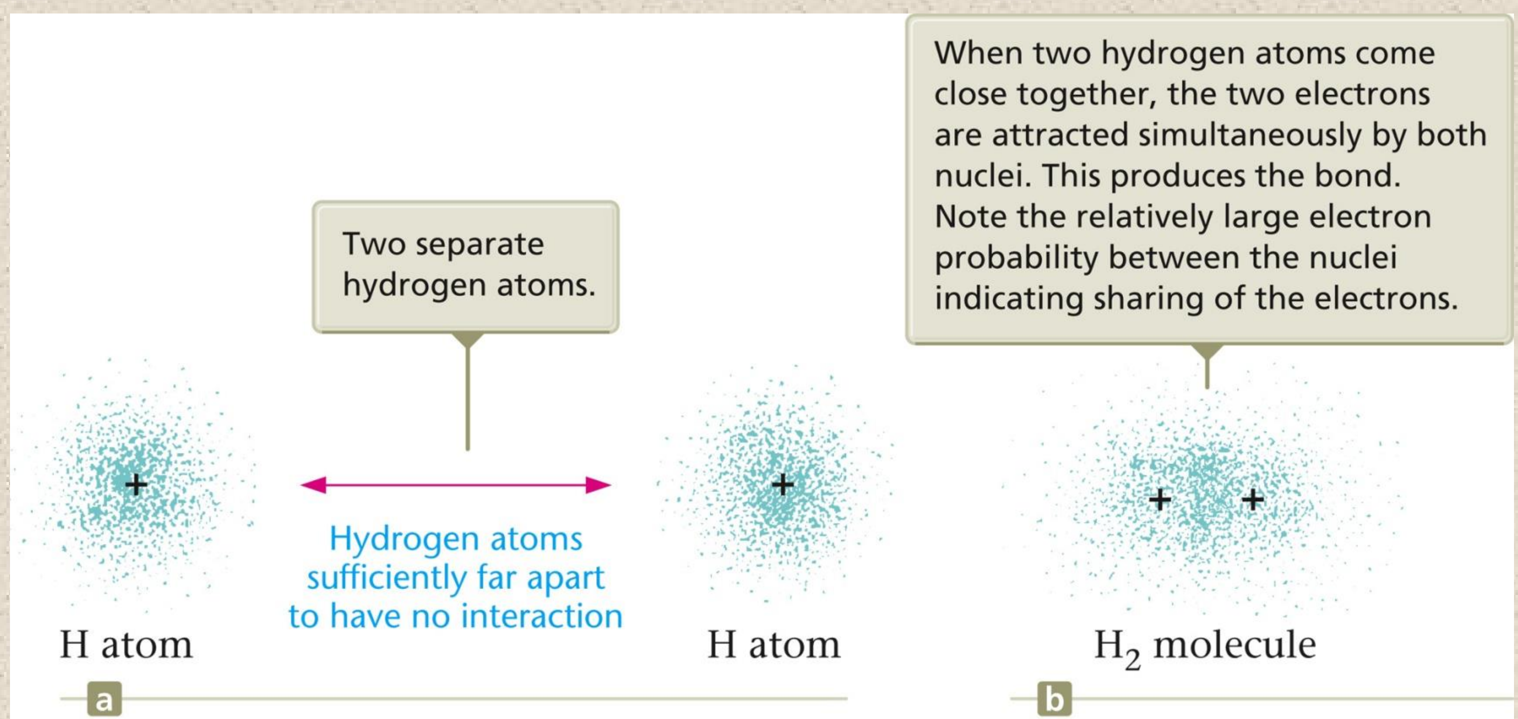
Types of Chemical Bonds

Covalent Bonding

- A covalent bond results when electrons are shared by nuclei.



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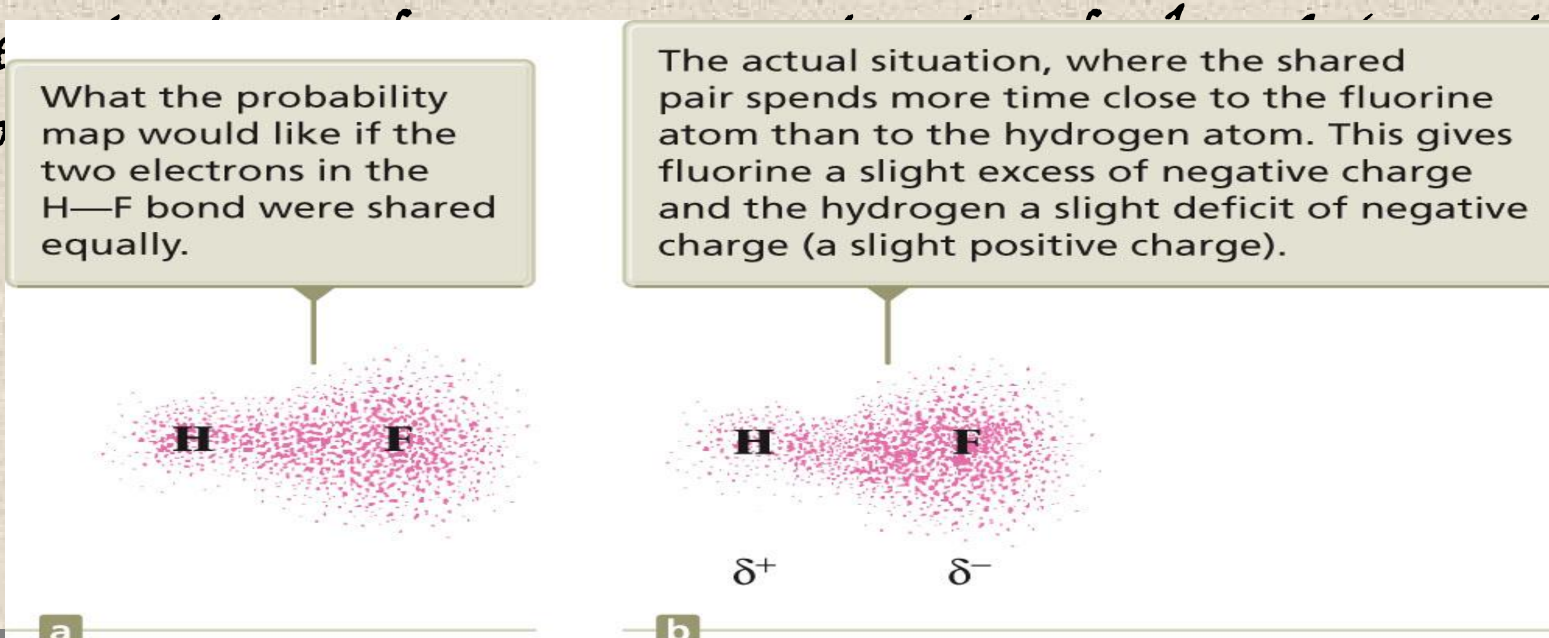
Types of Chemical Bonds

Polar Covalent Bond

- Unequal sharing of electrons between atoms in a molecule.
- One atom attracts the electrons more than the other atom.
- Re al
po

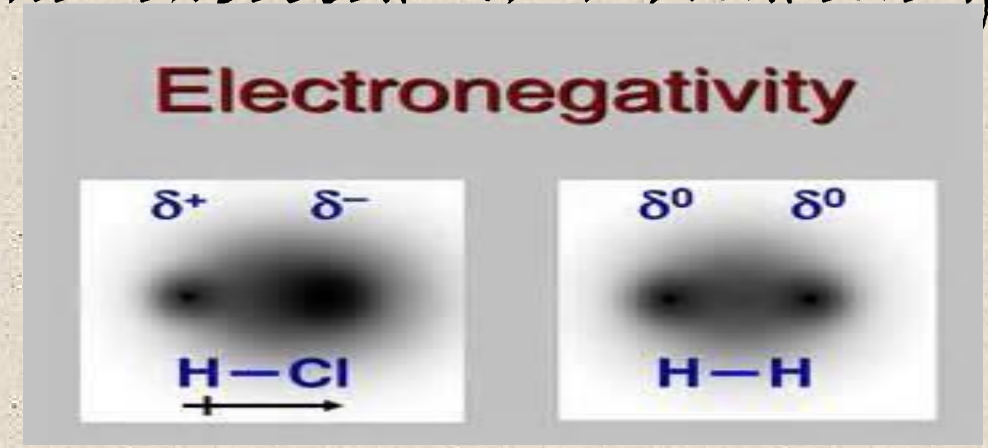


Return to TOC

Electronegativity

- The ability of an atom in a molecule to attract shared electrons to itself.

- For a molecule HX, the relative electronegativities of the H and X atoms are determined by comparing the measured H–X bond energy with the "expected" H–X bond energy.



Return to TOC

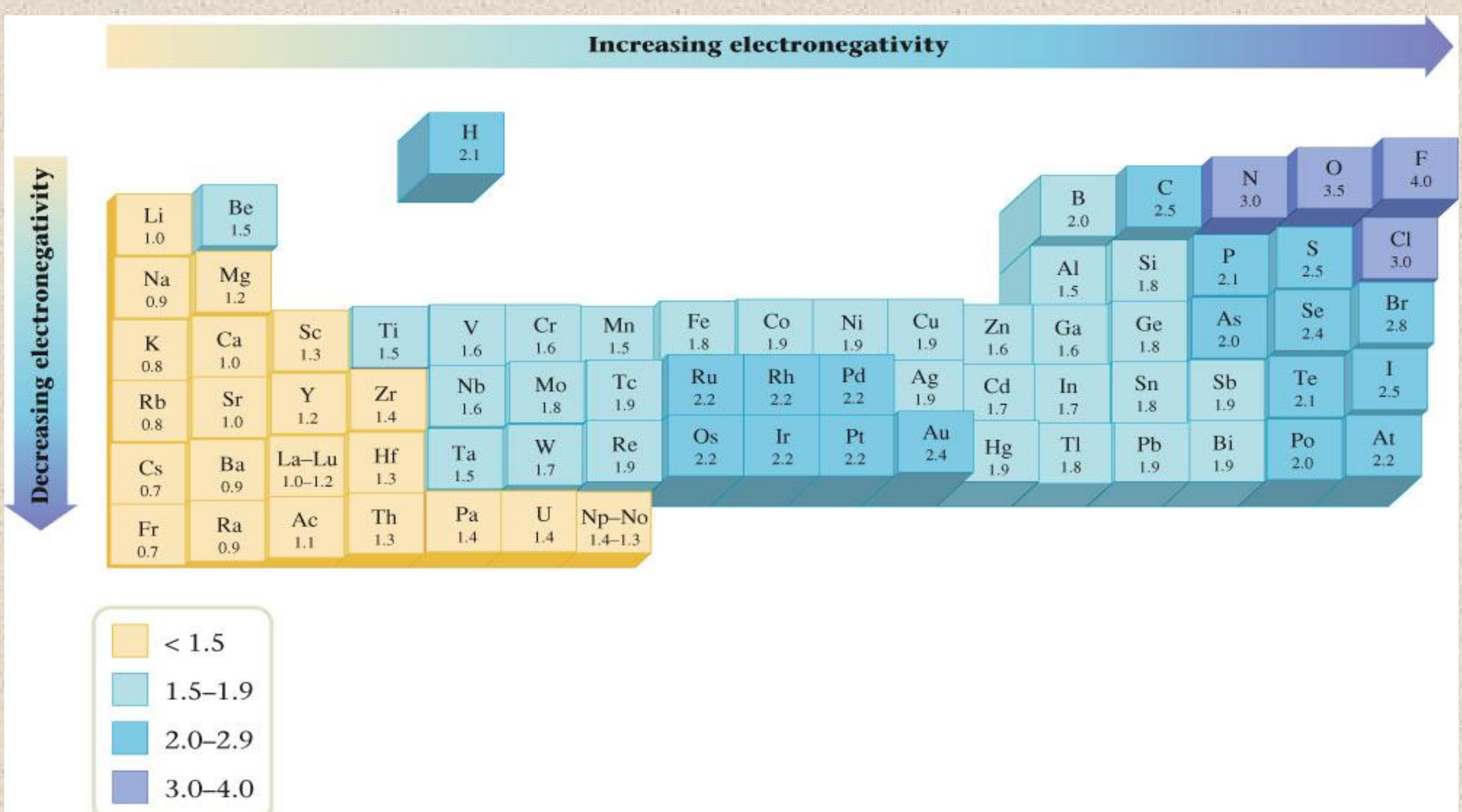
Electronegativity

- On the periodic table, electronegativity generally increases across a period and decreases down a group.

- The range of electronegativity values is from 4.0 for fluorine (the most electronegative) to 0.7 for cesium and francium (the least electronegative).

Electronegativity

Electronegativity Values for Selected Elements



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Electronegativity

Concept Check

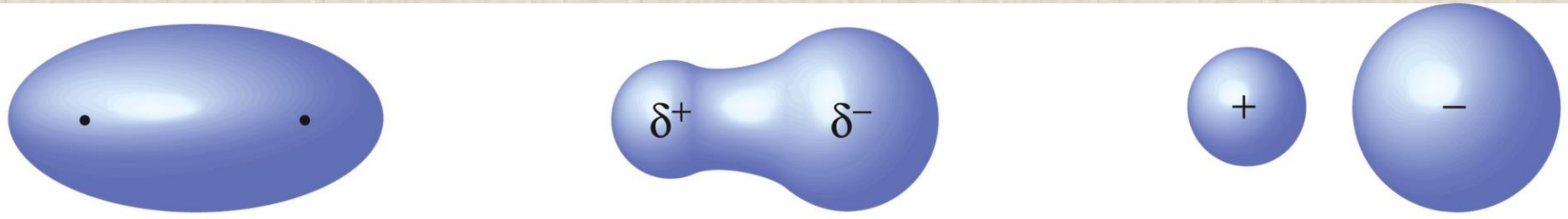
If lithium and fluorine react, which has more attraction for an electron? Why?

In a bond between fluorine and iodine, which has more attraction for an electron? Why?

Return to TOC

Electronegativity

- The polarity of a bond depends on the difference between the electronegativity values of the atoms forming the bond.



a A covalent bond formed between identical atoms.

b A polar covalent bond, with both ionic and covalent components.

c An ionic bond, with no electron sharing.

Electronegativity

Using Electronegativity to Determine Bond Polarity

Using the electronegativity values given in Figure 12.4, arrange the following bonds in order of increasing polarity: H—H, O—H, Cl—H, S—H, and F—H.

Solution

The polarity of the bond increases as the difference in electronegativity increases. From the electronegativity values in Figure 12.4, the following variation in bond polarity is expected (the electronegativity value appears in parentheses below each element).

Bond	Electronegativity Value	Difference in Electronegativity Values	Bond Type	Polarity
H—H	(2.1) (2.1)	$2.1 - 2.1 = 0$	Covalent	Increasing ↓
S—H	(2.5) (2.1)	$2.5 - 2.1 = 0.4$	Polar covalent	
Cl—H	(3.0) (2.1)	$3.0 - 2.1 = 0.9$	Polar covalent	
O—H	(3.5) (2.1)	$3.5 - 2.1 = 1.4$	Polar covalent	
F—H	(4.0) (2.1)	$4.0 - 2.1 = 1.9$	Polar covalent	

Therefore, in order of increasing polarity, we have

H—H S—H Cl—H O—H F—H

Least polar → Most polar

Return to TOC

Electronegativity

Exercise

Arrange the following bonds from most to least polar:

a) N–F O–F C–F
b) C–F N–O Si–F
c) Cl–Cl B–Cl S–Cl

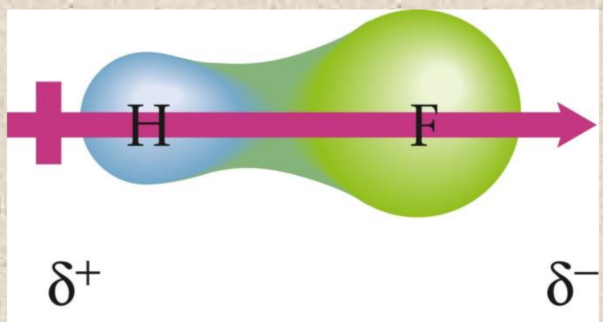
a) C–F, N–F, O–F
b) Si–F, C–F, N–O
c) B–Cl, S–Cl, Cl–Cl

Return to TOC

Bond Polarity and Dipole Moments

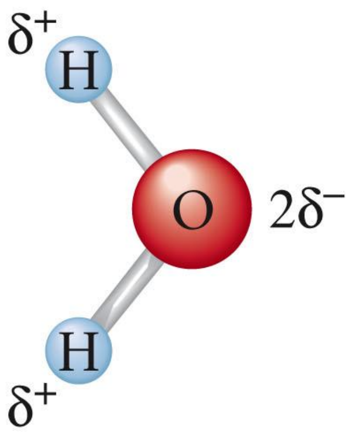
Dipole Moment

- Property of a molecule whose charge distribution can be represented by a center of positive charge and a center of negative charge.
- Use an arrow to represent a dipole moment.
  - Point to the negative charge center with the tail of the arrow indicating the positive center of charge.



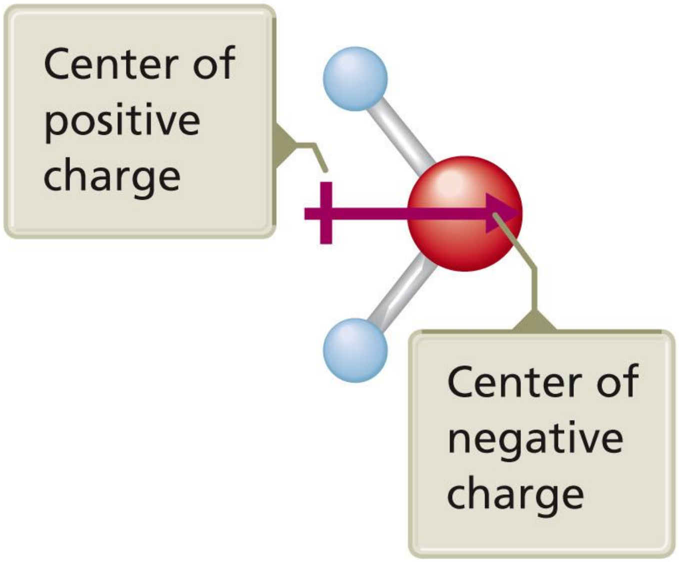
Bond Polarity and Dipole Moments

Dipole Moment in a Water Molecule



a

The charge distribution in the water molecule. The oxygen has a charge of $2\delta^-$ because it pulls δ^- of charge from each hydrogen atom ($\delta^- + \delta^- = 2\delta^-$).



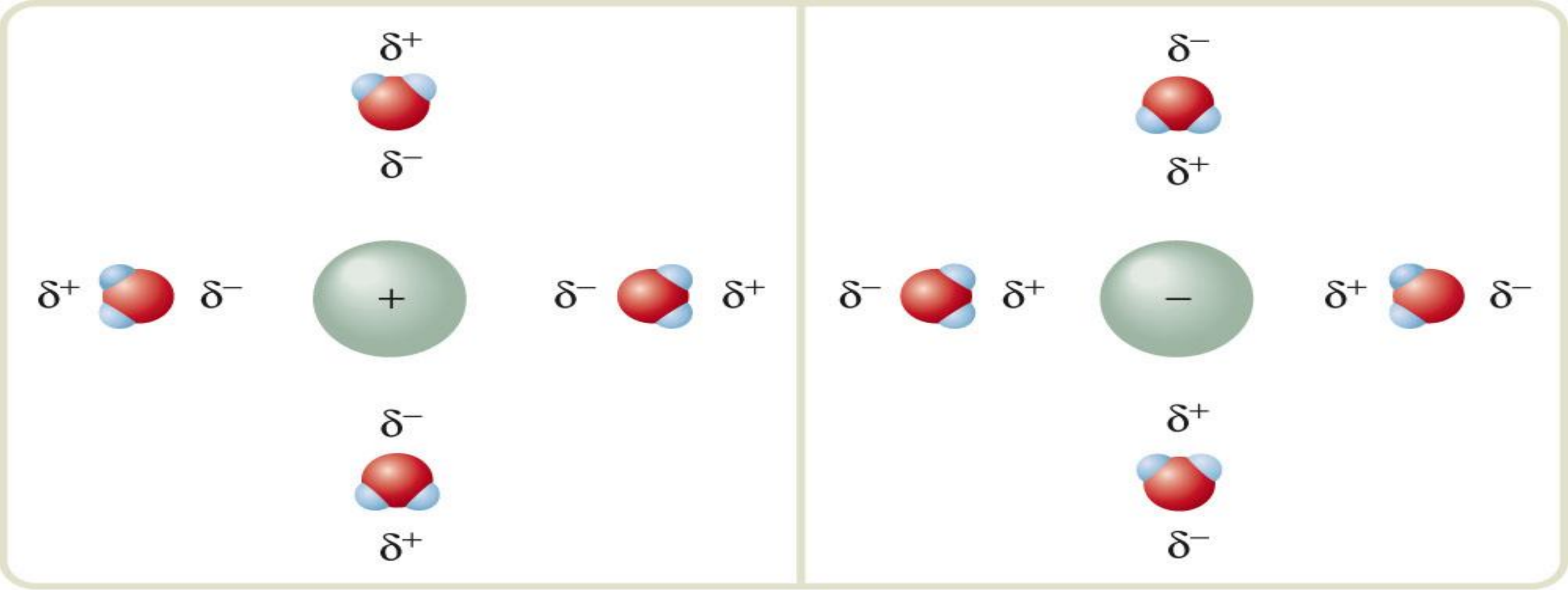
b

The water molecule behaves as if it had a positive end and a negative end, as indicated by the arrow.

Return to TOC

Bond Polarity and Dipole Moments

- The polarity of water affects its properties.
  - Permits ionic compounds to dissolve in it.



a
Polar water molecules are strongly attracted to positive ions by their negative ends.

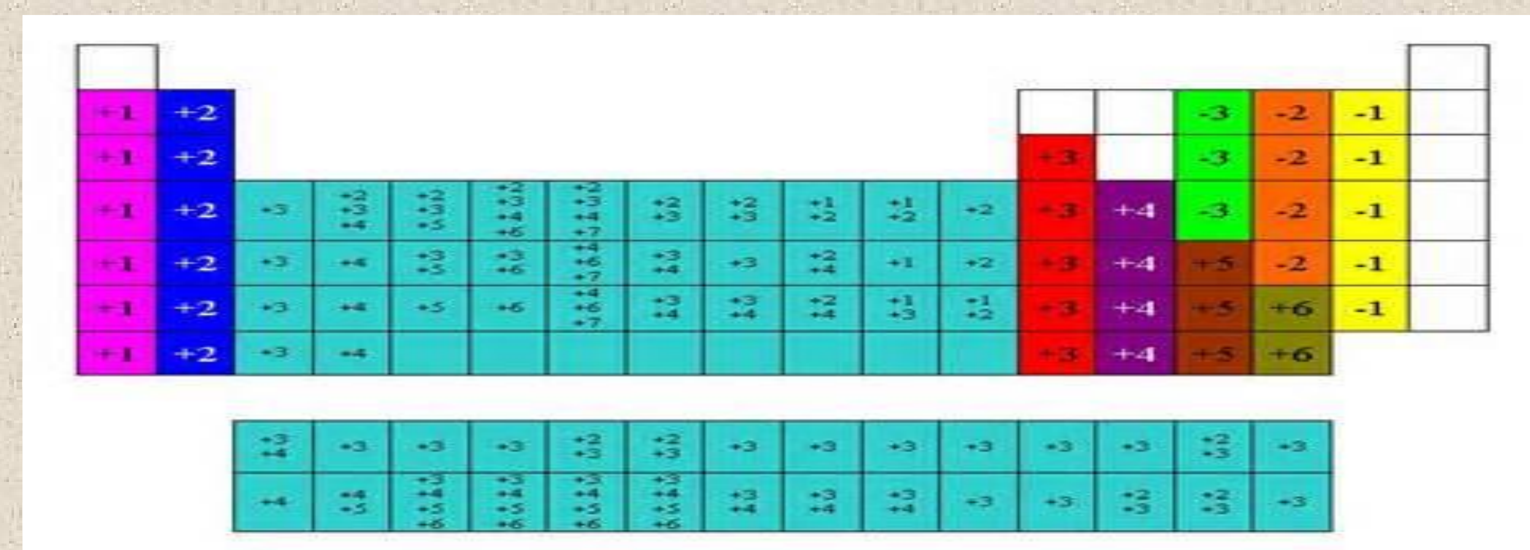
b
They are also strongly attracted to negative ions by their positive ends.

  - Causes water to remain liquid at higher temperature.

Return to TOC

Stable Electron Configurations and Charges on Ions

- Group 1 metals always form 1+ cations.
- Group 2 metals always form 2+ cations.
- Aluminum in Group 3 always forms a 3+ cation.
- Group 7 nonmetals form 1– anions.
- Group 6 elements always form 2– anions.



Return to TOC

Stable Electron Configurations and Charges on Ions

The Formation of Ions by Metals and Nonmetals

Table 12.2 The Formation of Ions by Metals and Nonmetals

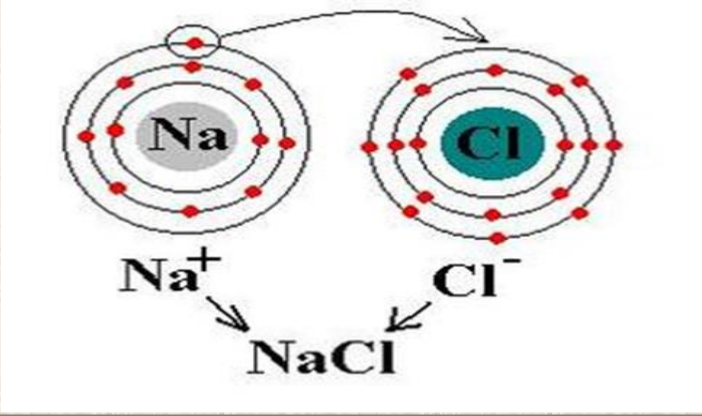
Group	Ion Formation	Electron Configuration: Atom		Electron Configuration: Ion
1	$Na \rightarrow Na^+ + e^-$	$[Ne]3s^1$	$\xrightarrow{e^- \text{ lost}}$	[Ne]
2	$Mg \rightarrow Mg^{2+} + 2e^-$	$[Ne]3s^2$	$\xrightarrow{2e^- \text{ lost}}$	[Ne]
3	$Al \rightarrow Al^{3+} + 3e^-$	$[Ne]3s^23p^1$	$\xrightarrow{3e^- \text{ lost}}$	[Ne]
6	$O + 2e^- \rightarrow O^{2-}$	$[He]2s^22p^4 + 2e^-$	$\rightarrow [He]2s^22p^6 =$	[Ne]
7	$F + e^- \rightarrow F^-$	$[He]2s^22p^5 + e^-$	$\rightarrow [He]2s^22p^6 =$	[Ne]

Return to TOC

Stable Electron Configurations and Charges on Ions

Electron Configurations of Ions

- Atoms in stable compounds usually have a noble gas electron configuration.
  - Metals lose electrons to reach noble gas configuration.
  - Nonmetals gain electrons to reach noble gas configuration.



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Stable Electron Configurations and Charges on Ions

Electron Configurations and Bonding

1. When a nonmetal and a Group 1, 2, or 3 metal react to form a binary ionic compound, the ions form so that the valence-electron configuration of the nonmetal achieves the electron configuration of the next noble gas atom.

   – The valence orbital of the metal are emptied to achieve the configuration of the previous noble gas.

Stable Electron Configurations and Charges on Ions

2. When two nonmetals react to form a covalent bond, they share electrons in a way that completes the valence-electron configurations of both atoms.

Return to TOC

Predicting Formulas of Ionic Compounds

- Chemical compounds are always electrically neutral.

Table 12.3 Common Ions with Noble Gas Configurations in Ionic Compounds

Group 1	Group 2	Group 3	Group 6	Group 7	Electron Configuration
Li^+	Be^{2+}				[He]
Na^+	Mg^{2+}	Al^{3+}	O^{2-}	F^-	[Ne]
K^+	Ca^{2+}		S^{2-}	Cl^-	[Ar]
Rb^+	Sr^{2+}		Se^{2-}	Br^-	[Kr]
Cs^+	Ba^{2+}		Te^{2-}	I^-	[Xe]

Return to TOC

Stable Electron Configurations and Charges on Ions

Concept Check

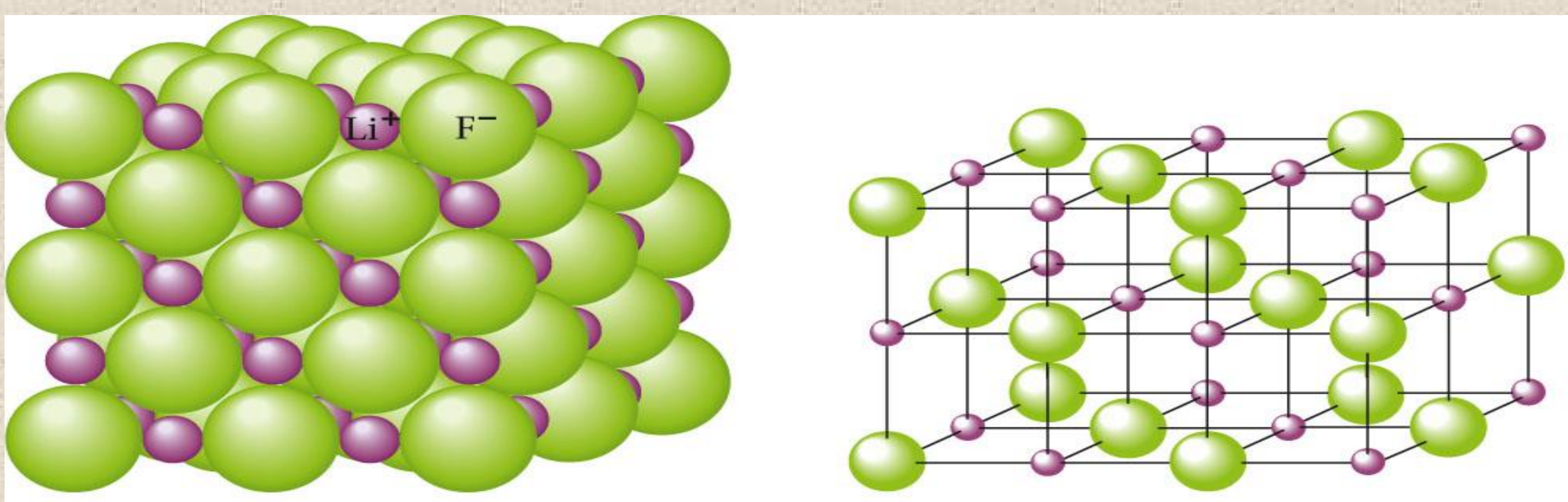
What is the correct electron configuration for the most stable form of the sulfur ion in an ionic compound?

a) $1s^22s^22p^63s^2$
b) $1s^22s^22p^63s^23p^2$
c) $1s^22s^22p^63s^23p^4$
d) $1s^22s^22p^63s^23p^6$

Return to TOC

Structures of Ionic Compounds

- Ions are packed together to maximize the attractions between ions.



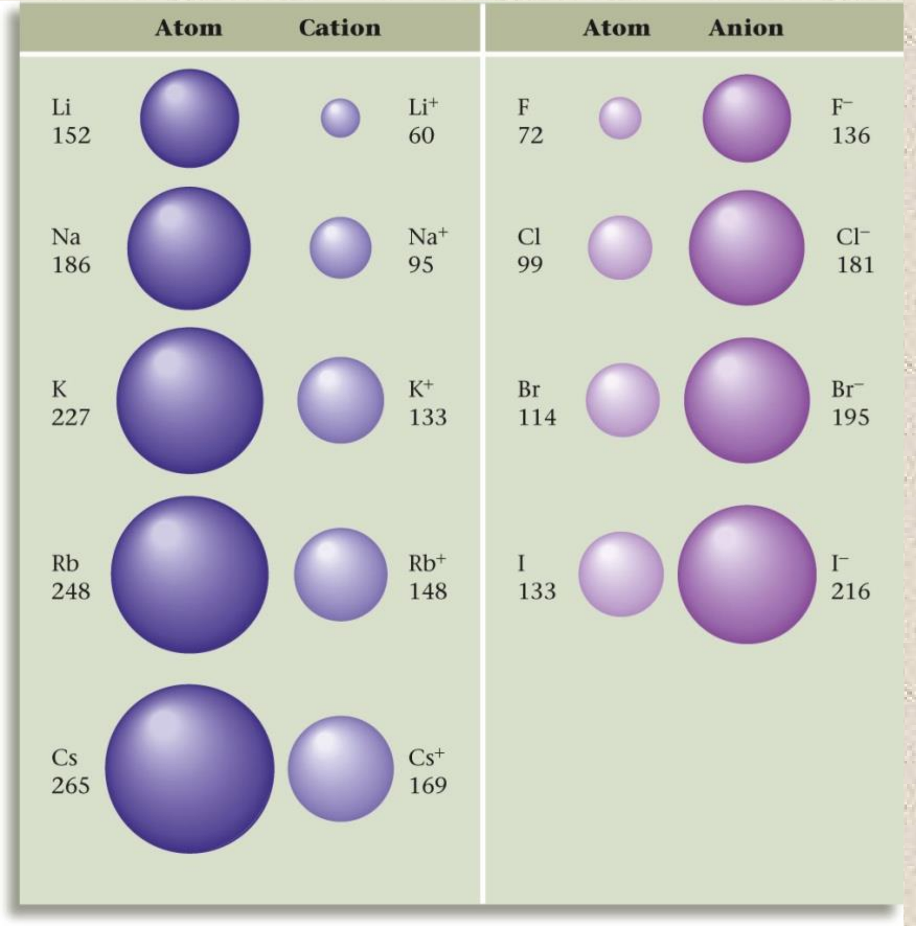
a
This structure represents the ions as packed spheres.

b
This structure shows the positions (centers) of the ions. The spherical ions are packed in the way that maximizes the ionic attractions.

Return to TOC

Structures of Ionic Compounds

- Cations are always smaller than the parent atom.
- Anions are always larger than the parent atom.



Concept Check

Which atom or ion has the smallest radius?

a) O^{2+}
b) O^{+}
c) O
d) O^{2-}

Return to TOC

Putting Ions Together

$Na^{+} + Cl^{-} = NaCl$

$Ca^{+2} + Cl^{-} = CaCl_2$

$Ca^{+2} + O^{-2} = CaO$

$Na^{+} + O^{-2} = Na_2O$

$Al^{+3} + S^{-2} = Al_2S_3$

$Ca^{+2} + N^{-3} = Ca_3N_2$

You try these!

$Li^{+} + Br^{-} =$ $LiBr$

$Mg^{+2} + F^{-} =$ MgF_2

$Al^{+3} + I^{-} =$ AlI_3

$NH_4^{+} + PO_4^{-3} = (NH_4)_3PO_4$

Not $NH_{43}PO_4$

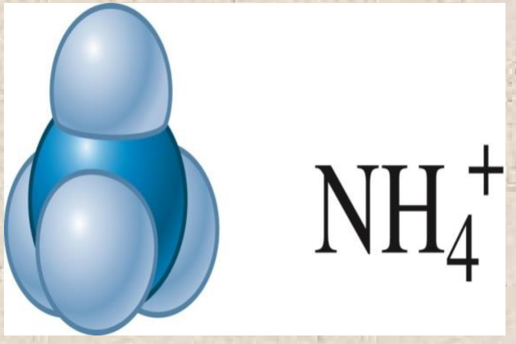
$Sr^{+2} + P^{-3} =$ Sr_3P_2

$K^{+} + Cl^{-} =$ KCl

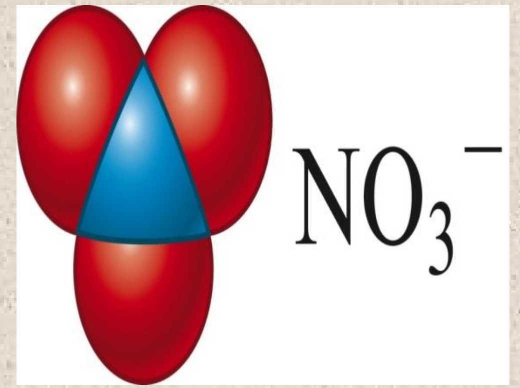
Ionic Bonding and Structures of Ionic Compounds

Ionic Compounds Containing Polyatomic Ions

- Polyatomic ions work in the same way as simple ions.
  - The covalent bonds hold the polyatomic ion together so it behaves as a unit.



cation	anion	compound
Ca^{+2}	Cl^{-1}	$CaCl_2$
Ba^{+2}	O^{-2}	BaO
K^{+1}	S^{-2}	K_2S
Fe^{+3}	Br^{-1}	$FeBr_3$
Cr^{+3}	O^{-2}	Cr_2O_3



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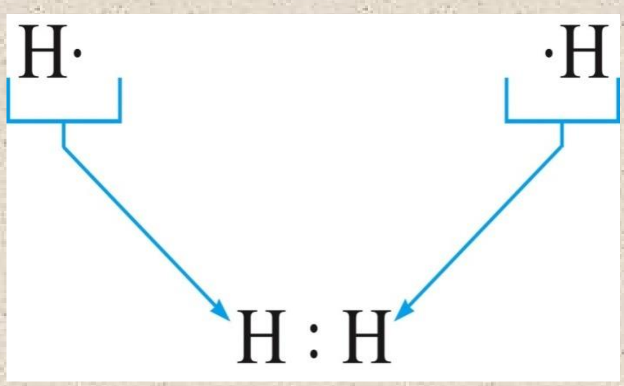
Lewis Structure

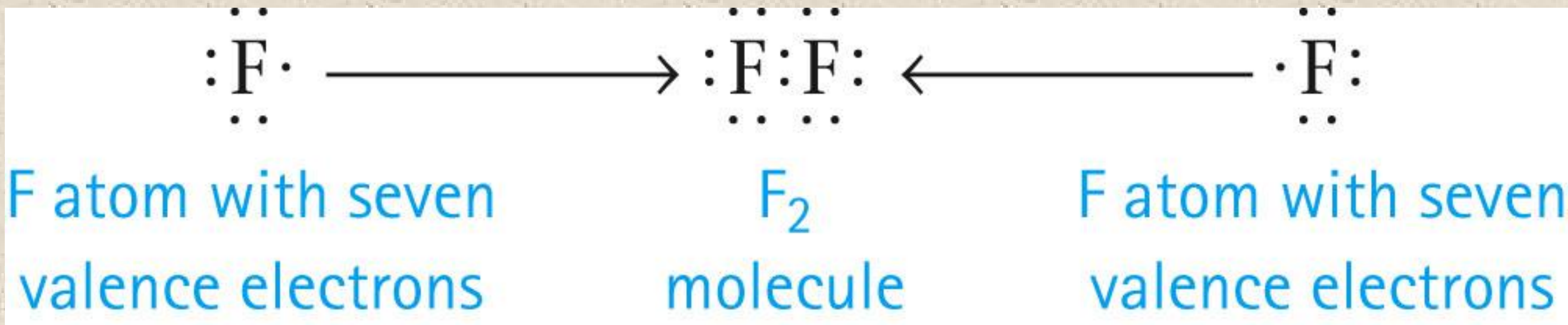
- Shows how valence electrons are arranged among atoms in a molecule.
- Most important requirement
  - Atoms achieve noble gas electron configuration (octet rule, duet rule).



Lewis Structures

Writing Lewis Structures

- Bonding pairs are shared between 2 atoms.
- Unshared pairs (lone pairs) are not shared and not involved in bonding.



Return to TOC

Rules to Write Dot Structures

1. Write a skeleton molecule with the lone atom in the middle (Hydrogen can never be in the middle)
2. Find the number of electrons needed (N)

   (8 x number of atoms, 2 x number of H atoms)
3. Find the number of electrons you have (valence e^-'s) (H)
4. Subtract to find the number of bonding electrons (N-H=B)
5. Subtract again to find the number of non-bonding electrons (H-B=NB)
6. Insert minimum number of bonding electrons in the skeleton between atoms only. Add more bonding if needed until you have B bonding electrons.
7. Insert needed non-bonding electrons around (not between) atoms so that all atoms have 8 electrons around them. The total should be the same as NB in 5 above.

Return to TOC

Let's Try it!

H O H

Water H_2O

1.S

2 x 2 = 4 for Hydrogen
1 x 8 = 8 for Oxygen
4+8=12 needed electrons

2.N

2 x 1 = 2 for Hydrogen
1 x 6 = 6 for Oxygen
You have 8 available electrons

3.H

12 - 8 = 4 bonding electrons

4.B

8 – 4 = 4 non-bonding electrons

5.NB

6.E

12 N
- 8 H
- 4 B
4 NB

H:O:H

H:O:H (with two dots above and two dots below the O)

Return to TOC

Let's Try it!

Ammonia NH_3

H
H N H

1.S
3 x 2 = 6 for Hydrogen
1 x 8 = 8 for Nitrogen
2.N
6+8=14 needed electrons

3.H
3 x 1 = 3 for Hydrogen
1 x 5 = 5 for Nitrogen
You have 8 available electrons

4.B
14 - 8 = 6 bonding electrons

5.NB
8 – 6 = 2 non-bonding electrons

6.E

H
··
H:N:H
••

14 N
- 8 H
- 6 B
2 NB

H
··
H:N:H

H
··
H:N:H
••

Let's Try it!

O C O

Carbon Dioxide CO_2

1.S

1 x 8 = 8 for Carbon
2 x 8 = 16 for Oxygen

2.N

8+16=24 needed electrons

24 N

3.H

1 x 4 = 4 for Carbon
2 x 6 = 12 for Oxygen
You have 16 available electrons

- 16 H

- 8 B

8 NB

4.B

24 - 16 = 8 bonding electrons

5.NB

16 – 8 = 8 non-bonding electrons

O::C::O

6.E

$\ddot{\underset{..}{\mathrm{O}}}$::C::$\ddot{\underset{..}{\mathrm{O}}}$

$\ddot{\underset{..}{\mathrm{O}}}$::C::$\ddot{\underset{..}{\mathrm{O}}}$

Return to TOC

Let's Try it!

Carbonate CO_3^{-2}

1.S

O
O C O

2.N

3 x 8 = 24 for Oxygen
1 x 8 = 8 for Carbon
24+8=32 needed electrons

3.H

3 x 6 = 18 for Oxygen
1 x 4= 4 for Carbon
You have 22 + 2 more available e^-'s

4.B

32 - 24 = 8 bonding electrons

5.NB

24 – 8 = 16 non-bonding electrons

6.E

$$\left[\begin{array}{c} \ddot{O}: \\ \ddot{O}::C:\ddot{O}: \end{array}\right]^{-2}$$

32 N
- 24 H
- 8 B
16 NB

O
O::C:O

:Ö:
O::C: O:

Return to TOC

Concept Check

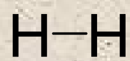
Draw a Lewis structure for each of the following molecules:

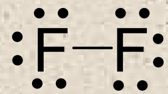
H_2

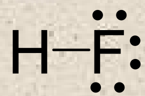
F_2

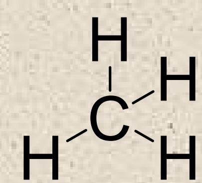
HF

CH_4









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Lewis Structures of Molecules with Multiple Bonds

- Single bond – covalent bond in which 1 pair of electrons is shared by 2 atoms.

$$H–H$$

- Double bond – covalent bond in which 2 pairs of electrons are shared by 2 atoms.

$$O=C=O$$

- Triple bond – covalent bond in which 3 pairs of electrons are shared by 2 atoms.

$$N \equiv N$$

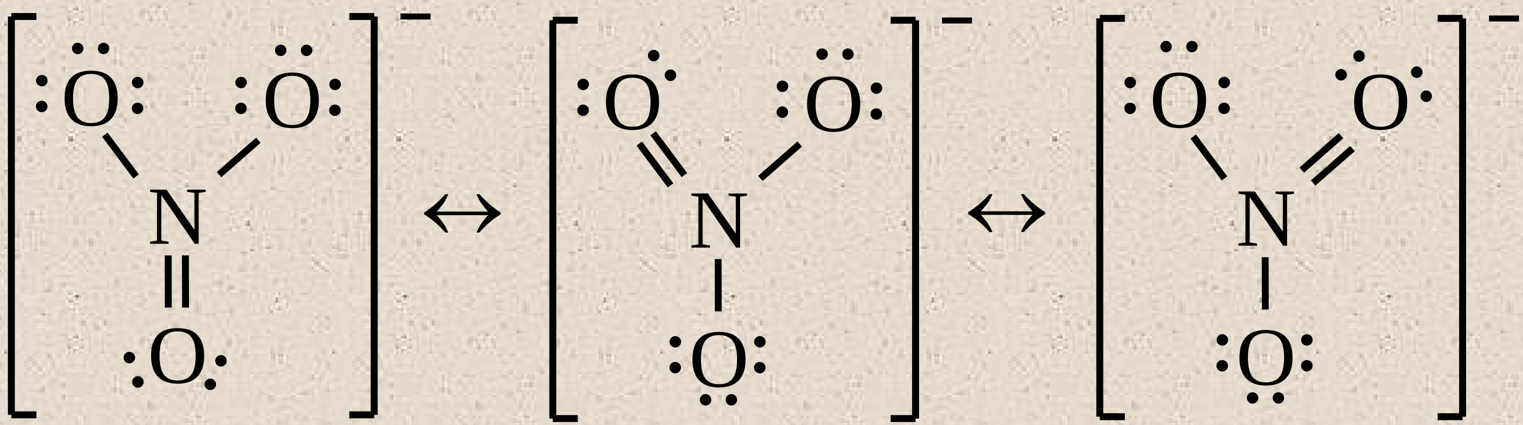
Return to TOC

Lewis Structures of Molecules with Multiple Bonds

Resonance

- A molecule shows resonance when more than one Lewis structure can be drawn for the molecule.

$NO_3^- = 24e^-$



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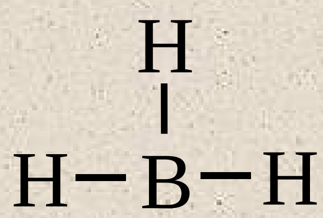
Lewis Structures of Molecules with Multiple Bonds

Some Exceptions to the Octet Rule

- Boron tends to form compounds in which the boron atom has fewer than eight electrons around it (it does not have a complete octet).

$BH_3 = 6e^-$



- Molecules containing odd numbers of electrons like NO and NO_2.

Return to TOC

Concept Check

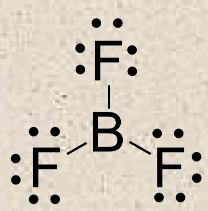
Draw a Lewis structure for each of the following molecules:

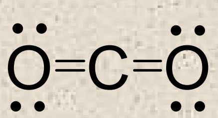
BF_3

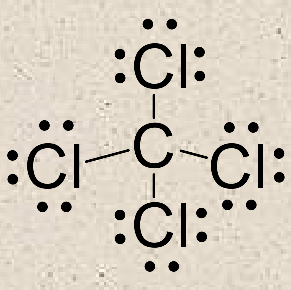
CO_2

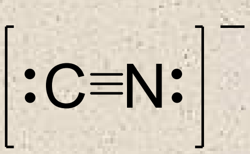
CCl_4

CN^-









Lewis Structures of Molecules with Multiple Bonds

Concept Check

Consider the following compounds:

CO_2 N_2 CCl_4

Which compound exhibits resonance?

a) $\boxed{CO_2}$

b) N_2

c) CCl_4

d) At least two of the above compounds exhibit resonance.

Return to TOC

Lewis Structures of Molecules with Multiple Bonds

Concept Check

Which of the following supports why Lewis structures are not a completely accurate way to draw molecules?

a) We cannot say for certain where an electron is located yet when drawing Lewis structures, we assume the electrons are right where we place them.

b) When adding up the number of valence electrons for a molecule, it is possible to get an odd number which would make it impossible to satisfy the octet rule for all atoms.

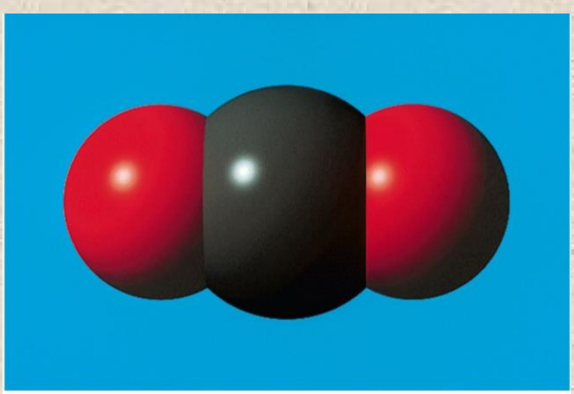
c) Both statements 1 and 2 above support why Lewis structures are not a completely accurate way to draw molecules.

d) Lewis structures are the most accurate way to draw molecules and are completely correct.

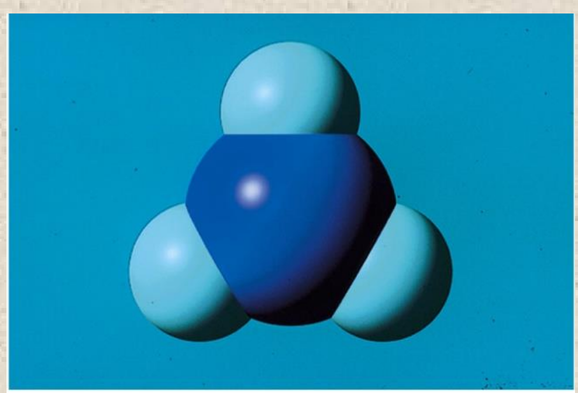
Return to TOC

Molecular Structure

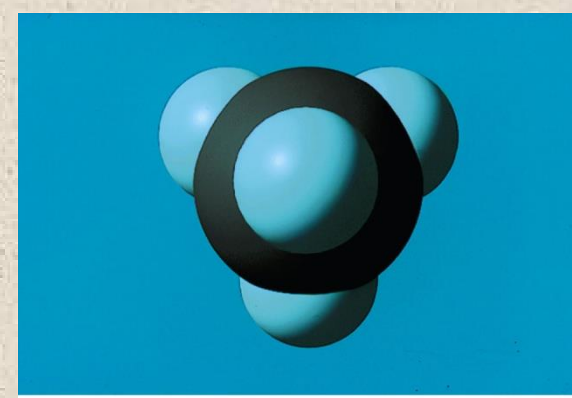
- Three dimensional arrangement of the atoms in a molecule.



a
Computer graphic of a linear molecule containing three atoms



b
Computer graphic of a trigonal planar molecule

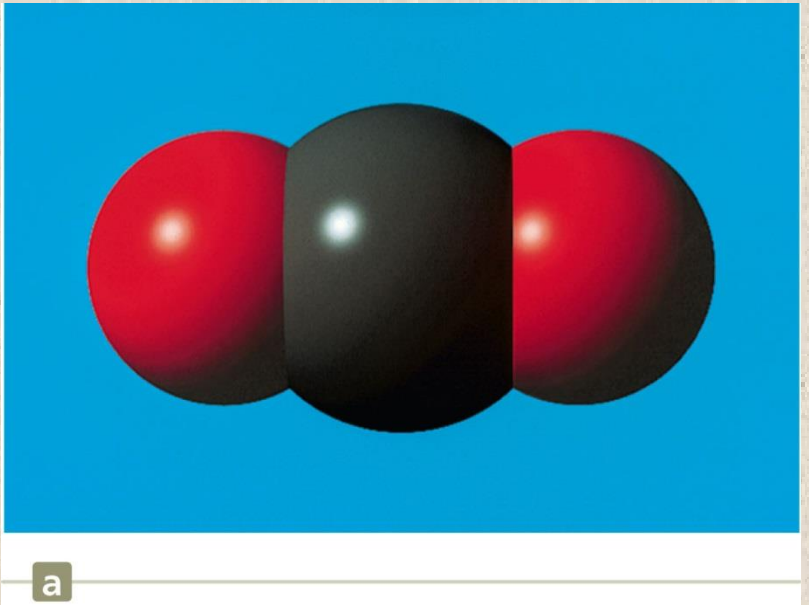


c
Computer graphic of a tetrahedral molecule

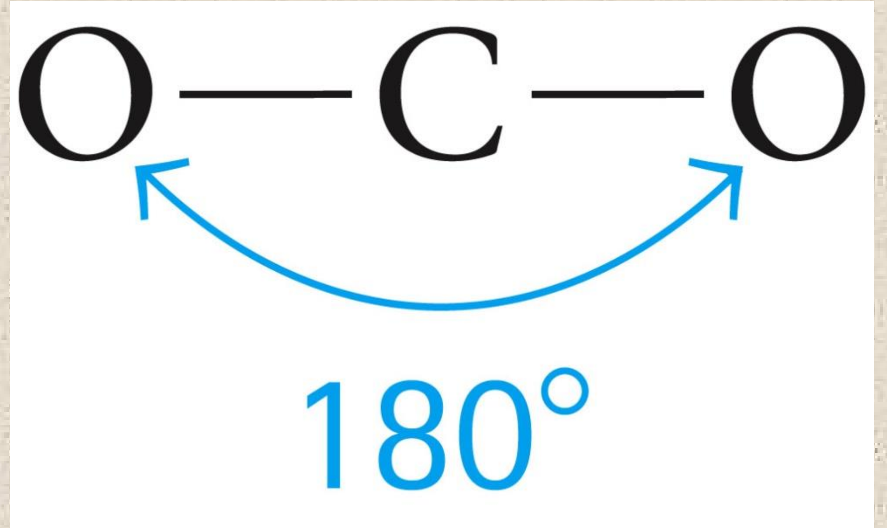
Bond Angle 180^{o} ~120^{o} ~109^{o}

Molecular Structure

- Linear structure – atoms in a line
  - Carbon dioxide



Computer graphic of a linear molecule containing three atoms



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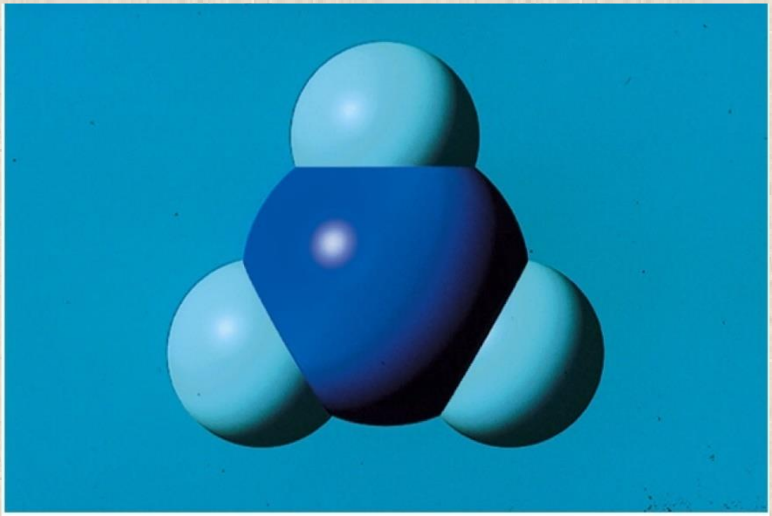
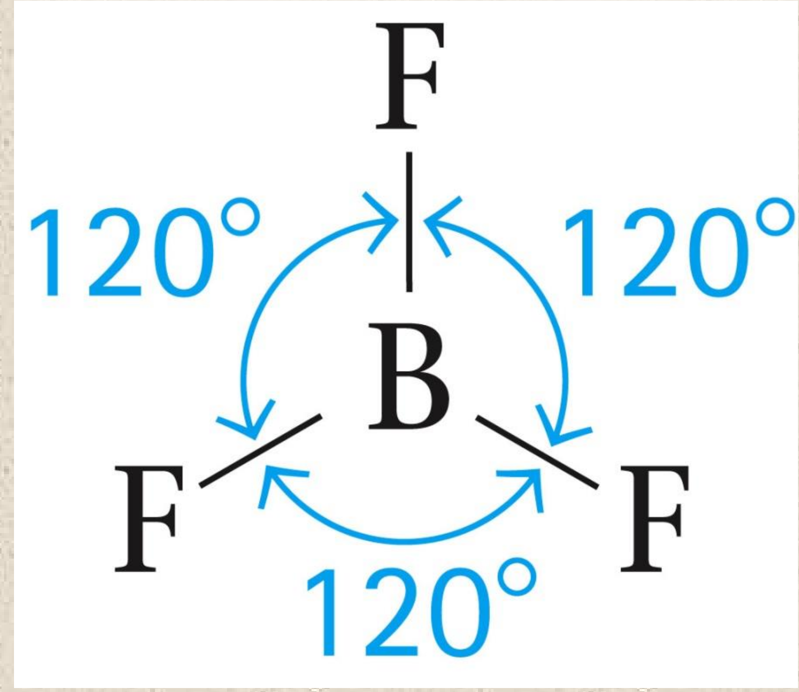
Molecular Structure

- Trigonal planar – atoms in a triangle
  - Boron trifluoride



b

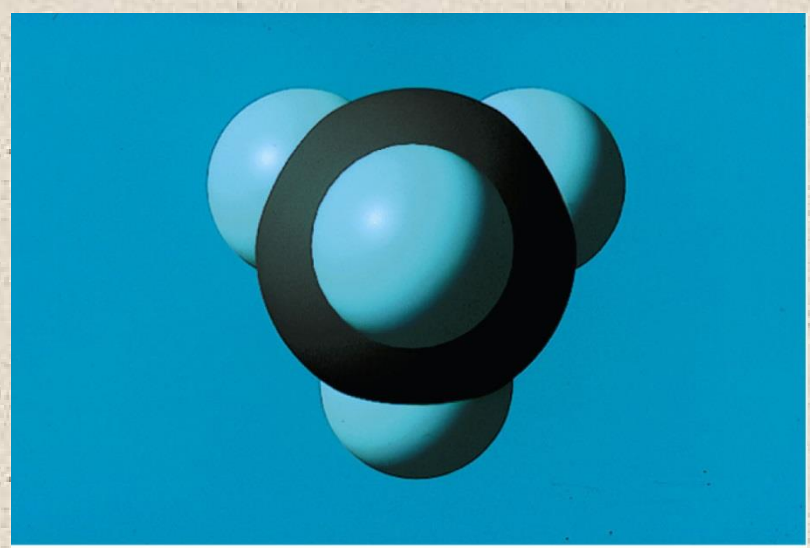
Computer graphic of a trigonal planar molecule



Return to TOC

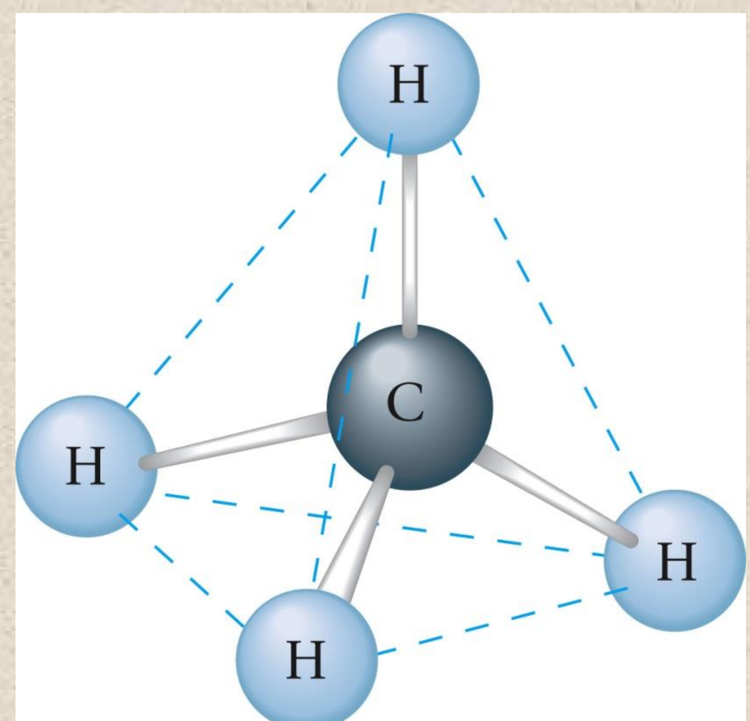
Molecular Structure

- Tetrahedral structure
  - Methane



c

Computer graphic of a tetrahedral molecule



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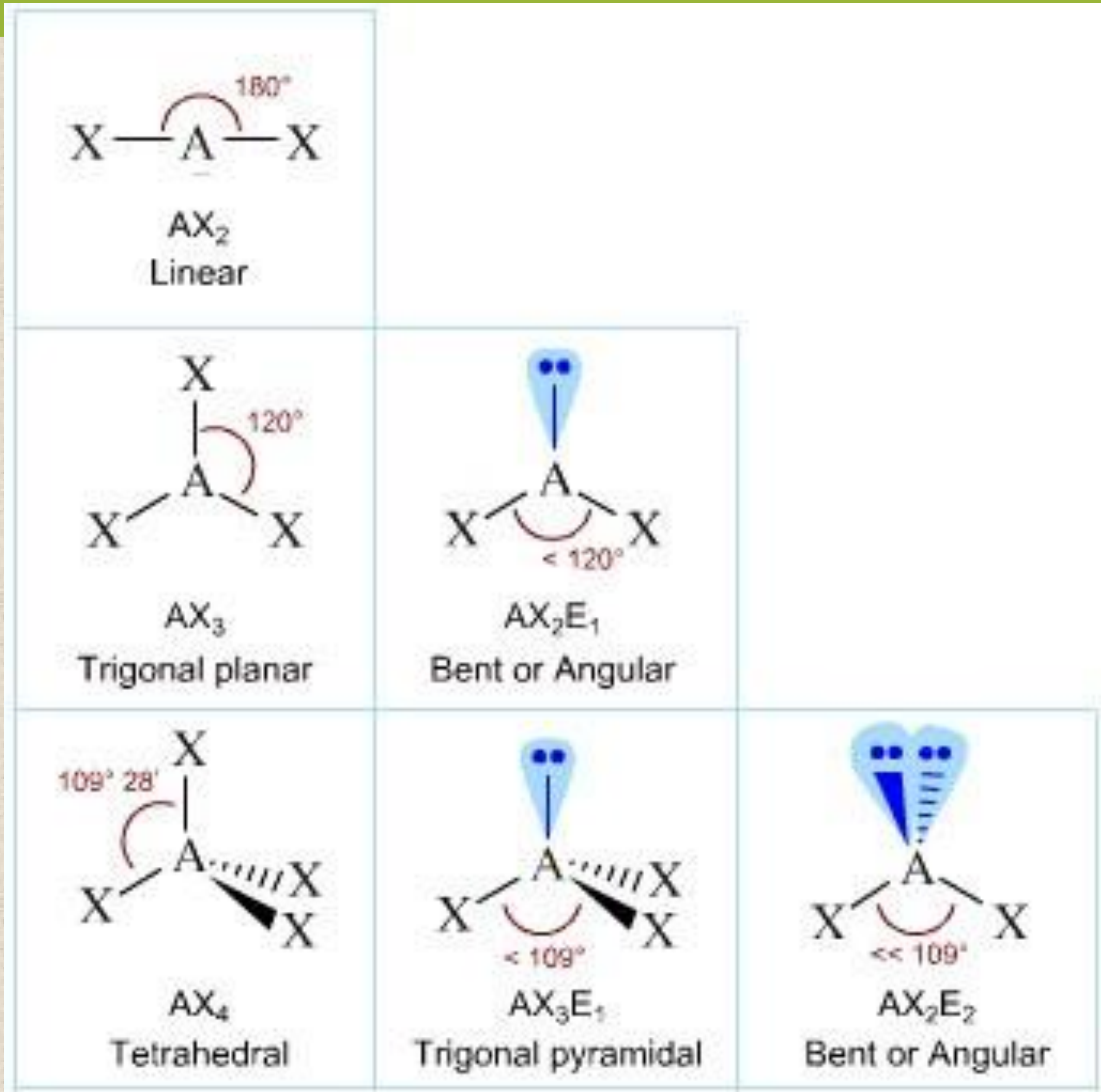
Molecular Structure

Molecular Shapes and bond angles from Lab Book Figure

2 Electron Grps = 180º
3 Electron Grps = ~120º
4 Electron Grps = ~109º



VSEPR Model

- VSEPR: Valence Shell Electron-Pair Repulsion.
- The structure around a given atom is determined principally by minimizing electron pair repulsions.

Molecular Structure

Two Pairs of Electrons

- $BeCl_2$
- 180°
- Linear

Return to TOC

Molecular Structure

Three Pairs of Electrons

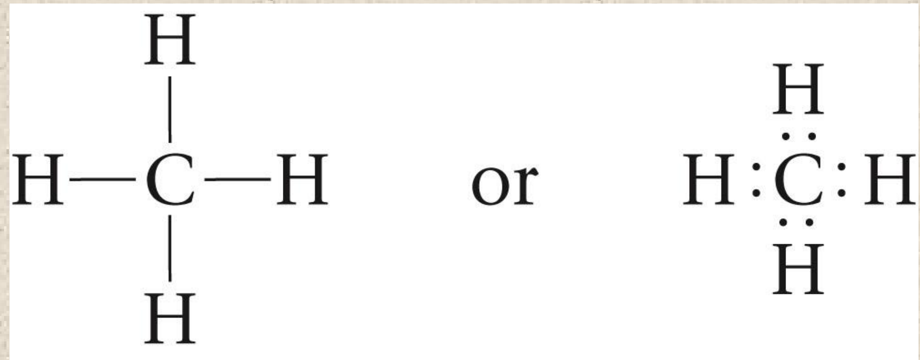
- BF_3
- 120°
- Trigonal planar

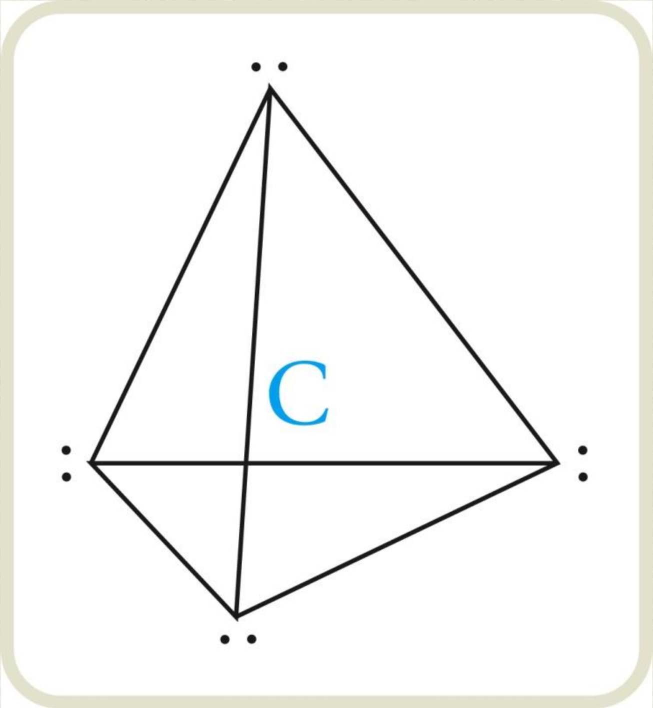
Return to TOC

Molecular Structure

Four Pairs of Electrons

- CH_4
- 109.5°
- Tetrahedral





Return to TOC

Molecular Structure

Steps for Predicting Molecular Structure Using the VSEPR Model

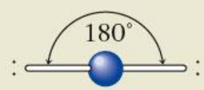
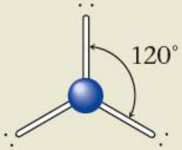
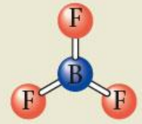
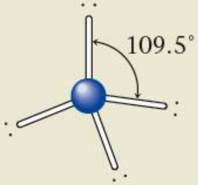
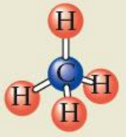
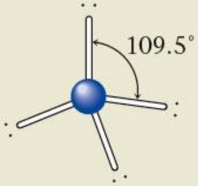
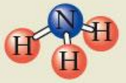
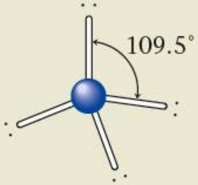
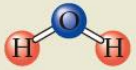
1. Draw the Lewis structure for the molecule.
2. Count the electron pairs and arrange them in the way that minimizes repulsion (put the pairs as far apart as possible).
3. Determine the positions of the atoms from the way electron pairs are shared (how electrons are shared between the central atom and surrounding atoms).
4. Determine the name of the molecular structure from positions of the atoms.

Return to TOC

Molecular Structure

Arrangements of Electron Pairs and the Resulting Molecular Structures for Two, Three, and Four Electron Pairs

Table 12.4 Arrangements of Electron Pairs and the Resulting Molecular Structures for Two, Three, and Four Electron Pairs

Number of Electron Pairs	Bonds	Electron Pair Arrangement	Ball-and-Stick Model	Molecular Structure	Partial Lewis Structure	Ball-and-Stick Model
2	2	Linear		Linear	A—B—A	
3	3	Trigonal planar (triangular)		Trigonal planar (triangular)	A | B A A	
4	4	Tetrahedral		Tetrahedral	A | A — B — A | A	
4	3	Tetrahedral		Trigonal pyramid	A — B̈ — A | A	
4	2	Tetrahedral		Bent or V-shaped	A — B — A (with lone pairs on B)	

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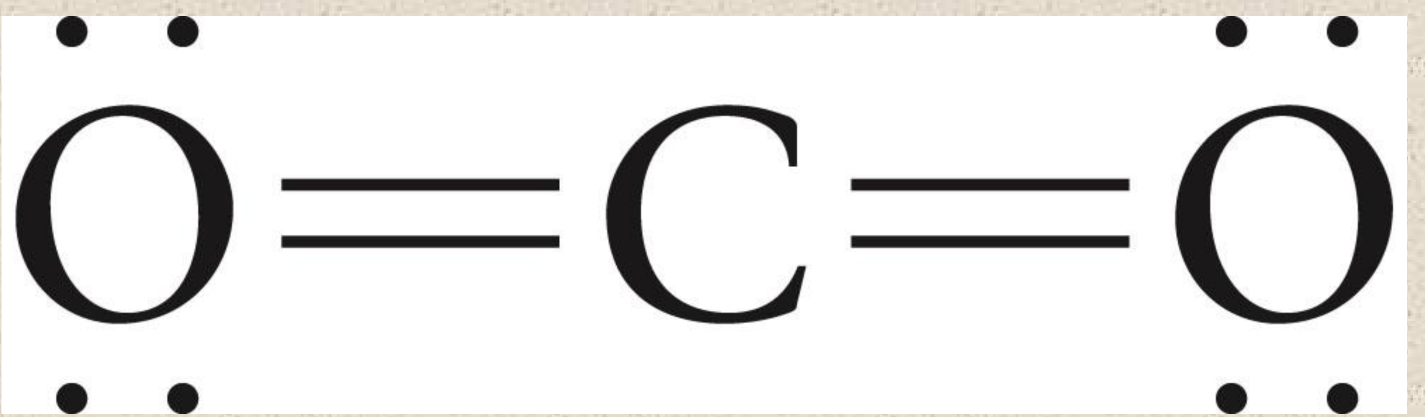
Molecular Structure

Molecules with Double Bonds

- When using the VSEPR model to predict the molecular geometry of a molecule, a double or triple bond is counted the same as a single electron pair.
  - CO_2



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Molecular Structure

Concept Check

Determine the molecular structure for each of the following molecules, and include bond angles:

- HCN
- PH_3
- SeO_2
- O_3

HCN – linear, 180^{o}

PH_3 – trigonal pyramid, 109.5^{o} (107^{o})

SeO_2 – bent, 120^{o}

O_3 – bent, 120^{o}

Return to TOC

CHAPTER SEVEN

EQUILIBRIUM CONCEPTS AND ACID-BASE EQUILIBRIUM

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Objectives of the Chapter:

- Explain the dynamic nature of a chemical equilibrium
- Predict the response of a stressed equilibrium using Le Châtelier's principle
- Identify acids, bases based on the three acid-base concepts
- Calculate the pH and pOH of a given solution
- Describe the composition and function of acid-base buffer
- Calculate the pH of the buffer before and after the addition of added acid or base

7.1. Introduction

- The convention for writing chemical equations involves placing reactant formulas on the left side of a reaction arrow and product formulas on the right side.

- By this convention, and the definitions of "**reactant**" and "**product**," a chemical equation represents the reaction in question as proceeding from **left to right.**

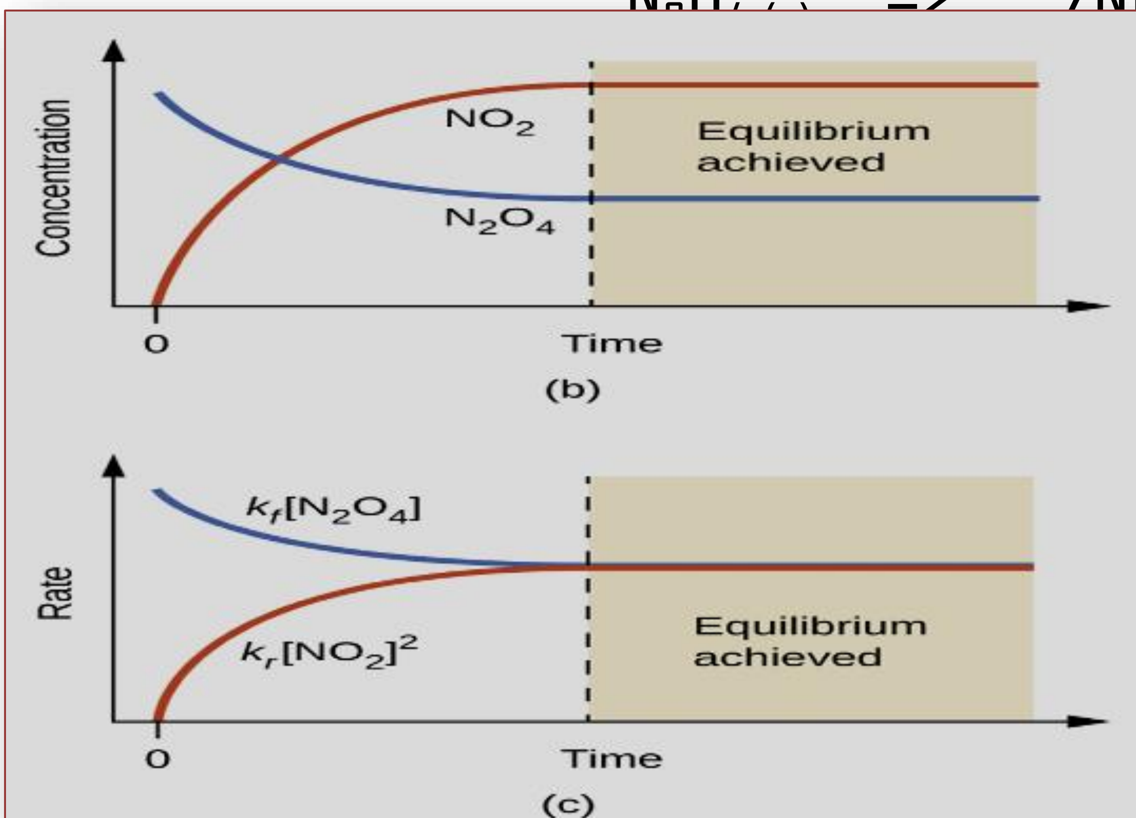
- **Reversible reactions**, however, may proceed in both forward (left to right) and reverse (right to left) directions.

- When the rates of the forward and reverse reactions are **equal**, the concentrations of the reactant and product species remain constant over time and the system is at **equilibrium.**

- The relative concentrations of reactants and products in **equilibrium** systems vary greatly; some systems contain mostly products at equilibrium, some contain mostly reactants, and some contain appreciable amounts of both.

Figure 7.1: *Illustrates fundamental equilibrium concepts using the **reversible** decomposition of colorless dinitrogen tetroxide to yield brown nitrogen dioxide, an elementary reaction described by the equation:*

$$N_2O_{4(g)} \Rightarrow 2NO_{2(g)}$$



(b) Changes in concentration over time as the decomposition reaction achieves equilibrium.

(c) At equilibrium, the forward and reverse reaction rates are equal.

- It's important to emphasize that chemical equilibria are dynamic; a reaction at equilibrium has not "**stopped**," but is proceeding in the **forward** and **reverse** directions at the same rate.

- Physical changes, such as **phase transitions**, are also reversible and may establish equilibria. As one example, consider the vaporization of bromine:

$$Br_2(l) \rightleftharpoons Br_2(g)$$

7.2. Chemical Equilibrium

- The status of a reversible reaction is conveniently assessed by evaluating its **reaction quotient (Q)**. For a reversible reaction described by:

$$mA + nB \rightleftharpoons xC + yD$$

- The reaction quotient is derived directly from the stoichiometry of the balanced equation as:

$$Q_c = \frac{[C]^x[D]^y}{[A]^m[B]^n}$$

- **Where**: The subscript *C* denotes the use of molar concentrations in the expression.

- If the reactants and products are gaseous, a reaction quotient may be similarly derived using partial pressures:

$$Q_p = \frac{P_C^x P_D^y}{P_A^m P_B^n}$$

Example 7.1: Write the concentration-based reaction quotient expression for each of the following reactions:

A. $3O_2(g) \rightleftharpoons 2O_3(g)$

$$Q_c = \frac{[O_3]^2}{[O_2]^3}$$

B. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$Q_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

C. $4NH_3(g) + 7O_2(g) \rightleftharpoons 4NO_2(g) + 6H_2O(g)$

$$Q_c = \frac{[NO_2]^4[H_2O]^6}{[NH_3]^4[O_2]^7}$$

- The **numerical** value of Q varies as a reaction proceeds towards equilibrium; therefore, it can serve as a useful indicator of the reaction's status.

- The constant value of Q exhibited by a system at equilibrium is called the **equilibrium constant,** K:

$$K = Q \qquad \text{at equilibrium}$$

$$K_c = Q_c = \frac{[C]^x [D]^y}{[A]^m [B]^n}$$

Example 7.2: Evaluating a Reaction Quotient

- Gaseous nitrogen dioxide forms dinitrogen tetroxide according to this equation:

$$2NO_{2(}g) \Rightarrow N_2O_4\,(g)$$

- When 0.10 mol NO_2 is added to a 1.0-L flask at 25°C, the concentration changes so that at equilibrium, $[NO_2] = 0.016$ M and $[N_2O_4] = 0.042$ M.

a. What is the value of the reaction quotient before any reaction occurs?
b. What is the value of the equilibrium constant for the reaction?

Solution

a. Before any reaction started or before any product is formed the concentrations of:

$$[NO_2] = \frac{0.1\,mol}{1.0\,L} = 0.1\,M \qquad [N_2O_4] = 0$$

Thus, $$K_c = Q_c = \frac{[N_2O_4]}{[NO_2]^2} = \frac{0}{(0.1\,M)^2} = 0$$

b. At equilibrium: The Equilibrium Constant is $1.6x10^2$

$$K_c = Q_c = \frac{[N_2O_4]}{[NO_2]^2} = \frac{(0.042\,M)}{(0.016\,M)^2} = 1.6x10^2$$

7.2.1. Homogeneous Equilibria

- A homogeneous equilibrium is one in which all reactants and products are present in the **same phase**.
  - By this definition, homogeneous equilibria take place in **solutions**.
  - These solutions are most commonly either liquid or gaseous phases, as shown by the examples below:

A. $C_2H_2(aq) + 2Br_2(aq) \rightleftharpoons C_2H_2Br_2(aq)$

$$K_c = \frac{[C_2H_2Br_2]}{[C_2H_2][Br_2]^2}$$

B. $HF(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + F^-(aq)$

$$K_c = \frac{[H_3O^-][F^-]}{[HF]}$$

- These examples all involve aqueous solutions, those in which water functions as the solvent.
- In the examples, water also functions as a reactant, but its concentration is *not* included in the reaction quotient.
- The reason for this omission is related to the more rigorous form of the *Q* (or *K*) expression in which *relative concentrations for liquids and solids are equal to 1 and needn't be included.*
- Consequently, reaction quotients include concentration or pressure terms only for gaseous and solute species.

- The equilibria below all involve **gas-phase** solutions:

A. $C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$ $$K_c = \frac{[C_2H_4][H_2]}{[C_2H_6]}$$

B. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

- For gas-phase solutions, the equilibrium constant may be expressed in terms of either the molar concentrations **(*Kc*)** or partial pressures **(*Kp*)** of the reactants and products.
- A relation between these **two *K*** values may be simply derived from the ideal gas equation and the definition of molarity:

$$PV = nRT \qquad P = \left(\frac{n}{V}\right)RT = MRT$$

For the gas-phase reaction:

$$mA + nB \rightleftharpoons xC + yD$$

$$K_p = \frac{(P_C)^x (P_D)^y}{(P_A)^m (P_B)^n}$$

$$K_p = \frac{([C]xRT)^x ([D]xRT)^y}{([A]xRT)^m ([B]xRT)^n}$$

$$K_p = \frac{[C]^x [D]^y}{[A]^m [B]^n} x \frac{(RT)^{x+y}}{(RT)^{m+n}} = K_C (RT)^{(x+y)-(m+n)}$$

$$K_p = K_C (RT)^{\Delta n}$$

Example 7.3: Write the equations relating ***Kc*** to ***Kp*** for each of the following reactions:

A. $C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$

$$\Delta n = 2 - 1 = 1$$

$$K_p = K_C(RT)^{\Delta n} = K_C(RT)$$

B. $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

$$\Delta n = 2 - 2 = 0$$

$$K_p = K_C(RT)^{\Delta n} = K_C(RT)^0 = K_C$$

C. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$\Delta n = 2 - (1 + 3) = -2$$

$$K_p = K_C(RT)^{\Delta n} = K_C(RT)^{-2} = \frac{K_C}{(RT)^2}$$

D. *Kc* is equal to 0.28 for the following reaction at 900 °C: What is *KP* at this temperature? $CS_2(g) + 4H_2(g) \rightleftharpoons CH_4(g) + 2H_2S(g)$

$$\Delta n = (2 + 1) - (1 + 4) = -2$$

$$K_p = K_C(RT)^{\Delta n} = K_C(RT)^{-2}$$

$$K_P = 0.28x(0.0821x1173)^{-2}$$

$$K_P = 3.02x10^{-5}$$

7.2.2. Heterogeneous Equilibria

- A heterogeneous equilibrium involves reactants and products in **two or more different phases**, as illustrated by the following examples:

A. $PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2Cl^-(aq)$ $$K_C = [Pb^{2+}][Cl^-]^2$$

B. $CaO(s) + CO_2(g) \rightleftharpoons CaCO_3(s)$ $$K_C = \frac{1}{[CO_2]} = \frac{1}{P_{CO_2}}$$

C. $C(s) + 2S(g) \rightleftharpoons CS_2(g)$ $$K_C = \frac{[CS_2]}{[S]^2} = \frac{P_{CS_2}}{(P_S)^2}$$

D. $Br_2(l) \rightleftharpoons Br_2(g)$ $$K_C = [Br_2]$$

7.3. Le Chatelier's Principle

- A system at equilibrium is in a state of dynamic balance, with forward and reverse reactions taking place at equal rates.

- If an equilibrium system is subjected to a change in conditions that affects these reaction rates differently (a stress), then the rates are no longer equal and the system is not at equilibrium.

- The system will subsequently experience a net reaction in the direction of greater rate (a shift) that will re-establish the equilibrium.

- This phenomenon is summarized by **LeChâtelier's** principle: if an equilibrium system is stressed, the system will experience a shift in response to the stress that re-establishes equilibrium.

- Reaction rates are affected primarily by **concentrations**, as described by the reaction's rate law, and **temperature**, as described by the Arrhenius equation.

- Consequently, changes in concentration and temperature are the two stresses that can shift an equilibrium.

7.3.1. Effect of a Change in Concentration

- *If an equilibrium system is subjected to a change in the concentration of a reactant or product species, the rate of either the forward or the reverse reaction will change.*

- *As an example, consider the equilibrium reaction:*

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \qquad K_C = 50 \text{ @ } 400^oc$$

$$Q_C = \frac{[HI]^2}{[H_2][I_2]} = K_C$$

- When this system is at equilibrium, the forward and reverse reaction rates are equal.

$$Rate_f = Rate_r$$

- If the system is stressed by adding reactant, either H2 or I2, the resulting increase in concentration causes the rate of the forward reaction to increase, exceeding that of the reverse reaction:

$$Rate_f > Rate_r$$

- This same shift will result if some product HI is removed from the system, which decreases the rate of the reverse reaction, again resulting in the same imbalance in rates.

7.3.2. Effect of a Change in Temperature

- When an equilibrium shifts in response to a temperature change, it is re-established with a different relative composition that exhibits a different value for the equilibrium constant.

- Predicting the shift an equilibrium will experience in response to a change in temperature is most conveniently accomplished by considering the enthalpy change of the reaction.

- For example, the decomposition of dinitrogen tetroxide is an endothermic (heat consuming) process:

$$N_2O_4\,(g) \;\; => \;\; 2NO_2(g) \qquad \Delta H=+57.20\ KJ$$

- For purposes of applying Le Chatelier's principle, heat (*q*) may be viewed as a reactant:

$$Heat \; + \; N_2O_4\,(g) \;\; => \;\; 2NO_2(g)$$

  - Raising the temperature of the system is causing to increasing the amount of a product, and so the equilibrium will shift to the right.
  - Lowering the system temperature will likewise cause the equilibrium to shift left.

7.3.3.Effect of Change in Pressure

- According to Le Chatelier's principle, if the pressure at equilibrium is increased then the reaction will proceed in that direction where the pressure is reduced.

- Since the pressure depends upon the number of moles, on increasing the pressure the reaction will proceed in that direction where the number of moles are reduced.

- If the pressure at equilibrium is decreased then the reaction will proceed in that direction where the

- For a general reaction, $mA + nB \rightleftharpoons xC + yD$
- The effect of pressure is decided by Δn. $\Delta n = (x+y) - (m+n)$
  - If Δ**n > 0**, that means the total moles of products is greater than the total moles of reactants. Lowering of pressure will favour the reaction in forward direction. That is, more products will be formed at equilibrium if the pressure is lowered.
  - If Δ**n < 0**, that means the total moles of products is less than the total moles of reactants. Increasing the pressure will favour the reaction in forward direction. That is, more products will be formed at equilibrium if the pressure is increased.
  - If Δ**n = 0**, then the change in pressure has no effect on the position of equilibrium.

For Example: In the formation of Ammonia

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

$$\Delta n = 2-(1+3) = -2$$

- *Therefore, an increase in pressure at equilibrium will favour the forward reaction.*

For the dissociation of dinitrogen tetroxide,

$$N_2O_4\,(g) \quad => \quad 2NO_2(g)$$

$$\Delta n = 2-(1) = 1$$

- *The decrease in pressure at equilibrium, favours the forward reaction.*

7.4. Equilibrium Calculations

A. Calculation of an Equilibrium Constant

Example 7.4: *Iodine molecules react reversibly with iodine ions to produce triiodide ions.*

$$I_2(aq) + I^-(g) \rightleftharpoons I^-_3(aq) \qquad K_C = \frac{[I_3^-]}{[I_2][I^-]}$$

If a solution with the concentrations of I_2 and I^- both equal to $1x10^{-3}M$ before reaction gives an equilibrium concentration of $6.61x10^{-4}M$, what is the equilibrium constant for the reaction?

Solution:

	$I_2(aq)$	+	$I^-(g)$	$\rightleftharpoons$	$I^-_3(aq)$
Initial (M)	$1x10^{-3}M$		$1x10^{-3}M$		0
Change (M)	$-x$		$-x$		$+x$
Equilibrium (M)	$1x10^{-3}M - x$		$1x10^{-3}M - x$		$+x$

- At equilibrium the concentration of I_2 is $6.61x10^{-4}$ M so that

$$1x10^{-3} - x = 6.61x10^{-4}\ M$$

$$x = 3.39x10^{-4}\ M$$

	$I_2(aq)$	+	$I^-(g)$	$\rightleftharpoons$	$I^-_3(aq)$
Initial (M)	$1x10^{-3}$M		$1x10^{-3}$M		0
Change (M)	$-3.39x10^{-4}$M		$-3.39x10^{-4}$M		$3.39x10^{-4}$M
Equilibrium (M)	$6.61x10^{-4}$M		$6.61x10^{-4}$M		$3.39x10^{-4}$M

$$K_C = \frac{[I_3^-]}{[I_2][I^-]}$$

$$K_C = \frac{3.39x10^{-4}M}{6.61x10^{-4}M\ \ x\, 6.61x10^{-4}M}$$

$$K_C = 776$$

B. Calculation of a Missing Equilibrium Concentration

Example: At 2000 ^{0}c, the value of the Kc for the reaction, $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, is $4.1x10^{-4}$. Calculate the equilibrium concentration of NO(g) in air at 1 atm pressure. The equilibrium concentrations of N2 and O2 at this pressure and temperature are 0.036 M and 0.0089 M respectively.

Solution:

$$K_C = \frac{[NO]^2}{[N_2][O_2]}$$

$$[NO]^2 = K_C[N_2][O_2]$$

$$[NO] = \sqrt{(4.1x10^{-4})(0.036)(0.0089)}$$

$$[NO] = 3.6x10^{-4}\,M$$

C. Calculation of Equilibrium Concentrations from Initial Concentrations

Example 7.6: Under certain conditions, the equilibrium constant **Kc** for the decomposition of **PCl_5(g)** in to **PCl_3 (g)** and **Cl_2 (g)** is **0.0211**. What are the equilibrium concentrations of **PCl_5**, **PCl_3** and **Cl_2** in a mixture that initially contained only **PCl_5** at a concentration of **1 M**?

Solution:

	$PCl_5(g)$	$\rightleftharpoons$ $PCl_3(g)$ +	$Cl_2(g)$
Initial (M)	1 M	0	0
Change (M)	$-x$	$+x$	$+x$
Equilibrium (M)	1 M $-x$	x	x

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = 0.0211 \qquad K_C = \frac{(x)(x)}{(1-x)} = 0.0211$$

$$0.0211 = \frac{x^2}{(1-x)} \qquad 0.0211(1-x) = x^2$$

$$x^2 + 0.0211x - 0.0211 = 0$$

- The quadratic equation **$ax^2+bx+c=0$** form can be rearranged to solve for **x**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad x = \frac{-0.0211 \pm \sqrt{(0.0211)^2 - 4(1)(-0.0211)}}{2(1)}$$

$$x = 0.135\ M$$

- The equilibrium concentrations are:
  - $[PCl_5]$ = 1.0-0.135= 0.87 M
  - $[PCl_3]$ = 0.135 M
  - $[Cl_2]$ = 0.135 M

Example 7.7: What are the concentrations at equilibrium of a 0.15 M solution of HCN?

$$HCN(g) \rightleftharpoons H^+(aq) + CN^-(aq) \qquad Kc = 4.9x10^{-10}$$

Solution:

$$K_C = \frac{[H^+][CN^-]}{[HCN]} \qquad K_C = \frac{x^2}{0.15 - x} = 4.9x10^{-10}$$

$$x <<< 0.15, \; then \; (0.15 - x) \cong 0.15$$

$$4.9x10^{-10} = \frac{x^2}{0.15} \qquad x^2 = 7.4x10^{-11}$$

$$x = 8.6x10^{-6} \; M$$

7.5. Concepts of Acid-Base

7.5.1. Arrhenius Concept

- An Arrhenius acid dissociates in water to form hydrogen ions (H^+), while an Arrhenius base dissociates in water to form hydroxide ions (OH^-).

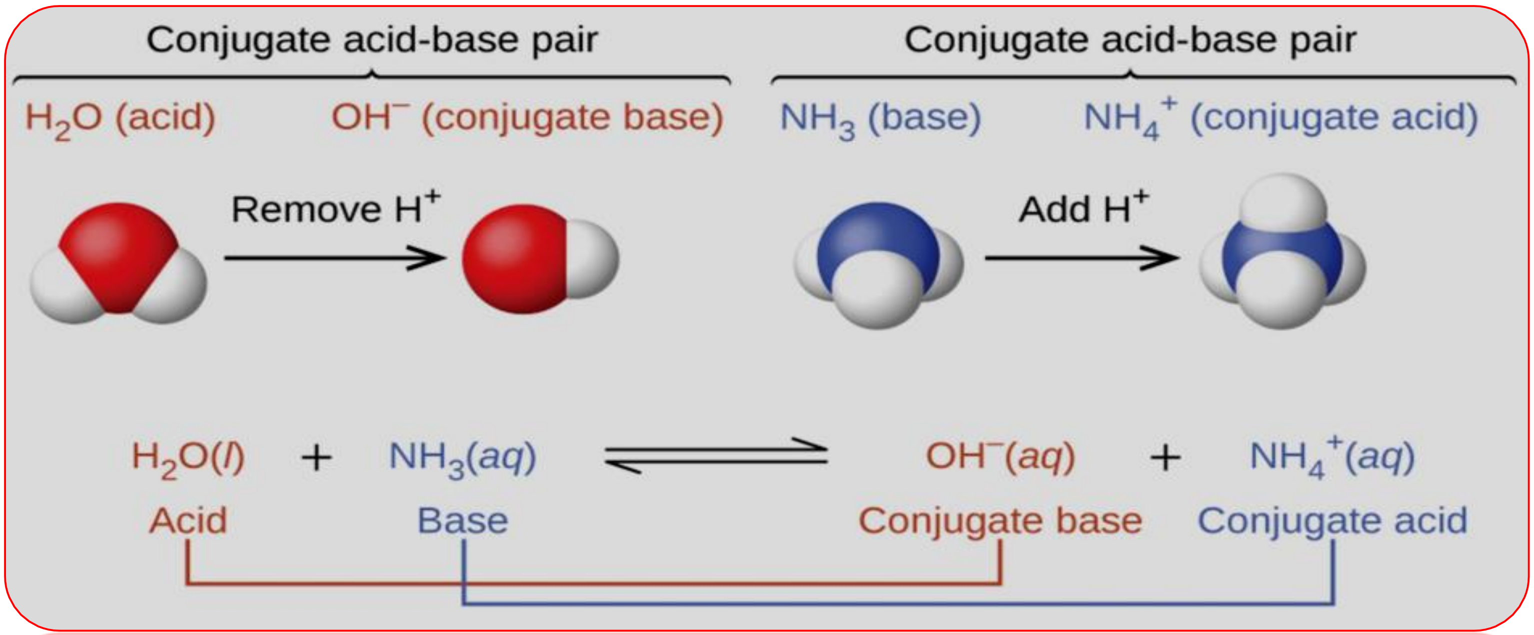
- *Limitation*:
  - The Arrhenius definitions of acidity and alkalinity are restricted to aqueous solutions.
  - Thus, the Arrhenius definition can only describe acids and bases in an aqueous environment.

7.5.2. BrØnsted-Lowery Concept

- An acid is a **proton donor**, any species that donates an H^+ ion.
  - An acid must contain H in its formula; HNO_3 and $H_2PO_4^-$ are two examples, all Arrhenius acids are Brønsted-Lowry acids.

- A base is a **proton acceptor**, any species that accepts an H^+ ion.
  - A base must contain a lone pair of electrons to bind the H^+ ion; a few examples are NH_3, CO_3^{2-}, F^-, as well as OH^-.
  - Brønsted-Lowry bases are not Arrhenius bases, but all Arrhenius bases contain the Brønsted-Lowry base OH^-.

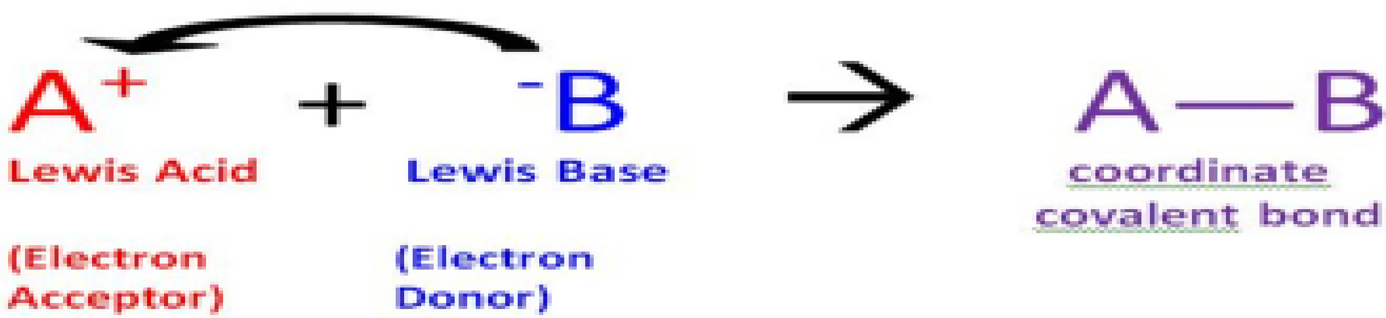
- The concept of **conjugate pairs** is useful in describing BrØnsted-Lowry acid base reactions (and other reversible reactions, as well).
  - When an acid donates H+, the species that remains is called the **conjugate base** of the acid because it reacts as a proton acceptor in the reverse reaction.
  - Likewise, when a base accepts H+, it is converted to its **conjugate acid**.



7.5.3. Lewis concept

- *A Lewis acid is defined as an **electron pair acceptor**, and a Lewis base as an **electron pair donor**.*

  - **Examples of Lewis Acids**: H^+, K^+, Mg^{2+}, Fe^{3+}, BF3, , $AlCl_3$, Br_2.
  - **Examples of Lewis Bases**: OH^-, F^-, H_2O, NH_3, SO_4^{2-}, H^-, CO, C_6H_6.

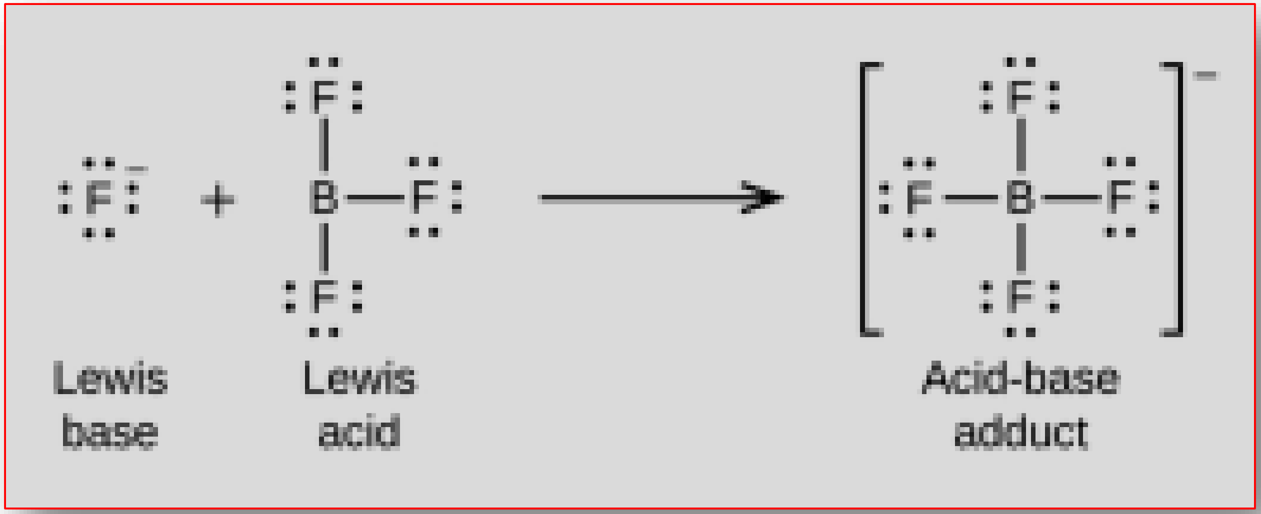


- **Lewis Acid**: a species that **accepts** an electron pair (i.e., an **electrophile**) and will have vacant orbitals, or the ability to form a vacant orbital.
- **Lewis Base**: a species that **donates** an electron pair (i.e., a **nucleophile**) and will have lonepair electrons.
- In a Lewis Acid-Base reaction a **bond** is formed where both electrons come from the Lewis base, and this is called a **Coordinate Covalent bond** (or **dative covalent bond**).
  - **Lewis Acids:** There are three general types of Lewis Acids, that is, chemical species that can accept electrons.
    A. Molecules with an incomplete octet
    B. Molecules with double bonds
    C. Metal ions with vacant d-orbitals.

A. Incomplete Octets

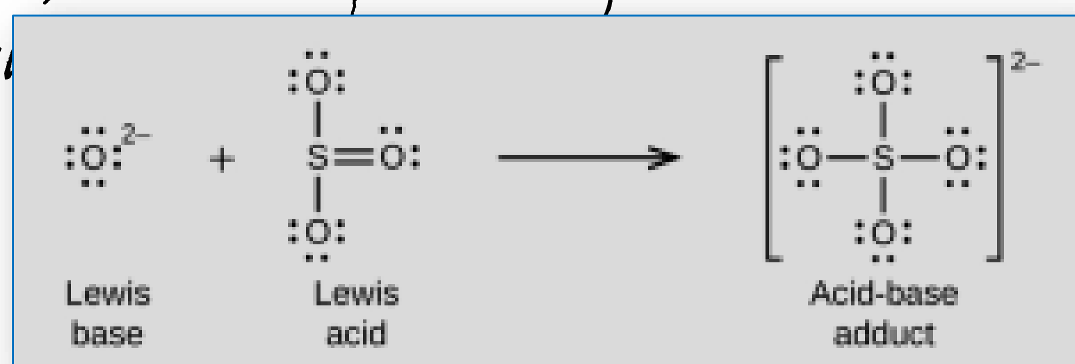
- **Boron Trifluoride** is an exception to the octet rule and has an incomplete octet, and can thus accept electrons to form a bond.

- In the process it goes from **sp^2** hybrid to **sp^3** hybrid.



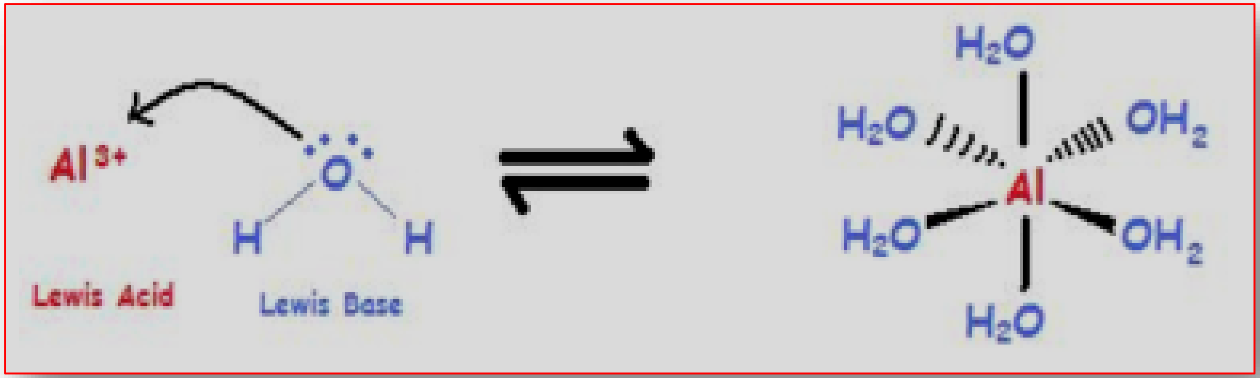
B. Double Bonds

- A double bond has **4 electrons** shared between **2 atoms** in a PI and **sigma** orbital.
- If you break the pi bond and move the two electrons to a non-bonding orbital on one atom, the other atom develops an incomplete octet and can function as a Lewis acid.
- This is shown next figure where the pi electrons move onto the oxygen, allowing the sulfur to receive electrons from a Lewis b



C. Metal Cations

- Many metals have vacant **d-orbitals** that can accept electrons and thus behave as Lewis Acids.
- In fact, most metals react with water when they dissolve in water to form what is called a complex ion.
- For example, aluminum (III) is not a free ion, but a complex with 6 water molecules.



7.6. pH and pOH

- pH/pOH is a measure of **acidity** or **basicity** of a solution.

- pH is defined as the negative logarithm of the **hydronium ion** $[H_3O^+]$ molar concentration in **mol/L.**

$$pH = -\log[H^+] = -\log[H_3O^+]$$

- The hydronium ion molarity yields the equivalent expression:

$$[H_3O^+] = 10^{-pH}$$

- **pOH** of a solution is defined as the negative logarithm of the **hydroxide ion** molar concentration in **mol/L**.

$$pOH = -\log[OH^-]$$

- The hydroxide ion molarity yields the equivalent expression:

$$[HO^-] = 10^{-pOH}$$

- From the definitions of pH and pOH and the ion-product of water,

$$pH + pOH = 14$$

- The pH and pOH of a solution is dimensionless.

7.6.1. Dissociation of Water

- **Ion-product** of water:

$$\mathbf{H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-}$$

- The dissociation constant for the above reaction is:

$$\mathbf{K_w = [H_3O^+][OH^-]}$$

$$-\log K_w = -\log([H_3O^+][OH^-])$$

$$-\log K_w = (-\log[H_3O^+]) + (-\log[OH^-])$$

$$pK_w = pH + pOH$$

- Where $\mathbf{K_w}$ is ion product of water, which is temperature dependent at 25^0C, $\mathbf{K_w}$ is $\mathbf{1 \times 10^{-14}}$.

- In **pure water**, or in a **neutral** solution of salts in water, the concentration of the $\mathbf{H_3O^+}$ and $\mathbf{OH^-}$ ions is **equal**.

$$[H_3O^+] = [OH^-] = \sqrt{Kw} = 1 \times 10^{-7}\ M,\ \text{at}\ 25^oC$$

- In **acidic** solutions, the concentration of the hydronium ions is greater than that of the hydroxyl ions, or:
    - $[H_3O^+] > 1.0 \times 10^{-7}$ M and
    - $[OH^-] < 1.0 \times 10^{-7}$ M.

- In **basic** or alkaline solutions:
    - $[OH^-]>[H_3O^+]$, or
    - $[OH^-] > 1.0 \times 10^{-7}$M.

Example 7.8: What is th **pH** of stomach acid, a solution of HCl with a hydronium ion concentration of $1.2x10^{-3}$ M?

Solution: $pH = -\log[H_3O^-]$ $pH = -\log(1.2x10^{-3}\ M)$

$$pH = 3 - \log 1.2$$

$$pH = 2.92$$

Example 7.9: Calculate the **hydronium ion** concentration of blood, the pH of which is **7.3**.

Solution: $pH = -\log[H_3O^-]$ $7.3 = -\log[H_3O^-]$

$$[H_3O^+] = 10^{-pH}$$

$$[H_3O^+] = 10^{-7.3} = anti\log of\ -7.3$$

$$[H_3O^+] = 5x10^{-8}\ M$$

7.7. Relative Strength of Acids and Bases

- The relative strength of an acid or base is the extent to which it ionizes when dissolved in water.

- If the **ionization** reaction is essentially **complete**, the acid or base is termed **strong**; if relatively little ionization occurs, the acid or base is **weak**.

Strong Acids	Strong Bases
Perchloric Acid, $HClO_4$	Lithium Hydroxide, LiOH
Hydrochloric Acid, HCl	Sodium Hydroxide, NaOH
Hydrobromic Acid, HBr	Potassium Hydroxide, KOH
Hydroiodic Acid, HI	Calcium Hydroxide, $Ca(OH)_2$
Nitric Acid, HNO_3	Strontium Hydroxide, $Sr(OH)_2$
Sulfuric Acid, H_2SO_4	Barium Hydroxide, $Ba(OH)_2$

- The relative strengths of acids or bases may be quantified by measuring their **equilibrium** constants in aqueous solutions.

- In solutions of the same concentration, stronger acids ionize to a greater extent, and so yield higher concentrations of hydronium ions than do weaker acids.

- The equilibrium constant for an **weak acid** is called the **acid-ionization constant**, K_a. For the reaction of an acid **HA**:

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

- The equilibrium constant for an **weak Base** is called the **Base-ionization constant**, K_b. For the reaction of an acid **B**:

$$B(aq) + H_2O(l) \rightleftharpoons BH^+(aq) + OH^-(aq)$$

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

- Another measure of the strength of an acid or base is its percent ionization.

- The percent or degree of ionization of a **acid or base** is defined in terms of the composition of an equilibrium mixture:

$$\%\ ionization = \alpha = \frac{[H_3O^+]}{[HA]} x100$$

$$\%\ ionization = \alpha = \frac{[OH^-]}{[B]} x100$$

Example 7.10: Calculate the $[H^+]$ in a 0.1 M HCNO solution. What is the degree of ionization of cyanic acid in this same solution? ($K_1=2\times10^{-4}$)

Solution: - $HCNO(aq) + H_2O(aq) \longleftrightarrow H_3O^+(aq) + CNO^-(aq)$

	HCNO	H_3O^+	CNO^-
Initial	0.1m	0 M	0 M
Change	-x	+x	+x
$[\ \ ]_{eq.}$	0.1 – x	x	x

$$K_a = \frac{[H_3O^+][CNO^-]}{[HCNO]} \qquad K_a = \frac{x^2}{0.1-x} = \frac{x^2}{0.1} = 2x10^{-4}$$

$$x = [H^+] = [CNO^-] = 0.0045M$$

$$\alpha = \frac{[H_3O^+]}{[HCNO]}x100 \qquad \alpha = \frac{0.0045M}{0.1M}x100 = 4.5\%$$

Example 7.11: Calculate the OH^- ion conc. in 0.4 M NH_3 solution. What is the degree of ionization of NH_3 solution in this same solution? ($K_b = 1.8\times10^{-5}$)

Solution: -

	$NH_{3\,(aq)} + H_2O_{(aq)}$	$\longleftrightarrow$	$NH_4^+{}_{(aq)}$	$+ OH^-_{(aq)}$
Initial	0.4 M		0 M	0 M
Change	-x		+x	+x
$[\ \ ]_{Eq}$	(0.4 – x) M		x M	x M

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} \qquad K_b = \frac{x^2}{0.4-x} = \frac{x^2}{0.4} = 1.8x10^{-5}$$

$$x = [OH^-] = 2.7x10^{-3}M$$

$$\alpha = \frac{[OH^-]}{[NH_3]}x100 \qquad \alpha = \frac{2.7x10^{-3}M}{0.4M}x100 = 0.675\%$$

7.7.1. p-Functions

- The p- value is the negative base (-10) logarithm of the molar concentration of a certain species:

$$\mathbf{pX = -log\,[X] = log\,1/[X]}$$

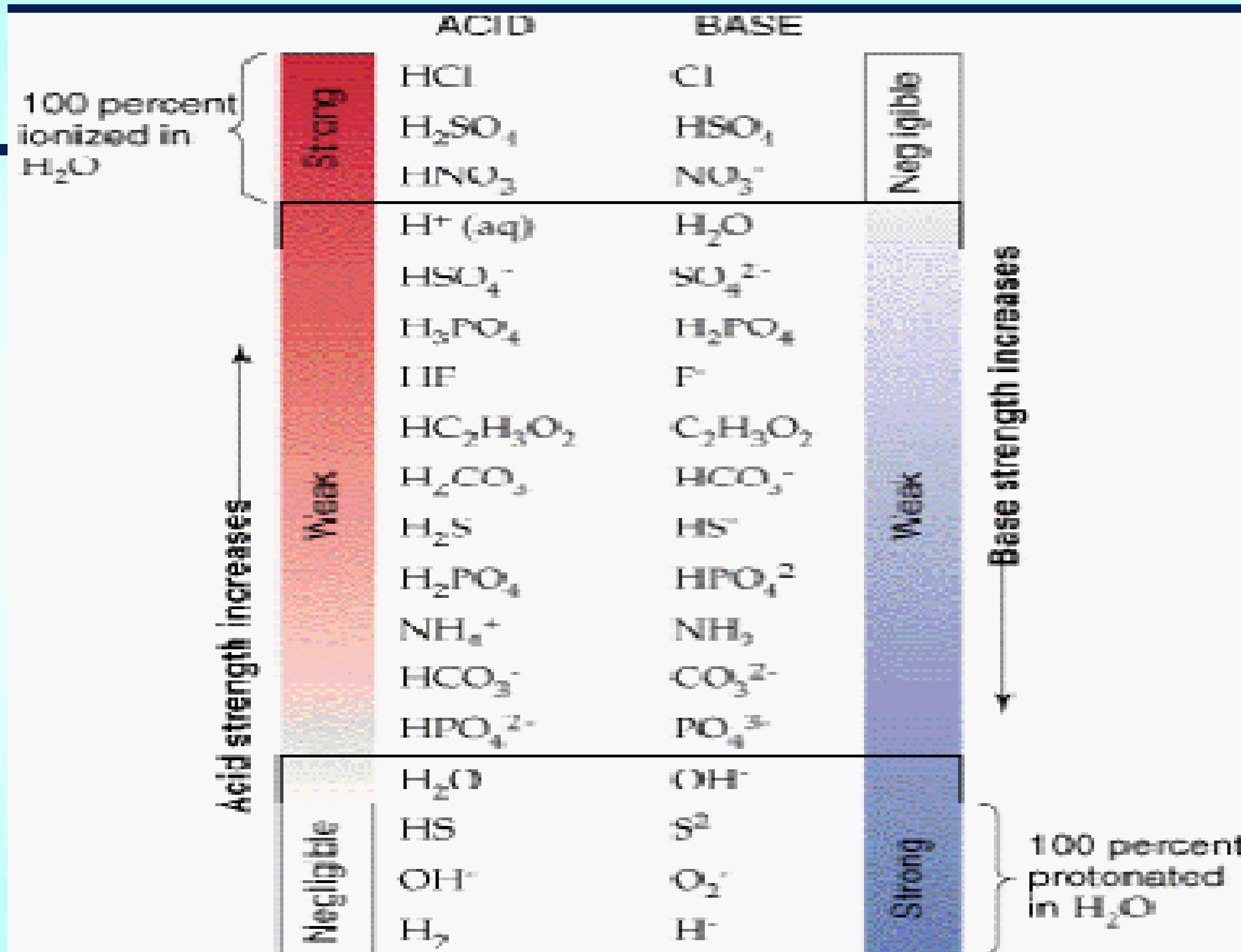
- We can also express equilibrium constants for the strength of acids and bases in a log form:

$$\mathbf{pK_a = -\,log(K_a)}$$

$$\mathbf{pK_b = -\,log\,(K_b)}$$

$$\mathbf{K_w = K_a * K_b}$$

7.7.2. Strength of Acids and Bases



Strength of Acids and Bases

TABLE 17.3 Ionization Constants of Some Weak Acids and Weak Bases in Water at 25 °C

	Ionization Equilibrium	Ionization Constant K	pK	
Acid		K_a =	pK_a =	Acid strength
Iodic acid	$HIO_3 + H_2O \rightleftharpoons H_3O^+ + IO_3^-$	1.6×10^{-1}	0.80	
Chlorous acid	$HClO_2 + H_2O \rightleftharpoons H_3O^+ + ClO_2^-$	1.1×10^{-2}	1.96	
Chloroacetic acid	$HC_2H_2ClO_2 + H_2O \rightleftharpoons H_3O^+ + C_2H_2ClO_2^-$	1.4×10^{-3}	2.85	
Nitrous acid	$HNO_2 + H_2O \rightleftharpoons H_3O^+ + NO_2^-$	7.2×10^{-4}	3.14	
Hydrofluoric acid	$HF + H_2O \rightleftharpoons H_3O^+ + F^-$	6.6×10^{-4}	3.18	
Formic acid	$HCHO_2 + H_2O \rightleftharpoons H_3O^+ + CHO_2^-$	1.8×10^{-4}	3.74	
Benzoic acid	$HC_7H_5O_2 + H_2O \rightleftharpoons H_3O^+ + C_7H_5O_2^-$	6.3×10^{-5}	4.20	
Hydrazoic acid	$HN_3 + H_2O \rightleftharpoons H_3O^+ + N_3^-$	1.9×10^{-5}	4.72	
Acetic acid	$HC_2H_3O_2 + H_2O \rightleftharpoons H_3O^+ + C_2H_3O_2^-$	1.8×10^{-5}	4.74	
Hypochlorous acid	$HOCl + H_2O \rightleftharpoons H_3O^+ + OCl^-$	2.9×10^{-8}	7.54	
Hydrocyanic acid	$HCN + H_2O \rightleftharpoons H_3O^+ + CN^-$	6.2×10^{-10}	9.21	
Phenol	$HOC_6H_5 + H_2O \rightleftharpoons H_3O^+ + C_6H_5O^-$	1.0×10^{-10}	10.00	
Hydrogen peroxide	$H_2O_2 + H_2O \rightleftharpoons H_3O^+ + HO_2^-$	1.8×10^{-12}	11.74	
Base		K_b =	pK_b =	Base strength
Diethylamine	$(C_2H_5)_2NH + H_2O \rightleftharpoons (C_2H_5)_2NH_2^+ + OH^-$	6.9×10^{-4}	3.16	
Ethylamine	$C_2H_5NH_2 + H_2O \rightleftharpoons C_2H_5NH_3^+ + OH^-$	4.3×10^{-4}	3.37	
Ammonia	$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	1.8×10^{-5}	4.74	
Hydroxylamine	$HONH_2 + H_2O \rightleftharpoons HONH_3^+ + OH^-$	9.1×10^{-9}	8.04	
Pyridine	$C_5H_5N + H_2O \rightleftharpoons C_5H_5NH^+ + OH^-$	1.5×10^{-9}	8.82	
Aniline	$C_6H_5NH_2 + H_2O \rightleftharpoons C_6H_5NH_3^+ + OH^-$	7.4×10^{-10}	9.13	

7.7.3. The Common Ion Effect

- In acid-base ionization of a single electrolyte there is formation of one anion and one cation of the solute material.

- What happens when a solution contains two different dissolved compounds?

Example 7.12: - Both CH_3COONa and CH_3COOH are in solution and they dissociate and ionize to produce CH_3COO^- ions:

$$CH_3COONa_{(s)} \xrightarrow{H_2O} CH_3COO^-_{(aq)} + Na^+_{(aq)}$$

$$CH_3COOH_{(aq)} \longleftrightarrow CH_3COO^-_{(aq)} + H^+_{(aq)}$$

- CH_3COONa is a strong electrolyte, so it dissociates completely in solution, but CH_3COOH, a weak acid, ionizes partially.

- According to *Le Chatelier's principle*, the addition of CH_3COO^- ions from CH_3COONa to a solution of CH_3COOH will **suppress** the ionization of CH_3COOH, that is, shifts the equilibrium to the **left** there by **decreasing** the H^+ ion concentration.

- Thus, a solution containing both CH_3COOH and CH_3COONa will be **less acidic** than a solution containing only CH_3COOH at the same concentration.

- The shift in equilibrium of the CH_3COOH ionization is caused by the additional acetate, CH_3COO^-, ions which comes from the salt.

- The shift in equilibrium caused by the addition of a compound having an ion in common with the dissolved substances is called the *Common ion effect*. It is a special case of Le Chatelier's principle.

7.8. Buffer solutions

- It is a solution that has the ability to resist change in pH up on the addition of small amounts of either acid or base.

  - Buffer solution is a solution of a weak acid/weak base and its salt.

- A buffer solution must contain an acid to react with OH^- ions that may be added to it and a base to react with any added H^+ ions.

- Furthermore, the acid and the base components of the buffer must not consume each other in a neutralization reaction.

Buffer solution

Types of buffer solution

a. **Acidic buffer:** CH_3COOH and CH_3COONa

b. **Basic buffer:** NH_4OH and NH_4Cl

c. **Salt or neutral buffer:** CH_3COONH_4

- Now, to find the pH, Let us consider

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)$$

$$K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]}$$

- Since CH_3COOH is parially ionized, so most of CH_3COO^- will come from the salt.

$$[H_3O^+] = K_a \frac{[CH_3COOH]}{[CH_3COO^-]}$$

- Hence, $[CH_3COO^-] = [Salt]$

- Taking log of both sides

$$-\log[H_3O^+] = -\log K_a - \log\frac{[Acid]}{[Salt]}$$

- **Similarly, for Basic Buffer**

$$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

$$pOH = pK_b + \log\frac{[Salt]}{[Base]}$$

Example 7.13: Calculate the pH of a buffer solution containing 0.1 M each of acetic acid and sodium acetate. (K_a=1.8×10^{-5})

- What will be the change in pH on adding

  (a) 0.01 moles of HCl to 1.0 L of solution?

  (b) 0.01 moles of NaOH to 1.0 L of solution?

- Assume that no change in volume occurs on the addition of HCl or NaOH.

Solution:

$$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

$$pH = -\log(1.8x10^{-5}) + \log\frac{0.1}{0.1}$$

$$pH = 4.745 + 0$$

$$pH = 4.745$$

a) 0.01 moles of HCl give 0.01 moles of H^+ which reacts with 0.01 moles of CH_3COO^- to form CH_3COOH. Therefore,

$$[CH_3COO^-] = 0.1 - 0.01 = 0.09M$$

$$[CH_3COOH] = 0.1 + 0.01 = 0.11M$$

$$pH = 4.745 + \log\frac{0.09}{0.11} \qquad pH = 4.745 - 0.087$$

$$pH = 4.658$$

- Change in pH = 4.745 – 4.658 = 0.087 units.
- Hence, pH will decrease!

(b) On adding 0.01 moles of NaOH to a litre of solution, 0.01 moles of OH^- will react with 0.01 moles of CH_3COOH. Therefore,

$$[CH_3COO^-] = 0.1 + 0.01 = 0.11M$$

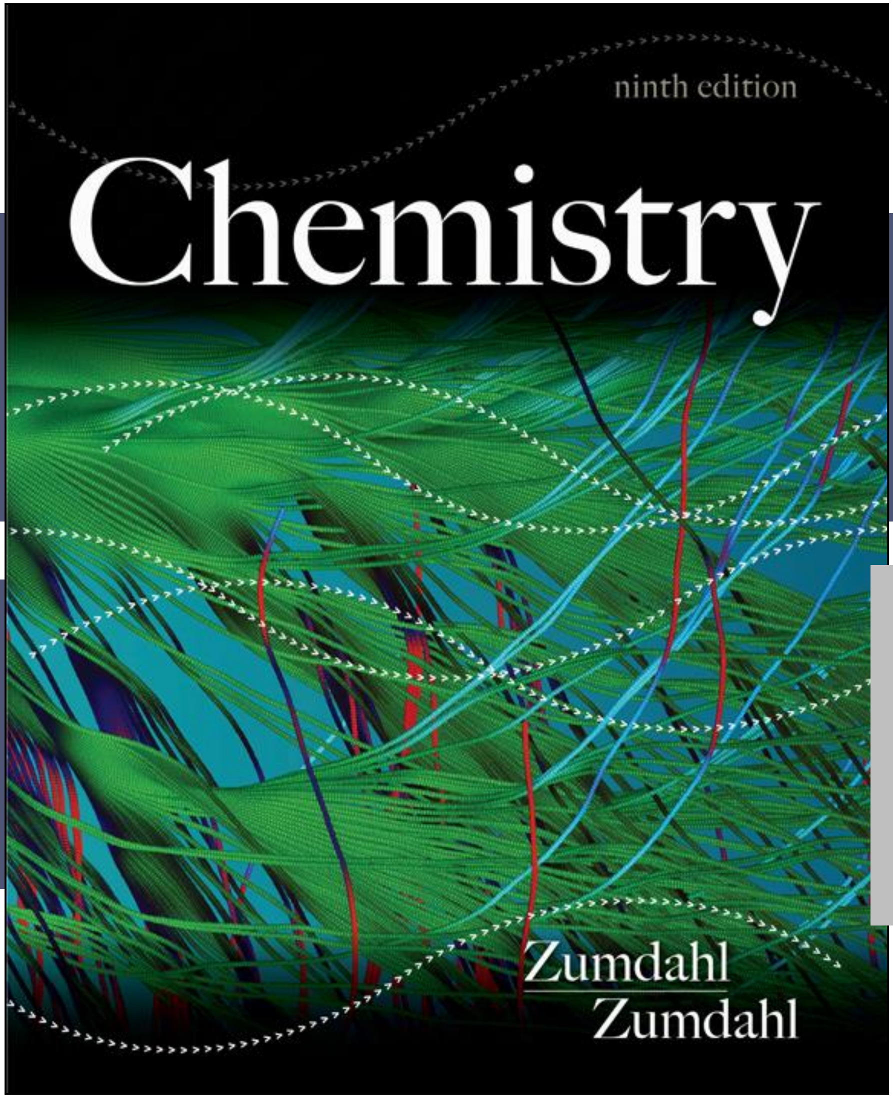
$$[CH_3COOH] = 0.1 - 0.01 = 0.09M$$

$$pH = 4.745 + \log\frac{0.11}{0.09}$$

$$pH = 4.745 + 0.086$$

$$pH = 4.831$$

- Change in pH = 4.831 – 4.745 = 0.086 units.
- Hence, pH will increase!



Chapter Eight

ORGANIC CHEMISTRY

Chapter 8

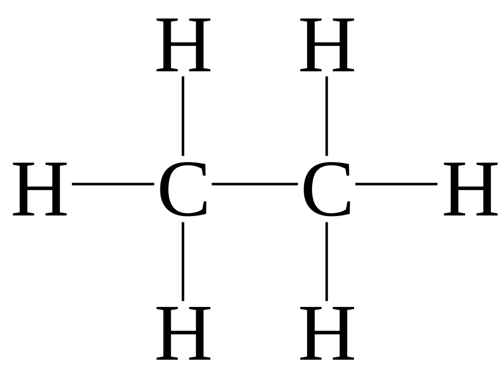
Organic Chemistry

- The study of carbon-containing compounds and their properties.
- The vast majority of organic compounds contain chains or rings of carbon atoms.

Hydrocarbons

- Compounds composed of carbon and hydrogen.
- Saturated: C—C bonds are all single bonds.

alkanes [C_nH_{2n+2}]



Alkanes

Alkanes have the general formula C_nH_{2n+2} where n = 1,2,3,…

- only single covalent bonds
- ***saturated hydrocarbons*** because they contain the **maximum** number of hydrogen atoms that can bond with the number of carbon atoms in the molecule

CH_4	C_2H_6	C_3H_8
methane	ethane	propane

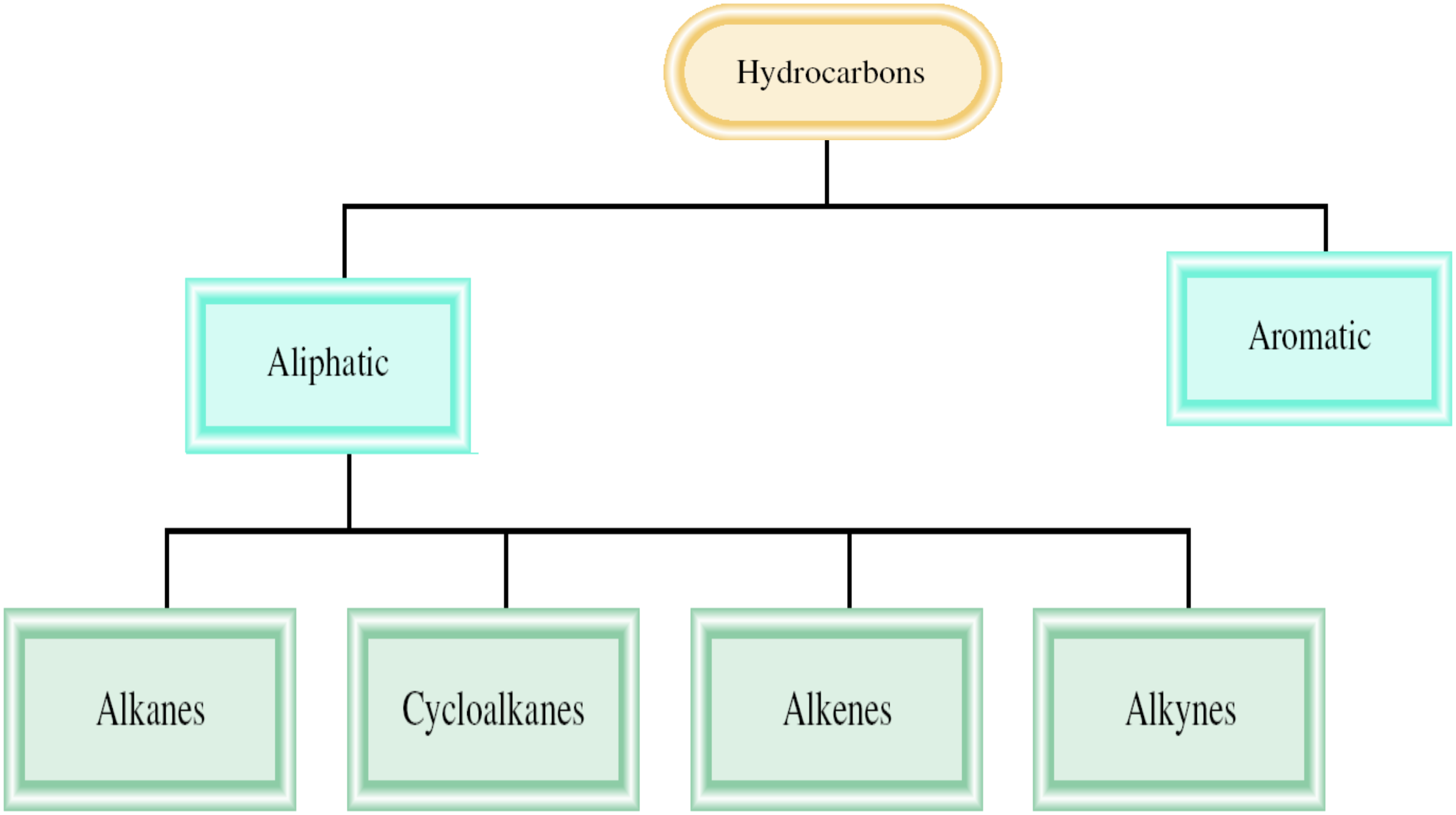
First Ten Normal Alkanes

Table 22.1 | Selected Properties of the First Ten Normal Alkanes

Name	Formula	Molar Mass	Melting Point (°C)	Boiling Point (°C)	Number of Structural Isomers
Methane	CH_4	16	−182	−162	1
Ethane	C_2H_6	30	−183	−89	1
Propane	C_3H_8	44	−187	−42	1
Butane	C_4H_{10}	58	−138	0	2
Pentane	C_5H_{12}	72	−130	36	3
Hexane	C_6H_{14}	86	−95	68	5
Heptane	C_7H_{16}	100	−91	98	9
Octane	C_8H_{18}	114	−57	126	18
Nonane	C_9H_{20}	128	−54	151	35
Decane	$C_{10}H_{22}$	142	−30	174	75

Classification of Hydrocarbons

Alkanes: Saturated Hydrocarbons

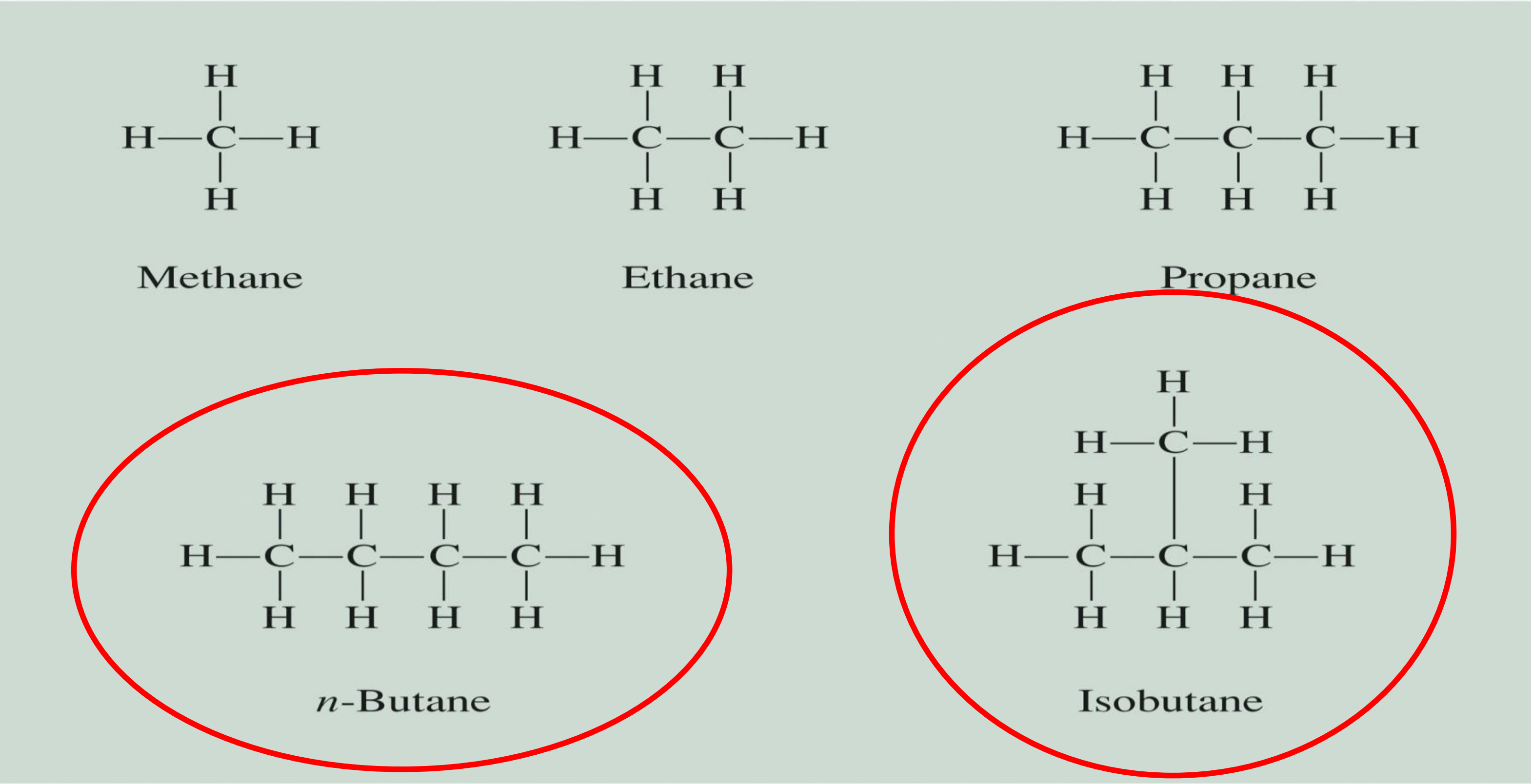


Isomerism in Alkanes

- **Structural isomerism** – occurs when two molecules have the same atoms but different bonds.
  - Butane and all succeeding members of the alkanes exhibit structural isomerism.

Section 8.1

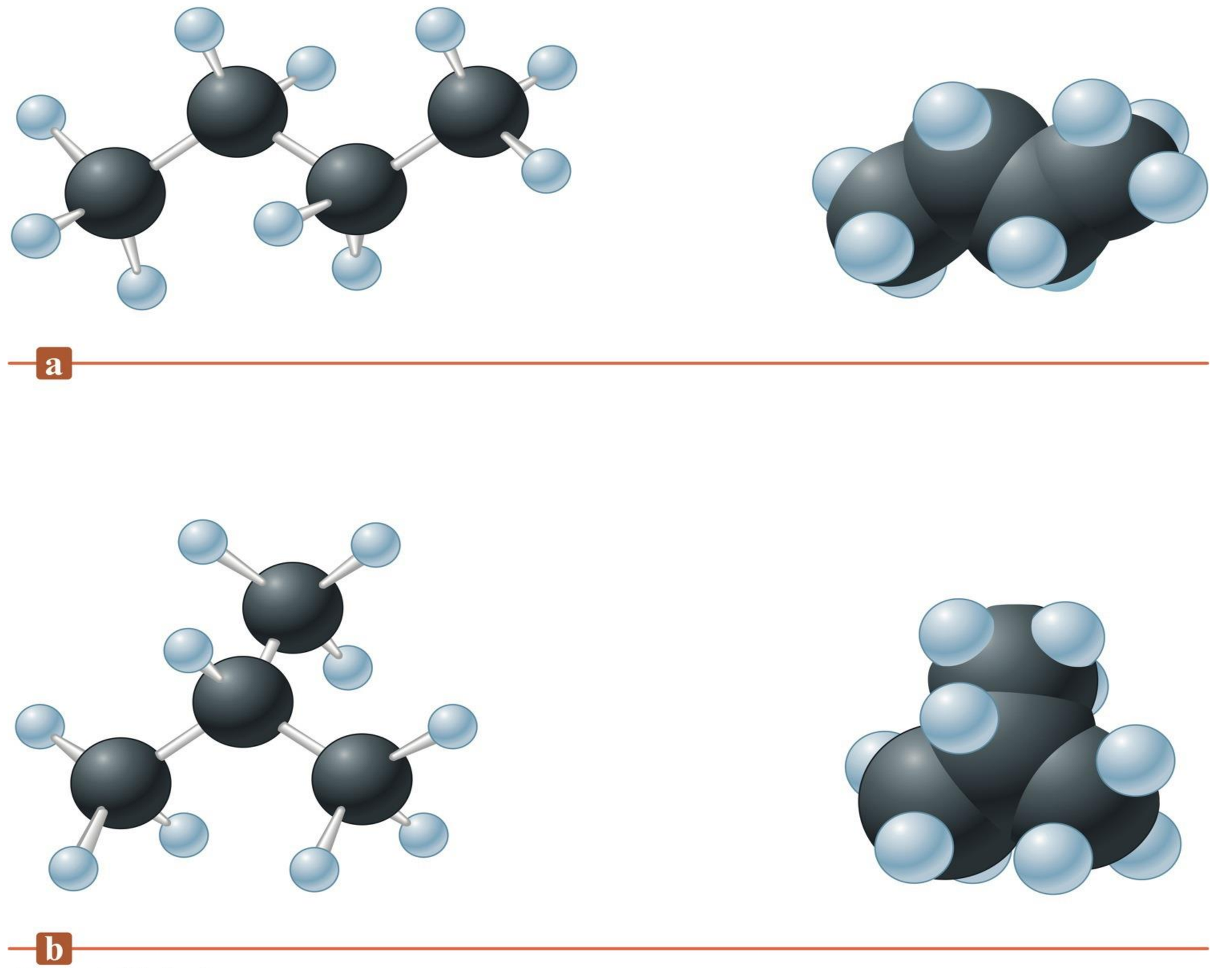
Alkanes: Saturated Hydrocarbons



- ***Structural isomers*** are molecules that have the same molecular formula but different structures.

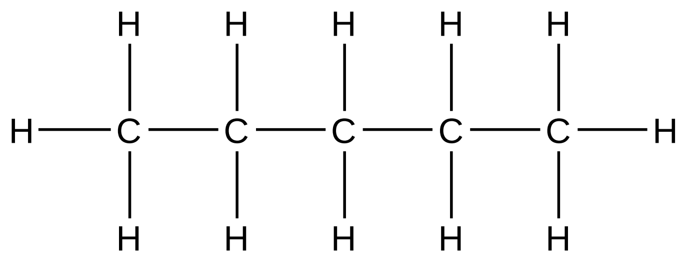
Alkanes: Saturated Hydrocarbons

Butane

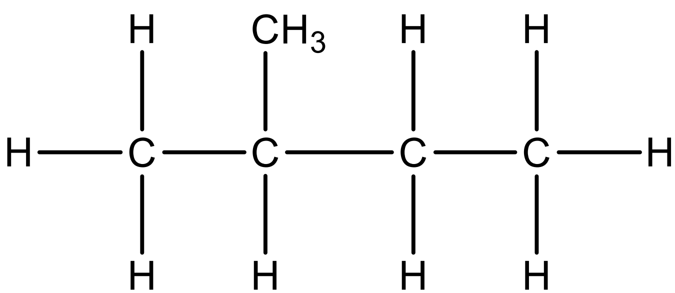


Structural Isomers of Pentane

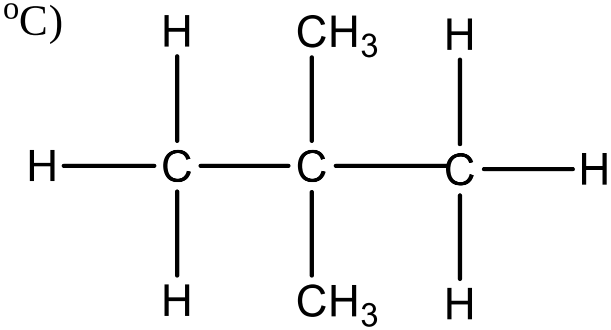
- n-Pentane, Isopentane, and neopentane



n-pentane
(b.p. 36.1 °C)



Isopentane 2-methylbutane
(b.p. 27.9 °C)



Neopentane
2, 2-dimethylpropane
(b.p. 9.5 °C)

Rules for Naming Alkanes

1. For alkanes beyond butane, add –*ane* to the Greek root for the number of carbons.

   $CH_3–CH_2–CH_2–CH_2–CH_2–CH_3$ => hex**ane**

2. Alkyl substituents: drop the –*ane* and add –*yl*.
   - C_2H_6 is eth**ane**
   - C_2H_5 is eth**yl**

3. Positions of substituent groups are specified by numbering the longest chain sequentially. The numbering is such that substituents are at lowest possible number along chain.

$$
\begin{array}{cccccc}
 & & CH_3 & & & \\
 & & | & & & \\
CH_3- & CH_2- & CH- & CH_2- & CH_2- & CH_3 \\
1 & 2 & 3 & 4 & 5 & 6
\end{array}
$$

3-methylhexane

4. Location and name are followed by root alkane name. Substituents in alphabetical order and use *di–*, *tri–*, etc.

$$\begin{array}{cccccc} & & CH_3 & CH_3 & & \\ & & | & | & & \\ CH_3- & CH_2- & CH- & CH- & CH_2- & CH_3 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{array}$$

3,4-dimethylhexane

The Most Common Alkyl Substituents and Their Names

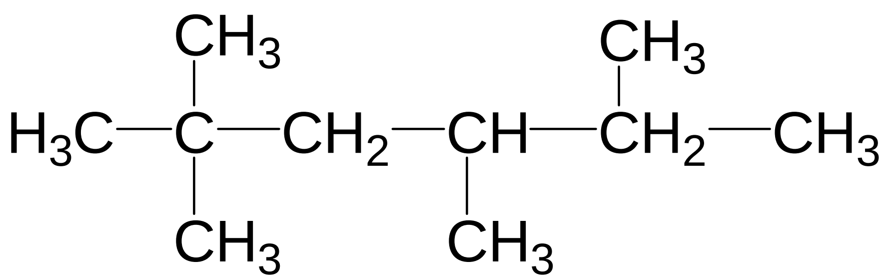
Table 22.2 | The Most Common Alkyl Substituents and Their Names

Structure*	Name†
$-CH_3$	Methyl
$-CH_2CH_3$	Ethyl
$-CH_2CH_2CH_3$	Propyl
$CH_3\overset{\vert}{C}HCH_3$	Isopropyl
$-CH_2CH_2CH_2CH_3$	Butyl
$CH_3\overset{\vert}{C}HCH_2CH_3$	*sec*-Butyl
$-CH_2-\overset{\overset{H}{\vert}}{\underset{\underset{CH_3}{\vert}}{C}}-CH_3$	Isobutyl
$-\overset{\overset{CH_3}{\vert}}{\underset{\underset{CH_3}{\vert}}{C}}-CH_3$	*tert*-Butyl

*The bond with one end open shows the point of attachment of the substituent to the carbon chain.
†For the butyl groups, *sec*- indicates attachment to the chain through a secondary carbon, a carbon atom attached to *two* other carbon atoms. The designation *tert*- signifies attachment through a tertiary carbon, a carbon attached to *three* other carbon atoms.

EXERCISE!

Name each of the following:

a) 

Answer: 2,2,4,5-tetramethylhexane

Combustion Reactions of Alkanes

- At a high temperature, alkanes react vigorously and exothermically with oxygen.
- Basis for use as fuels.

$$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(g)$$

Substitution Reactions of Alkanes

- Primarily where halogen atoms replace hydrogen atoms.

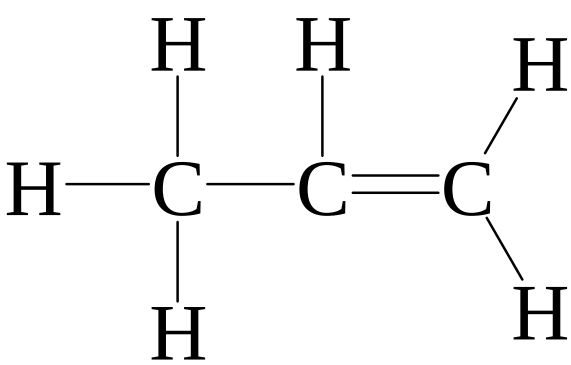
$$CH_4 + Cl_2 \xrightarrow{hv} CH_3Cl + HCl$$

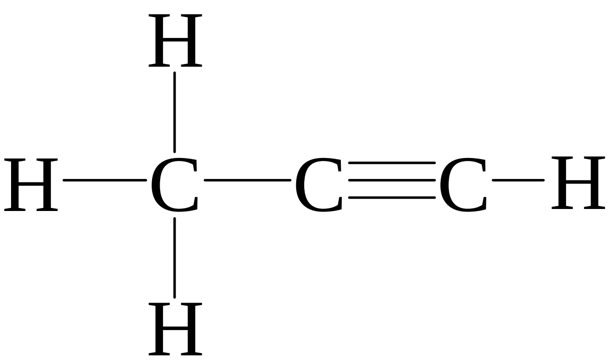
$$CH_3Cl + Cl_2 \xrightarrow{hv} CH_2Cl_2 + HCl$$

$$CH_2Cl_2 + Cl_2 \xrightarrow{hv} CHCl_3 + HCl$$

$$CHCl_3 + Cl_2 \xrightarrow{hv} CCl_4 + HCl$$

- **Unsaturated**: contains carbon–carbon multiple bonds.





Dehydrogenation Reactions of Alkanes

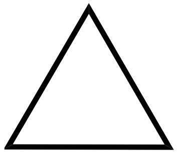
- Hydrogen atoms are removed and the product is an unsaturated hydrocarbon.

$$CH_3CH_3 \xrightarrow[500°C]{Cr_2O_3} \underbrace{CH_2{=}CH_2}_{\text{Ethylene}} + H_2$$

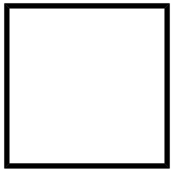
Cyclic Alkanes

- Carbon atoms can form rings containing only C—C single bonds.
- General formula: C_nH_{2n}

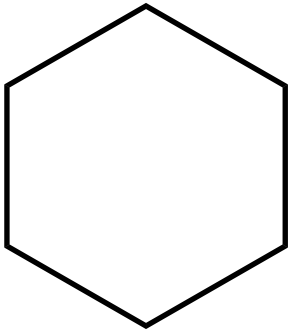
C_3H_6



C_4H_8

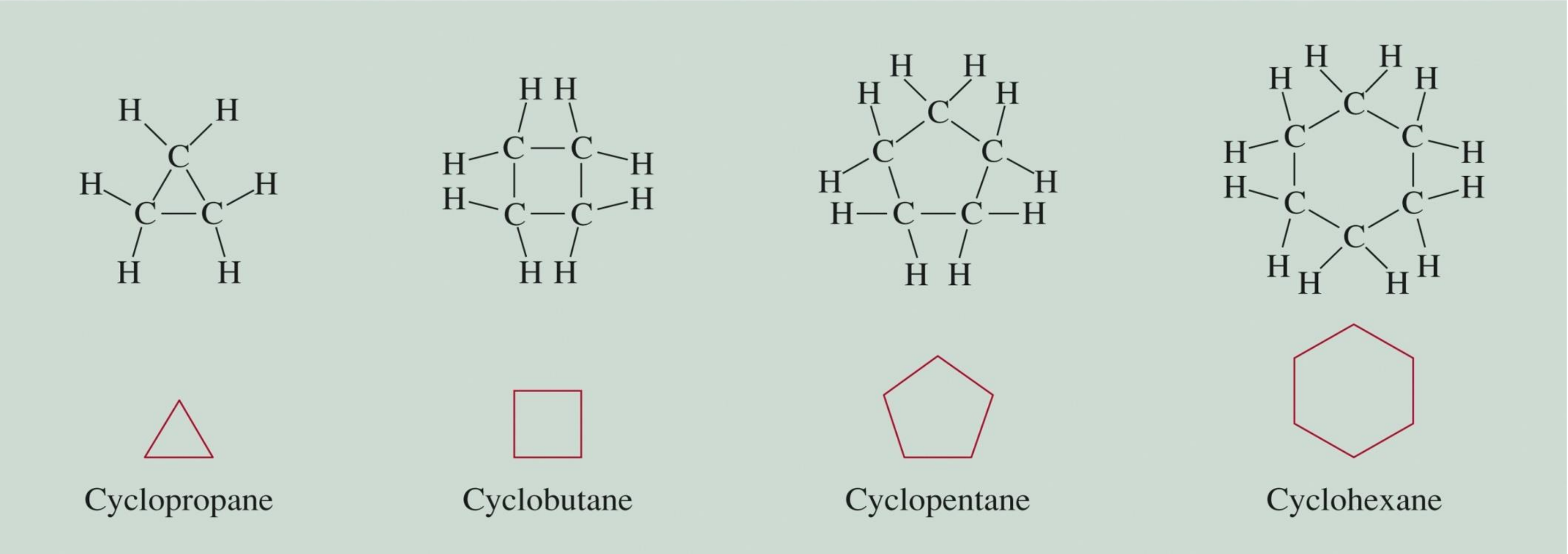


C_6H_{12}

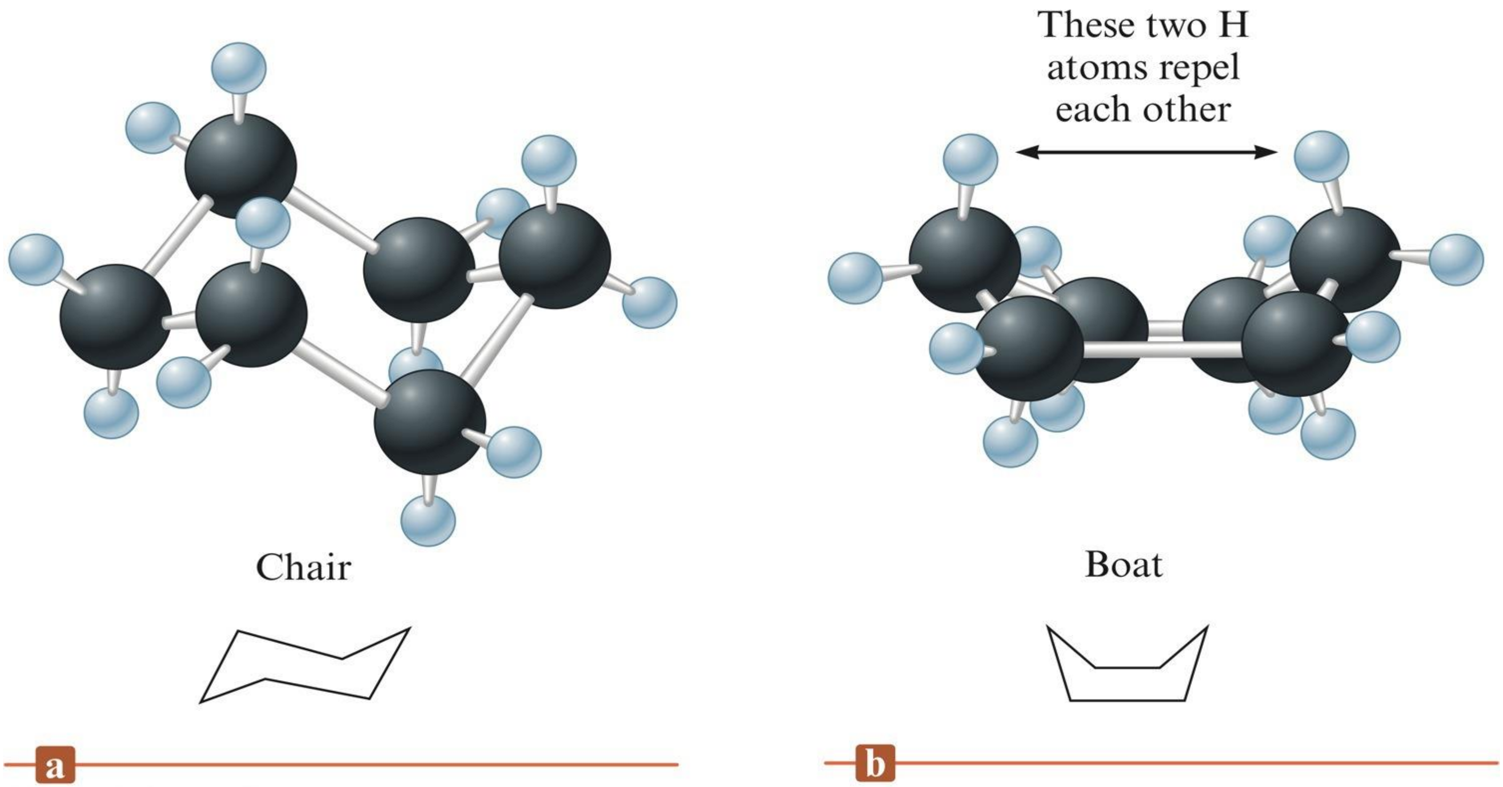


Cycloalkanes

- Alkanes whose carbon atoms are joined in rings are called **cycloalkanes.**
- They have the general formula C_nH_{2n} where $n = 3,4,\ldots$



The Chair and Boat Forms of Cyclohexane



- **Alkenes**: hydrocarbons that contain at least one carbon–carbon double bond. **[C_nH_{2n}]**

  $CH_3–CH=CH_2$ propene

- **Alkynes**: hydrocarbons containing at least one carbon–carbon triple bond. **[C_nH_n]**

  $CH_3–CH_2–C≡C–CH_3$ 2–pentyne

Rules for Naming Alkenes

1. Root hydrocarbon name ends in ***–ene***.

   C_2H_4 is eth**ene**

2. With more than 3 carbons, double bond is indicated by the lowest–numbered carbon atom in the bond.

   $CH_2{=}CH{-}CH_2{-}CH_3$

   1 2 3 4

   1–butene

Rules for Naming Alkynes

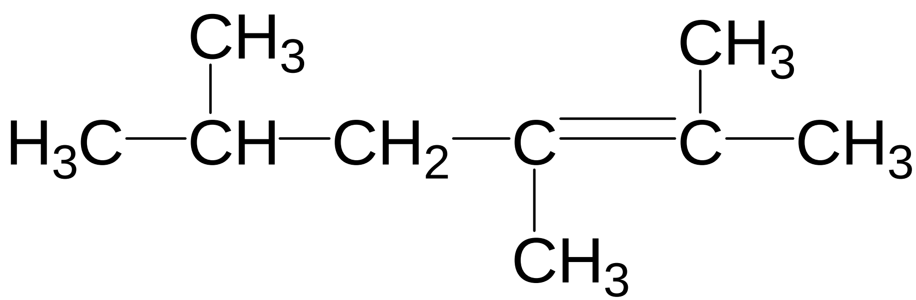
- Same as for alkenes except use –*yne* as suffix.

$$CH_3{-}CH_2{-}C{\equiv}C{-}CH_2{-}CH_2{-}CH_2{-}CH_3$$

3–oct**yne**

EXERCISE!

Name each of the following:

a) 

2,3,5-trimethyl-2-hexene

Addition Reactions

- Pi Bonds (which are weaker than the C—C bonds), are broken, and new σ bonds are formed to the σ atoms being added.

$$CH_2 = CHCH_3 + H_2 \xrightarrow{\text{catalyst}} CH_3CH_2CH_3$$

Halogenation Reactions

- Addition of halogen atoms of alkenes and alkynes.

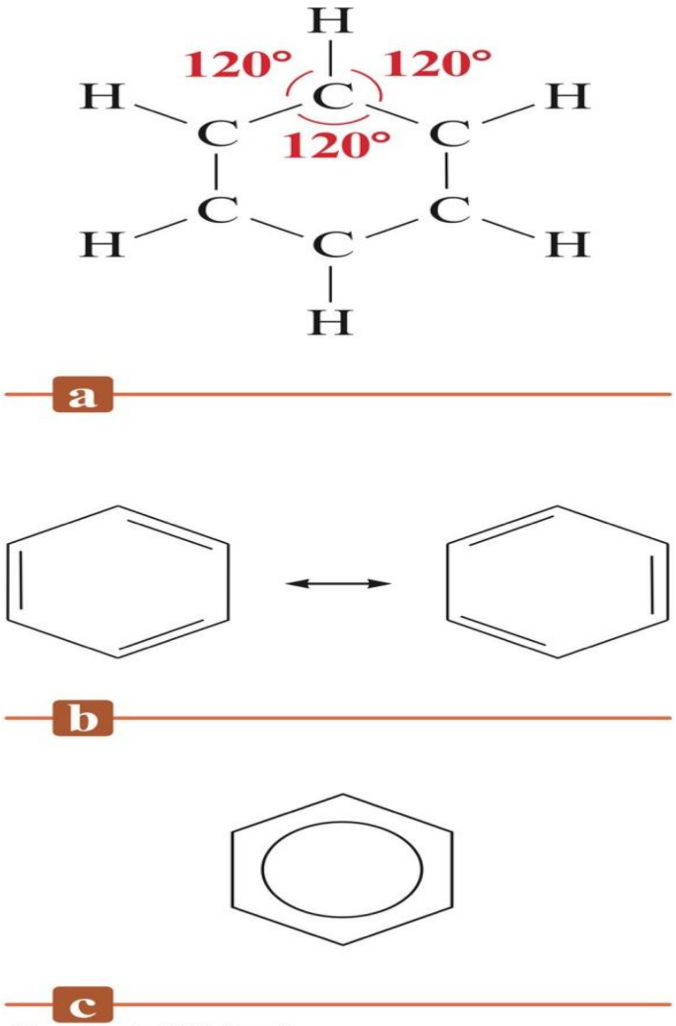
$$CH_2{=}CHCH_2CH_2CH_3 + Br_2 \longrightarrow CH_2BrCHBrCH_2CH_2CH_3$$

1-Pentene → 1,2-Dibromopentane

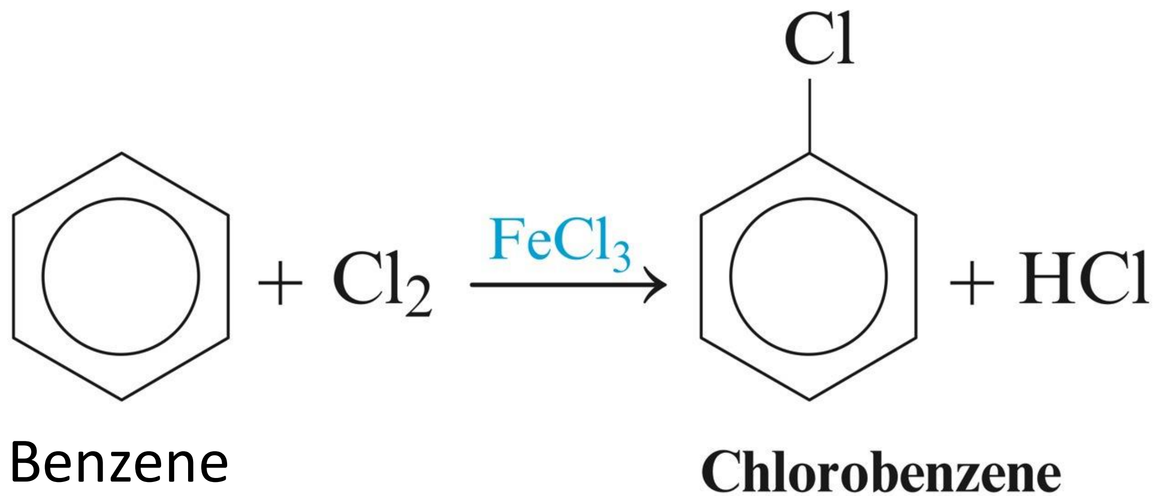
Aromatic Hydrocarbons

- A special class of cyclic unsaturated hydrocarbons.
- Simplest of these is benzene (C_6H_6).
- The delocalization of the π electrons makes the benzene ring behave differently from a typical unsaturated hydrocarbon.

Benzene (Aromatic Hydrocarbon)

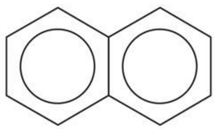
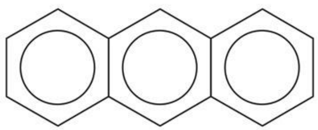
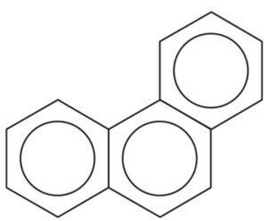
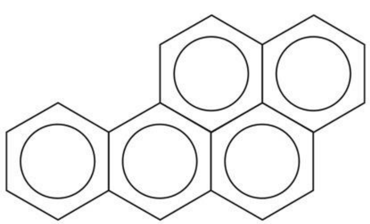


- Unsaturated hydrocarbons generally undergo rapid addition reactions, but benzene does not.
- Benzene undergoes substitution reactions in which hydrogen atoms are replaced by other atoms.



More Complex Aromatic Systems

Table 22.3 | More Complex Aromatic Systems

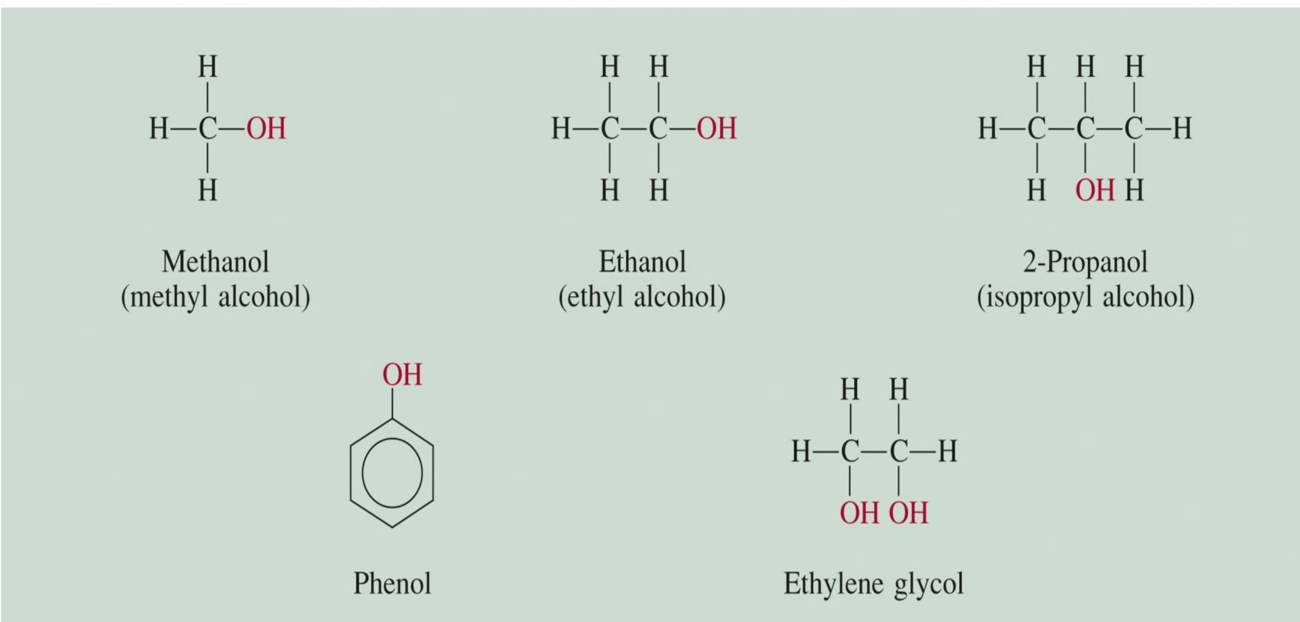
Structural Formula	Name	Use of Effect
	Naphthalene	Formerly used in mothballs
	Anthracene	Dyes
	Phenanthrene	Dyes, explosives, and synthesis of drugs
	3,4-Benzpyrene	Active carcinogen found in smoke and smog

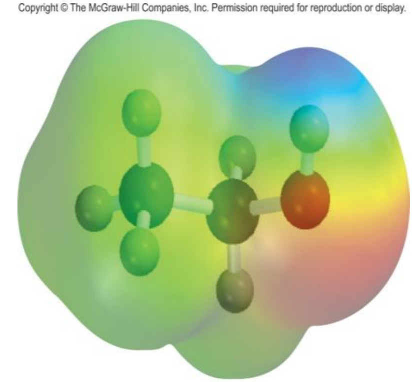
Hydrocarbon Derivatives

- Molecules that are fundamentally hydrocarbons but have additional atoms or groups of atoms called **functional groups**.

Functional Group Chemistry

- **Alcohols**: contain the hydroxyl functional group and have the general formula **R–OH.**



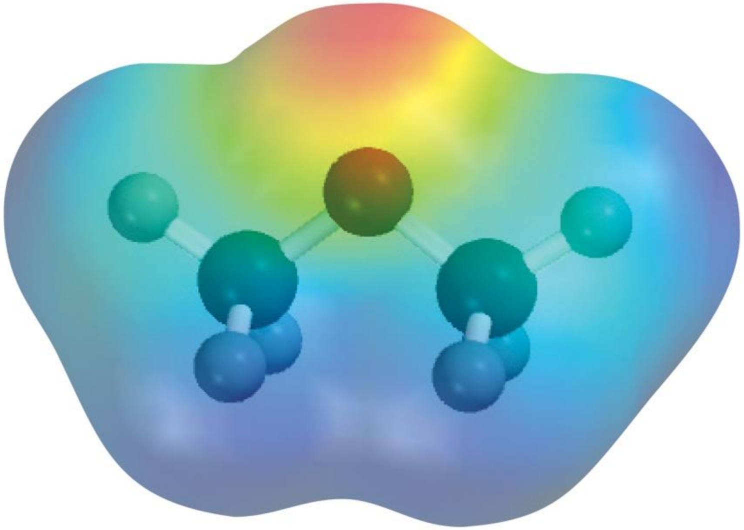


C_2H_5OH

- ***Ethers*** have the general formula R−O−R′.

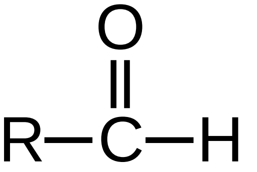
<u>Condensation Reaction</u>

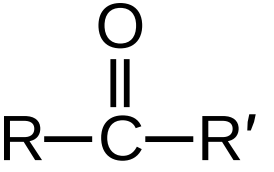
$$CH_3OH + HOCH_3 \xrightarrow[\text{catalyst}]{H_2SO_4} CH_3OCH_3 + H_2O$$

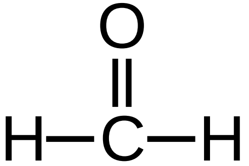


CH_3OCH_3

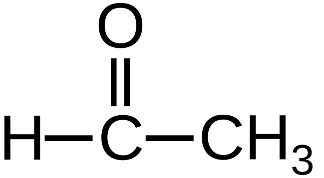
Aldehydes and Ketones contain the carbonyl ($>C=O$) functional group.

- **aldehydes** have the general formula 

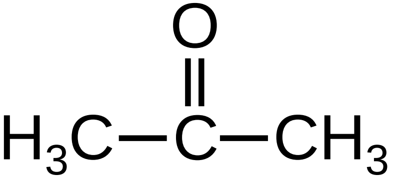
- **ketones** have the general formula 



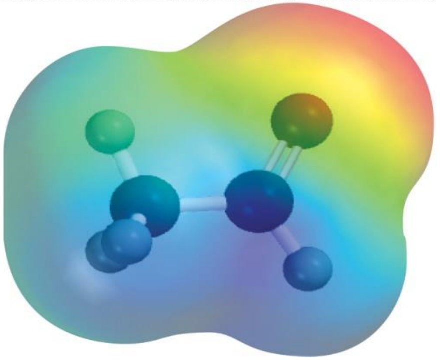
formaldehyde



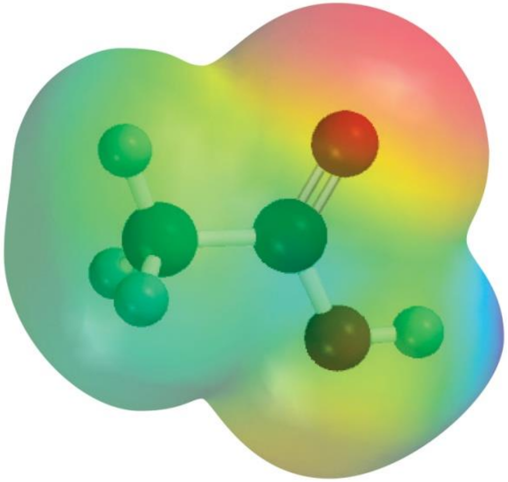
acetaldehyde



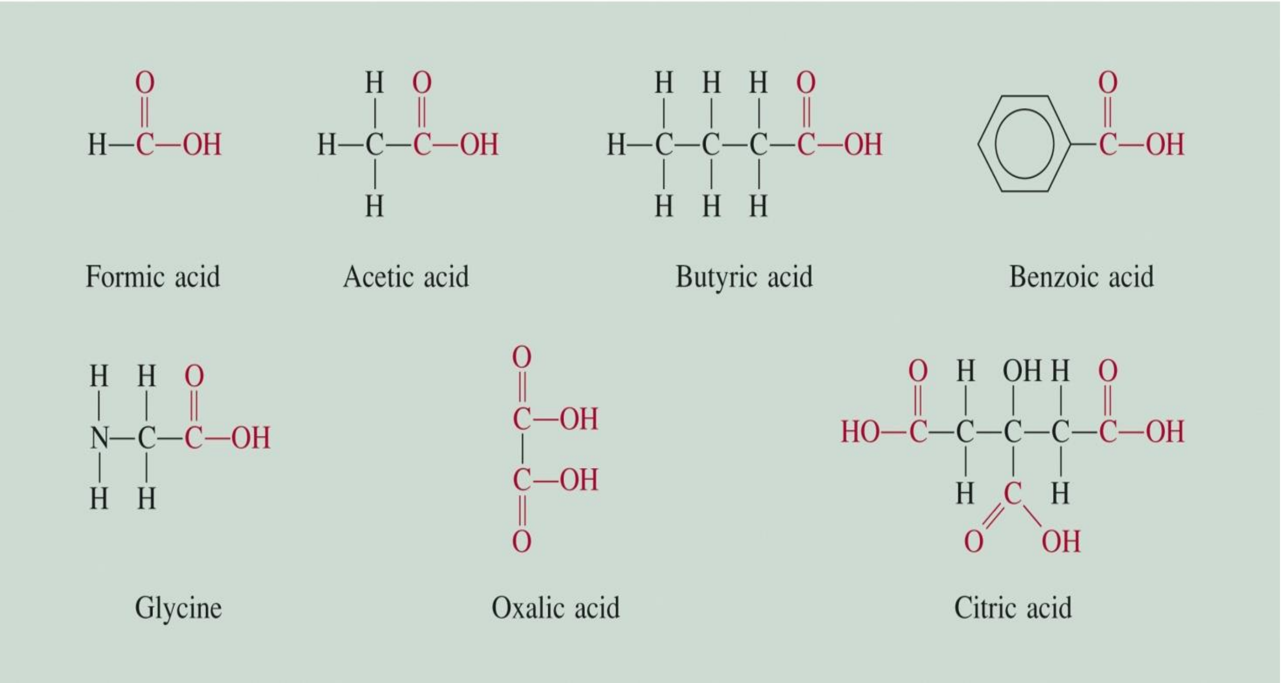
acetone



- ***Carboxylic acids*** contain the carboxyl (−COOH) functional group.



CH_3COOH



- ***Esters*** have the general formula R′COOR, where R is a hydrocarbon group.

$$\underset{\text{acetic acid}}{CH_3COOH} + \underset{\text{ethanol}}{HOCH_2CH_3} \longrightarrow \underset{\text{ethyl acetate}}{CH_3-\overset{\overset{\large O}{\|}}{C}-O-CH_2CH_3} + H_2O$$

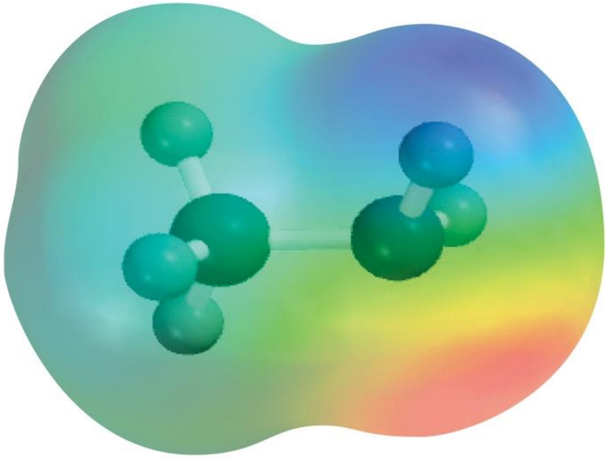
Characteristic odors and flavors

- **Amines** are organic bases with the general formula R_3N.

$$CH_3NH_2 + H_2O \longrightarrow RNH_3^+ + OH^-$$

Neutralization

$$CH_3CH_2NH_2 + HCl \longrightarrow CH_3CH_2NH_3^+Cl^-$$



CH_3NH_2

The Common Functional Groups

Table 22.4 | The Common Functional Groups

Class	Functional Group	General Formula*	Example
Halohydrocarbons	—X (F, Cl, Br, I)	R—X	CH_3I Iodomethane (methyl iodide)
Alcohols	—OH	R—OH	CH_3OH Methanol (methyl alcohol)
Ethers	—O—	R—O—R′	CH_3OCH_3 Dimethyl ether
Aldehydes	$-\overset{\displaystyle O}{\overset{\|}{C}}-H$	$R-\overset{\displaystyle O}{\overset{\|}{C}}-H$	CH_2O Methanal (formaldehyde)
Ketones	$-\overset{\displaystyle O}{\overset{\|}{C}}-$	$R-\overset{\displaystyle O}{\overset{\|}{C}}-R'$	CH_3COCH_3 Propanone (dimethyl ketone or acetone)
Carboxylic acids	$-\overset{\displaystyle O}{\overset{\|}{C}}-OH$	$R-\overset{\displaystyle O}{\overset{\|}{C}}-OH$	CH_3COOH Ethanoic acid (acetic acid)
Esters	$-\overset{\displaystyle O}{\overset{\|}{C}}-O-$	$R-\overset{\displaystyle O}{\overset{\|}{C}}-O-R'$	$CH_3COOCH_2CH_3$ Ethyl acetate
Amines	$-NH_2$	$R-NH_2$	CH_3NH_2 Aminomethane (methylamine)

*R and R′ represent hydrocarbon fragments.


Thank You